

Part I

From massless to massive neutrinos

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Chapter 1

Introduction

1.1 History

The phenomenon of radioactivity was discovered in 1892 by Becquerel. Bohr was the first to realize that beta decay is a process in which the electron is ejected from the nucleus. In 1914, Chadwick made the crucial discovery that the primary beta spectrum is continuous. Until well into the twenties, this result was believed to have other explanations. In 1929, in a letter to a conference [1, 2], Pauli wrote, “Dear Radioactive Ladies and Gentlemen, . . . as a desperate remedy to save the principle of energy conservation in beta decay, . . . I propose the idea of a neutral particle of spin half.” So, the neutrino was born into the world of theoretical physics. Chadwick discovered the neutron in 1932, which was realized to be the particle in the nucleus which emitted the electron and the neutrino in the process of beta decay. Fermi wrote the four-Fermi Hamiltonian for beta decay using the neutrino, electron, neutron and the proton. A new field of theoretical physics came into existence — the field of Weak Interactions. In 1956, Reines and Cowan discovered the neutrino. As more and more particles were discovered and found to participate in weak processes, weak interactions acquired legitimacy as a new force of nature and neutrinos became an integral part of this interaction. With the invention of new accelerators and sophisticated detectors and after the work of many dedicated experimentalists, we now know that neutrinos come in three varieties. Each neutrino comes with its own family of quarks and leptons. Observations involving neutrinos have played a key role in our understanding of weak interactions in the form of the so-called standard model and it is believed that further observation involving them may hold the key to our understanding

of physics beyond the standard model.

In the past few decades it has also been realized that our understanding of the Universe depends critically on our understanding of the neutrino. For instance the neutrino is the most abundant form of matter in the Universe next to radiation, and through its role in nucleosynthesis, it is responsible for the origin of heavy elements that form the basis of life. There are also speculations that neutrinos may have played a role in the formation of galaxies that provided the stage for the evolution of life. Thus the neutrino is clearly an important particle. This book is an attempt to summarize the present status of our knowledge of the neutrino and some popular speculations about its yet undiscovered properties.

Let us start with a brief overview of the various properties of the neutrino. It is known to have a spin of half in units of $\hbar$ and is believed to have no electric charge. Whether it has a mass or not is one of the key questions of present day particle physics and will constitute one of the chief obsessions of some of the latter chapters in this book. It is however known from various laboratory experiments that, if it has a mass, it is very much smaller than any of the masses of other known particles such as quarks or leptons in the same family.

As mentioned earlier, now there are known to be three kinds of neutrinos. The existence of the muon and tau neutrino was suspected from the properties of the corresponding charged partners: for instance it was found that the muon does not decay to an electron and a photon which is a priori allowed and neither does a muon decay to $e^+e^-e^-$. The simplest way to understand this is to postulate that there exists a separate neutrino corresponding to the muon. The muon neutrino was discovered in 1962 by the group of Lederman, Schwartz and Steinberger. The third charged lepton, the tau, was discovered by the group of Perl in the mid-seventies and by similar reasoning, a tau neutrino is believed to exist, although no direct detection of it has yet been made. The families of fermions can therefore be written as follows:

$$\begin{aligned} &(u, d, e, \nu_e), \\ &(c, s, \mu, \nu_\mu), \\ &(t, b, \tau, \nu_\tau). \end{aligned} \tag{1.1}$$

A prime focus of experiments now is to ascertain whether there exists

a fourth type of neutrino in nature. Measurements of the width of the Z -boson have definitely ruled out a fourth neutrino if it is lighter than 40 GeV or so. If there exists a fourth type of neutrino, by analogy it would imply the existence of a fourth type of matter, which of course will be a profound discovery.

1.2 Four-Fermi interaction

Using the neutrino hypothesis of Pauli, Fermi [3] was the first to write down the field theoretical form of an interaction involving the neutron, the proton, electron and neutrino fields that would describe weak interaction. If we denote these fermion fields by ψ_i , where $i = n, p, e$ and ν_e , the so-called *four-Fermi interaction* Hamiltonian density is written as

$$\mathcal{H}_{\text{weak}} = \frac{G_F}{\sqrt{2}} \bar{\psi}_p \gamma_\mu \psi_n \bar{\psi}_e \gamma^\mu \psi_{\nu_e} . \tag{1.2}$$

The form of the weak Hamiltonian in low energy weak decays has been confirmed over the year to basically have the same form as in Eq. (1.2) although a great deal more is known about its Lorentz structure than is given in Eq. (1.2). Also, many new four-Fermi terms describing other kinds of weak processes are now known to be part of $\mathcal{H}_{\text{weak}}$. Let us give a very brief and qualitative summary of these new features.

1.2.1 Modern form of four-Fermi interaction

The main observations that demand some modification of Fermi's original form of the four-Fermi interaction are as follows:

- The fundamental fields entering the Hamiltonian are not hadronic fields like the proton or the neutron, but rather are fields describing quarks.
- The space-time structure of the currents have not only the vector currents (V) as in Eq. (1.2), but have also axial vector currents (A). The fact that the currents are of the $V - A$ form was noted by Marshak and Sudarshan, and by Feynman and Gell-Mann [4].
- The weak interactions consist not only of *charged current processes* like the beta decay in which electric charge of leptons and quarks

change, but also of processes where the electric charge remain unchanged among the leptons and hadrons. Such processes are called *neutral current processes*.

Using the above considerations, the weak interaction Hamiltonian can be written in a compact form as follows, using a charged current J^μ and a neutral current K^μ :

$$\mathcal{H}_{\text{weak}} = \frac{4G_F}{\sqrt{2}} \left[J^\mu(x) J_\mu^\dagger(x) + \rho K^\mu(x) K_\mu(x) \right]. \quad (1.3)$$

In terms of quark and lepton fields can, the charged current, J^μ can be written as

$$J_\mu(x) = \bar{u}\gamma_\mu L V d + \bar{\nu}\gamma_\mu L \ell \quad (1.4)$$

where the chirality projection operators are defined as

$$L = \frac{1}{2}(1 - \gamma_5), \quad R = \frac{1}{2}(1 + \gamma_5). \quad (1.5)$$

The fermion fields in Eq. (1.4) are

$$u = \begin{pmatrix} u \\ c \\ t \end{pmatrix}, \quad d = \begin{pmatrix} d \\ s \\ b \end{pmatrix}, \quad \nu = \begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix}, \quad \ell = \begin{pmatrix} e \\ \mu \\ \tau \end{pmatrix}, \quad (1.6)$$

and V is a matrix which denotes the mixing effect between different generations of quarks. It is called the Cabibbo-Kobayashi-Maskawa matrix [5], or CKM matrix for short. Present information about the magnitudes of various elements of V is given below [6]:

$$V = \begin{pmatrix} 0.9741 \text{ to } 0.9756 & 0.219 \text{ to } 0.226 & 0.0025 \text{ to } 0.0048 \\ 0.219 \text{ to } 0.226 & 0.9732 \text{ to } 0.9748 & 0.038 \text{ to } 0.044 \\ 0.004 \text{ to } 0.014 & 0.037 \text{ to } 0.044 & 0.9990 \text{ to } 0.9993 \end{pmatrix}. \quad (1.7)$$

There is also a phase in the matrix which is responsible for CP violation. In Ch. 2, we will see that in the standard electroweak theory of Glashow, Weinberg and Salam, V is a unitary matrix. Apart from unphysical phases which can be absorbed in the definitions of the quark fields, this unitary matrix can be parametrized by four parameters, e.g.:

$$V = \begin{pmatrix} c_{12}c_{13} & -s_{12}c_{13} & s_{13}e^{-i\delta} \\ s_{12}c_{23} + c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & -s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & c_{12}s_{23} + s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}, \quad (1.8)$$

where, e.g., $c_{12} = \cos \theta_{12}$ and $s_{12} = \sin \theta_{12}$. The phase δ in this matrix signals CP violation.

The first term in the interaction of Eq. (1.3) gives rise to a host of new weak interaction phenomena such as:¹

$$\begin{aligned}\mu^- &\rightarrow e^- \hat{\nu}_e \nu_\mu \\ c &\rightarrow s e^+ \nu_e \\ s &\rightarrow u e^- \hat{\nu}_e.\end{aligned}\tag{1.9}$$

Whereas we have described above weak processes in terms of quarks, the actual observation involves a color-singlet hadron containing the quark. For instance, the last process in Eq. (1.9) is really inferred from $\Lambda \rightarrow p e^- \hat{\nu}_e$.

In Eq. (1.3), K_μ is the neutral current, which has the following form:

$$\begin{aligned}K_\mu(x) = & \sum_q [\epsilon_L(q) \bar{q} \gamma_\mu L q + \epsilon_R(q) \bar{q} \gamma_\mu R q] \\ & + \frac{1}{2} \sum_\nu \bar{\nu} \gamma_\mu L \nu + \frac{1}{2} \sum_\ell \bar{\ell} \gamma_\mu [g_V(\ell) - \gamma_5 g_A(\ell)] \ell.\end{aligned}\tag{1.10}$$

In Ch. 2, we will discuss the presently available information on various parameters appearing in the neutral currents. This involves the parameters in Eq. (1.10) as well as the ρ parameter defined in Eq. (1.3), which measures the relative strengths of neutral and charged current interactions.

Several characteristics of the Hamiltonian in Eq. (1.3) are worth noting. In the neutral current K_μ , there is no term which connects two different species of quarks or leptons. The quarks and leptons of a particular generation are often called *flavor*, and the possible currents that could connect two different flavors are called *flavor changing neutral currents* (FCNC). At present extensive search is going on for flavor changing hadronic as well as leptonic neutral currents. The experiments look for processes [7] such as $K_L \rightarrow \mu^+ \mu^-$, $K_L \rightarrow e^+ e^-$, $K_L \rightarrow \mu^\pm e^\mp$, $K^+ \rightarrow \pi^+ \nu \hat{\nu}$, $\mu \rightarrow e \gamma$, $\mu^\pm \rightarrow e^- e^+ e^\pm$, $\mu + \text{Nucl} \rightarrow e + \text{Nucl}$ etc. The present limits are very stringent and imply upper bounds on the strength of couplings expressed in the form of Eq. (1.3) to be $10^{-4} G_F$ for Kaon

¹As explained in the prechapter called "Notations", we will use the hat to denote antiparticles.

decays to $10^{-6}G_F$ for muon decays. These selection rules imply that in the leptonic sector, one can define new quantum numbers called *generational lepton numbers* L_e , L_μ and L_τ . The known weak interactions appear to obey the conservation of all these numbers. Note that due to the presence of the mixing matrix V , no analogous kind of selection rule can be written down for the hadronic (or quark) sector.

1.2.2 Fierz transformation

In the weak interaction Hamiltonian of Eq. (1.3), there are two kinds of terms, viz, the terms coming from charged-current interactions and those coming from neutral current interactions. However, four-Fermi interactions of these two kinds can be transformed into each other. This is achieved through the Fierz transformations [8, 9], which we now discuss. Sometimes, this helps a lot in calculations, or in understanding the main results without getting into details.

Since any fermionic field ψ is a four-component object, there can be sixteen independent fermionic bilinears of the form $\bar{\psi}F\psi$. Thus, any F can be written in terms of 16 basis bilinears, which we choose in the following way:

$$\begin{aligned}\Gamma^1 &\equiv 1, \\ \Gamma^{2,3,4,5} &\equiv \gamma^\mu, \\ \Gamma^{6 \text{ to } 11} &\equiv \sigma^{\mu\nu}, \quad (\mu < \nu) \\ \Gamma^{12,13,14,15} &\equiv i\gamma^\mu\gamma_5, \\ \Gamma^{16} &\equiv \gamma_5.\end{aligned}\tag{1.11}$$

The Γ matrices with lower indices can be defined by taking all the Lorentz indices down. Then it is easily verified that

$$\text{Tr}(\Gamma^r\Gamma_q) = 4\delta_q^r.\tag{1.12}$$

The most general four-fermion interaction that is Lorentz invariant and parity conserving can be written as

$$\sum_{r=1}^{16} C_r [\Gamma_r]_{1,2} [\Gamma^r]_{3,4},\tag{1.13}$$

where we have used the shorthand

$$[\Gamma_r]_{1,2} \equiv \bar{\psi}_1 \Gamma_r \psi_2,\tag{1.14}$$

for two fermion fields ψ_1 and ψ_2 . Notice that, in Eq. (1.13) as well as in the rest of this section, repeated indices of Γ are not assumed to be automatically summed over, unless there is an explicit summation sign.

However, in Eq. (1.13), fermions ψ_1 and ψ_3 are created, whereas ψ_2 and ψ_4 are annihilated, so there is no reason why it cannot be written in the form

$$\sum_{q=1}^{16} C'_q [\Gamma_q]_{1,4} [\Gamma^q]_{3,2}. \quad (1.15)$$

The forms given in Eqs. (1.13) and (1.15) are therefore equivalent, and the task is to find the relation between the coefficients C_r and C'_q . Since this equivalence should be valid for arbitrary fermion fields, we can extract the coefficients of $(\bar{\psi}_1)_A (\psi_2)_B (\bar{\psi}_3)_C (\psi_4)_D$ from each expression, where A, B, C, D are Dirac indices. This enables us to write

$$\sum_{r=1}^{16} C_r (\Gamma_r)_{AB} (\Gamma^r)_{CD} = - \sum_{q=1}^{16} C'_q (\Gamma_q)_{AD} (\Gamma^q)_{CB}, \quad (1.16)$$

where the minus sign comes because the fermion fields on the right side involve one interchange compared to those on the left side. Multiplying both sides now by $(\Gamma_{r'})_{DC} (\Gamma^{r''})_{BA}$ and summing over all the Dirac indices, we obtain

$$C_r = \sum_q \Lambda_{r,q} C'_q, \quad (1.17)$$

where

$$\Lambda_{r,q} = -\frac{1}{16} \text{Tr} (\Gamma_q \Gamma_r \Gamma^q \Gamma^r). \quad (1.18)$$

It is now straightforward to calculate the matrix Λ . However, a more useful form is obtained by noting that Eq. (1.16) can also be written as

$$\sum_i \tilde{C}_i (\mathcal{O}_i)_{AB} (\mathcal{O}^i)_{CD} = - \sum_j \tilde{C}'_j (\mathcal{O}_j)_{AD} (\mathcal{O}^j)_{CB}, \quad (1.19)$$

where now the index i runs over the five different rows of Eq. (1.11), i.e., corresponding to the scalar (S), vector (V), tensor (T), axial vector (A) and pseudo-scalar (P) interactions. In this case, one can write

$$\tilde{C}_i = \sum_j \tilde{\Lambda}_{ij} \tilde{C}'_j, \quad (1.20)$$

instead of Eq. (1.17). Obviously, the elements of the matrix $\tilde{\Lambda}$ are simply related to those of the matrix Λ . For example,

$$\tilde{\Lambda}_{SS} = \Lambda_{1,1} = -\frac{1}{16} \text{Tr } 1 = -\frac{1}{4}. \quad (1.21)$$

Now consider the element $\tilde{\Lambda}_{VS}$. This means that we want to see the scalar contribution, with the ordering of Eq. (1.13), coming from all four components of the vector interaction with the ordering of Eq. (1.15). Thus,

$$\tilde{\Lambda}_{VS} = -\frac{1}{16} \text{Tr } (1 \gamma_\mu 1 \gamma^\mu) = -1. \quad (1.22)$$

Similarly, one can proceed to see that the matrix $\tilde{\Lambda}$ is given by

$$\tilde{\Lambda} = -\frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 4 & -2 & 0 & 2 & -4 \\ 6 & 0 & -2 & 0 & 6 \\ 4 & 2 & 0 & -2 & -4 \\ 1 & -1 & 1 & -1 & 1 \end{pmatrix}. \quad (1.23)$$

This is the Fierz transformation matrix.

So far, we talked about parity conserving interactions only. In a discussion of weak interactions, parity violation will of course be present. This is possible if the terms shown in Eq. (1.13) appear in conjunction with terms of the form

$$\sum_{r=1}^{16} D_r [\Gamma_r]_{1,2} [\Gamma^r \gamma_5]_{3,4}. \quad (1.24)$$

One can similarly use a different ordering of the fermion fields in this case and find the relation between the coefficients in the two cases. However, this can also be read off from Eq. (1.23) directly once we note that Eq. (1.24) can be written in the form of Eq. (1.13) with a different field ψ'_4 which is related to ψ_4 by the relation $\psi'_4 = \gamma_5 \psi_4$.

□ **Exercise 1.1** Show that the $V - A$ and $V + A$ interactions are both invariant under the Fierz transformations, i.e.,

$$\begin{aligned} [\gamma_\mu(1 - \gamma_5)]_{1,2} [\gamma^\mu(1 - \gamma_5)]_{3,4} &= [\gamma_\mu(1 - \gamma_5)]_{1,4} [\gamma^\mu(1 - \gamma_5)]_{3,2} \\ [\gamma_\mu(1 + \gamma_5)]_{1,2} [\gamma^\mu(1 + \gamma_5)]_{3,4} &= [\gamma_\mu(1 + \gamma_5)]_{1,4} [\gamma^\mu(1 + \gamma_5)]_{3,2} \end{aligned} \quad (1.25)$$

1.2.3 Problems with the four-Fermi interaction

While the four-Fermi interaction $\mathcal{H}_{\text{weak}}$ given in Eq. (1.3) provides an excellent description of observed low energy weak interaction phenomena, it is fraught with difficulties when considered as a field theory. There are many ways to see it. For instance, if one calculates higher order weak interaction corrections to any lowest order weak process, the contributions are divergent and the divergence at the L -th loop goes as Λ^{2L} , Λ being the cutoff for the theory. Therefore, in a strict sense, the lowest order calculations are not reliable.

Even at the lowest order, the total cross section for neutrino-electron scattering, for example, comes out to be proportional to $G_F^2 s$, where s is the Mandelstam variable defined by $s = (p_{\nu_e} + p_e)^2$ in terms of the momenta of the incoming neutrino and electron. With increasing energy this cross section grows without limit. Since $\nu_e e$ scattering occurs in the s -wave, the amplitude for this process should obey the s -wave unitarity bound, viz.,

$$\sigma_{\text{tot}}^s \leq \frac{16\pi}{s} . \quad (1.26)$$

This leads to a contradiction, implying that the four-Fermi description of weak interaction must breakdown above a certain energy, which in old days was called the weak-interaction cutoff, Λ_{weak} . Examination of various weak processes produced values of Λ_{weak} between 300 GeV down to 4 GeV [10]. This presented a crisis for four-Fermi description of weak interaction.

The basic problem was connected with the fact that the four-Fermi Lagrangian is not renormalizable. So search for a renormalizable weak interaction Lagrangian began in the late sixties. It culminated in the discovery of gauge theories, whose basic ingredients we describe below.

1.3 Symmetries and forces

Symmetries have played a fundamental role in our understanding of particle physics. Starting with the Poincaré symmetry group of space-time transformations to the isospin invariance group of $SU(2)$ in nuclear physics and the $SU(3)$ symmetry group of Gell-Mann and Neéman for

hadron physics, our understanding of physics has always deepened with the identification of an invariance group in the system.

1.3.1 Global symmetries

There are two distinct kinds of symmetries for physical systems: *global symmetries* where the same symmetry transformation is applied to a field at all space-time points, and *local symmetries* where the symmetry transformations at different space-time points are unrelated. The isospin symmetry of nuclear physics introduced by Heisenberg, which provided a lot of insight into nuclear energy levels and forces, is an example of global symmetry. With the discovery of many baryon and meson states, the global $SU(3)$ symmetry was proposed as a generalization of isospin to classify the various hadronic states and mass spectra.

Under the isospin symmetry the up (u) and the down (d) quarks form a doublet whereas all other quarks are invariant. The proton and the neutron are made out of u and d quarks alone: $p \equiv uud$ and $n \equiv udd$. The mesons $\pi^\pm$ and π^0 are made out of u and d in combination with their antiparticles.

Under $SU(3)$, the three quarks u , d and s form a triplet. Again, baryons made out of these three quarks in various combinations, and mesons made out of various combinations of u , d , s and $\bar{u}$, $\bar{d}$, $\bar{s}$, have been identified, thus confirming the validity of the $SU(3)$ symmetry in particle physics.

The existence of exact global symmetries always implies relations between masses and coupling constants among particles, provided the symmetry is realized in the so-called Wigner-Weyl mode. This just means that the ground state of the system is invariant under the symmetry transformations. If the isospin symmetry were realized in this way, this would imply the proton and the neutron have the same mass. Since that is not true ($m_n - m_p = 1.3 \text{ MeV}$), the isospin symmetry must therefore be approximate. Similarly, the $SU(3)$ symmetry is also approximate. Study of approximate $SU(3)$ symmetry led, in fact, to the concept of quarks as constituents of hadrons, which was a major leap forward in our understanding of particle interactions.

1.3.2 Local symmetries

Let us now turn to local symmetries. Their importance was realized long ago in connection with electromagnetic theory, and it is the cornerstone of modern particle physics.

The demands of a local symmetry are much more stringent and can only be met provided some new spin-1 fields – to be called the *gauge fields*, are introduced into the theory. These gauge fields, to be denoted by A_μ , have interaction with all fields that transform non-trivially under the local symmetry and their exchange gives rise to the forces. Thus, requiring invariance under a local symmetry always leads to forces — a result which takes the mystery out of the forces and puts them back into the nature of invariances.

To see in detail how it works, consider a local symmetry group G and let some matter fields transform as a certain representation under this group. If we denote these matter fields by a column vector $\psi(x)$, then under a gauge transformation

$$\psi(x) \rightarrow U(x)\psi(x), \quad (1.27)$$

where $U(x)$ is a matrix representation of an element of the group G . The ordinary kinetic energy term for $\psi(x)$ will not be invariant under this transformation. To write down a term that is invariant under Eq. (1.27), we replace the ordinary space-time derivative $\partial_\mu\psi$ in the Lagrangian by the covariant derivative

$$D_\mu\psi = \partial_\mu\psi - igA_\mu\psi, \quad (1.28)$$

and write, for the fermions

$$\mathcal{L}_{\text{kin}} = \bar{\psi}i\gamma^\mu D_\mu\psi, \quad (1.29)$$

which is gauge invariant. Here, $A_\mu(x)$ contains spin-1 fields and is matrix-valued in the group space, i.e.,

$$A_\mu(x) = \sum_a T^a A_\mu^a(x), \quad (1.30)$$

where T^a are the generators of the group G in the appropriate representation. From the requirement that Eq. (1.29) is invariant under the

gauge transformation of Eq. (1.27), we obtain that $A_\mu(x)$ must transform like

$$A_\mu(x) \rightarrow U A_\mu(x) U^{-1} + \frac{1}{ig} (\partial_\mu U) U^{-1}. \quad (1.31)$$

□ **Exercise 1.2** Show that for scalar fields, the Lagrangian given by

$$\mathcal{L}_{\text{kin}} = (D_\mu \phi)^\dagger (D^\mu \phi) \quad (1.32)$$

is invariant under the gauge transformation, where the covariant derivative is defined as in Eq. (1.28), with ϕ replacing ψ .

On substituting Eq. (1.28) to Eqs. (1.29) and (1.32), we see that the principle of gauge invariance has generated interaction terms between matter fields ψ , ϕ and the gauge fields A_μ . Exchange of gauge fields then generates forces, as mentioned earlier.

There is a good side and a bad side to this. The good side is that the coupling g is universal to all matter fields as long as G is a non-Abelian group. Therefore, this has the potential to explain universality of couplings to different particles observed in weak interactions (see Eq. (1.3)). The problem however is that gauge invariance under Eq. (1.31) demands that the terms involving the gauge fields only are of the following form:

$$\mathcal{L}_{(\text{pure gauge})} = -\frac{1}{4} \text{tr } F_{\mu\nu} F^{\mu\nu} \quad (1.33)$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu]. \quad (1.34)$$

Notice that there is no mass term in Eq. (1.33), because it is forbidden by gauge invariance. So the gauge field must be massless. Therefore, its exchange can only lead to long range forces. But we know that weak and strong interactions are short range. Therefore, if we want to generate weak interaction via a gauge principle, we must somehow give mass to the gauge boson. This was achieved with the discovery of the Higgs mechanism described below. The short range nature of strong interactions is understood on the basis of a different principle called confinement.

1.3.3 Spontaneous breaking of symmetries

The time-evolution of a physical system depends on two things: (i) the Hamiltonian (or the Lagrangian) and (ii) the initial condition. In quantum field theories with symmetries, two analogous conditions determine the way in which a symmetry manifests itself. First, the Lagrangian must be invariant under the symmetry transformations; secondly, the vacuum state must be invariant under the same. When this happens, the symmetry is said to be realized in the Wigner-Weyl mode. In this case, natural expectations such as equality of masses of particles in a given irreducible representation, equality of S -matrix elements among the appropriate physical states — are satisfied.

It was discovered around 1960, through the work of Nambu and Goldstone, that a Lagrangian may be invariant under a symmetry transformation but the vacuum state may not be. If the symmetry is global, the spectrum of the theory contains a massless particle known as the Nambu-Goldstone boson. It was realized through the works of Higgs, Kibble, Guralnik, Hagen, Brout and Englert [11] that if a gauge symmetry is spontaneously broken, no such massless particle results. Rather, the gauge boson corresponding to broken generators pick up mass. Since this will be the principal technique in the construction of the standard model as well as its extensions, we outline the practical way of implementing this in gauge theories.

Let G be the local symmetry group and T^a be the generators of the group. Consider a set of scalar fields ϕ_a belonging to a representation of G . To discuss spontaneous breaking of G , consider the potential of the scalar fields, $V(\phi)$, which is invariant under G . This invariance implies that

$$\sum_{i,j} \frac{\partial V}{\partial \phi_i} (T^a)_{ij} \phi_j = 0. \quad (1.35)$$

Condition for the minimum is given by

$$\left. \frac{\partial V}{\partial \phi_i} \right|_{\phi=\langle\phi\rangle} = 0. \quad (1.36)$$

Differentiating Eq. (1.35) with respect to ϕ_k and setting $\phi_k = \langle\phi_k\rangle$, we

get

$$\sum_{i,j} \frac{\partial^2 V}{\partial \phi_k \partial \phi_i} \bigg|_{\phi=\langle \phi \rangle} (T^a)_{ij} \langle \phi_j \rangle = 0, \quad (1.37)$$

by using Eq. (1.36). Now, the mass matrix of the scalar fields is given by

$$\mathcal{M}_{ki}^2 = \frac{\partial^2 V}{\partial \phi_k \partial \phi_i} \bigg|_{\phi=\langle \phi \rangle}. \quad (1.38)$$

Thus, if $\sum_j (T^a)_{ij} \langle \phi_j \rangle$ is not a null vector, i.e., if the generator T^a does not annihilate the vacuum, then Eq. (1.37) shows that $\sum_j (T^a)_{ij} \langle \phi_j \rangle$ is a zero mass eigenvector of the mass matrix. This gives rise to a Nambu-Goldstone mode.

The G -invariant kinetic energy for ϕ is given by Eq. (1.32). On substituting $\phi = \langle \phi \rangle$ (and the fact that $\partial_\mu \langle \phi \rangle = 0$) in that equation, we obtain

$$\begin{aligned} \mathcal{L}_{\text{kin}}(\phi) \big|_{\phi=\langle \phi \rangle} &= \sum_{a,b} (T^a \langle \phi \rangle)^\dagger (T^b \langle \phi \rangle) A_\mu^a A^{b\mu} \\ &\equiv \sum_{a,b} M_{ab}^2 A_\mu^a A^{b\mu}. \end{aligned} \quad (1.39)$$

This is precisely the term that makes the gauge bosons massive after symmetry breaking. But note that if for a particular generator $T^a \langle \phi \rangle = 0$, the corresponding gauge field has no mass. Thus, only the gauge fields corresponding to spontaneously broken generators, i.e., the ones for which $T^a \langle \phi \rangle \neq 0$, pick up mass. This is the *Higgs mechanism*.

It may be useful for the reader to see a simple Abelian example of Higgs mechanism. Consider a $U(1)$ local symmetry with the scalar field ϕ transforming non-trivially under the gauge group as follows:

$$\phi \rightarrow e^{ie\lambda(x)} \phi. \quad (1.40)$$

If we demand that the spin-1 gauge field transforms under the $U(1)$ transformation as

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \lambda(x), \quad (1.41)$$

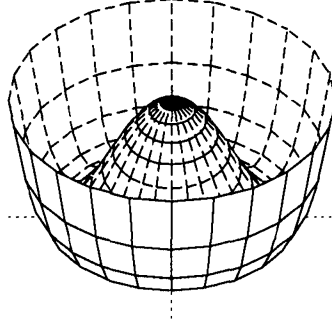


Figure 1.1: The potential which gives rise to broken-symmetry solutions.

then the gauge invariant Lagrangian for the system can be written as

$$\mathcal{L} = \mathcal{L}_0 - V(\phi), \quad (1.42)$$

where

$$\mathcal{L}_0 = (D_\mu \phi)^* D^\mu \phi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \quad (1.43)$$

with the covariant derivative defined by

$$D_\mu \phi = \partial_\mu \phi - ie A_\mu \phi. \quad (1.44)$$

The potential $V(\phi)$ must be gauge invariant, i.e., it is a function of $\phi^* \phi$. The most general renormalizable potential then consists of two terms:

$$V(\phi) = -\mu^2 \phi^* \phi + \lambda (\phi^* \phi)^2. \quad (1.45)$$

Now, λ must be positive in order that the potential has a lower bound. But nothing restricts the sign of μ^2 . If it happens to be positive as well, then the shape of the potential looks like Fig. 1.1. Minimization of this potential gives

$$\langle \phi \rangle = \sqrt{\frac{\mu^2}{2\lambda}}. \quad (1.46)$$

Of course, the first derivative vanishes at $\phi = 0$ as well, but Fig. 1.1 clearly shows that the minimum corresponds to the nonzero value. Let us now redefine ϕ as

$$\phi(x) = \varphi(x) + \frac{v}{\sqrt{2}} \quad (1.47)$$

where $v = \sqrt{\mu^2/\lambda}$ defines the minimum and φ is the fluctuation around this minimum. This way, we get $\langle \varphi \rangle = 0$, so that φ can be expanded in terms of creation and annihilation operators as is usual in quantum field theory. We then see that the kinetic energy term for ϕ leads to a term $\frac{1}{2}e^2v^2A_\mu A^\mu$ in the Lagrangian, which is a mass term for A^μ . Thus, the process of spontaneous symmetry breaking gives mass to a gauge boson.

□ **Exercise 1.3** Instead of Eq. (1.47), use

$$\phi(x) = \frac{\eta(x) + v}{\sqrt{2}} e^{i\zeta(x)}, \quad (1.48)$$

where $\eta(x)$ and $\zeta(x)$ are real scalar fields which replace the complex field $\phi(x)$. Show that, in this representation, the field $\zeta(x)$ remains massless after the symmetry breaking. Moreover, by performing a gauge transformation $\phi(x) \rightarrow \phi(x)e^{-i\zeta(x)}$, show that the field $\zeta(x)$ vanishes from the Lagrangian altogether.

At this point, we have presented the three basic ingredients needed for gauge model building, which we now summarize [12]:

- Choice of the gauge group G .
- Assignment of the fermions to suitable representation of the gauge group.
- Choice of Higgs bosons and their vacuum expectation values to break the gauge symmetry down to $U(1)_Q$. This choice must also be good enough to reproduce the quark and lepton masses in a phenomenologically acceptable way.

□ **Exercise 1.4** Consider an $SU(2)$ gauge theory with a scalar multiplet transforming like a doublet. In the vacuum, one component of the doublet scalar field acquires a non-zero value. Verify that all the three gauge bosons of $SU(2)$ acquire equal masses.

□ **Exercise 1.5** Repeat the same problem with the vev in one component of a triplet scalar multiplet. Do all the gauge bosons acquire mass in this case? If not, why not?

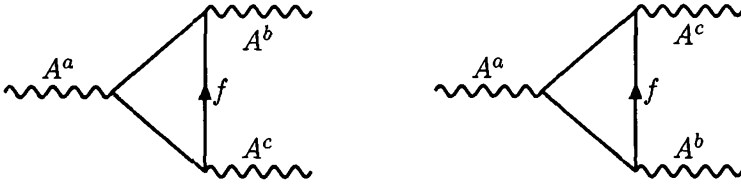


Figure 1.2: Triangle graphs giving rise to anomalies.

1.4 Renormalizability and anomalies

One might wonder about the necessity of going through the elaborate machinery of Higgs mechanism to give mass to gauge fields instead of simply adding to a gauge invariant theory an explicit mass term for the gauge bosons that need to pick up mass to fit observations. In fact, historically, these kinds of attempts were made in the late sixties and early seventies. It was however realized that these theories are plagued with divergences, which proliferate as one goes to higher loops. Such theories therefore have no predictive power. It was shown by 't Hooft [13] in 1971 that if gauge boson mass is generated by the Higgs mechanism, then the theory is renormalizable. It is therefore very important to remember in building gauge models that, the gauge symmetries that need to be broken (so that the corresponding gauge bosons pick up mass), must be broken by the Higgs mechanism. Furthermore, to keep the theory renormalizable, all terms in the Lagrangian must have mass dimensions less than or equal to four.

Another important criterion for renormalizability is the absence of triangle anomalies. To understand this, we recall that in proving renormalizability of a theory, gauge invariance is essential. Gauge invariance means that the currents corresponding to gauge symmetries must be conserved to all orders in perturbation theory. It was however pointed out by Adler, Bell and Jackiw [14] that if a theory involves chiral interactions (i.e., interactions involving γ_5 currents) of fermions, triangular one-loop graphs (see Fig. 1.2) in general destroy the current conservation which was true at the tree level. This is called the *axial anomaly*. If such anomalies are not canceled, then the theory loses its renormalizability. One must, therefore, impose the constraint of anomaly cancellation on

gauge theories. This imposes restrictions on the nature of fermion spectrum. In general, if in the space of all fermions, T^a denotes the coupling matrix of fermions to the current J^a_μ , then the condition for anomaly cancellation is

$$\text{Tr} \left(T^a \{ T^b, T^c \} \right)_{\text{L}} - \text{Tr} \left(T^a \{ T^b, T^c \} \right)_{\text{R}} = 0 \, , \tag{1.49}$$

where the subscripts L and R denote left and right chirality states of fermions. This constraint plays an important role in understanding fermion spectrum, charge quantization etc and helps reduce the arbitrariness of gauge theories.

Chapter 2

The standard model and the neutrino

From the early days of particle physics, the universality of Fermi coupling led many people to suspect that there may exist an underlying symmetry in weak interaction which is not manifest in the mass spectrum of weakly interacting particles. From the nature of the charged current connecting two different particles in beta decay and other weak decays observed in the late fifties and early sixties, it was suspected that the underlying symmetry must have an $SU(2)$ part to it. In 1961, Glashow [1] proposed $SU(2) \times U(1)$ as a possible local symmetry of weak interactions. In 1964, Salam and Ward [2] used this symmetry group to construct a theory of both electrons and muons. Then, Weinberg in 1967 [3] and Salam in 1968 [4] proposed the spontaneously broken $SU(2)_L \times U(1)_Y$ theory of leptons in its modern version. The correct extension of this model to include quarks was done in the seventies using an earlier suggestion of Glashow, Iliopoulos and Maiani [5]. A crucial question regarding the model was its renormalizability, and this question remained until 1971, when 't Hooft [6] proved that all spontaneously broken gauge theories that include only interactions of mass dimension less than or equal to four are renormalizable. This not only solved the problems associated with the four-Fermi theory of weak interaction concerning its high energy behavior as well as reliability of the tree-level calculations but it also opened up the possibility of calculating radiative corrections to weak amplitudes and checking against experiments. Soon afterwards, in 1973, Gross, Wilczek and Politzer [7] showed that the unbroken $SU(3)_c$ group of strong interactions is asymptotically free, i.e., that strong interactions become weaker at higher energies, thus explaining Bjorken scaling observed in deep inelastic lepton-nucleon scattering. Thus was born the standard model of electro-weak and strong interactions based on the

gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$. Since neutrinos have no strong interactions, we will not concern ourselves with the $SU(3)_c$ part of the group but instead will discuss the $SU(2)_L \times U(1)_Y$ model.

2.1 Gauge interactions in the standard model

Let us first discuss the spontaneous breaking of the $SU(2)_L \times U(1)_Y$ symmetry. The three gauge bosons of $SU(2)_L$ are denoted by $W_\mu^\pm, W_\mu^0$, whereas the $U(1)_Y$ gauge boson is denoted by B_μ . To use the Higgs mechanism to break this group down to $U(1)_Q$, we use a doublet Higgs representation ϕ :

$$\phi = \begin{pmatrix} \phi_+ \\ \phi_0 \end{pmatrix}, \quad (2.1)$$

with the Higgs potential given by

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2. \quad (2.2)$$

For $\mu^2 > 0$, the minimum of $V(\phi)$ occurs at

$$\langle \phi_0 \rangle = \sqrt{\frac{\mu^2}{2\lambda}} \equiv \frac{v}{\sqrt{2}}. \quad (2.3)$$

This gives mass to the charged gauge bosons $W^\pm$:

$$M_W = \frac{1}{2} g v. \quad (2.4)$$

In the neutral gauge boson sector, only one particular combination of W^0 and B , acquires mass:

$$M_Z = \frac{M_W}{\cos \theta_W}. \quad (2.5)$$

This combination is given by

$$Z = \cos \theta_W W^0 - \sin \theta_W B, \quad (2.6)$$

where

$$\tan \theta_W = g'/g, \quad (2.7)$$

where this angle θ_W is known as the Weinberg angle. There is one gauge boson, the combination orthogonal to Z , which remains massless corresponding to an unbroken $U(1)$ symmetry. This will be identified with the electromagnetic potential of the photon. The associated gauge generator, the electric charge, is given by

$$Q = I_{3L} + \frac{Y}{2} . \quad (2.8)$$

In terms of the gauge coupling constants, the electric charge of the positron is given by

$$e = g \sin \theta_W . \quad (2.9)$$

In order to discuss the weak interaction of fermions, we assign quarks and leptons to the following representations of $SU(2)_L \times U(1)_Y$:

$$\begin{aligned} q_L &= \begin{pmatrix} u_L \\ d_L \end{pmatrix} & : & \quad (2, \frac{1}{3}) \\ u_R & & : & \quad (1, \frac{4}{3}) \\ d_R & & : & \quad (1, -\frac{2}{3}) \\ \psi_L &= \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix} & : & \quad (2, -1) \\ e_R & & : & \quad (1, -2) . \end{aligned} \quad (2.10)$$

Similar assignments are repeated for all three families of quarks and leptons with (u, d, e, ν_e) replaced by (c, s, μ, ν_μ) and (t, b, τ, ν_τ) respectively. Eq. (2.10) implies the following charged $W^\pm$ and neutral Z interaction with the quarks and leptons:

$$\mathcal{L}_{\text{weak}} = \frac{g}{\sqrt{2}} \left(J^\mu W_\mu^+ + J^{\mu\dagger} W_\mu^- \right) + \frac{g}{\cos \theta_W} K^\mu Z_\mu , \quad (2.11)$$

where

$$J^\mu = (\bar{u}^0 \quad \bar{c}^0 \quad \bar{t}^0) \gamma^\mu \mathbf{L} \begin{pmatrix} d^0 \\ s^0 \\ b^0 \end{pmatrix} + (\bar{\nu}_e \quad \bar{\nu}_\mu \quad \bar{\nu}_\tau) \gamma^\mu \mathbf{L} \begin{pmatrix} e \\ \mu \\ \tau \end{pmatrix} \quad (2.12)$$

The subscript '0' on quarks implies that they are not mass eigenstates. The mass eigenstates are determined only after quarks acquire masses

in the process of spontaneous symmetry breaking. The neutral current K_μ is given by

$$K^\mu = \sum_{f^0} \bar{f}^0 \gamma^\mu \left[I_{3L} \mathbf{L} - \sin^2 \theta_W Q \right] f^0, \quad (2.13)$$

where $f^0 = \nu_e, e, u^0$ or d^0 or the corresponding objects in the higher generations, Q is the charge of f^0 and I_{3L} is the value of the neutral generator of $SU(2)_L$ for the left-chiral projection of the fermion field.

It is now clear that exchange of $W^\pm$ leads to charged current weak processes such as beta decay, muon decay etc. whereas exchange of Z leads to new type of interactions called neutral current weak interactions. Thus, the advent of gauge theories endowed the neutrino with new dynamical properties which were not known prior to the 1970's. The discovery of neutrino neutral currents was a major triumph of gauge theories. An important property of the neutral currents is that K_μ is flavor diagonal prior to mass generation. We will see that in the standard model, this property is preserved by symmetry breaking. The details of this will be discussed in the next section.

In order to complete the discussion of charged current weak interactions, let us discuss the origin of quark and lepton masses in the standard model. The gauge invariance prevents adding bare masses for them in the Lagrangian. They arise from the following Yukawa interactions allowed by gauge symmetry:

$$-\mathcal{L}_Y = \sum_{a,b} \left[h_{ab}^{(u)} \bar{q}_{aL} \hat{\phi} u_{bR} + h_{ab}^{(d)} \bar{q}_{aL} \phi d_{bR} + h_{ab}^{(\ell)} \bar{\psi}_{aL} \phi \ell_{bR} \right] + \text{h.c.} \quad (2.14)$$

Here, a, b stand for generation indices and

$$\hat{\phi} = i\tau_2 \phi^*. \quad (2.15)$$

□ **Exercise 2.1** Under an $SU(2)$ transformation, a doublet transforms as

$$\phi \rightarrow \exp(i \frac{\boldsymbol{\tau}}{2} \cdot \boldsymbol{\theta}) \phi,$$

where $\boldsymbol{\tau}$ denote the Pauli matrices. Show that the components of ϕ^* do not transform like a doublet, but those of $i\tau_2 \phi^*$ do.

On substituting the non-zero vacuum expectation values for ϕ_0 given in Eq. (2.3), the following mass terms for up and down quarks as well as

the charged leptons are generated:

$$-\mathcal{L}_{\text{mass}} = \sum_{a,b} \left[\bar{\mathbf{u}}_{aL} M_{ab}^{(u)} \mathbf{u}_{bR} + \bar{\mathbf{d}}_{aL} M_{ab}^{(d)} \mathbf{d}_{bR} + \bar{\ell}_{aL} M_{ab}^{(\ell)} \ell_{bR} \right] + \text{h.c.} \quad (2.16)$$

where

$$M_{ab}^{(f)} = h_{ab}^{(f)} v / \sqrt{2} \quad \text{with } f = u, d \text{ or } \ell. \quad (2.17)$$

By an appropriate choice of the quark and lepton basis, the coupling matrices $h^{(u)}$ and $h^{(\ell)}$ can be chosen diagonal so that we have $\mathbf{u}_a^{(0)} = \mathbf{u}_a$ and $\ell_a^{(0)} = \ell_a$. The $M^{(d)}$ is however, a complex non-diagonal matrix in this basis and can be diagonalized by the following biunitary transformation:

$$V_L M^{(d)} V_R^\dagger = D^{(d)}. \quad (2.18)$$

The down-type quark mass eigenstates are then related to the $\mathbf{d}_a^{(0)}$ states as

$$(\mathbf{d}_{L,R})_a = \sum_b (V_{L,R})_{ab} (\mathbf{d}_{L,R}^{(0)})_b. \quad (2.19)$$

In terms of the mass eigenstates, i.e., the physical states, the weak charged current can be written as

$$J^\mu = (\bar{u} \quad \bar{c} \quad \bar{t}) \gamma^\mu \mathbf{L} V_L^\dagger \begin{pmatrix} d \\ s \\ b \end{pmatrix} + (\bar{\nu}_e \quad \bar{\nu}_\mu \quad \bar{\nu}_\tau) \gamma^\mu \mathbf{L} \begin{pmatrix} e \\ \mu \\ \tau \end{pmatrix}. \quad (2.20)$$

This V_L can be identified with the matrix V introduced in Eq. (1.4). Thus, a very important implication of the standard model is that the quark mixing matrix is unitary. This property of unitarity has been used in giving the values of the elements of this matrix in Eq. (1.7). It is also important to note here that the neutral current in Eq. (2.13) is not affected by this transition to physical basis and remains flavor diagonal in the mass eigenbasis.

The W and Z masses are predicted in terms of a single parameter $\sin^2 \theta_W$ at the tree level as follows:

$$M_W^0 = \left(\frac{\pi \alpha^0}{\sqrt{2} G_\mu^0 \sin^2 \theta_W} \right)^{1/2} \quad (2.21)$$

$$M_Z^0 = M_W^0 / \cos \theta_W, \quad (2.22)$$

where α is the fine structure constant, G_μ is the Fermi constant as evaluated from the muon decay, and the superscripts 0 on these and other quantities stand for the values at the tree level of calculation. Once the radiative corrections are included, new parameters such as the t -quark mass or the Higgs mass enter and we have the radiatively corrected value:

$$\begin{aligned} M_W &= \frac{A_0}{\sin \theta_W (1 - \Delta r)^{1/2}} \\ M_Z &= M_W / \cos \theta_W . \end{aligned} \quad (2.23)$$

where $A_0 = \sqrt{\pi\alpha/\sqrt{2}G_\mu} \simeq 37.2802$ GeV. The correction Δr includes the radiative corrections relating α , $\alpha(M_Z)$, G_F , M_W and M_Z and has been calculated by various authors [8]. It is known to depend quadratically on m_t and logarithmically on the Higgs mass. For $m_t = 180 \pm 7$ GeV and $M_H = 300$ GeV for instance, $\Delta r \simeq 0.0376 \pm 0.0025 \pm 0.0007$ [9]. Eq. (2.23) can then be used to obtain the value of $\sin^2 \theta_W$ from precise measurements of M_W and M_Z .

- **Exercise 2.2** Write down the tree-level amplitude for muon decay, $\mu^- \rightarrow e^- + \nu_\mu + \bar{\nu}_e$, using the charged current interaction of Eq. (2.20). In the low-energy approximation where the momentum transfer carried by the W -boson can be neglected, compare this amplitude with Eq. (1.3) to show that

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2} . \quad (2.24)$$

From this, using Eq. (2.9), deduce the tree-level expression for M_W given in Eq. (2.22).

2.2 Neutral current interactions of neutrinos

The weak neutral current interaction for the first generation of quarks and leptons following from Eq. (2.13) is:

$$\mathcal{L} = \frac{g}{\cos \theta_W} K_\mu(x) Z^\mu(x) \quad (2.25)$$

where, for the fermions in the first generation, we may use the standard notation, introduced in Eq. (1.10), and write

$$K_\mu(x) = \sum_q [\epsilon_L(q) \bar{q} \gamma_\mu L q + \epsilon_R(q) \bar{q} \gamma_\mu R q] \\ + \frac{1}{2} \sum_\nu \bar{\nu} \gamma_\mu L \nu + \frac{1}{2} \sum_\ell \bar{\ell} \gamma_\mu (g_V^\ell - \gamma_5 g_A^\ell) \ell, \quad (2.26)$$

where standard model predicts,

$$\begin{aligned} \epsilon_L(u) &= \frac{1}{2} - \frac{2}{3} \sin^2 \theta_W, & \epsilon_R(u) &= -\frac{2}{3} \sin^2 \theta_W, \\ \epsilon_L(d) &= -\frac{1}{2} + \frac{1}{3} \sin^2 \theta_W, & \epsilon_R(d) &= \frac{1}{3} \sin^2 \theta_W. \end{aligned} \quad (2.27)$$

and

$$g_V(e) = -\frac{1}{2} + 2 \sin^2 \theta_W, \quad g_A(e) = -\frac{1}{2}. \quad (2.28)$$

□ **Exercise 2.3** The definition of the covariant derivative in the $SU(2)_L \times U(1)_Y$ model is given by

$$D_\mu = \partial_\mu + i g \frac{\mathbf{T}}{2} \cdot \mathbf{W}_\mu + i g' \frac{Y}{2} B_\mu, \quad (2.29)$$

where $\mathbf{T}/2$ represents the three generators of $SU(2)_L$ in the appropriate representation. Use this and Eq. (2.10) to write down the neutral currents in terms of W_μ^0 and B_μ . Then use the definition of Z_μ in terms of W_μ^0 and B_μ , and the orthogonal combination for A_μ , to deduce the neutral current in Eq. (2.26).

For low momenta, Eq. (2.25) leads to the effective Hamiltonian:

$$\mathcal{H}_{\text{weak}} = \frac{4\rho G_F}{\sqrt{2}} K_\mu(x) K^\mu(x), \quad (2.30)$$

where

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W}, \quad (2.31)$$

which can be set equal to unity in view of the mass relations given in Eq. (2.23). The Hamiltonian in Eq. (2.30) leads to neutral current processes such as $\nu_\mu e \rightarrow \nu_\mu e$ or $\nu q \rightarrow \nu q$.

Table 2.1: Experimental values of various neutral current parameters and their comparison with predictions of the standard model. (From Ref. [9]).

Quantity	Experimental value	Standard model prediction
$\epsilon_L(u)$	0.332 ± 0.016	0.345 ± 0.0003
$\epsilon_L(d)$	-0.438 ± 0.012	-0.429 ± 0.0004
$\epsilon_R(u)$	-0.178 ± 0.013	-0.156
$\epsilon_R(d)$	$-0.026^{+0.075}_{-0.048}$	-0.078
$g_A(e)$	-0.507 ± 0.014	-0.507 ± 0.0004
$g_V(e)$	-0.041 ± 0.015	-0.036 ± 0.0003

- **Exercise 2.4** Consider the decay $Z \rightarrow f\bar{f}$, where f is some fermion with $m_f \ll M_Z$. Let

$$\langle f\bar{f} | K_\mu | 0 \rangle = \bar{u}\gamma_\mu(a - b\gamma_5)v, \quad (2.32)$$

where K_μ is the neutral current. Show that

$$\Gamma(Z \rightarrow f\bar{f}) = \frac{\sqrt{2}}{3\pi} (a^2 + b^2) G_F^2 M_Z^3. \quad (2.33)$$

Using the proper values of a and b for neutrinos, show that the rate for $Z \rightarrow \nu\bar{\nu}$ is 165 MeV.

The weak neutral currents were first observed experimentally in 1972 in an experiment at CERN [10] in $\nu_\mu e$ scattering and was confirmed shortly thereafter by experiments at Fermilab [11]. In the subsequent years a great variety of neutral current processes involving ν -quark, ν -lepton, e -quark have been observed and thoroughly studied [12]. On the theoretical front, the radiative corrections to the neutral current processes were carried out [8], thus enabling a detailed comparison between theory and experiment. Various authors [13, 14] have carried out a detailed comparison between theory and experiment and have obtained the following value of $\sin^2 \theta_W$ which provides the best fit to all available neutral current data (see Table 2.1):

$$\begin{aligned} \sin^2 \theta_W &= 0.229 \pm 0.0064 \\ \rho &= 0.998 \pm 0.0086. \end{aligned} \quad (2.34)$$

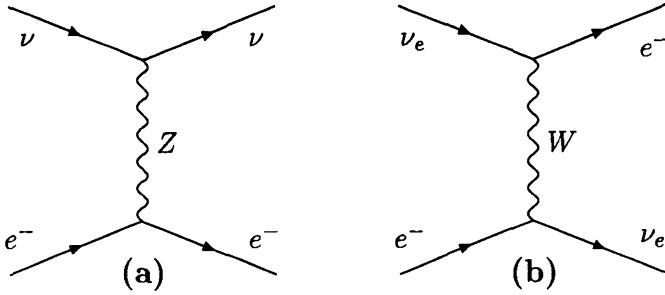


Figure 2.1: Tree-level diagrams that contribute to the ν_e scattering in the standard model.

However recent measurements at LEP [15] and SLC [16] of the Z -pole observables have led to a much more precise determination of $\sin^2 \theta_W$. The global fit to all data in conjunction with the CDF/D0 [17] value for $m_t = 180 \pm 12 \pm \text{GeV}$ yields [9]

$$\sin^2 \theta_W = 0.2315 \pm 0.0002 \pm 0.0003. \quad (2.35)$$

2.3 Neutrino-electron scattering in the standard model

In this section, we present the predictions of the standard model for neutrino scattering cross-sections off electrons. For the sake of simplicity, we will restrict ourselves to low energies, i.e., when the momentum q carried by the intermediate vector bosons satisfy $|q^2| \ll M_W^2$.

2.3.1 $\nu_e e$ and $\bar{\nu}_e e$ scattering

The $\nu_e e$ scattering receives contribution both from charged and neutral current interactions, as shown in Fig. 2.1. Since we will work in the low energy limit, we will neglect the momentum dependence of the W and Z -propagators and work in the four-Fermi interaction approximation.

For the neutral current diagram given in Fig. 2.1a, the Feynman amplitude is given by

$$\mathcal{M}_{\text{nc}} = \sqrt{2}G_F \left[\bar{u}'(k')\gamma^\lambda L u'(k) \right] \left[\bar{u}(p')\gamma_\lambda (g_L L + g_R R) u(p) \right], \quad (2.36)$$

where the neutrino spinors have been denoted by a prime, and

$$g_L = 2 \sin^2 \theta_W - 1, \quad g_R = 2 \sin^2 \theta_W, \quad (2.37)$$

which can be read off from the neutral current interaction given in Eqs. (2.26) and (2.28). In writing down the matrix element, a factor of 2 comes in because each current can come from either factor of K_μ in the interaction Lagrangian.

The contribution of the charged current diagram of Fig. 2.1b, on the other hand, can be written as

$$\begin{aligned} \mathcal{M}_{cc} &= 2\sqrt{2}G_F \left[\bar{u}'(k')\gamma^\lambda \mathbf{L} u(p) \right] \left[\bar{u}(p')\gamma_\lambda \mathbf{L} u'(k') \right] \\ &= 2\sqrt{2}G_F \left[\bar{u}'(k')\gamma^\lambda \mathbf{L} u'(k) \right] \left[\bar{u}(p')\gamma_\lambda \mathbf{L} u(p) \right], \end{aligned} \quad (2.38)$$

using the Fierz transformation rules in the last step. Adding up the two contributions, we thus obtain

$$\mathcal{M} = \sqrt{2}G_F \left[\bar{u}'(k')\gamma^\lambda \mathbf{L} u'(k) \right] \left[\bar{u}(p')\gamma_\lambda \{ (g_L + 2)\mathbf{L} + g_R \mathbf{R} \} u(p) \right]. \quad (2.39)$$

Averaging over the initial electron spin and summing over the final electron spin, we thus obtain

$$\frac{1}{2} \sum_{\text{spin}} |\mathcal{M}|^2 = G_F^2 N^{\lambda\rho} L_{\lambda\rho}, \quad (2.40)$$

where

$$\begin{aligned} N^{\lambda\rho} &= \text{tr} \left[\not{k} \gamma^\rho \mathbf{L} \not{k}' \gamma^\lambda \mathbf{L} \right] \\ &= 2 \left[k^\rho k'^\lambda + k'^\rho k^\lambda - g^{\lambda\rho} k \cdot k' + i\varepsilon^{\alpha\rho\beta\lambda} k_\alpha k'_\beta \right], \end{aligned} \quad (2.41)$$

and $L_{\lambda\rho}$ is the corresponding trace with the electron bilinears. Upon contraction, one obtains

$$N^{\lambda\rho} L_{\lambda\rho} = 16 \left[(g_L + 2)^2 (k \cdot p)^2 + g_R^2 (k' \cdot p)^2 - (g_L + 2) g_R m_e^2 k \cdot k' \right], \quad (2.42)$$

using momentum conservation, $k + p = k' + p'$, to ensure that $k \cdot p = k' \cdot p'$ and $k \cdot p' = k' \cdot p$.

□ **Exercise 2.5** Verify Eq. (2.42) by evaluating $L_{\lambda\rho}$ explicitly and using the form of $N^{\lambda\rho}$ given above.

To proceed now, we decide to work in the frame in which the initial electron is at rest and choose the axes so that the incoming neutrino is traveling along the positive z -direction. Then

$$p^\lambda = (m_e, 0, 0, 0), \quad k^\lambda = (\omega, 0, 0, \omega). \quad (2.43)$$

In this frame, let the energies of the final electron and the neutrino be E' and ω' respectively. Starting from the general formula for the cross-section given in Eq. (2.57) and performing the integration over the final electron momentum, we now obtain

$$\sigma_E(\nu_e e) = \frac{1}{16\pi^2 m_e \omega} \int \frac{d^3 k'}{4E'\omega'} \delta(m_e + \omega - E' - \omega') \cdot \frac{1}{2} \sum_{\text{spin}} |\mathcal{M}|^2, \quad (2.44)$$

where the subscript E on σ denotes the elastic cross section. It is straight forward to check that the argument of the δ -function of energy vanishes when

$$\omega' = \frac{m_e \omega}{m_e + \omega(1 - \zeta)}, \quad (2.45)$$

where ζ is the cosine of the angle the outgoing neutrino makes with the z -axis. Also, in the frame used by us,

$$N^{\lambda\rho} L_{\lambda\rho} \simeq 16m_e^2 \left[(g_L + 2)^2 \omega^2 + g_R^2 \omega'^2 \right], \quad (2.46)$$

where we have dropped the last term of Eq. (2.42), which is negligible if we assume $\omega \gg m_e$.

The integration over the azimuthal angle as well as the magnitude of $\mathbf{k}'$ can now be performed and one obtains

$$\frac{d\sigma_E(\nu_e e)}{d\zeta} = \frac{G_F^2 m_e^2}{2\pi} \frac{(g_L + 2)^2 \omega^2 + g_R^2 \omega'^2}{[m_e + \omega(1 - \zeta)]^2}, \quad (2.47)$$

where ω' is given by Eq. (2.45).

This gives the differential cross-section, but it is customary to express it in terms of the variable

$$y \equiv \frac{E' - m_e}{\omega} = 1 - \frac{\omega'}{\omega}. \quad (2.48)$$

In terms of this variable, we get the following differential cross section when $m_e \ll \omega$:

$$\frac{d\sigma_E(\nu_e e)}{dy} = \frac{G_F^2 m_e \omega}{2\pi} \left[(2 \sin^2 \theta_W + 1)^2 + 4 \sin^4 \theta_W (1 - y)^2 \right], \quad (2.49)$$

putting in the values for g_L and g_R at this stage. On integrating over y we get

$$\sigma_E(\nu_e e) = \frac{G_F^2 m_e \omega}{2\pi} \left[(2 \sin^2 \theta_W + 1)^2 + \frac{4}{3} \sin^4 \theta_W \right]. \quad (2.50)$$

Using the value $G_F^2 = 5.29 \times 10^{-38} \text{ cm}^2/\text{GeV}^2$, one obtains

$$\sigma_E(\nu_e e) = 0.9 \times 10^{-43} \left(\frac{\omega}{10 \text{ MeV}} \right) \text{ cm}^2. \quad (2.51)$$

This cross section has recently been measured by a Maryland-LosAlamos-Irvine collaboration [18] at Los Alamos for neutrino energies up to 50 MeV. They measure total cross section

$$\langle \sigma \rangle = (3.01 \pm 0.46(\text{stat}) \pm 0.38(\text{syst})) \times 10^{-43} \text{ cm}^2 \quad (2.52)$$

for a mean neutrino energy of 31.7 MeV, in excellent agreement with the prediction of the standard model. Specially important is the fact that it provides evidence for the interference between the charged and neutral current amplitudes as predicted by the standard model.

□ **Exercise 2.6** In deriving the $\nu_e e$ elastic cross-section, we neglected the last term of Eq. (2.42). Show that, if this term is not neglected, one obtains

$$\frac{d\sigma_E(\nu_e e)}{dy} = \frac{G_F^2 m_e \omega}{2\pi} \left[(g_L + 2)^2 + g_R^2 (1 - y)^2 - (g_L + 2) g_R \frac{y m_e}{\omega} \right] \quad (2.53)$$

Turning now to $\hat{\nu}_e e$ scattering, there are also both W and Z -contributions in this case, as in Fig. 2.2. The differential cross section is given here by

$$\frac{d\sigma_E(\hat{\nu}_e e)}{dy} = \frac{G_F^2 m_e \omega}{2\pi} \left[(2 \sin^2 \theta_W + 1)^2 (1 - y)^2 + 4 \sin^4 \theta_W \right] \quad (2.54)$$

and integration over y yields the total cross section

$$\sigma_E(\hat{\nu}_e e) = \frac{G_F^2 m_e \omega}{2\pi} \left[\frac{1}{3} (2 \sin^2 \theta_W + 1)^2 + 4 \sin^4 \theta_W \right]. \quad (2.55)$$

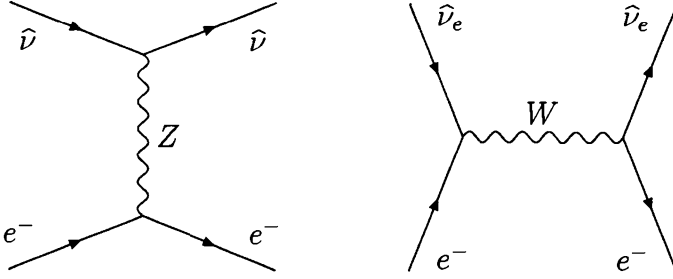


Figure 2.2: Tree-level diagrams that contribute to the $\hat{\nu}_e e$ scattering in the standard model.

Putting in numbers, we get

$$\sigma(\hat{\nu}_e e) = 0.378 \times 10^{-43} \left(\frac{\omega}{10 \text{ MeV}} \right) \text{ cm}^2 \quad (2.56)$$

This cross section was measured [19].

□ **Exercise 2.7** Consider the scattering of two particles of mass m and μ , with 4-momentum given by p and k , to two particles with 4-momentum given by p' and k' . The general formula for the differential cross section for this process is given by

$$d\sigma = \frac{(2\pi)^4 \delta^4(p + k - p' - k')}{4[p \cdot k - m^2 \mu^2]^{1/2}} \frac{|\mathcal{M}|^2}{S} \frac{d^3 p'}{(2\pi)^3 2p'_0} \frac{d^3 k'}{(2\pi)^3 2k'_0}, \quad (2.57)$$

where $\mathcal{M}$ is the matrix element of the transition and S is the symmetry factor, which is $n!$ if there are n identical particles in the final state. Use this to deduce the $\hat{\nu}_e e$ cross section given above.

2.3.2 $\nu_\mu e$ and $\hat{\nu}_\mu e$ scattering

For these scatterings, there is no W -contributions and only the neutral current contributes. The Feynman amplitude is now given by Eq. (2.36). Following the same arguments, here we get,

$$\begin{aligned} \frac{d\sigma_E(\nu_\mu e)}{dy} &= \frac{G_F^2 m_e \omega}{2\pi} \left[(2 \sin^2 \theta_W - 1)^2 + 4(1 - y)^2 \sin^4 \theta_W \right], \\ \frac{d\sigma_E(\hat{\nu}_\mu e)}{dy} &= \frac{G_F^2 m_e \omega}{2\pi} \left[(2 \sin^2 \theta_W - 1)^2 (1 - y)^2 + 4 \sin^4 \theta_W \right]. \end{aligned} \quad (2.58)$$

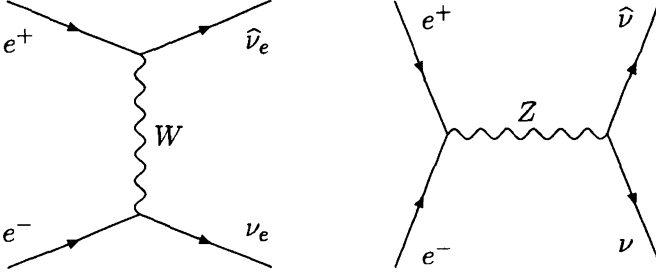


Figure 2.3: Tree-level diagrams contributing to the process $e^+e^- \rightarrow \nu\bar{\nu}$.

They lead to

$$\begin{aligned}\sigma_E(\nu_\mu e) &= \frac{G_F^2 m_e \omega}{2\pi} \left[(2 \sin^2 \theta_W - 1)^2 + \frac{4}{3} \sin^4 \theta_W \right] \\ \sigma_E(\bar{\nu}_\mu e) &= \frac{G_F^2 m_e \omega}{2\pi} \left[\frac{1}{3} (2 \sin^2 \theta_W - 1)^2 + 4 \sin^4 \theta_W \right].\end{aligned}\quad (2.59)$$

Numerically, we obtain,

$$\begin{aligned}\sigma_E(\nu_\mu e) &= 0.15 \times 10^{-43} \left(\frac{\omega}{10 \text{ MeV}} \right) \text{ cm}^2 \\ \sigma_E(\bar{\nu}_\mu e) &= 0.14 \times 10^{-43} \left(\frac{\omega}{10 \text{ MeV}} \right) \text{ cm}^2.\end{aligned}\quad (2.60)$$

Experimental studies [20] have verified these theoretical expectations with good accuracy.

2.3.3 Neutrino pair production

Before concluding this section, we would also like to present the cross sections for production of neutrinos in e^+e^- collision. Again, as in the case of $\nu_e e$ scattering, $e^+e^- \rightarrow \nu_e \bar{\nu}_e$ receives contribution from both W and Z exchange graphs, as shown in Fig.2.3. If we write a general coupling

$$\mathcal{H}_{\text{weak}} = \frac{4G_F}{\sqrt{2}} \bar{\nu}_\ell \gamma^\mu \mathbf{L} \nu_\ell \bar{e} \gamma_\mu \frac{h_V - \gamma_5 h_A}{2} e, \quad (2.61)$$

we get

$$\sigma(e^+e^- \rightarrow \nu\bar{\nu}) = \frac{G_F^2 s}{12\pi} (h_V^2 + h_A^2). \quad (2.62)$$

For the case of $e^+e^- \rightarrow \nu_e \hat{\nu}_e$, the parameters h_V and h_A are given by

$$h_V = \frac{1}{2} + 2 \sin^2 \theta_W, \quad h_A = \frac{1}{2}, \quad (2.63)$$

leading to

$$\sigma(e^+e^- \rightarrow \nu_e \hat{\nu}_e) = \frac{G_F^2 s}{12\pi} \left(\frac{1}{2} + 2 \sin^2 \theta_W + 4 \sin^4 \theta_W \right) \quad (2.64)$$

For the case of muon neutrinos, since only the neutral current contributes, h_V and h_A are equal to $g_V^{(e)}$ and $g_A^{(e)}$ respectively. We then get

$$\sigma(e^+e^- \rightarrow \nu_\mu \hat{\nu}_\mu) = \frac{G_F^2 s}{12\pi} \left(\frac{1}{2} - 2 \sin^2 \theta_W + 4 \sin^4 \theta_W \right). \quad (2.65)$$

A typical estimate for these cross sections is $5 \times 10^{-43} \text{ cm}^2$ at the electron energy of 10 MeV.

2.4 Neutrino-nucleon scattering in the standard model

2.4.1 Quasi-elastic $\nu_e N$ and $\hat{\nu}_e N$ scattering

In the low energy approximation ($\omega \ll m_N$ where m_N is the mass of the nucleon), the following interactions are possible:

$$\begin{aligned} \nu_\ell + n &\rightarrow \ell^- + p \\ \hat{\nu}_\ell + p &\rightarrow \ell^+ + n, \end{aligned} \quad (2.66)$$

where ℓ can be either e or μ . The matrix element for the first process, for example, can be written as

$$\begin{aligned} \mathcal{M} &= \langle \ell^-(k'), p(p') | \mathcal{H}_{\text{weak}} | \nu_\ell(k), n(p) \rangle \\ &= \frac{4G_F \cos \theta_c}{\sqrt{2}} \left[\bar{u}_{(\ell)} \gamma^\mu L u_{(\nu_\ell)} \right] \\ &\quad \times \left[\bar{u}_{(p)} \gamma_\mu \left(\frac{1}{2} \left\{ F_V(q^2) + F_A(q^2) \gamma_5 \right\} \right) u_{(n)} \right]. \end{aligned} \quad (2.67)$$

Here, $\cos \theta_c$ is the element V_{11} of the mixing matrix V introduced in Eq. (1.4). In the nucleon matrix element, F_V and F_A are form factors which

appear because the proton and the neutron are not elementary particles. In general, other form factors can appear in the matrix element, coupled to tensors of the form $\sigma_{\mu\nu}q^\nu$, $\sigma_{\mu\nu}\gamma_5q^\nu$ etc, but their contribution to the final cross section is small in the regime of our interest. Although all these form factors depend on q^2 , where q^μ is the momentum carried by the internal W -boson, we use their values at $q^2 = 0$ in view of the fact that we are examining the regime of small q^2 . The cross sections therefore involve the vector and the axial weak charges of the neutron and the proton $g_V = F_V(0)$ and $g_A = F_A(0)$ respectively. The conserved vector current hypothesis says that $g_V = 1$, whereas g_A has been measured to be 1.26. For low energy neutrinos ($\omega \ll m_N$), the $\nu_e n$ cross section is given by

$$\begin{aligned}\sigma(\nu_e n) &= \frac{G_F^2 \omega^2}{\pi} [g_V^2 + 3g_A^2] \\ &\simeq 9.75 \times 10^{-42} \left(\frac{\omega}{10 \text{ MeV}} \right)^2 \text{ cm}^2.\end{aligned}\quad (2.68)$$

The same expression also applies to $\hat{\nu}_e p$ scattering, which is of more practical importance in astrophysical discussions. It is worth pointing out that these cross sections are of vital importance in designing neutrino detectors in various cases. For instance in the detection of neutrinos from supernovas or from the sun, one is interested in detecting neutrinos with energy in the range of 1 to 50 MeV. To get a better signal, one needs to rely on ν_e -nucleon scattering rather than ν_e -electron scattering. Since free nucleons cannot be used (unless in the case of water detectors), one has to study ν_e -nucleus scattering, which is in general more suppressed compared to free nucleon case. In the case of water detectors, a massive apparatus is required since the mean free path in water at an energy of 10 MeV is 1.5×10^{18} cm for $\hat{\nu}_e$ and 3.2×10^{19} cm for ν_e . Furthermore, nuclei have different threshold for neutrino absorption which is also important in the choice of detector material. Presently used targets are ^{37}Cl , Water, ^{71}Ga , ^{97}Mo . Use of several others such as ^{115}In , ^7Li etc are in the planning stage.

2.4.2 Deep inelastic scattering of neutrinos off nucleons

In §2.4.1, we considered the kinematical region of $q^2 \ll m_N^2$. Here, in contrast, we consider the region where $|q^2| \gg m_N^2$, but still $|q^2| \ll M_W^2$.

The processes will now be of the generic form

$$\nu_\ell(k) + N(p) \rightarrow \ell^-(k') + X(p'), \quad (2.69)$$

where the momenta of different particles are shown in parentheses, and X is a hadronic state for which there can be a large number of possibilities, depending on the value of q^2 . We will consider the inclusive cross section, i.e., the cross section summed over all possible final hadronic states. The matrix element will now have a leptonic part which is the same as that occurring in Eq. (2.67), but the hadronic part will depend on the final state and hence must be parameterized differently. For this, we write the spin-summed square of the matrix element as

$$|\mathcal{M}|^2 = \left(\frac{4G_F}{\sqrt{2}} \right)^2 L_{\mu\nu} W^{\mu\nu}, \quad (2.70)$$

where $L_{\mu\nu}$ and $W^{\mu\nu}$ come from the leptonic and the hadronic parts of the matrix element respectively. The leptonic part of the matrix element is known, from which we can write

$$\begin{aligned} L_{\mu\nu} &= \text{tr} [\not{k} \gamma_\mu \not{L} \not{k}' \gamma_\nu \not{L}] \\ &= 2 \left[k_\mu k'_\nu + k_\nu k'_\mu - g_{\mu\nu} k \cdot k' - i \varepsilon_{\mu\nu\lambda\rho} k^\lambda k'^\rho \right], \end{aligned} \quad (2.71)$$

neglecting the charged lepton mass. For $W^{\mu\nu}$, on the other hand, we must use a parameterization since the matrix element itself is not known. Thus, we must write the most general tensor which depends on two vectors p and p' , or alternatively on p and $q = p - p'$. This is given by

$$\begin{aligned} W^{\mu\nu} &= -g^{\mu\nu} W_1 + \frac{p^\mu p^\nu}{m_N^2} W_2 - \frac{i \varepsilon^{\mu\nu\lambda\rho} p_\lambda q_\rho}{2m_N^2} W_3 + \frac{q^\mu q^\nu}{m_N^2} W_4 \\ &\quad + \frac{(p^\mu q^\nu + p^\nu q^\mu)}{2m_N^2} W_5 + \frac{i(p^\mu q^\nu - p^\nu q^\mu)}{2m_N^2} W_6. \end{aligned} \quad (2.72)$$

Here m_N is the nucleon mass, and W_1 to W_6 are form factors. All these form factors, being Lorentz scalars, can depend only on q^2 and $p \cdot q$, since these are the only kinematical scalar variables (note that $p^2 = m_N^2$, which is not a variable). Customarily, one chooses the independent variables to be

$$Q^2 \equiv -q^2, \quad \nu \equiv \frac{p \cdot q}{m_N}. \quad (2.73)$$

Thus, Q^2 is positive, and in the rest frame of the initial nucleon target, ν is the energy carried by the intermediate W -boson. More useful are the dimensionless *Bjorken scaling variables* x and y , defined by

$$x \equiv \frac{-q^2}{2p \cdot q} = \frac{Q^2}{2m_N \nu}, \quad y \equiv \frac{p \cdot q}{p \cdot k}. \quad (2.74)$$

Putting the expression of Eqs. (2.71) and (2.72) into Eq. (2.70), it is easy to see that the terms involving W_4 , W_5 and W_6 are proportional to the charged lepton mass, and hence we neglect them. Then we are left with

$$\begin{aligned} L_{\mu\nu} W^{\mu\nu} = & 4k \cdot k' W_1 + \frac{2}{m_N^2} [2p \cdot k p \cdot k' - m_N^2 k \cdot k'] W_2 \\ & + \frac{2}{m_N^2} [p \cdot k q \cdot k' - q \cdot k p \cdot k'] W_3. \end{aligned} \quad (2.75)$$

In the lab frame, using $p^\mu = (m_N, 0, 0, 0)$ and $k^\mu = (\omega, 0, 0, \omega)$, we obtain

$$\begin{aligned} L_{\mu\nu} W^{\mu\nu} = & 4m_N \omega xy W_1 + 2\omega [2(\omega - \nu) - m_N xy] W_2 \\ & + 2\omega (2\omega - \nu) xy W_3. \end{aligned} \quad (2.76)$$

The differential cross section can now be calculated. It will be a function of W_1 , W_2 and W_3 . One obtains

$$\begin{aligned} \frac{d^2\sigma}{dx dy} = & \frac{G_F^2 m_N \omega}{\pi(1 + Q^2/M_W^2)} \\ & \times \left[xy^2 F_1 + \left(1 - y - \frac{m_N xy}{2\omega}\right) F_2 + y\left(1 - \frac{y}{2}\right) x F_3 \right], \end{aligned} \quad (2.77)$$

where

$$F_1 = m_N W_1, \quad F_2 = \nu W_2, \quad F_3 = \nu W_3. \quad (2.78)$$

For $\hat{\nu}N$ interactions, the right-chirality projection operator R will appear instead of L in the expression for $L_{\mu\nu}$, so that the sign of the term containing the antisymmetric tensor in Eq. (2.71) will be reversed. In the expression for the cross section, this will reverse the sign of the F_3 term.

Experimental determination of the cross section gives information about the form factors, or *structure functions* as they are called. These

are then analyzed to obtain the structure of the nucleon, e.g., the distribution of quarks in the nucleon. A detailed discussion of this procedure [21] is related to strong interaction physics and has little connection with neutrino physics which is the topic of discussion of this book.

2.5 Neutrino mass in the standard model

So far we have discussed the dynamical properties of the neutrino. Let us now turn to a discussion of its mass, which is the basis for a great deal of the static properties that the neutrino could acquire, such as mixings and oscillations, magnetic moments, decays etc. In the standard model only one helicity state of the neutrino per generation is present. Therefore, it could not have a Dirac mass, which requires both the helicity states. Neutrinos can in general have an alternative type of mass term called Majorana mass terms, which are of the form $\nu_L^T C^{-1} \nu_L$, where C is the Lorentz conjugation matrix. This form requires just one helicity state of a particle and uses the opposite helicity state of the antiparticle (see Ch. 4 for detailed discussion). However, since the ν_L is part of the $SU(2)_L$ doublet field and has lepton number +1, the above neutrino mass term transforms as an $SU(2)_L$ triplet: it is not gauge invariant. Furthermore, this type of mass breaks lepton number by two units. On the other hand, a quick look at the standard model Lagrangian convinces one that the model has exact lepton number symmetry even after symmetry breaking; therefore such terms can never arise in perturbation theory. Neutrinos are therefore massless to all orders in perturbation theory.

One can ask whether non-perturbative effects can induce neutrino mass in the standard model. The only known source of lepton number violation is the weak instanton effects. We mentioned the anomaly cancellation condition in Eq. (1.49), which must be satisfied by all gauge currents in order to ensure renormalizability of a theory. The same need not be true if one or more of the generators in Eq. (1.49) pertain to some global symmetry of the theory. In particular, if X is the generator of a conserved global current, we can consider expressions like

$$a_X = \text{Tr} \left(X \{T^a, T^b\} \right)_L - \text{Tr} \left(X \{T^a, T^b\} \right)_R, \quad (2.79)$$

where T^a, T^b are generators of the gauge group. If any of these co-

efficients do not vanish, the associated current will have a non-zero divergence induced by instanton effects, which will be of the form $\partial_\mu j_X^\mu = c \mathbf{W}^{\mu\nu} \cdot \widetilde{\mathbf{W}}_{\mu\nu} + c' B^{\mu\nu} \widetilde{B}_{\mu\nu}$, where $\mathbf{W}^{\mu\nu}$ and $B^{\mu\nu}$ are the field-strength tensors for the $SU(2)_L$ and $U(1)_Y$ gauge fields. In the standard model, lepton number current conservation is in fact broken non-perturbatively through the anomaly. But it turns out that the anomaly contribution to the baryon number current nonconservation has also an identical form, so that the $B - L$ current is conserved to all orders in the gauge couplings. As a consequence, nonperturbative effects from the gauge sector cannot induce $B - L$ violation. Since the neutrino mass operator described above violates also $B - L$, this proves that neutrino masses remain zero even in the presence of nonperturbative effects.

- **Exercise 2.8** Calculate the quantities shown in Eq. (2.79) for different choices of gauge generators, taking X as the lepton number L . Next, do the same thing identifying X as the baryon number B . From this, show that all such quantities vanish if $X = B - L$.

Thus, we conclude that neutrino mass vanishes in the standard model. Because of this, its mixings, its magnetic moment etc also vanish. There is, however, one interesting property of neutrino induced by higher order weak interactions in the standard model, viz the charge radius of the neutrino. Even though the neutrino has zero charge and therefore has no interactions with the photon to lowest order, it can interact with the photon through higher order effects induced at the loop level. This interaction must however respect electromagnetic gauge invariance. If Γ_μ is the effective interaction vertex, then for neutral particles like the neutrino, Ward identity implies $\Gamma^\mu q_\mu = 0$ where q is the momentum of the photon at the vertex. Moreover, Γ cannot involve any right handed neutrino, since the latter are absent in the Standard model. Therefore, the interaction in momentum space will have the form [see Ch. 13 for details]:

$$\bar{\nu} \gamma_\mu \mathbb{L} \nu (q^2 g^{\mu\lambda} - q^\mu q^\lambda) \epsilon_\lambda^* . \quad (2.80)$$

Assuming, for the moment, that the form factor associated with this interaction is momentum independent, we get the following interaction in co-ordinate space:

$$\bar{\nu} \gamma_\mu \mathbb{L} \nu \partial_\lambda F^{\mu\lambda} . \quad (2.81)$$

§2.5. <i>Neutrino mass in the standard model</i>	41
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Since this is non-vanishing only for off-shell photons, this kind of electromagnetic effect will only appear in processes such as $\nu_e e$ or $\nu_e q$ elastic scatterings. This kind of virtual coupling is called the charge radius of the neutrino since the strength of this coupling measures the $\langle r^2 \rangle$ of the neutrino. Since the coupling involves off-shell photon, there is no unique gauge invariant definition of neutrino charge radius [22].

Chapter 3

Massive neutrinos

3.1 Introduction

We saw in Ch.2 that the standard model predicts massless neutrinos, making them essentially different from other fermions such as the charged leptons (e, μ, τ) and the quarks (u, d, s, c, t, b), which are known to have masses. Standard model has also been extremely successful in explaining the various low weak energy processes involving charged and neutral current interactions of neutrinos. For a long time therefore, there was a strong prejudice among some theorists that neutrinos are indeed massless. Many scenarios of physics beyond the standard model that became extremely popular in fact were closely tied to the masslessness of the neutrino.

The situation has now changed drastically. There is convincing evidence for neutrino masses from several experiments that we describe below and go into details in subsequent chapters. The landscape of particle physics has been fundamentally altered for ever by this discovery. For one thing, the neutrinos are now in the same footing as quarks — a fact anticipated by people who believed in quark-lepton symmetry more literally than just as a token indicated by the weak interactions; but more importantly, it is giving new and interesting clues to the direction of new physics beyond the standard model.

In this chapter, we give an overview [1, 2] of the evidences and also long-standing theoretical motivations for nonzero neutrino masses that preceded the discovery of neutrino masses. We then summarize the issues raised by the experimental evidences and briefly indicate possible new directions for new physics suggested by them.

3.2 Theoretical motivations for neutrino mass

Even a cursory look at the standard model reveals that the masslessness of the neutrinos in this model is somewhat concocted. Why doesn't one, e.g., introduce a right handed neutrino field (ν_R) in the fundamental fermion content, as one does for all other fermions? If one did, the ν_R could have paired with the ν_L through the Higgs mechanism to produce a mass term for the neutrinos. [For details, see §7.2.1].

In fact, there is no fundamental reason why one does not do that. In other words, the ν_R is not introduced just because we want to predict massless neutrinos in the standard model. Things are arranged so as to produce that result. If one wants otherwise, one can just as well arrange things in some other way.

Contrast this issue with that of the masslessness of the photon. The latter is massless because of a conserved gauge symmetry which, in turn, governs the dynamics of the electromagnetic interaction. For the masslessness of the neutrinos, we see no such symmetry principle in the standard model. Hence the masslessness is unsatisfactory from a theoretical point of view.

Secondly, one of the major aims of Particle Physics is unification of the fundamental interactions. This provides extra motivations for massive neutrinos since, as we will see in Ch. 9, many of the interesting unification models predict neutrino mass at some level or other. Supersymmetry, which is a symmetry between fermions and bosons, constitutes another interesting motivation of venturing beyond the standard model. We will see in Ch. 10 that the supersymmetrized standard model naturally predicts massive neutrinos, unless a lepton number symmetry is imposed artificially on it.

The theoretical question is not only how one can extend the standard model to find models with massive neutrinos, but also how one can understand the smallness of the neutrino masses compared to the masses of the charged fermions. Of course, even the charged fermion masses vary widely. The top-quark mass is roughly five orders of magnitude larger than the electron mass, and the standard model does not even attempt to explain that. Why should then we be seeking a reason for the ν_e being at least five orders of magnitude lighter than the electron?

The point here is that the two examples of mass ratios quoted in the previous paragraph pose two different questions. The ratio m_t/m_e

can be properly understood only through a proper understanding of the separation of the different fermion families. But the smallness of the neutrino mass remains a question even within a single family. Consider, for example, the first family. The electron mass is 0.511 MeV, and the current masses for u and d quarks are in the 5–10 MeV range. Thus they all conform within about an order of magnitude, whereas ν_e is at least 5 orders lighter. Similarly, in the second generation, we get $m_\mu = 105$ MeV, $m_s \simeq 150$ MeV, $m_c \simeq 1500$ MeV, again conforming to one another within about an order of magnitude, but ν_μ is at least 3 orders lighter than m_μ . A good theory of neutrino mass should also throw some light on the reason for this smallness.

Another big question concerning neutrino mass is whether the massive neutrinos are Dirac particles or Majorana particles. They can of course be Dirac particles, like all charged fermions have to be. However, there is also the possibility that the neutrinos are their own antiparticles since they do not have any electric charge. This question is discussed in Ch. 4.

It is worth pointing out here that the standard model without right handed neutrinos has got the desirable property that anomaly cancellations imply charge quantization [3] if there is only one family of fermions. This property is lost in presence of right handed neutrinos, as well as of more than one fermion generation. However, it has been pointed out [4] that if one includes right handed neutrinos and assumes that the neutrinos are Majorana particles, then anomaly cancellation implies charge quantization regardless of the number of generations. One could perhaps use this as an argument for neutrinos being both massive as well as Majorana type. Majorana nature of neutrinos might also help us understand the smallness of neutrino masses through the see-saw mechanism, which will be discussed in various chapters later in the book.

3.3 Questions related to neutrino mass

In this book, we will study physics issues related to neutrino mass. However, the question of neutrino mass cannot be discussed in isolation. It is intimately related to some other issues.

The most important of these issues is that of neutrino mixing. The electron neutrino, for example, is defined to be the object that couples

to the electron via the charged weak current. Similarly, ν_μ connects to the muon and ν_τ to the tau. If neutrinos get mass, there is no fundamental reason why these objects should describe physical particles with a specified mass eigenvalue. Rather, in general these *flavor eigenstates* ν_e, ν_μ, ν_τ would be superpositions of the *mass eigenstates*:

$$\nu_{\ell L} = \sum_{\alpha} U_{\ell\alpha} \nu_{\alpha L} . \quad (3.1)$$

In other words, the mass eigenstates ν_α are mixtures of flavor eigenstates so that a given physical neutrino can couple to more than one charged lepton via the charged current. This is analogous to the mixing of quarks and called *neutrino mixing*. The mixing matrix U is called PMNS matrix in the context of neutrinos, after the names of Pontecorvo, Maki, Nakagawa and Sakata [5] who first dealt with this matrix.

In presence of neutrino mixing, the generational lepton numbers (i.e., the e -number, the μ -number and the τ -number) cannot remain as valid global symmetries. Unlike the predictions from the standard model, one should see processes that violate these numbers, even in processes where neutrinos do not appear in the initial and final states. Some such issues have been discussed in Ch. 15.

Another possible consequence of neutrino mixing is that, like the quark sector mixing matrix V defined in Eq. (1.4), the mixing matrix in the leptonic sector can also be complex in general. This would imply CP-violating phenomena in the leptonic sector, which will also be discussed in Ch. 15.

If the neutrinos are Majorana particles, then they must not have any additive quantum number, local or global. Thus, total lepton number symmetry must be broken. To be more precise, we should say that the symmetry that has to be broken is $B - L$, which is the only true global symmetry of the standard model in presence of neutrino mixing. One should then see processes violating $B - L$. One such process, called the neutrinoless double beta decay, has been discussed in detail in Ch. 14.

One can then ask whether $B - L$ is violated spontaneously. If it is, Goldstone theorem would assert the existence of a massless pseudo-scalar particle. Since the breaking of $B - L$ is related to Majorana masses for the neutrinos, the related Goldstone boson is called Majoron. The physical implications of the existence of such a particle is a rich and interesting topic by itself. In Ch. 7, we show how Majorons can arise in

realistic models.

Another related issue is that of neutrino stability. If neutrinos are massless, they cannot decay. But if they have mass, all but the lightest neutrino should be unstable since in general, there would be no symmetry to prevent their decay. Only the lightest one, being the lightest fermion, would be stable. We discuss some specific decay mechanisms for unstable neutrinos in Ch. 7 and Ch. 13. The decays of the heavier neutrinos in turn can profoundly affect the cosmological scenarios. This question has been dealt with in Ch. 8 and Ch. 18 in some detail.

3.4 Tests of neutrino mass

Broadly speaking, there are two kinds of experiments in which neutrino masses can be detected. We discuss them one by one.

3.4.1 Kinematic tests

These are tests on processes which are allowed even in the standard model with $m_\nu = 0$. Take any such known process involving neutrinos in the final state. Calculate the rate as a function of neutrino mass. Try to see whether the observed rate differs significantly from the calculated rate with $m_\nu = 0$.

Obviously, sensitivity of such experiments increase with m_ν/Q , where Q is the difference between the total energy in the initial state and the sum of masses of all particles in the final state except the neutrinos. Some examples follow. Details will be discussed in Ch. 12.

- Nuclear β -decay: One can look at the energy distribution of electrons emitted in a β -decay process. The shape of the curve can be calculated assuming $m_{\nu_e} = 0$. If, however the mass is not zero, the observed count will fall short of the calculated one. The lowest known Q value (18.6 keV) for this process occurs for ^3H β -decay. Extensive experimental studies of tritium decay have been conducted at various laboratories around the world to look for deviations from the Kurie plot, which will signal a non-vanishing neutrino mass.
- Pion decay: One can look for the muon energy in $\pi^+ \rightarrow \mu^+ \nu_\mu$ (or its charge conjugate decay). Obviously this energy depends on the

ν_μ mass. Accurate measurement of the muon energy would then determine the mass of the ν_μ .

- Tau decay: There are various decay modes of the tau. Since all the observed ones are consistent with the standard model, they all conserve the tau number L_τ . Thus, all these modes have a ν_τ in the final state. One can use the kinematics of the final state to find the mass of the ν_τ .

3.4.2 Exclusive tests

This constitutes looking for processes which are forbidden if the neutrinos are massless. Observation of even one event of any such process would then be an unmistakable signature for neutrino mass. But often the specific tests involve extra assumptions about the nature of neutrino mass.

The most common of such assumptions is that of neutrino mixing. Although it is by no means a logical necessity, it is so hard to make gauge models of neutrino mass without having neutrino mixing that it is natural to accept the presence of mixing. Various processes might show the effect of this. Some examples follow.

- Neutrino oscillation: Suppose at $t = 0$, a ν_e is emitted in a particle interaction. If neutrino mixing is present, this ν_e is a superposition of various mass eigenstates with different masses. As time goes on, these different components evolve differently so that, after some time, the original beam becomes a different combination of the mass eigenstates. As a result, it might look somewhat like ν_μ or ν_τ , for example. It can then have charged current interactions with μ or τ . Such a phenomenon is called neutrino oscillation. Various evidences of neutrino oscillation have been obtained. A brief summary of these evidences will be presented in §3.5, and the phenomenon will be treated in detail in Ch. 5.
- Neutrino decays: The possibility was mentioned in §3.3. Specific processes which might have interesting consequences include the radiative decay of a heavier neutrino (ν_α) to a lighter one ($\nu_{\alpha'}$):

$$\nu_\alpha \rightarrow \nu_{\alpha'} + \gamma, \quad (3.2)$$

which has been discussed in Ch. 13. Another interesting possibility is to have three neutrinos in the final state, which may or may not be of the same kind:

$$\nu_\alpha \rightarrow \nu_{\alpha_1} \nu_{\alpha_2} \nu_{\alpha_3} . \quad (3.3)$$

This process receives some importance in the context of Ch. 8.

In models where the global $B - L$ symmetry is broken spontaneously, one obtains a Goldstone boson J which is usually called the Majoron. In such models, the mode

$$\nu_\alpha \rightarrow \nu_{\alpha'} + J \quad (3.4)$$

is also interesting. This has been discussed in some detail in Ch. 7.

Besides the above processes, one can look at others which do not crucially depend on neutrino mixing. The most important of such processes are the following:

- Electromagnetic properties: In Ch. 2, it has been mentioned that a massless neutrino has only one electromagnetic form factor, its charge radius. Observation of any other electromagnetic property, like a magnetic moment, would imply a massive neutrino. This question has been discussed in Ch. 13.
- Neutrinoless double β -decay: This process, abbreviated as $\beta\beta_{0\nu}$, is essentially two neutrons in a nucleus converting into two protons by emitting two electrons; or, in the quark picture,

$$d + d \rightarrow u + u + e + e . \quad (3.5)$$

Unlike the previous processes, it does not depend on neutrino mixing. However, it involves a violation of lepton number by 2 units. If the neutrinos are all Dirac particles, total lepton number is conserved and this process is forbidden. Only with Majorana neutrinos is such a process possible. A detailed discussion of this process, including the experimental efforts to measure its rate, is given in Ch. 14.

3.5 Evidences of neutrino mass

Curiously enough, direct or kinematic searches have not revealed any indication for neutrino mass. There are only upper limits from such experiments:

$$m_{\nu_e} \leq 2.2 \text{ eV} , \quad (3.6)$$

$$m_{\nu_\mu} \leq 170 \text{ keV} , \quad (3.7)$$

$$m_{\nu_\tau} \leq 18.2 \text{ MeV} . \quad (3.8)$$

However, several different observations of neutrino oscillations have now been made. Two of the very first ones involve solar and cosmic ray neutrinos; more recently oscillations have been observed using terrestrially produced neutrinos e.g., electron neutrinos from the reactors in the Kamland experiment and muon neutrinos from accelerators in the K2K experiment. The positive evidences for neutrino oscillations give two parameters: the mass difference squared Δm^2 and mixing angle θ . Below we summarize the results.

From the existing data on atmospheric neutrinos it is found that the oscillation scenario that fit the data best is $\nu_\mu \rightarrow \nu_\tau$ oscillation for the mass and mixing parameters

$$\begin{aligned} \Delta m_{\text{atm}}^2 &\simeq (1.5 \text{ to } 3) \times 10^{-3} \text{ eV}^2 , \\ \sin^2 2\theta_{\text{atm}} &\simeq 0.9 \text{ to } 1 . \end{aligned} \quad (3.9)$$

Ch. 5 contains detailed discussions on this issue.

The second evidence for neutrino oscillation comes from the experiments that have observed a deficit in the flux of neutrinos from the Sun as compared to the predictions of the standard solar model championed by Bahcall and his collaborators and more recently studied by many groups. The global analysis of all solar neutrino data seem to favor large mixing among neutrinos:

$$\begin{aligned} \Delta m_{\odot}^2 &\simeq 4.7 \times 10^{-5} \text{ to } 1.5 \times 10^{-4} \text{ eV}^2 , \\ \sin^2 2\theta_{\odot} &\simeq 0.64 \text{ to } 0.94 , \end{aligned} \quad (3.10)$$

where the symbol ' $\odot$ ' indicates the sun. The details of these considerations are given in Ch. 6.

The Kamland data finds a depletion of the Reactor neutrino flux by 34% over an average distance of about 200 km, which provides evidence for oscillation of ν_e 's corresponding to a mass difference square in the same range as the solar one. It also provides evidence for the mixing angle being large.

Similarly, the K2K experiment using a multi-GeV muon neutrino beam from an accelerator also sees evidence of a depletion of flux as would be suggested by the observed atmospheric neutrino deficit. The statistics in these experiments is limited but nonetheless they do provide evidence for $\nu_\mu \rightarrow \nu_\tau$ oscillation.

Oscillation experiments only depend on the difference of mass squares of the different neutrinos and the mixing angles. Therefore, in order to have a complete picture of neutrino masses, we need other experiments. Neutrinoless double beta decay experiments can provide such complementary data, because they measure the following combination of masses and mixing angles:

$$\langle m \rangle_{\beta\beta} = \sum_i U_{ei}^2 m_i, \quad (3.11)$$

where U is the neutrino mixing matrix. It is therefore sensitive to the overall scale of the neutrino mass. As we discuss in great detail in Ch. 14, this process occurs only if the neutrino mass violates lepton number conservation.

Chapter 4

Dirac versus Majorana masses

The charged leptons and the quarks — i.e., all known fundamental fermions other than the neutrinos — are Dirac particles as a consequence of electric charge conservation. In other words, they obey the Dirac equation and are described by four component complex spinors. If the neutrinos were massless, it was well known that they could have been described by two component complex spinors, alternatively called Weyl spinors. Since neutrinos are now known to have mass, they cannot be Weyl spinors. Instead, it is tempting to think that they are like any other fermion, and should therefore be Dirac spinors.

However, there is an important difference between neutrinos and other fundamental fermions, viz, that the neutrinos do not carry any electric charge. This brings in a new theoretical possibility that the neutrinos might also be Majorana spinors [1, 2]. In this chapter we discuss what a Majorana spinor is, spelling out its differences with a Dirac spinor.

4.1 Two-component spinor field

To start with, consider a Dirac field, e.g., the electron. Its states can be described by four basic spinors. Two of these can be taken as the left and the right helicity states of the electron, e_L and e_R . As for the other two, we can use $\hat{e}_L$ and $\hat{e}_R$, the two helicity states of the antiparticle positron.

Now suppose that in our frame of reference, we find an electron moving in the z -direction. The z -component of its spin is $-\frac{1}{2}$. Thus, spin and momentum are anti-parallel, so we have a left-handed electron e_L . How does this particle appear to another observer who is running

faster than the electron in the z -direction? To him, the electron is moving in the negative z -direction, which is also the direction of its spin. Therefore, he sees a right-handed object. But we have two right-handed objects, e_R and $\hat{e}_R$, in our repertory. One can then ask which one of these two states is this other observer seeing.

The answer is easy here. The state $\hat{e}_R$ has opposite electric charge compared to the state e_L . As we know, electric charge is a Lorentz invariant quantity. By boosting to a different Lorentz frame, one thus cannot see a different charge on a particle. What is e_L in one frame cannot be $\hat{e}_R$ in another. The boosted observer, therefore, sees an e_R .

We now reopen the question in the context of a neutrino. We have a ν_L in our frame. If the neutrino is massive, its speed must be less than that of light, so we can imagine a boosted observer who runs faster. As before, this observer sees some right handed object. What is this object?

We must recall at this point that we have direct experimental evidence of only the left handed state of the neutrino and right handed state of the antineutrino. Thus, we have the states ν_L and $\hat{\nu}_R$ in our repertory. If we want to mimic the situation of the electron, we must postulate two more states, ν_R and $\hat{\nu}_L$, in this collection. The boosted observer will see a ν_R when we see a ν_L , just as in the electron case. In this case, the neutrino will be a Dirac particle with four complex degrees of freedom, as is apparent from the presence of the four basis spinor states.

But can we not do without postulating these two new spinor states? After all, the boosted observer sees a right-handed object, and we do have a right-handed object, viz, $\hat{\nu}_R$. Can't the boosted observer see the state $\hat{\nu}_R$? Unlike the case of electrons, ν_L and $\hat{\nu}_R$ have the same electric charge (viz, zero), so nothing goes wrong there. The only quantum number defined in the context of the standard model and which is different for ν_L and $\hat{\nu}_R$ is the lepton number (L). But the lepton number symmetry is a global symmetry. Unlike the electromagnetic $U(1)$ symmetry, it does not govern the dynamics. Rather, it is a consequence of the dynamics and the field contents of the standard model. So there is nothing sacred about the lepton number symmetry. If this is broken, there is thus no reason why ν_L and $\hat{\nu}_R$ cannot be the boosted counterparts of one another. These two spinors can thus constitute the left and right handed projections of the same fermionic field. Since just two

basis spinors are needed here, we are talking about a two-component massive fermion field. This was originally proposed by Majorana [1] in 1937. Therefore, such a field is called a Majorana field.

One might ask, how can one boost a certain particle and obtain its antiparticle? The point here is that the words *particle* and *antiparticle* are defined with respect to some conserved quantum number. If there is no conserved quantum number to distinguish the *particle* and the *antiparticle* states, the particle is identical to its own antiparticle. A Majorana neutrino is its own antineutrino.

The difference between a Majorana particle and a Weyl particle must be clearly understood. Both are two component spinors, but for two completely different reasons. A Weyl particle is massless. Thus, the ν_L moves at the speed of light. No observer can overtake it and view it as a right-handed object. So a right handed counterpart of ν_L is not necessary to obtain a Lorentz covariant picture. Similarly, a $\hat{\nu}_R$ does not require its left handed counterpart. But they can have different lepton numbers (or other quantum numbers) to distinguish themselves from each other.

A Majorana neutrino, on the other hand, has mass. But the ν is the same as $\hat{\nu}$. So the right handed counterpart of ν_L can be equivalently called ν_R or $\hat{\nu}_R$. Similarly, ν_L is the same as $\hat{\nu}_L$. That is why only ν_L and $\hat{\nu}_R$ suffice. They can be Lorentz transformed to each other and thus the neutrino cannot have any additive quantum number. This self-conjugacy is the reason why a Majorana particle has half as many degrees of freedom as a Dirac particle.

An analogy might clarify the case further. Consider the field of a charged pion. It is a complex scalar field since π^+ and π^- are obviously different. But the neutral pion π^0 is its own antiparticle, and thus is described by a real scalar field. A complex field amounts to two real fields, viz, the real and the imaginary parts of it. Thus, the neutral pion has one real degree of freedom whereas the charged pion field has two. Similarly, a Majorana particle has half as many degrees of freedom as a Dirac particle.

4.2 Mathematical definition of a Majorana field

First consider a free Dirac field. The field operator $\psi(x)$ can be written as

$$\psi(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E_p}} \sum_{s=\pm\frac{1}{2}} \left(f_s(\mathbf{p}) u_s(\mathbf{p}) e^{-ip \cdot x} + \hat{f}_s^\dagger(\mathbf{p}) v_s(\mathbf{p}) e^{ip \cdot x} \right) \quad (4.1)$$

in terms of the operator $f_s(\mathbf{p})$ that annihilates a single particle state of momentum $\mathbf{p}$ and spin component s in the direction of the momentum. Similarly $\hat{f}_s(\mathbf{p})$ annihilates, i.e., $\hat{f}_s^\dagger(\mathbf{p})$ creates an antiparticle state. The spinors $u_s(\mathbf{p})$ and $v_s(\mathbf{p})$ are the plane wave solutions of positive and negative energies respectively. They satisfy the equations

$$\begin{aligned} (\gamma^\mu p_\mu - m) u_s(\mathbf{p}) &= 0, \\ (\gamma^\mu p_\mu + m) v_s(\mathbf{p}) &= 0, \end{aligned} \quad (4.2)$$

where the γ^μ are a set of four matrices satisfying the relations

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \quad (4.3)$$

and

$$\gamma_\mu^\dagger = \gamma_0 \gamma_\mu \gamma_0. \quad (4.4)$$

If we write down $\psi^*(x)$, it will involve $f_s^\dagger(\mathbf{p})$ and $\hat{f}_s(\mathbf{p})$, which will create a particle or annihilate an antiparticle.

Since the particle and the antiparticle are identical in the case of a Majorana neutrino, we feel that $\psi(x)$ must in some sense be related to $\psi^*(x)$. However, if we try to impose

$$\psi(x) = \psi^*(x), \quad (4.5)$$

that would be wrong in general, for the following reason.

A physically meaningful equation must be Lorentz covariant. It is easy to see that the above equation is not. Under a Lorentz transformation that changes the co-ordinates as

$$x'^\mu = x^\mu + \omega^\mu{}_\nu x^\nu, \quad (4.6)$$

the spinor field changes by the rule

$$\psi'(x') = \exp\left(-\frac{i}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\psi(x) \quad (4.7)$$

where

$$\sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu] \quad (4.8)$$

involves the commutator of the Dirac γ -matrices. Eq. (4.7) gives

$$\psi'^*(x') = \exp\left(\frac{i}{4}\sigma_{\mu\nu}^*\omega^{\mu\nu}\right)\psi^*(x), \quad (4.9)$$

which, in general, is not the same transformation law as Eq. (4.7) because the $\sigma_{\mu\nu}$ matrices are not purely imaginary. Therefore, even if we impose Eq. (4.5) in one frame of reference, it will not be valid in another frame and will therefore lose its validity as a physical condition.

We therefore define a *conjugate* field

$$\hat{\psi}(x) \equiv \gamma_0 C \psi^*(x) \quad (4.10)$$

in terms of an yet undefined matrix C such that $\hat{\psi}(x)$ transforms the same way under Lorentz transformation as $\psi(x)$, i.e.,

$$\begin{aligned} \hat{\psi}'(x') &= \exp\left(-\frac{i}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\hat{\psi}(x) \\ &= \exp\left(-\frac{i}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\gamma_0 C \psi^*(x). \end{aligned} \quad (4.11)$$

However, using Eq. (4.7) directly on Eq. (4.10), we obtain

$$\begin{aligned} \hat{\psi}'(x') &= \gamma_0 C \psi'^*(x') \\ &= \gamma_0 C \exp\left(\frac{i}{4}\sigma_{\mu\nu}^*\omega^{\mu\nu}\right)\psi^*(x). \end{aligned} \quad (4.12)$$

Equating the above two expressions, we demand that the matrix C satisfies the relation

$$\gamma_0 C \exp\left(\frac{i}{4}\sigma_{\mu\nu}^*\omega^{\mu\nu}\right) = \exp\left(-\frac{i}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\gamma_0 C \quad (4.13)$$

for any arbitrary $\omega^{\mu\nu}$, which implies

$$\gamma_0 C \sigma_{\mu\nu}^* = -\sigma_{\mu\nu} \gamma_0 C. \quad (4.14)$$

The specific form of the matrix C depends on the representation of the γ -matrices and will be discussed in §4.3. At this point, it is enough to say that such a matrix C , and therefore $\hat{\psi}$, can be defined. So we can modify Eq. (4.5) by replacing the ψ^* on the right hand side by $\hat{\psi}$. That will surely be a Lorentz-covariant relation to define a Majorana field. Since $\hat{\psi}$ involves ψ^* and therefore $f_s^\dagger(\mathbf{p})$ and $\hat{f}_s(\mathbf{p})$, its identification with ψ serves the purpose of identifying particles with antiparticles.

The true definition of a Majorana field is a little more general. We demand

$$\psi(x) = e^{i\theta} \hat{\psi}(x) \quad (4.15)$$

since a phase can always be absorbed in the definition of the fermion field $\psi(x)$. We can always choose $\theta = 0$ by suitably defining the fermion field, but the freedom of the phase is sometimes convenient.

It can now be easily shown that the plane wave expansion of a Majorana field operator is

$$\psi(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E_p}} \sum_{s=\pm\frac{1}{2}} \left(f_s(\mathbf{p}) u_s(\mathbf{p}) e^{-ip \cdot x} + \lambda f_s^\dagger(\mathbf{p}) v_s(\mathbf{p}) e^{ip \cdot x} \right). \quad (4.16)$$

Notice that $\lambda f_s^\dagger(\mathbf{p})$ appears in the place where $\hat{f}_s^\dagger(\mathbf{p})$ appears for a Dirac field. This signifies that the antiparticle is the same as the particle except for a phase λ . Thus λ is related to the angle θ of Eq. (4.15).

To find this relation and also to verify that Eq. (4.15) is indeed satisfied by Eq. (4.16), we start from the latter and evaluate $\hat{\psi}(x)$:

$$\begin{aligned} \hat{\psi}(x) &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E_p}} \sum_{s=\pm} \gamma_0 C \left(f_s^\dagger(\mathbf{p}) u_s^*(\mathbf{p}) e^{ip \cdot x} + \lambda^* f_s(\mathbf{p}) v_s^*(\mathbf{p}) e^{-ip \cdot x} \right) \\ &= \lambda^* \int \frac{d^3p}{\sqrt{(2\pi)^3 2E_p}} \sum_{s=\pm} \left(\lambda f_s^\dagger(\mathbf{p}) v_s(\mathbf{p}) e^{ip \cdot x} + f_s(\mathbf{p}) u_s(\mathbf{p}) e^{-ip \cdot x} \right) \\ &= \lambda^* \psi(x). \end{aligned} \quad (4.17)$$

We have used the fact that $|\lambda|^2 = 1$, along with the spinor relations

$$\begin{aligned} \gamma_0 C u_s^*(\mathbf{p}) &= v_s(\mathbf{p}), \\ \gamma_0 C v_s^*(\mathbf{p}) &= u_s(\mathbf{p}), \end{aligned} \quad (4.18)$$

which will be proven after we define C explicitly. We thus see that

$$\lambda = e^{i\theta} . \quad (4.19)$$

In the literature, λ is often called the creation phase factor since it goes with the creation operator in Eq. (4.16). However, we must emphasize that it is only a convention to put the phase in the creation part. It can just as well be put in the annihilation part.

□ **Exercise 4.1** The Lagrangian for a free Majorana field is

$$\mathcal{L} = \frac{1}{2} (\bar{\psi} i \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi) . \quad (4.20)$$

Show that the propagator of a Majorana field is given by the same expression as that of a Dirac field. [Hint: the propagator of a scalar field is the same irrespective of whether the field is real or complex.]

4.3 Different representations of Dirac matrices

The Dirac matrices γ^μ are defined through the Hamiltonian

$$H = \gamma^0 (\boldsymbol{\gamma} \cdot \mathbf{p} + m) , \quad (4.21)$$

and they satisfy the anticommutation relations of Eq. (4.3) in order that Eq. (4.21) reproduces the relativistic energy momentum relation $E^2 = \mathbf{p}^2 + m^2$. Since the Hamiltonian is a hermitian operator, it also follows that γ^0 and $\gamma^0 \gamma^i$ must be hermitian, so that γ^i is anti-hermitian, which has already been succinctly expressed as Eq. (4.4). Apart from these constraints, the γ -matrices are arbitrary and there are different equivalent conventions to choose them, as we will shortly show with some examples.

As far as the matrix C is concerned, first of all we notice that in Eq. (4.10), if we demand that $\hat{\psi}$ is properly normalized when ψ is, then $\gamma_0 C$ must be a unitary matrix. This implies that C is unitary,

$$C^\dagger = C^{-1} , \quad (4.22)$$

using the properties of γ_0 . Then, we note that Eq. (4.14) implies

$$C \sigma_{\mu\nu}^* C^{-1} = - \gamma_0 \sigma_{\mu\nu} \gamma_0 . \quad (4.23)$$

Now, it is easy to see that the above equation can be satisfied if we take

$$C^{-1}\gamma_\mu C = -\gamma_\mu^\top \quad (4.24)$$

as the defining relation for C . Finally, we demand that

$$\widehat{\bar{\psi}} = \psi. \quad (4.25)$$

Using Eq. (4.24), this can be shown to imply $CC^* = -1$, which can be transformed into the form

$$C^\top = -C, \quad (4.26)$$

using Eq. (4.22). In other words, the matrix C is antisymmetric.

With these properties, we can go back to the definition of conjugate fields in Eq. (4.10) and rewrite it as

$$\widehat{\bar{\psi}} \equiv \widehat{\psi}^\dagger \gamma_0 = \psi^\top C^\dagger \gamma_0^\dagger \gamma_0 = \psi^\top C^{-1}, \quad (4.27)$$

using Eq. (4.4), as well as $(\gamma_0)^2 = 1$ which follows from Eq. (4.3). This is an alternative form in which the conjugate field is often defined.

- **Exercise 4.2** Using the definition in Eq. (4.10) for the conjugate fermion field and the properties of the matrix C , show that

$$\begin{aligned} \bar{\psi}_L \psi_R &= \widehat{\bar{\psi}}_L \widehat{\psi}_R, \\ \bar{\psi}_L \gamma_\mu \psi_L &= -\widehat{\bar{\psi}}_R \gamma_\mu \widehat{\psi}_R. \end{aligned} \quad (4.28)$$

- **Exercise 4.3** Starting from the definition of a u -spinor in Eq. (4.2) and using the properties of the matrix C given above, show that $\gamma_0 C u^*$ satisfies the equation of a v -spinor.

4.3.1 Dirac representation

For the γ -matrices, Dirac himself chose

$$\gamma^0 = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad (4.29)$$

where $\mathbb{1}$ is the 2×2 unit matrix and the σ^i 's are the Pauli matrices. In this representation, the plane wave solutions are given by

$$u_s(\mathbf{p})e^{-ip \cdot x} \quad \text{or} \quad v_s(\mathbf{p})e^{ip \cdot x}, \quad (4.30)$$

which will be normalized by the relations

$$u_s^\dagger(\mathbf{p})u_{s'}(\mathbf{p}) = v_s^\dagger(\mathbf{p})v_{s'}(\mathbf{p}) = 2E_p\delta_{ss'}. \quad (4.31)$$

Using the explicit forms of the γ -matrices in the equations for the u -type and v -type spinors given in Eq. (4.2), we can find the solution in these representation. These are:

$$\begin{aligned} u_s(\mathbf{p}) &= \sqrt{E+m} \begin{pmatrix} \chi_s \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \chi_s \end{pmatrix}, \\ v_s(\mathbf{p}) &= \sqrt{E+m} \begin{pmatrix} \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \chi'_s \\ \chi'_s \end{pmatrix}, \end{aligned} \quad (4.32)$$

in the normalization defined in Eq. (4.31), where

$$\chi_{+\frac{1}{2}} = -\chi'_{-\frac{1}{2}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_{-\frac{1}{2}} = \chi'_{+\frac{1}{2}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (4.33)$$

We now try to identify the matrix C in this representation. Notice that the defining relation in Eq. (4.24), or the associated properties given in Eqs. (4.22) and (4.26), still leaves the arbitrariness of an overall phase in the matrix C . We have to fix it with the relations between the spinors given in Eq. (4.18). With the choice of the spinors given above, we obtain

$$C = i\gamma_2\gamma_0 = \begin{pmatrix} 0 & i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix} \quad (4.34)$$

in this representation.

□ **Exercise 4.4** Verify that the definition of the matrix C and the spinors u and v satisfy Eq. (4.18). Use the fact that

$$\sigma_i^* = -\sigma_2\sigma_i\sigma_2 \quad (4.35)$$

for the Pauli matrices.

4.3.2 Majorana representation

There is another representation which is particularly illuminating when we are talking of Majorana particles. Recall that we said in the previous section that we cannot take $\psi = \psi^*$ because the $\sigma_{\mu\nu}$ matrices are

not imaginary in general. If they are so, however, in a particular representation of the Dirac matrices, then Eq. (4.5) becomes an acceptable condition. In such a representation, a Majorana particle is just a spinor whose components are all real. The analogy of Dirac and Majorana spinors with complex and real scalars become more apparent in such a representation.

Such a representation exists. It is given by

$$\begin{aligned}\gamma^0 &= \begin{pmatrix} 0 & \sigma^2 \\ \sigma^2 & 0 \end{pmatrix} \quad , \quad \gamma^1 = \begin{pmatrix} i\sigma^3 & 0 \\ 0 & i\sigma^3 \end{pmatrix} \\ \gamma^2 &= \begin{pmatrix} 0 & -\sigma^2 \\ \sigma^2 & 0 \end{pmatrix} \quad , \quad \gamma^3 = \begin{pmatrix} -i\sigma^1 & 0 \\ 0 & -i\sigma^1 \end{pmatrix} .\end{aligned}\quad (4.36)$$

Notice that all the γ -matrices are imaginary. By Eq. (4.8), it assures that the $\sigma_{\mu\nu}$'s are also purely imaginary. The matrix C can be defined in this representation as

$$C = -\gamma^0. \quad (4.37)$$

Since $(\gamma^0)^2 = 1$, it is easily seen that Eq. (4.14) is satisfied.

□ **Exercise 4.5** Check that the definitions of the matrix C given in Eqs. (4.34) and (4.37) satisfy the definition in Eq. (4.24).

4.3.3 Other representations

In general, knowing one representation of the gamma matrices allows one to write down other ones, denoted by a prime, in the form

$$\gamma'_\mu = U\gamma_\mu U^\dagger, \quad (4.38)$$

where U is a unitary matrix. It is easy to see that the primed γ -matrices also satisfy the anticommutation relations of Eq. (4.3). In fact, it can be shown that if two sets of γ -matrices satisfy Eq. (4.3), they can always be related by a similarity transformation as in Eq. (4.38). The proof of this last statement is rather involved, and we are not even trying to indicate how the proof goes.

Starting from the Dirac representation, one can use

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & \sigma_2 \\ \sigma_2 & -1 \end{pmatrix} \quad (4.39)$$

to define the Majorana representation. The plane wave solutions in the primed representation can be defined as

$$u'_s(\mathbf{p}) = U u_s(\mathbf{p}), \quad v'_s(\mathbf{p}) = U v_s(\mathbf{p}), \quad (4.40)$$

which can easily be shown to satisfy Eq. (4.2) with the primed γ -matrices. Similarly, in the primed representation, we can take the conjugation matrix to be

$$C' = U C U^\top, \quad (4.41)$$

which satisfies Eq. (4.24). Since C is antisymmetric in the Dirac representation, this shows that it is antisymmetric in other representations as well.

Using Eqs. (4.40) and (4.41), it is trivial to check that if the relations between the spinors given in Eq. (4.18) are true in one representation, they are valid in the primed representation as well. Thus, these relations are valid in general, a fact we used before in deriving Eq. (4.17).

□ **Exercise 4.6** Show that the definition of the conjugate field given in Eq. (4.10) is independent of any representation of the γ -matrices.

4.4 Majorana neutrinos and discrete symmetries of space-time

A Majorana neutrino is its own antiparticle. Therefore, it is expected that it possesses special properties under discrete symmetries like $\mathcal{C}$, $\mathcal{CP}$ and $\mathcal{CPT}$. We examine these properties one by one [3].

4.4.1 Properties under $\mathcal{C}$

Under the charge conjugation operation $\mathcal{C}$, a free fermion field transforms as

$$\mathcal{C}\psi(\mathbf{x}, t)\mathcal{C}^{-1} = \eta_C^* \gamma_0 \mathcal{C}\psi^*(\mathbf{x}, t) = \eta_C^* \hat{\psi}(\mathbf{x}, t), \quad (4.42)$$

where η_C is a phase and the conjugate field $\hat{\psi}$ has been defined in Eq. (4.10). While the above equation is true for any fermion field, for a Majorana field we can write

$$\mathcal{C}\psi(\mathbf{x}, t)\mathcal{C}^{-1} = (\eta_C \lambda)^* \psi(\mathbf{x}, t) \quad (4.43)$$

using Eq. (4.17). Plugging in the plane wave expansion of a Majorana field given in Eq. (4.16), we obtain

$$\begin{aligned}\mathcal{C}f_s(\mathbf{p})\mathcal{C}^{-1} &= (\eta_C\lambda)^*f_s(\mathbf{p}) \\ \mathcal{C}f_s^\dagger(\mathbf{p})\mathcal{C}^{-1} &= (\eta_C\lambda)^*f_s^\dagger(\mathbf{p}).\end{aligned}\tag{4.44}$$

Using the unitarity of the operator $\mathcal{C}$, we observe that the consistency of these two equations demands that

$$(\eta_C\lambda)^* = \eta_C\lambda.\tag{4.45}$$

If the vacuum is $\mathcal{C}$ -symmetric, we can apply the second one of Eq. (4.44) on the vacuum to obtain

$$\mathcal{C}|\mathbf{p}, s\rangle = \eta_C\lambda|\mathbf{p}, s\rangle,\tag{4.46}$$

where

$$|\mathbf{p}, s\rangle = f_s^\dagger(\mathbf{p})|0\rangle\tag{4.47}$$

is a one-particle state with momentum $\mathbf{p}$ and spin component s . Calling the $\mathcal{C}$ eigenvalue of the one-particle state as $\tilde{\eta}_C$, we thus obtain from Eq. (4.46) the relation

$$\tilde{\eta}_C = \eta_C\lambda,\tag{4.48}$$

which is real due to Eq. (4.45).

A free Majorana particle is thus an eigenstate of the charge conjugation operator. The same cannot be said about a physical Majorana neutrino, because the interactions of a neutrino necessarily violate $\mathcal{C}$ by large amounts, so that it does not make any sense to talk about an eigenstate of $\mathcal{C}$. However, if instead of neutrinos one talks about some other Majorana fermions whose interactions conserve $\mathcal{C}$ (at least to a good extent), the particle would be an eigenstate of $\mathcal{C}$ (to the same good extent).

- **Exercise 4.7** If the vacuum is $\mathcal{C}$ -symmetric, show that the operator $\mathcal{C}$ acting on the state $|\mathbf{p}, s\rangle$ of a Dirac particle produces an antiparticle with the same momentum and same spin. Thus, the $\mathcal{C}$ operation does not change the helicity of the particle.

4.4.2 Properties under $\mathcal{CP}$

To discuss how a Majorana neutrino behaves under $\mathcal{CP}$, let us introduce the shorthand

$$\Xi \equiv \mathcal{CP} \quad (4.49)$$

for convenience. The transformation of a free fermion field under parity can be given as

$$\mathcal{P}\psi(\mathbf{x}, t)\mathcal{P}^{-1} = \eta_P \gamma_0 \psi(-\mathbf{x}, t), \quad (4.50)$$

η_P being the parity phase of the field. Combining with Eq. (4.42), we get

$$\Xi\psi(\mathbf{x}, t)\Xi^{-1} = \eta_\Xi^* \gamma_0 \hat{\psi}(-\mathbf{x}, t) \quad (4.51)$$

where η_Ξ is a phase factor. For a Majorana field, we thus obtain

$$\Xi\psi(\mathbf{x}, t)\Xi^{-1} = (\eta_\Xi \lambda)^* \gamma_0 \psi(-\mathbf{x}, t). \quad (4.52)$$

Using the plane wave expansion of the Majorana field of Eq. (4.16) and changing the dummy variable $\mathbf{p}$ to $-\mathbf{p}$, we can write

$$\begin{aligned} \gamma_0 \psi(-\mathbf{x}, t) = \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E_p}} \sum_{s=\pm\frac{1}{2}} \left(f_s(-\mathbf{p}) \gamma_0 u_s(-\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x}} \right. \\ \left. + \lambda f_s^\dagger(-\mathbf{p}) \gamma_0 v_s(-\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}} \right). \end{aligned} \quad (4.53)$$

We can now use the spinor relations

$$\gamma_0 u_s(-\mathbf{p}) = u_s(\mathbf{p}), \quad \gamma_0 v_s(-\mathbf{p}) = -v_s(\mathbf{p}), \quad (4.54)$$

whose validity can be explicitly checked in the Dirac representation and then generalized to other representations by the use of Eqs. (4.38) and (4.40). Comparing with Eq. (4.52), we then obtain

$$\begin{aligned} \Xi f_s(\mathbf{p}) \Xi^{-1} &= (\eta_\Xi \lambda)^* f_s(-\mathbf{p}) \\ \Xi f_s^\dagger(\mathbf{p}) \Xi^{-1} &= -(\eta_\Xi \lambda)^* f_s^\dagger(-\mathbf{p}). \end{aligned} \quad (4.55)$$

The consistency of these equations demands the condition

$$\eta_\Xi \lambda = -(\eta_\Xi \lambda)^*. \quad (4.56)$$

The free one-particle states then satisfy

$$\Xi |\mathbf{p}, s\rangle = \tilde{\eta}_\Xi |-\mathbf{p}, s\rangle, \quad (4.57)$$

where

$$\tilde{\eta}_\Xi = \eta_\Xi \lambda, \quad (4.58)$$

which are necessarily imaginary because of Eq. (4.56).

Once again, the above analysis strictly makes sense for physical particles only if $\mathcal{CP}$ is conserved, and can be used as a good approximation if $\mathcal{CP}$ is only slightly violated. In the quark sector, $\mathcal{CP}$ is indeed violated by a small amount compared to the amount of $\mathcal{C}$ or $\mathcal{P}$ violation. If the same is true in the leptonic sector, we can talk about Majorana neutrinos as $\mathcal{CP}$ eigenstates, as we will do for the most part in this book. However, it

- **Exercise 4.8** Consider the decay $Z \rightarrow \nu\bar{\nu}$. For Majorana neutrinos, the final state contains two identical fermions. Show that they must be in the 3P_1 state (i.e., $L = 1$, $S = 1$ state). Since at the tree level, this decay is $\mathcal{CP}$ -conserving and since the $\mathcal{CP}$ eigenvalue of Z is $+1$, derive from this that the $\mathcal{CP}$ eigenvalues of Majorana neutrinos must be $\pm i$.

4.4.3 Properties under $\mathcal{CPT}$

For the sake of convenience, we now introduce the notation

$$\Theta \equiv \mathcal{CPT}. \quad (4.59)$$

Then, for any fermion field,

$$\Theta\psi(x)\Theta^{-1} = -\eta_\Theta^* \gamma_5^\top \psi^*(-x). \quad (4.60)$$

where η_Θ can be called the $\mathcal{CPT}$ phase of the fermion field. For a Majorana neutrino, we can use Eqs. (4.10) and (4.17) to write

$$\Theta\psi(x)\Theta^{-1} = -(\eta_\Theta\lambda)^* \gamma_5^\top C^{-1} \gamma_0 \psi(-x). \quad (4.61)$$

We can now use the plane wave expansion of a Majorana field to obtain the Θ -transformation properties of the creation and annihilation operators. In order to simplify the right side of Eq. (4.61), we can use the spinor relations

$$\begin{aligned} \gamma_5^\top C^{-1} \gamma_0 u_s(\mathbf{p}) &= (-1)^{s-\frac{1}{2}} u_{-s}^*(\mathbf{p}), \\ \gamma_5^\top C^{-1} \gamma_0 v_s(\mathbf{p}) &= (-1)^{s+\frac{1}{2}} v_{-s}^*(\mathbf{p}), \end{aligned} \quad (4.62)$$

which can be proved directly in the Dirac representation and can be shown to be valid in any other representation. This gives

$$\begin{aligned}
 \gamma_5^\top C^{-1} \gamma_0 \psi(-x) &= \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E_p}} \sum_{s=\pm\frac{1}{2}} (-1)^{s-\frac{1}{2}} \left(f_s(\mathbf{p}) u_{-s}^*(\mathbf{p}) e^{ip \cdot x} \right. \\
 &\quad \left. - \lambda f_s^\dagger(\mathbf{p}) v_{-s}^*(\mathbf{p}) e^{-ip \cdot x} \right) \\
 &= \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E_p}} \sum_{s=\pm\frac{1}{2}} (-1)^{s+\frac{1}{2}} \left(f_{-s}(\mathbf{p}) u_s^*(\mathbf{p}) e^{ip \cdot x} \right. \\
 &\quad \left. - \lambda f_{-s}^\dagger(\mathbf{p}) v_s^*(\mathbf{p}) e^{-ip \cdot x} \right), \quad (4.63)
 \end{aligned}$$

where in the last step, we have changed the dummy variable s to $-s$. To deal with the left side of Eq. (4.61), we need to remember that Θ is an anti-linear operator, i.e., for a complex number a and an operator A ,

$$\Theta a A \Theta^{-1} = a^* \Theta A \Theta^{-1} = -a^* \Theta A \Theta^\dagger. \quad (4.64)$$

Using this, we get

$$\begin{aligned}
 \Theta f_s(\mathbf{p}) \Theta^{-1} &= \eta_\Theta^* \lambda^* (-1)^{s-\frac{1}{2}} f_{-s}(\mathbf{p}) \\
 \Theta f_s^\dagger(\mathbf{p}) \Theta^{-1} &= \eta_\Theta^* \lambda (-1)^{s+\frac{1}{2}} f_{-s}^\dagger(\mathbf{p}), \quad (4.65)
 \end{aligned}$$

Taking now the hermitian conjugate of the second equation in Eq. (4.65) and using Eq. (4.64), we obtain

$$\Theta f_s(\mathbf{p}) \Theta^{-1} = \eta_\Theta \lambda^* (-1)^{s+\frac{1}{2}} f_{-s}(\mathbf{p}). \quad (4.66)$$

Comparing this with the first equation in Eq. (4.65), we get

$$\eta_\Theta^* = -\eta_\Theta, \quad (4.67)$$

i.e., the phase η_Θ is $\pm i$. Applying Eq. (4.65) on the CPT-symmetric vacuum, we then get

$$\Theta |\mathbf{p}, s\rangle = \tilde{\eta}_\Theta^s |\mathbf{p}, -s\rangle, \quad (4.68)$$

where

$$\tilde{\eta}_\Theta^s = \eta_\Theta \lambda (-1)^{s-\frac{1}{2}}. \quad (4.69)$$

There is one sense in which this discussion about CPT is more fundamental than the previous ones concerning $\mathcal{C}$ and $\mathcal{CP}$. The point is that while $\mathcal{C}$ and $\mathcal{CP}$ are violated by some interactions, there are strong reasons to believe that all particle interactions conserve CPT . So, whatever be the CPT properties of a free Majorana field, we can say that the same properties are present in the physical particle.

Thus, Eq. (4.61) can be taken as the definition of a Majorana particle with no sacrifice of rigor. We can thus say that a Majorana neutrino is a CPT eigenstate. In a world where neither $\mathcal{C}$ nor $\mathcal{CP}$ is conserved, this is the only way of saying that it is its own antiparticle [4].

4.5 Majorana basis of mass terms

As far as CPT is concerned, a Majorana spinor is a self-contained item. It has a left-handed part and a right-handed one which are CPT conjugates of each other. It has half as many components as a Dirac spinor. Therefore, we can think of using this minimal representation as the basis of any mass term.

We first show that a Dirac spinor can be thought of to be made up of two such 2-component spinors. For this purpose, let us take a field ψ_L and its CPT conjugate $\hat{\psi}_R$. They can form one Majorana spinor. Similarly, there is χ_L and $\hat{\chi}_R$, which can do the same. However, suppose for some reason mass terms such as

$$\bar{\psi}_L \hat{\psi}_R \quad \text{and} \quad \bar{\chi}_L \hat{\chi}_R \quad (4.70)$$

are absent. On the other hand, there are cross terms like

$$\frac{1}{2} m \bar{\psi}_L \hat{\chi}_R + \frac{1}{2} m' \bar{\chi}_L \hat{\psi}_R + \text{h.c.} \quad (4.71)$$

We can summarize these statements by putting the mass terms $\mathcal{L}_{\text{mass}}$ in a matrix form:

$$-\mathcal{L}_{\text{mass}} = \frac{1}{2} (\bar{\psi}_L \quad \bar{\chi}_L) \begin{pmatrix} 0 & m \\ m' & 0 \end{pmatrix} \begin{pmatrix} \hat{\psi}_R \\ \hat{\chi}_R \end{pmatrix} + \text{h.c.} \quad (4.72)$$

We will later show that m' must equal m in this case. In that case, we can identify

$$\psi_L \rightarrow \Psi_L \quad , \quad \chi_L \rightarrow \Psi_R \quad , \quad \hat{\psi}_R \rightarrow \hat{\Psi}_R \quad , \quad \hat{\chi}_R \rightarrow \hat{\Psi}_L \quad , \quad (4.73)$$

and then Ψ is a 4-component Dirac field, because its mass term from Eq. (4.72) reads

$$\frac{1}{2}m \left(\bar{\Psi}_L \Psi_R + \bar{\hat{\Psi}}_L \hat{\Psi}_R \right) + \text{h.c.} = m \bar{\Psi}_L \Psi_R + \text{h.c.}, \quad (4.74)$$

by using the definition of $\hat{\Psi}$. The last equation is the expected form of the mass term of a Dirac fermion.

□ **Exercise 4.9** The kinetic energy term of a Majorana field was given in Eq. (4.20). Show that, in the construction of a Dirac spinor from two Majorana spinors, the kinetic energy terms of the basis Majorana spinors add up to give the proper kinetic term for a Dirac spinor.

In general, for a number of 2-component spinors ψ_{aL} , $\hat{\psi}_{aR}$ we can write

$$-\mathcal{L}_{\text{mass}} = \frac{1}{2} \sum_{a,b} \bar{\psi}_{aL} M_{ab} \hat{\psi}_{bR} + \text{h.c.} \quad (4.75)$$

and then M_{ab} is called the mass matrix in the Majorana basis. This basis is very useful for considering neutrino masses, as we will see in Part II of the book.

It is straightforward to show that written in this form, M_{ab} must be a symmetric matrix. For that, we use the definition of $\hat{\psi}$ from Eq. (4.10) or Eq. (4.27) and the definition of the matrix C from Eq. (4.24) to write

$$\begin{aligned} -\mathcal{L}_{\text{mass}} &= \frac{1}{2} \sum_{a,b} \bar{\psi}_a M_{ab} R \hat{\psi}_b + \text{h.c.} \\ &= \frac{1}{2} \sum_{a,b} \hat{\psi}_a^\top C^{-1} M_{ab} R \gamma_0 C \psi_b^* + \text{h.c.} \\ &= -\frac{1}{2} \sum_{a,b} \hat{\psi}_a^\top C^{-1} M_{ab} R C \gamma_0^\top \psi_b^* + \text{h.c.} \\ &= \frac{1}{2} \sum_{a,b} \psi_b^\dagger \gamma^0 \left(C^{-1} R C \right)^\top M_{ab} \hat{\psi}_a + \text{h.c.} \end{aligned} \quad (4.76)$$

In going to the last step, we have used the following property of two fermion fields ψ , ψ' and a 4×4 matrix Γ :

$$\begin{aligned} \psi^\top \Gamma \psi' &= \sum_{A,B} \psi_A \Gamma_{AB} \psi'_B \\ &= - \sum_{A,B} \psi'_B \Gamma_{AB} \psi_A = -\psi'^\top \Gamma^\top \psi, \end{aligned} \quad (4.77)$$

The spinor indices A, B have been put in explicitly in the intermediate steps for the sake of clarity. The negative sign appears in the middle of this operation because of the interchange of the orders of two anti-commuting fermion fields. Now, since Eq. (4.24) implies

$$C^{-1}\gamma_5 C = \gamma_5^T, \quad (4.78)$$

it follows that $(C^{-1}RC)^T = R$, so that we can write Eq. (4.76) as

$$\begin{aligned} -\mathcal{L}_{\text{mass}} &= \frac{1}{2} \sum_{a,b} \bar{\psi}_{bL} M_{ab} \hat{\psi}_{aR} + \text{h.c.} \\ &= \frac{1}{2} \sum_{a,b} \bar{\psi}_{aL} M_{ba} \hat{\psi}_{bR} + \text{h.c.} \end{aligned} \quad (4.79)$$

changing the dummy indices a, b in the last step. Comparing this with Eq. (4.75), we conclude that

$$M_{ab} = M_{ba}. \quad (4.80)$$

This justifies why we must have $m' = m$ in Eq. (4.72), a fact that we mentioned earlier without proof.

From the discussion above, it is clear that the Dirac spinor is a very special form of two 2-component spinors. In general the mass matrix of Eq. (4.72) could have non-zero diagonal elements as well, i.e., the terms in Eq. (4.70) need not be forbidden. In such general cases, the diagonalization would yield two different mass eigenvalues. The physical particles would be two Majorana spinors. We will encounter this situation quite often in our discussion of the models of neutrino mass. Only when the diagonal elements are zero, a Dirac fermion results.

When the number of basis Majorana fields are not 2 but $2\mathcal{N}$, say, the mass matrix will obviously be a $2\mathcal{N}$ -dimensional symmetric square matrix. Suppose, however, that we can define a set of basis fields denoted by ψ_L, χ_L , where each of these symbols denote a collection of $\mathcal{N}$ fields, such that the entire mass matrix divides into four blocks out of which the diagonal ones are zero, as in Eq. (4.72). In place of the element m of Eq. (4.72), we will now get an $\mathcal{N} \times \mathcal{N}$ matrix M_D . In order to maintain the symmetric nature of the mass matrix, m' of Eq. (4.72) should now be replaced by M_D^T . The diagonal $\mathcal{N} \times \mathcal{N}$ blocks are zero, as already stated. Diagonalization of the mass matrix would then yield $\mathcal{N}$ Dirac

neutrinos. In this simple case, it is enough to diagonalize just the matrix M_D and not the entire $2\mathcal{N} \times 2\mathcal{N}$ mass matrix in the Majorana basis.

In general, the elements other than M_D are non zero, and diagonalization yields Majorana neutrinos. However, in view of the previous discussion, the matrix M_D is often called the Dirac mass matrix. The terms going in the diagonal blocks of the matrix are called the Majorana mass terms. However, it is important to realize that these names do not imply anything about whether the mass eigenstates are Majorana or Dirac.

The condition for appearance of Dirac neutrinos can be stated in various ways. Diagonalization of the matrix of Eq. (4.72) yields two eigenvalues m and $-m$. Thus, we can say that for a general mass matrix M_{ab} in the 2-component basis, we will obtain a Dirac spinor for each pair of equal and opposite eigenvalues. Alternatively, one can say that if we find some symmetry that forbids the mass terms such as in Eq. (4.70) for two different 2-component fields, we obtain a Dirac spinor in the spectrum. For charged fermions, electromagnetic gauge symmetry ensures this. For neutrinos, it has to be some global symmetry like the lepton number.

In the example of the identification of states shown in Eq. (4.73), if ψ_L is the left handed electron neutrino ν_{eL} , we take $\hat{\chi}_R$ to be a right handed neutrino state, absent in the standard model. In this case, the Dirac mass term conserves lepton number, which is equivalent to the electron number L_e for first generation fermions. However, there is also the possibility that $\hat{\chi}_R$, instead of being a separate fermion absent in the standard model, could be the antiparticle of $\nu_{\mu L}$. In this case, the conserved global quantum number turns out to be $L_e - L_\mu$. This possibility was suggested by Zeldovich and independently by Konopinsky and Mahmoud [5].

4.6 The two-component basis in a different notation

The chiral nature of weak interactions and the near masslessness of the neutrino allows a different two-component notation sometimes used in the literature. For the sake of completeness, we summarize this notation in this section. The basic mathematics is almost identical to that of §4.5.

It is useful to start with the $SL(2, \mathbb{C})$ group, which is the set of complex linear transformations S with unit determinant. If we take $z = (z_1, z_2)^T$ as the two dimensional complex basis, then

$$z' = Sz. \quad (4.81)$$

The unit determinant constraint implies that

$$S_{11}S_{22} - S_{12}S_{21} = 1. \quad (4.82)$$

As is well-known, this condition enables us to identify S with the Lorentz transformations as follows. Define $\sigma_\mu = (1, \boldsymbol{\sigma})$ as a Lorentz 4-vector. Then

$$\sigma_\mu P^\mu = P^0 - \boldsymbol{\sigma} \cdot \mathbf{P}. \quad (4.83)$$

Note that

$$\det(\sigma_\mu P^\mu) = P^\mu P_\mu. \quad (4.84)$$

If we then require that $\sigma_\mu P^\mu$ transforms under $SL(2, \mathbb{C})$ as

$$(\sigma_\mu P^\mu)' = (S^\dagger)^{-1}(\sigma_\mu P^\mu)S^{-1} \quad (4.85)$$

so that products of the form $z^\dagger(\sigma_\mu P^\mu)z$ remain invariant, we obtain

$$\det(\sigma_\mu P^\mu)' = \det(\sigma_\mu P^\mu). \quad (4.86)$$

By Eq. (4.84), this implies the invariance of $P^\mu P_\mu$, which is the essence of Lorentz transformations. Thus, one can identify the $SL(2, \mathbb{C})$ transformations with the Lorentz transformations. A further check of this can be made from Eq. (4.82) which implies that the matrix S has six independent parameters (three complex ones), as in the case of Lorentz transformations.

The fundamental representations of $SL(2, \mathbb{C})$ are two dimensional and are called two-component spinors, like z in Eq. (4.81). We can identify them with the two-component chiral fermions χ , i.e.,

$$\chi \equiv \begin{pmatrix} \chi_1 \\ \chi_2 \end{pmatrix}. \quad (4.87)$$

Under Lorentz transformation, $\chi \rightarrow \chi' = S\chi$. Note that χ^* is inequivalent to χ and transforms as $\chi^* \rightarrow \chi'^* = S^*\chi^*$. We will call χ a (2,1) representation and χ^* a (1,2) representation. Then $\sigma_\mu P^\mu$ transforms as a (2,2) representation under $SL(2, \mathbb{C})$. Components of χ will be denoted by χ_α with a lower index, and those of χ^* by $\bar{\chi}^a$ with an upper index.

We can now proceed to construct Lorentz invariant bilinears, which will enable us to define Majorana and Dirac neutrinos. The unit determinant property of S implies that, if χ and ψ are two 2-component spinors, then $\epsilon^{\alpha\beta}\chi_\alpha\psi_\beta$ is invariant under $SL(2, \mathbb{C})$, where ϵ is the antisymmetric tensor defined by $\epsilon^{11} = \epsilon^{22} = 0$, $\epsilon^{12} = -\epsilon^{21} = 1$. Note that the anti-commuting property of spinor fields allows one to write $\epsilon^{\alpha\beta}\chi_\alpha\chi_\beta$ also as a non-vanishing Lorentz-invariant bilinear. These invariants can be written as $i\chi^\top\sigma_2\psi$ or $i\chi^\top\sigma_2\chi$ in the matrix notation. Similar invariants exist for $\bar{\chi}$ and $\bar{\psi}$.

A 4-component Dirac spinor can in general be written as

$$\Psi = \begin{pmatrix} \chi \\ -i\sigma_2\psi^* \end{pmatrix}. \quad (4.88)$$

Thus, if we choose $\gamma_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, a conventional Dirac mass term $\bar{\Psi}\Psi$ becomes

$$\bar{\Psi}\Psi = i\psi^\top\sigma_2\chi - i\chi^\dagger\sigma_2\psi^*. \quad (4.89)$$

Note that such a mass term is invariant under the $U(1)$ transformation under which

$$\psi \rightarrow e^{i\theta}\psi, \quad \chi \rightarrow e^{-i\theta}\chi. \quad (4.90)$$

This $U(1)$ can be identified with the lepton number and if χ represents a lepton, ψ represents an antilepton.

A general mass term involving many two component spinors can be written as

$$-\mathcal{L}_{\text{mass}} = \frac{i}{2} \sum_{a,b} M_{ab} \psi_a^\top \sigma_2 \chi_b + \text{h.c.}, \quad (4.91)$$

where now the indices a and b label different spinors. It is then easy to check that $M = M^\top$. Furthermore, it is also easy to see how the creation

phase factor introduced in Eq. (4.16) emerges in this framework. To see this, let us assume that a free 2-component neutrino is described by the following Lagrangian:

$$\mathcal{L} = \nu^\dagger \sigma^\mu i \partial_\mu \nu - \frac{im}{2} e^{i\delta} \nu^\top \sigma_2 \nu + \frac{im}{2} e^{-i\delta} \nu^\dagger \sigma_2 \nu^*. \quad (4.92)$$

This leads to the following field equation:

$$i\sigma^\mu \partial_\mu \nu + im e^{-i\delta} \sigma_2 \nu^* = 0. \quad (4.93)$$

If we want to rewrite the Lagrangian of Eq. (4.92) in the 4-component notation such that the Lagrangian looks like in Eq. (4.20), then we need to write

$$\psi = \begin{pmatrix} \nu \\ -i\sigma_2 \nu^* e^{-i\delta} \end{pmatrix}, \quad (4.94)$$

and

$$\gamma_\mu = \begin{pmatrix} 0 & \bar{\sigma}_\mu \\ \sigma_\mu & 0 \end{pmatrix}, \quad (4.95)$$

where $\bar{\sigma}_\mu = (1, -\sigma)$. This representation is called the Chiral representation. The conjugation matrix in this representation is given by

$$C = i\gamma_2\gamma_0 = \begin{pmatrix} i\sigma^2 & 0 \\ 0 & -i\sigma^2 \end{pmatrix}. \quad (4.96)$$

Thus, using Eq. (4.94), we obtain

$$\hat{\psi} = \gamma_0 C \psi^* = e^{i\delta} \psi. \quad (4.97)$$

This is the same as in Eq. (4.15). Thus, the 2-component formalism makes it clear that the phase arbitrariness in the conjugation operation arises when the mass matrices are complex. This can happen in gauge theories if either the Yukawa couplings or the vacuum expectation values of the Higgs fields are complex. This will also provide the origin of CP-violation in the leptonic sector, when all phases present cannot be removed by redefinition of complex 2-component neutrino fields.

- **Exercise 4.10** Show that the choice of γ_μ of Eq. (4.95) can be obtained from the Dirac representation through Eq. (4.38) through a suitable choice of the matrix U .

4.7 Feynman rules involving Majorana neutrinos

In courses on Quantum Field Theory, one usually learns Feynman rules involving Dirac fermions. Some of them remain unaltered if Majorana fermions are involved instead. For example, in Ex. 4.1, we mentioned that the propagator of a Majorana fermion has the same form as that of a Dirac fermion. What we meant is that, irrespective of whether the field ψ denotes a Dirac or a Majorana fermion, one obtains

$$\langle 0 | \mathcal{T}(\psi_A(x) \bar{\psi}_B(y)) | 0 \rangle = \int \frac{d^4 p}{(2\pi)^4} e^{ip \cdot (x-y)} [iS_F(p)]_{AB}, \quad (4.98)$$

with

$$S_F(p) = \frac{\not{p} + m}{p^2 - m^2 + i\epsilon}. \quad (4.99)$$

In Eq. (4.98), A, B are indices for spinor components, and $\mathcal{T}$ is the time-ordering operator.

However, the word ‘propagator’ means two point Green’s functions in general, and other kinds of propagators can be written down for Majorana fermions. For example, consider the two-point function $\langle 0 | \mathcal{T}(\psi_A(x) \psi_B(y)) | 0 \rangle$. If the fermion is Dirac type, this is obviously zero since matrix elements of the form $\langle 0 | \psi_A \psi_B | 0 \rangle$ vanish for Dirac fermions. However, a Majorana field operator can create and annihilate a particle, so such matrix elements are not zero. Indeed, for a Majorana field, using Eqs. (4.17) and (4.27), we can write

$$\bar{\psi} = \lambda \widehat{\bar{\psi}} = \lambda \psi^\top C^{-1}, \quad (4.100)$$

so that

$$\psi^\top = \lambda^* \bar{\psi} C. \quad (4.101)$$

Taking the matrix element of both sides, we see that

$$\begin{aligned} \langle 0 | \mathcal{T}(\psi_A(x) \psi_B(y)) | 0 \rangle &= \lambda^* C_{DB} \langle 0 | \mathcal{T}(\psi_A(x) \bar{\psi}_D(y)) | 0 \rangle \\ &= \lambda^* \int \frac{d^4 p}{(2\pi)^4} e^{ip \cdot (x-y)} [iS_F(p)C]_{AB}. \end{aligned} \quad (4.102)$$

In an obvious notation, we can write the corresponding momentum space propagator as

$$S_{\psi\psi}(p) = \lambda^* S_F(p) C. \quad (4.103)$$

Similarly, using the matrix elements of Eq. (4.100), we find

$$\begin{aligned} \langle 0 | \mathcal{T}(\bar{\psi}_A(x) \bar{\psi}_B(y)) | 0 \rangle &= \lambda (C^{-1})_{DA} \langle 0 | \mathcal{T}(\psi_D(x) \bar{\psi}_B(y)) | 0 \rangle \\ &= \lambda \int \frac{d^4 p}{(2\pi)^4} e^{ip \cdot (x-y)} [i C^{-1} S_F(p)]_{AB} \end{aligned} \quad (4.104)$$

or, in momentum space,

$$S_{\bar{\psi}\bar{\psi}}(p) = \lambda C^{-1} S_F(p). \quad (4.105)$$

□ **Exercise 4.11** Because of the anticommutation property of fermion fields, one must expect

$$\langle 0 | \mathcal{T}(\psi_A(x) \psi_B(y)) | 0 \rangle = - \langle 0 | \mathcal{T}(\psi_B(y) \psi_A(x)) | 0 \rangle. \quad (4.106)$$

Show that this property is indeed inherent in the expression obtained in Eq. (4.102). [Hint: The matrix C is antisymmetric, and Eq. (4.24) implies that $\gamma_\mu C$ is a symmetric matrix.]

We can similarly find the Feynman rules for vertices involving Majorana fermions. The crucial thing to remember is that the plane wave expansion of a Majorana field, given in Eq. (4.16), contains both creation and annihilation operators for the Majorana particles. Thus, a particle and equally well be created or annihilated at a vertex. For example, suppose we have an interaction term in the Lagrangian of the form $\bar{\Psi}\psi\phi$, where Ψ is a Dirac fermion and ψ is a Majorana fermion. This term will contribute to a vertex where the Ψ -particle is created and ψ -particle is annihilated, but also to the vertex where both particles are created.

Feynman rules for a specific situation can be worked out easily, and we help out here by giving one example. Suppose a Majorana particle is in the initial state of a process, and it is annihilated at a vertex where the interaction is $\bar{\Psi}\psi\phi$. Obviously the Feynman rule for the external line here would be $u(k)$, where k is the momentum of the particle. This is just like the corresponding rule for a Dirac particle. However, if the Majorana particle is created along with the Ψ particle and appears in the final state of a process, Eq. (4.16) tells us that the Feynman rule for it should be $\lambda v(k)$, where λ is the creation phase factor.

4.8 Diagonalization of fermion mass matrices

To find the eigenvalues of the Hamiltonian operator in quantum mechanics, we diagonalize it. For this, we use the theorem of matrix algebra that if a matrix A is normal (i.e., it commutes with its hermitian conjugate) then there exists a unitary matrix U such that $UAU^\dagger$ is diagonal. Hamiltonian operator for any system is hermitian and hence normal.

There are two reasons why this method of unitary transformation is insufficient for diagonalization of fermion mass matrices. First, the mass matrices in gauge theory are not necessarily normal. Second, even if they turn out to be normal in a specific situation, a unitary transformation does not guarantee that the diagonalized matrix will have only non-negative numbers along the diagonal. The non-negativity is essential in order to interpret the diagonal elements as masses of physical particles.

To this end, we prove another theorem of matrix algebra. Obviously, $A^\dagger A$ is hermitian for any matrix A . It can be diagonalized by a unitary transformation:

$$U^\dagger A^\dagger A U = D^2. \quad (4.107)$$

Writing the matrix elements explicitly, we see that

$$(D^2)_{aa} = \sum_b |(AU)_{ba}|^2, \quad (4.108)$$

which are real and non-negative. Define a diagonal matrix D whose elements are the positive square roots of those of D^2 . If we now define the matrix

$$H \equiv U D U^\dagger, \quad (4.109)$$

this H would be a hermitian matrix satisfying the equation

$$A^\dagger A = H^2. \quad (4.110)$$

It is now trivial to show that AH^{-1} is a unitary matrix which we will call U' . Then $A = U'H = VDU^\dagger$, where $V = U'U$ is a unitary matrix. Thus, $V^\dagger AU = D$, which shows that:

for any matrix A , one can find two unitary matrices U and V such that $V^\dagger AU$ is diagonal with non-negative elements.

In the above proof, we assumed that H is invertible. For a square matrix A , this implies by Eq. (4.110) $\det A \neq 0$, i.e., A is invertible. However, the result is valid even for singular matrices. Succinctly, we can say that any matrix can be non-negatively diagonalized by a *biunitary transformation*.

- **Exercise 4.12** Show that, for any $n \times n$ matrix A which has one zero eigenvalue, one can define two matrices Q_L and Q_R such that

$$Q_L^\top A Q_R = \begin{pmatrix} 0 & 0 \\ 0 & A' \end{pmatrix}, \quad (4.111)$$

where A' is a $(n-1) \times (n-1)$ non-singular matrix. Generalize this result for matrices for which the zero eigenvalue has a degeneracy k , and use this to find a proof that A can be diagonalized by a biunitary transformation even if it is singular.

In physical terms, a matrix makes sense when it connects two vectors. A biunitary transformation then means that these two vectors must be *rotated* by different amounts in order to obtain the eigenvalues. It makes sense when the two vectors sandwiching the matrix are different. But this is always the case for a fermion mass matrix, which connects left chiral fermions to right chiral ones. Thus, a biunitary transformation can always be applied. Of course, it is important to realize that U and V mentioned above are not unique. If $A = VDU^\dagger$, then A is also equal to $\tilde{V}D\tilde{U}^\dagger$, where $\tilde{U} = KU$ and $\tilde{V} = KV$ for some diagonal unitary matrix K . But this arbitrariness relates only to the overall phases of the physical fermions states and not to their mass eigenvalues.

One special case of our interest is that of symmetric matrices. From $A = VDU^\dagger$, we get $A^\top = U^*DV^\top$. If $A = A^\top$, we thus see that we can choose $V = U^*$. In other words, for any symmetric matrix A , there exists a unitary matrix U such that $U^\top A U$ is diagonal with non-negative elements.

The last statement must be used with some caution. Consider a case when a Majorana mass matrix is real for some reason like CP-conservation. A symmetric real matrix is hermitian. It is therefore tempting to think that U is just an orthogonal matrix, since real hermitian matrices can be diagonalized by orthogonal transformation.

The point to remember is that, if A is real hermitian, $O^\top A O$ is diagonal for some orthogonal matrix O , but the diagonal elements are not necessarily non-negative. However, suppose it has a negative element

at the n^{th} place, then we can define $U = OK$, where K is a diagonal matrix whose n^{th} diagonal entry is i and the others are 1. Then $U^{\text{T}}AU$ is non-negative diagonal, but this U is not an orthogonal matrix. In the CP-conserving case, it can be shown that the entries in K^2 are proportional to the CP-eigenvalues of the mass eigenstates.

- **Exercise 4.13** In the general biunitary transformation $V^{\dagger}AU = D$, we chose $V = U^*$ for symmetric matrices A . Argue that a more general prescription consists of choosing $V = U^*\Lambda^*$, where Λ is a diagonal matrix of phases along the diagonal. Show that if we apply it for the symmetric mass matrices of Majorana neutrinos,, the creation phase of the α^{th} eigenstate will be given by Λ_{α} .

Chapter 5

Neutrino oscillations

In almost all experiments that are performed with neutrinos, the neutrinos are produced by charged current weak interactions. If neutrino mixing exists, the charged current can produce any physical neutrino in conjunction with a charged lepton, as shown, e.g., in Eq. (7.9). Thus the neutrino beam produced is a superposition of different particle eigenstates. As the beam propagates, different components of this beam evolve differently, so that the probability of finding different eigenstates in the beam varies with time. This consequence of neutrino mixing, first suggested by Pontecorvo [1], is called neutrino oscillation.

The physics is exactly similar to that of strangeness oscillation in neutral kaons. Kaons are produced in strong interactions, which produces either K^0 or $\widehat{K}^0$. But none of these is a physical particle. Rather, they are superpositions of the physical states K_L and K_S , which do not have any well-defined value of strangeness as K^0 and $\widehat{K}^0$ do. As the original K^0 (or $\widehat{K}^0$) beam evolves, the proportion of K^0 and $\widehat{K}^0$ changes in the superposition. In fact, since K_S has a very short lifetime, the K_S component of the original beam decays quickly. Thus, no matter whether one starts with a K^0 beam or $\widehat{K}^0$, after a while one obtains almost a pure K_L beam, which contains about the same amount of K^0 and $\widehat{K}^0$.

5.1 Theory of neutrino oscillations

In this section, we discuss the mathematical formulation of neutrino propagation through the vacuum. Quite apart from its theoretical beauty, the formalism is also important because many oscillation experiments have either been performed or being planned. Thanks to the

experiments done so far, we now have several positive evidences for neutrino oscillations, with far reaching implications for neutrino physics. We will discuss them in §5.2.3.

5.1.1 Oscillation formula for mono-energetic neutrinos

Consider a neutrino beam created in a charged current interaction along with the antilepton $\widehat{\ell}$. By definition, the neutrino created is called ν_ℓ . In general, this is not a physical particle, but rather is a superposition of the physical fields ν_α with different masses m_α :

$$|\nu_\ell\rangle = \sum_{\alpha} U_{\ell\alpha} |\nu_\alpha\rangle, \quad (5.1)$$

where U is a unitary matrix which signifies neutrino mixing.

For a simple-minded approach to the propagation of this state, we assume that the 3-momentum $\mathbf{p}$ of the different components in the beam are the same. However, since their masses are different, the energies of all these components cannot be equal. Rather, for the component ν_α , the energy is given by the relativistic energy-momentum relation

$$E_\alpha = \sqrt{\mathbf{p}^2 + m_\alpha^2}. \quad (5.2)$$

After a time t , the evolution of the initial beam of Eq. (5.1) gives

$$|\nu_\ell(t)\rangle = \sum_{\alpha} e^{-iE_\alpha t} U_{\ell\alpha} |\nu_\alpha\rangle. \quad (5.3)$$

In writing this, we assume that the neutrinos ν_α are stable particles. If that is not the case, the entire analysis below is modified, as is discussed in §5.4.

Since all E_α 's are not equal if the masses are not, Eq. (5.3) represents a different superposition of the physical eigenstates ν_α compared to Eq. (5.1). In general, this state has not only the properties of a ν_ℓ , but also of other flavor states. The amplitude of finding a $\nu_{\ell'}$ in the original ν_ℓ beam is

$$\begin{aligned} \langle \nu_{\ell'} | \nu_\ell(t) \rangle &= \sum_{\alpha, \beta} \langle \nu_\beta | U_{\beta\ell'}^\dagger e^{-iE_\alpha t} U_{\ell\alpha} | \nu_\alpha \rangle \\ &= \sum_{\alpha} e^{-iE_\alpha t} U_{\ell\alpha} U_{\ell'\alpha}^* \end{aligned} \quad (5.4)$$

using the fact that the mass eigenstates are orthonormal:

$$\langle \nu_\beta | \nu_\alpha \rangle = \delta_{\alpha\beta}. \quad (5.5)$$

At $t = 0$, as expected, the amplitude is just $\delta_{\ell\ell'}$, using the unitarity of the matrix U . At any time t , the probability of finding a $\nu_{\ell'}$ in an originally ν_ℓ beam is

$$\begin{aligned} P_{\nu_\ell \nu_{\ell'}}(t) &= |\langle \nu_{\ell'} | \nu_\ell(t) \rangle|^2 \\ &= \sum_{\alpha, \beta} \left| U_{\ell\alpha} U_{\ell'\alpha}^* U_{\ell\beta}^* U_{\ell'\beta} \right| \cos \left[(E_\alpha - E_\beta)t - \varphi_{\ell\ell'\alpha\beta} \right], \end{aligned} \quad (5.6)$$

where

$$\varphi_{\ell\ell'\alpha\beta} = \arg(U_{\ell\alpha} U_{\ell'\alpha}^* U_{\ell\beta}^* U_{\ell'\beta}). \quad (5.7)$$

In all practical situations, neutrinos are extremely relativistic, so that we can approximate the energy-momentum relation as

$$E_\alpha \approx |\mathbf{p}| + \frac{m_\alpha^2}{2|\mathbf{p}|}, \quad (5.8)$$

and can also replace t by the distance x traveled by the beam. Thus we obtain

$$P_{\nu_\ell \nu_{\ell'}}(x) = \sum_{\alpha, \beta} \left| U_{\ell\alpha} U_{\ell'\alpha}^* U_{\ell\beta}^* U_{\ell'\beta} \right| \cos \left(\frac{2\pi x}{L_{\alpha\beta}} - \varphi_{\ell\ell'\alpha\beta} \right), \quad (5.9)$$

where, writing $|\mathbf{p}| = E$ for the sake of brevity, we defined

$$L_{\alpha\beta} \equiv \frac{4\pi E}{\Delta m_{\alpha\beta}^2}, \quad (5.10)$$

where

$$\Delta m_{\alpha\beta}^2 \equiv m_\alpha^2 - m_\beta^2. \quad (5.11)$$

The quantities $|L_{\alpha\beta}|$ are called the *oscillation lengths*, which give a distance scale over which the oscillation effects can be appreciable.

Again, in Eq. (5.9), notice that if the distance x is an integral multiple of all $L_{\alpha\beta}$, we obtain $P_{\nu_\ell \nu_{\ell'}} = \delta_{\ell\ell'}$, as in the original beam. But at distances where that condition is not satisfied, we can see nontrivial effects, which are sought for in the experiments.

- **Exercise 5.1** In deriving the oscillation probabilities in this section, we assumed that the 3-momentum of all constituent eigenstates are equal. Show that, when Eq. (5.8) is used, the same probabilities are obtained if instead we assume that the energies of all constituent eigenstates are equal.
- **Exercise 5.2** Consider N generations of neutrinos. If the energies are such that the oscillatory terms average to zero, show that

$$P_{\nu_e \nu_e} \geq \frac{1}{N}. \quad (5.12)$$

It is convenient to analyze the oscillation data in terms of the simplest assumption, viz., there is oscillation between two Dirac neutrinos only. In this case, the matrix U takes a particularly simple form:

$$U = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}. \quad (5.13)$$

Eq. (5.9) now reduces to

$$\begin{aligned} P_{\text{conv}}(x) &= \sin^2 2\theta \sin^2 \left(\frac{\Delta m^2}{4E} x \right) \\ P_{\text{surv}}(x) &= 1 - \sin^2 2\theta \sin^2 \left(\frac{\Delta m^2}{4E} x \right) \end{aligned} \quad (5.14)$$

where $\Delta m^2 = |m_1^2 - m_2^2|$, and the subscripts ‘conv’ and ‘surv’ on the left sides of these equations denote the conversion and the survival probabilities of a particular flavor of neutrino. The experimental data thus restricts Δm^2 as a function of $\sin^2 2\theta$ from the limits of the observed probabilities.

- **Exercise 5.3** For two generations of Majorana neutrinos, the most general form for the mixing matrix is

$$U = \begin{pmatrix} \cos \theta & e^{-i\varrho} \sin \theta \\ -e^{i\varrho} \sin \theta & \cos \theta \end{pmatrix}. \quad (5.15)$$

Show that the phase ϱ does not appear in oscillation probabilities, so that the probabilities are still given by Eq. (5.14).

5.1.2 Oscillation formula for three flavors

In the real world, there are three neutrinos, so the two-generation treatment given above holds strictly only in extreme cases, e.g. when one

of the neutrinos do not mix with the other two. Similar formulas can be derived for the realistic case when all three neutrinos mix with one another. We can start from general formulas like that given in Eq. (5.9) and see what they imply for the three generation case.

To keep things simple, we will assume the CP conserving case, when the mixing matrix U is real. Separating the $\alpha = \beta$ terms from the other terms in Eq. (5.9), we can write the oscillation probability as

$$P_{\nu_\ell \nu_{\ell'}}(x) = \sum_{\alpha} \left(U_{\ell\alpha} U_{\ell'\alpha} \right)^2 + 2 \sum_{\alpha > \beta} U_{\ell\alpha} U_{\ell'\alpha} U_{\ell\beta} U_{\ell'\beta} \cos \left(\frac{\Delta m_{\alpha\beta}^2}{2E} x \right). \quad (5.16)$$

This can be rewritten as

$$\begin{aligned} P_{\nu_\ell \nu_{\ell'}}(x) &= \left(\sum_{\alpha} U_{\ell\alpha} U_{\ell'\alpha} \right)^2 \\ &\quad - 4 \sum_{\alpha > \beta} U_{\ell\alpha} U_{\ell'\alpha} U_{\ell\beta} U_{\ell'\beta} \sin^2 \left(\frac{\Delta m_{\alpha\beta}^2}{4E} x \right) \\ &= \delta_{\ell\ell'} - 4 \sum_{\alpha > \beta} U_{\ell\alpha} U_{\ell'\alpha} U_{\ell\beta} U_{\ell'\beta} \sin^2 \left(\frac{\Delta m_{\alpha\beta}^2}{4E} x \right), \end{aligned} \quad (5.17)$$

using the orthogonality of the mixing matrix in the last step.

This form is still general in the sense that it applies for any number of generations. For three generations, simplified forms can be obtained in some limiting cases. Consider, for example, the case where one of the mass squared differences is very small. To be specific, let us assume that

$$(\Delta m_{12}^2/2E)x \ll 1. \quad (5.18)$$

In this case, we will neglect this mass squared difference altogether. Of course then $\Delta m_{31}^2 = \Delta m_{32}^2$. It is then easy to see that Eq. (5.17) reduces to the form [2]

$$P_{\nu_\ell \nu_{\ell'}}(x) = \delta_{\ell\ell'} - 4U_{\ell 3}U_{\ell' 3} \left(\delta_{\ell\ell'} - U_{\ell 3}U_{\ell' 3} \right) \sin^2 \left(\frac{\Delta m_{32}^2}{4E} x \right). \quad (5.19)$$

Note that in case of a conversion probability, i.e., when $\ell \neq \ell'$, the result looks very much like that for the two-flavor case.

For three generations, the most general mixing matrix ignoring CP violation is given by

$$U = \begin{pmatrix} c_{12}c_{13} & -s_{12}c_{13} & s_{13} \\ s_{12}c_{23} + c_{12}s_{23}s_{13} & c_{12}c_{23} - s_{12}s_{23}s_{13} & -s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13} & c_{12}s_{23} + s_{12}c_{23}s_{13} & c_{23}c_{13} \end{pmatrix}, \quad (5.20)$$

where $c_{\alpha\beta} = \cos \theta_{\alpha\beta}$ and $s_{\alpha\beta} = \sin \theta_{\alpha\beta}$. Using this, we obtain that the survival probability for electron neutrinos [2] is given by

$$P_{\nu_e\nu_e}(x) = 1 - \sin^2 2\theta_{13} \sin^2 \left(\frac{\Delta m_{32}^2}{4E} x \right). \quad (5.21)$$

In other words, the survival probability is given by the same expression used for the two-flavor cases, with the mixing angle being θ_{13} and the mass squared difference Δm_{32}^2 .

Another interesting limiting case is obtained for

$$(\Delta m_{32}^2/2E)x \gg 1, \quad (\Delta m_{31}^2/2E)x \gg 1. \quad (5.22)$$

In this case, the oscillatory terms with these arguments average out when one integrates over the energy spectrum of the initial beam [see §5.2.2 for more on energy integration]. One then obtains

$$\begin{aligned} P_{\nu_\ell\nu_\ell}(x) = & \delta_{\ell\ell'} - 2U_{\ell 3}U_{\ell' 3}(\delta_{\ell\ell'} - U_{\ell 3}U_{\ell' 3}) \\ & - 4U_{\ell 1}U_{\ell' 1}U_{\ell 2}U_{\ell' 2} \sin^2 \left(\frac{\Delta m_{21}^2}{4E} x \right). \end{aligned} \quad (5.23)$$

The survival probability for ν_e 's, e.g., is given by [2]

$$P_{\nu_e\nu_e}(x) = \cos^4 \theta_{13} \left[1 - \sin^2 2\theta_{12} \sin^2 \left(\frac{\Delta m_{21}^2}{4E} x \right) \right] + \sin^4 \theta_{13}. \quad (5.24)$$

If θ_{13} happens to be very small, this result coincides with the survival probability obtained in Eq. (5.14) for the two-flavor case.

5.1.3 More sophisticated derivations

In §5.1.1, we derived the oscillation probabilities assuming that the neutrino state produced as a definite value of 3-momentum, which is shared

by all the constituent eigenstates. This, of course, cannot be strictly true in any quantum theory because of the uncertainty principle. In particular, if $\mathbf{p}$ is very well determined, the production point of the neutrino has a large uncertainty. If this uncertainty is larger than any of the oscillation lengths, the quantity x , which was the distance traveled by the beam, becomes meaningless. Therefore, a correct treatment should use a wave packet of the neutrino, which would have a spread in momentum.

However, one can argue that in cases of practical interest, the wave packet treatment would make no difference in the prediction of the probabilities [3]. To show this, we restrict ourselves to the 2-generation case for the sake of notational simplicity. The oscillation length in this case will be denoted by L . Let us say that the uncertainty in the production point is δx . As argued above, we must demand $\delta x \ll L$ so that the arguments of the oscillatory factor in Eq. (5.14) makes sense. The uncertainty principle would then imply that the momentum uncertainty δp satisfies

$$\frac{\delta p}{p} \gg \frac{1}{Lp} = \frac{\Delta m^2}{4\pi E^2}, \quad (5.25)$$

where we have used the definition of the oscillation length L from Eq. (5.10) for the last step. As seen from Table 5.1 on page 86, the typical values of E are in the range of 1 MeV or higher, whereas one typically searches for values of Δm^2 which are smaller than a few eV^2 . Thus, the right side of Eq. (5.25) is of order 10^{-12} or less. This implies that, although the momentum uncertainty of the initial wave packet cannot be zero, it can be very small. Hence the formulas for oscillation probabilities derived by assuming a fixed momentum hold with very good accuracy [3].

In more recent times, several authors [4] have performed a field theoretical derivation of the oscillation formula, where the entire process, from the production of the neutrinos to their eventual detection, is considered as a single Feynman graph, where the propagating neutrino appears as an intermediate state. This method allows one to understand issues like that of coherence in a rigorous way. However, the final formula for neutrino oscillation probability obtained is the same as that in Eq. (5.9).

5.2 Experimental searches

5.2.1 Basic strategies

The goal of neutrino oscillation experiments is to look for the effects when the distance x in Eq. (5.9) is far from any integral multiple of all $L_{\alpha\beta}$. If it is so, there will be two kinds of consequences.

Firstly, we will obtain

$$P_{\nu_\ell\nu_\ell}(x) < 1, \quad (5.26)$$

so that we will conclude that some of the ν_ℓ of the original beam have disappeared. Looking for such effects are thus usually called *disappearance experiments*. In a typical experiment of this sort, one takes a $\hat{\nu}_e$ beam, let it travel for a certain distance, and let it hit a target. Owing to charged current weak interaction, a $\hat{\nu}_e$ -beam will produce positrons by inverse beta-decay processes:

$$\hat{\nu}_e + X \rightarrow e^+ + Y, \quad (5.27)$$

where X and Y are two nuclei. If there is no neutrino oscillation, one can calculate the flux of positrons thus produced. If the observed flux is less, the implication is that some $\hat{\nu}_e$ has disappeared from the original beam, thus providing evidence for neutrino oscillation.

Of course such disappearance must be compensated by the appearance of other flavor, i.e., by the fact that

$$P_{\nu_\ell\nu_{\ell'}}(x) > 0 \quad \text{for } \ell' \neq \ell. \quad (5.28)$$

This raises the possibility of a second kind of experiments, called *appearance experiments* where, for example, one starts with a $\hat{\nu}_\mu$ beam, let it travel through some distance, and let it hit a target to see if it can produce any e^+ . In the absence of neutrino oscillation, it would be impossible.

From this discussion, it might seem that the appearance experiments are much simpler, since the observation of even one event of the kind sought for would provide evidence of neutrino oscillation, whereas in a disappearance experiment one must meticulously observe a significant deviation of a certain probability from unity. But there is another side to it. An appearance experiment can look only for a specific channel,

Table 5.1: Order of magnitude estimates of the figure of merit of neutrino oscillation experiments.

Source	Typical values of		
	x in cm	E in MeV	$\overline{m}^2$ in eV ²
Reactor	10^2	1	10^{-2}
Meson factory	10^2	10	10^{-1}
Long baseline reactor	10^7	10	10^{-6}
Accelerators	10^3	10^3	1
Atmosphere	10^7	10^4	10^{-3}
Solar core	10^{11}	1	10^{-11}

e.g., a ν_μ oscillating to a ν_e or something like that. If ν_μ mixes very little with ν_e but very strongly with some other state, the $\nu_\mu \rightarrow \nu_e$ appearance experiment would be in bad shape. On the other hand, suppose we are doing a disappearance experiment on a ν_e beam. If there is an observable effect, it will show no matter whether the ν_e oscillates to ν_μ , or ν_τ , or to anything else.

The last point is more interesting for another reason. Later in this book, we discuss many models of neutrino mass which predict some *sterile* neutrinos, i.e., neutral fermions which do not have interactions mediated by the gauge bosons of the standard model. If the ν_e , for example, mixes significantly with one of these sterile neutrinos, an appearance experiment would be fruitless, since the cross section for the production of a charged lepton must be extremely small if one uses a sterile neutrino. On the other hand, a disappearance experiment can still register the depletion in the probability of a ν_e beam. In short, we can say that an appearance experiment measures oscillations $\nu_\ell \rightarrow \nu_{\ell'}$ for specific neutrinos, whereas a disappearance experiment is sensitive to $\nu_\ell \rightarrow \nu_X$ oscillations where ν_X can be any flavor, including the sterile ones.

Sensitivities of all oscillation experiments depend on a particular parameter [5],

$$\overline{m}^2 \equiv \frac{E}{x}, \tag{5.29}$$

which we will call the *figure of merit* of the experiment. Notice that the elements of the mixing matrix U and the mass eigenvalues m_α , m_β come from parameters in the fundamental Lagrangian and we have no handle over them. Thus, the amplitudes of the cosine terms in Eq. (5.9) are fixed for all experiments, and so are the values of $m_\alpha^2 - m_\beta^2$ and $\varphi_{\ell\ell'\alpha\beta}$ in the argument of the cosine. However, one can vary the momentum of the produced neutrinos as well as the distance through which they are let to travel, which appear only through their ratio $\bar{m}^2$. Thus, Eq. (5.9) can be rewritten as

$$P_{\nu_\ell\nu_{\ell'}}(x) = \sum_{\alpha,\beta} \left| U_{\ell\alpha} U_{\ell'\alpha}^* U_{\ell\beta}^* U_{\ell'\beta} \right| \cos \left(\frac{\Delta m_{\alpha\beta}^2}{2\bar{m}^2} - \varphi_{\ell\ell'\alpha\beta} \right), \quad (5.30)$$

using the figure of merit. From this formula, it is clear that if the figure of merit of a particular experiment is much bigger than the values of $|\Delta m_{\alpha\beta}^2|$, any oscillation effect would be hard to observe. A neutrino oscillation experiment is thus sensitive to the values of the mass-square difference satisfying the relation

$$\bar{m}^2 \lesssim |\Delta m_{\alpha\beta}^2|. \quad (5.31)$$

The values of $\bar{m}^2$ are listed in Table 5.1 for different types of neutrino sources. These include laboratory sources as well as atmospheric and solar neutrinos. In this chapter, we will discuss the experimental results of laboratory and atmospheric experiments. The implication of neutrino oscillation on solar neutrinos will be discussed in Ch. 6.

5.2.2 Effect of energy spread

In §5.1.1, we derived neutrino oscillation formula for mono-energetic neutrinos. To understand the nature of the results, we first have to realize that in a given experiment, the neutrinos can never be mono-energetic. Here we are not talking about the uncertainty of energy of a single neutrino in the incoming beam. Rather, we are talking about the spread of energies among different neutrinos in the beam. The spread can be represented by defining an energy spectrum function $\Phi(E)$ of the incoming beam, normalized by the relation

$$\int dE \Phi(E) = 1. \quad (5.32)$$

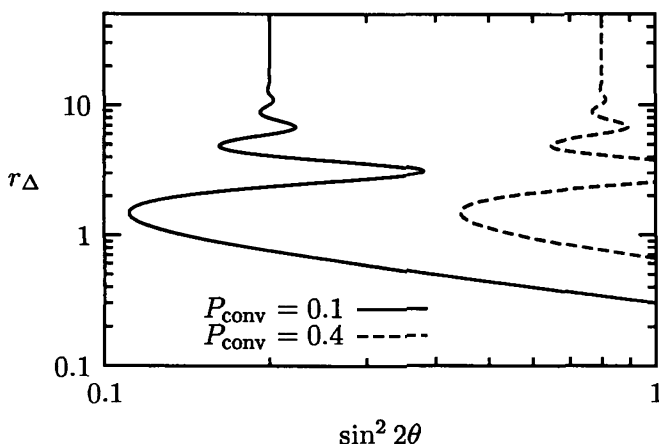


Figure 5.1: Contour of equal conversion probability using Eq. (5.35), for two different values. A Gaussian energy spectrum has been assumed, with a standard deviation of $0.2 \langle E \rangle$.

The function $\Phi(E)$ should be known from the design of the experiment. The conversion probability obtained in the real experiment should be given by

$$P_{\text{conv}}(x) = \sin^2 2\theta \int dE \Phi(E) \sin^2 \left(\frac{\Delta m^2}{4E} x \right). \quad (5.33)$$

Suppose now that an experiment has been performed and one obtained a non-zero conversion probability, say $P_{\text{conv}} = P_0$. To understand what it means, it is convenient to define [6] the dimensionless ratio

$$r_\Delta = \frac{\Delta m^2}{4 \langle E \rangle} x, \quad (5.34)$$

where $\langle E \rangle$ is the mean energy. Using Eq. (5.33), we can then write

$$\begin{aligned} P_0 &= \sin^2 2\theta \int dE \Phi(E) \sin^2 (r_\Delta \langle E \rangle / E) \\ &= \frac{1}{2} \sin^2 2\theta \int dE \Phi(E) [1 - \cos (2r_\Delta \langle E \rangle / E)]. \end{aligned} \quad (5.35)$$

If $r_\Delta \gg 1$, it is instructive to look at the second form of this expression, where the cosine term will have huge fluctuations and will average to

zero. This means that we will have

$$\sin^2 2\theta = 2P_0 \quad \text{for } r_\Delta \gg 1. \quad (5.36)$$

This shows that, in a plot of $\sin^2 2\theta$ along the x -axis and r_Δ on the y -axis, the contours of equal conversion probability are vertical at the high end of the plot, as can be seen in Fig. 5.1.

On the other hand, if $r_\Delta \ll 1$, we can use the first form of the expression in Eq. (5.35), replace the sine function by its argument, and obtain

$$P_0 = r_\Delta^2 \langle E \rangle^2 \sin^2 2\theta \int dE \Phi(E) (1/E)^2, \quad (5.37)$$

which can be written as [6]

$$r_\Delta^2 \sin^2 2\theta = \frac{P_0}{\langle E \rangle^2 \langle 1/E^2 \rangle} \quad \text{for } r_\Delta \ll 1. \quad (5.38)$$

Since the right hand side is a constant in a given experiment, it shows that in a log-log plot of r_Δ vs $\sin^2 2\theta$, the lower part therefore has a slope of $-\frac{1}{2}$, as seen in Fig. 5.1. Moreover, for a variable such as E which takes only positive values, it is easy to show that $\langle E \rangle^2 \langle 1/E^2 \rangle > 1$, because high values dominate $\langle E \rangle$ whereas low values dominate $\langle 1/E^2 \rangle$. Therefore the lowest value of r_Δ , obtained in the contour plot when $\sin^2 2\theta = 1$, is smaller than $\sqrt{P_0}$, by a factor which depends on the details of the energy spectrum [6]. Higher spread in the incoming neutrino spectrum allows one to reach lower values of r_Δ .

- **Exercise 5.4** Take the energy spectrum used for the plot in Fig. 5.1. Calculate the right hand side of Eq. (5.38), and verify that it agrees with the plot.

5.2.3 Results

All early experiments of neutrino oscillations produced null results. From them, one could only put bounds on Δm^2 and mixing. For example, suppose a disappearance experiment was performed and one obtained only the limit $P_{\text{conv}} < 0.1$. It would mean that the region above and to the right of the lower line in Fig. 5.1 would be ruled out.

In the recent years, some positive indication of neutrino oscillation have been obtained. The first experiment to report positive indications

was the Los Alamos LSND experiment [7]. In this experiment, 800 MeV protons were dumped on a target to produce pions. Because of charge conservation, π^+ was produced predominantly. These pions decayed: $\pi^+ \rightarrow \mu^+ + \nu_\mu$. The μ^+ 's produced in this decay subsequently decayed to $e^+ + \nu_e + \bar{\nu}_\mu$. Thus, they had ν_e , ν_μ and $\bar{\nu}_\mu$ in equal amounts, though not with the same energy distribution.

They performed two different experiment, both of the appearance type. In the first experiment [7] they searched for $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ oscillations, using $\bar{\nu}_\mu$ from decays of stopped pions. They obtained a conversion probability of $(3.1^{+1.1}_{-1.0} \pm 0.5) \times 10^{-3}$. Later, they used pion decays in flight and searched for $\nu_\mu \rightarrow \nu_e$ oscillations [8]. By CPT invariance, the mixing angle and the mass squared difference should be the same in both cases, and indeed they found a conversion probability $(2.6 \pm 1.0 \pm 0.5) \times 10^{-3}$, which is consistent with the earlier result. In the parameter space, they yield overlapping allowed regions in the two experiments.

Two other experiments have searched for neutrino oscillation in the overlapping Δm^2 and $\sin^2 \theta$ with negative results. These are the experiments by the E776 group at Brookhaven [9] and by the KARMEN [10] group at Rutherford Laboratory. The KARMEN experiment has very strongly constrained the parameter range allowed by the LSND data [10]. Currently the MiniBooNE experiment [11] at Fermilab is under way to probe the entire parameter region allowed by the LSND.

It should be noted that experiments such as LSND cannot probe values of Δm^2 much smaller than 10^{-2} eV^2 . On the other hand, data on atmospheric and solar neutrinos definitely indicated that the interesting region lies at smaller values of Δm^2 . Several strategies can be employed to probe smaller values. As seen from the discussion of §5.2.2, experiments really determine the quantity r_Δ , defined in Eq. (5.34), as a function of the mixing angle. If we perform experiments with smaller values of $\langle E \rangle$, the same value of r_Δ allows us to probe smaller values of Δm^2 [6]. Hence, reactor sources were preferred in later experiments, where the average energy is in the region of a few MeV. Alternatively, if one can set up an experiment with a large distance between the source and the detector, i.e., x is large in Eq. (5.34), small values of Δm^2 are probed. Such experiments are called *long baseline experiments*. A big energy spread in the incoming neutrino beam also allows one to reach small values of r_Δ [6], as is obvious from Eq. (5.38).

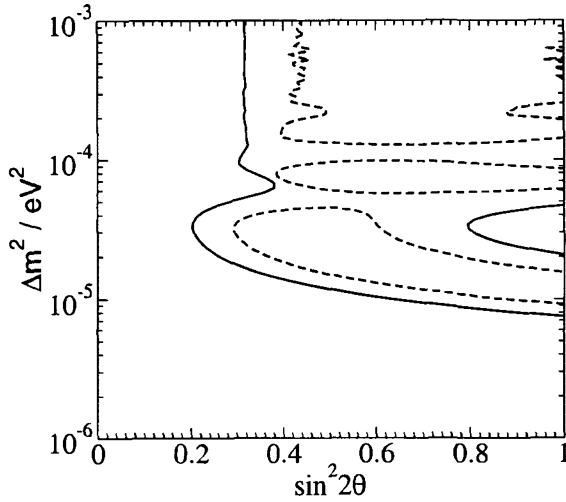


Figure 5.2: The parameter space allowed by Kamland data. The region between the solid lines is allowed by the total rate at 95% CL. The regions enclosed by the dashed lines are allowed by the energy spectrum data. (Courtesy Abhijit Bandyopadhyay.)

The CHOOZ experiment [12] was the first step in these directions. They used a disappearance experiment using electron antineutrinos from a reactor. The average energy was 3 MeV and the source to detector distance 1 km. With these values, they could probe as low as $\Delta m^2 = 8 \times 10^{-4} \text{ eV}^2$ for $\sin^2 2\theta = 1$, but they still obtained a null result.

A big advancement was made by the Kamland experiment, whose first results were published in 2003 [13]. It is also disappearance experiment like CHOOZ, in that they use electron antineutrinos as the initial beam, although from multiple reactors. The big difference with CHOOZ is that it is a large baseline experiment because the average distance between the sources and the detector is about 200 km. They observe a depletion of the flux by 34%, which provides evidence for oscillation. Because the conversion rate is high, the mixing angle coming out from this analysis is quite large, same order as the maximal mixing of 45° . The analysis of this data have been performed by several groups [13, 14], and allowed region from their analysis is presented in Fig. 5.2.

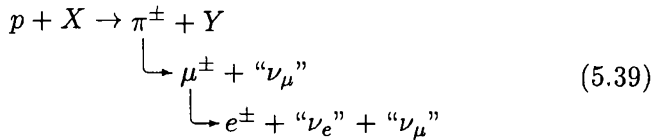
Similarly, the K2K experiment using a multi-GeV muon neutrino beam from an accelerator also sees evidence of a depletion of flux

[15]. The statistics in these experiments is limited, but nonetheless they do provide evidence for $\nu_\mu \rightarrow \nu_\tau$ oscillation. The lowest values in the allowed region is roughly $\sin^2 2\theta = 0.2$ for large Δm^2 , and $\Delta m^2 = 5 \times 10^{-4} \text{ eV}^2$ for $\sin^2 2\theta = 1$.

5.3 Atmospheric neutrinos

The interaction of cosmic ray protons with the nuclei in upper atmosphere leads to production of both electron and muon type neutrinos. Since the early 1990s, these neutrinos have been observed and analyzed in detail in several experiments [16, 17, 18, 19]. These observations provide sensitive information on the nature of neutrino oscillations. As we discuss below, the present observations can be interpreted as positive indications of oscillations between different neutrino species. In order to see how one arrives at this conclusion, let us first summarize the observations and theoretical expectations for the case of massless neutrinos, for which there are no oscillations.

The atmospheric neutrinos are supposed to originate in the following chain of reactions:



Here $\pi^\pm$'s could be replaced by $K^\pm$'s. The notation " ν_μ ", for example, means that we have not distinguished between ν_μ 's and $\bar{\nu}_\mu$'s as they are not distinguished in the experiments. Naive counting argument then says that the ratio r of ν_e -type to ν_μ -type events in the underground detectors (not distinguishing between particles and antiparticles) is 0.5. The experiments count the number of e - and μ -type events produced in the collisions of the neutrinos with the water detector. The events may either be fully contained or partially contained. We will consider mostly the contained events.

In this class of events, the neutrino-nucleus interaction vertex is located inside the detector as are all final state particles. The charged lepton energies for these events range from a few hundred MeV to 1.2 GeV. Using the contained events of electron- and muon- type, one con-

structs the following double ratio:

$$R \equiv \frac{(\mu/e)}{(\mu/e)_{\text{MC}}} \tag{5.40}$$

where MC stands for Monte-Carlo predictions. The denominator gives the theoretical expectation for (μ/e) ratio assuming neutrinos are massless and all their interactions are given by the standard model. An advantage of considering μ/e is that the cosmic ray flux uncertainties do not effect this result. If there are no neutrino oscillations, R should be 1 [20].

Early experiments obtained the following values for this double ratio:

Kamiokande [16]	$.60 \pm .06 \pm .05$
IMB [17]	$.54 \pm .05 \pm .12$
Sudan2 [18]	$.69 \pm .19 \pm .09$
FREJUS [19]	$.87 \pm .21$
NUSEX [19]	$.99 \pm .40$

The first two experiments use water Cerenkov detectors whereas the next three use iron-calorimeters. The last three have much lower statistics than the first two. In the water Cerenkov detectors, the muons are distinguished from the electrons in two ways: by the size of the rings of Cerenkov light or by observing decay products from the muons.

The first three sets of data indicate that there has either been a depletion of muon neutrinos or an excess of electron-type events due to some non-standard property of the neutrinos. To confirm this indication, super-Kamiokande collaboration started taking extensive data in the late 1990s. From a data of 4353 events, they observed [21] that the double ratio defined in Eq. (5.40) is considerably smaller than 1. Over the next few years, they increased the data set [22] by about a factor of 3, and the earlier results have been refined. The results are as follows:

Super-Kamiokande [21]	sub-GeV	$0.63 \pm 0.03 \pm 0.05$
	multi-GeV	$0.65 \pm 0.05 \pm 0.08$
Super-Kamiokande [22]	sub-GeV	$0.638^{+0.016}_{-0.016} \pm 0.050$
	multi-GeV	$0.658^{+0.030}_{-0.028} \pm 0.078$

Because of their large statistics, these are now considered to be the standard results, irrespective of the results of other experiments. We now discuss the implication of these results.

The simplest way to understand the atmospheric neutrino puzzle is to assume that the neutrinos oscillate into other species, thereby leading to observed values of the fluxes in the underground than what would be expected on the basis of the standard model. If $\nu_e \rightarrow \nu_\mu$ oscillation is to provide the solution to the solar neutrino puzzle which will be discussed in Ch. 6, then the most appealing scenario for the atmospheric neutrino puzzle is to assume that the atmospheric ν_μ oscillates into ν_τ , leaving the ν_e flux unchanged. The observed energy spectra for ν_μ and ν_e by the Kamiokande as well as super-Kamiokande collaboration seem to indicate support for this hypothesis. We will therefore consider [23] $\nu_\mu \rightarrow \nu_\tau$ oscillation to explain the atmospheric neutrino deficit. Analysis of the most recent data then indicate that the oscillation parameters are in the range [22]

$$\begin{aligned}\Delta m_{\nu_\mu \nu_\tau}^2 &\simeq (1.5 \text{ to } 3) \times 10^{-3} \text{ eV}^2, \\ \sin^2 2\theta_{\mu\tau} &> 0.9 \text{ to } 1, \end{aligned} \tag{5.41}$$

at 90% confidence level, with the best values given by

$$\begin{aligned}\Delta m_{\nu_\mu \nu_\tau}^2 &= 2.5 \times 10^{-3} \text{ eV}^2, \\ \sin^2 2\theta_{\mu\tau} &= 1. \end{aligned} \tag{5.42}$$

5.4 Oscillation with unstable neutrinos

The analysis of neutrino oscillation experiments becomes quite different from that described in §5.1.1 if the neutrinos are unstable [24]. For clarity of presentation, we restrict ourselves to the case of two neutrino flavors which we will call ν_e and ν_μ . The mass eigenstates ν_1 and ν_2 are related as usual to these flavor states through Eq. (5.1), with the mixing matrix U given in Eq. (5.13). Let ν_2 be the heavier of the two neutrinos, and let us say that it decays via the mode $\nu_2 \rightarrow \nu_1 X$, where X can be anything. Let the lifetime of this decay be Γ^{-1} .

Starting from the initial beam

$$|\nu_{\text{in}}\rangle = \cos \alpha |\nu_e\rangle + \sin \alpha |\nu_\mu\rangle, \tag{5.43}$$

one then obtains at time t the state

$$\begin{aligned}
 |\psi(t)\rangle &= e^{-iE_1 t} \cos(\theta + \alpha) |\nu_1\rangle + e^{-iE_2 t - \Gamma t/2} \sin(\theta + \alpha) |\nu_2\rangle \\
 &+ \sum_k c_k |\nu_1(k)X\rangle, \quad (5.44)
 \end{aligned}$$

where $|\nu_1(k)X\rangle$ denotes the final states of ν_2 decay where ν_1 has a momentum k . From this, we obtain

$$\begin{aligned}
 P_{\nu_{in}\nu_e}(t) &= \cos^2(\theta + \alpha) \cos^2 \theta + e^{-\Gamma t} \sin^2(\theta + \alpha) \sin^2 \theta \\
 &+ \frac{1}{2} e^{-\Gamma t/2} \sin 2\theta \sin 2(\theta + \alpha) \cos \frac{\Delta m^2 t}{2E} \\
 P_{\nu_{in}\nu_\mu}(t) &= \cos^2(\theta + \alpha) \sin^2 \theta + e^{-\Gamma t} \sin^2(\theta + \alpha) \cos^2 \theta \\
 &- \frac{1}{2} e^{-\Gamma t/2} \sin 2\theta \sin 2(\theta + \alpha) \cos \frac{\Delta m^2 t}{2E}. \quad (5.45)
 \end{aligned}$$

The interesting point is that, even for fast decays, i.e., $\Gamma t \gg 1$, we get

$$\begin{aligned}
 P_{\nu_{in}\nu_e}(t) &= \cos^2(\theta + \alpha) \cos^2 \theta \\
 P_{\nu_{in}\nu_\mu}(t) &= \cos^2(\theta + \alpha) \sin^2 \theta, \quad (5.46)
 \end{aligned}$$

so that none of these probabilities is zero. Despite their fast decays, the ν_μ 's can survive after a long journey.

Since the sum of all probabilities must be unity, we get

$$\sum_k P_{\nu_{in} \rightarrow \nu_1 X} = (1 - e^{-\Gamma t}) \sin^2(\theta + \alpha). \quad (5.47)$$

The ν_1 's produced in the decay of ν_2 are incoherent with respect to the ν_1 's in the original beam. Therefore,

$$\begin{aligned}
 \sum_k P_{\nu_{in} \rightarrow \nu_e X} &= (1 - e^{-\Gamma t}) \sin^2(\theta + \alpha) \cos^2 \theta \\
 \sum_k P_{\nu_{in} \rightarrow \nu_\mu X} &= (1 - e^{-\Gamma t}) \sin^2(\theta + \alpha) \sin^2 \theta. \quad (5.48)
 \end{aligned}$$

To point out the difference of this case with that of stable neutrinos, let us take the specific example of an initial ν_e beam, i.e., the one for which $\alpha = 0$. Specializing now to the case of fast decay, we obtain

$$P_{\nu_e \nu_e}(t) = \cos^4 \theta, \quad (5.49)$$

which can be very small if θ is close to $\pi/2$. However, without decay, this probability would be given by Eq. (5.14). Averaging over the distance, we obtain in that case

$$P_{\nu_e \nu_e}(t) = 1 - \frac{1}{2} \sin^2 2\theta, \quad (5.50)$$

which is greater than $\frac{1}{2}$ for all values of θ . Thus, the physical implications of the instability of neutrinos can be dramatic.

5.5 Neutrino oscillations in matter

Wolfenstein [25] pointed out that the patterns of neutrino oscillation might be significantly affected if the neutrinos travel through a material medium rather than through the vacuum. The basic reason for this is simple. Normal matter has electrons but no muons or taus at all. Thus, if a ν_e beam goes through matter, it encounters both charged and neutral current interactions with the electrons. But a ν_μ or a ν_τ interacts with the electron only via the neutral current, so their interaction is different in magnitude than that of the ν_e .

Interactions modify the effective mass that a particle exhibits while traveling through a medium. An well-known example is that of the photon, which is massless in the vacuum but develops an effective mass in a medium. As a result, electromagnetic waves do not travel with speed c through a medium. The effective masses of neutrinos are similarly modified in a medium by their interactions. Since ν_e has different interaction than the other neutrinos, the modification is different for ν_e than for the other flavored neutrinos.

This fact can have dramatic consequences if the neutrinos mix in the vacuum. In this case, a physical eigenstate can have components of ν_e , ν_μ , ν_τ and other possible states. When such a state travels through a medium, the modulation of its ν_e component is different from the same modulation in the vacuum. This leads to changes in the oscillation probabilities compared to their values in the vacuum.

5.5.1 Uniform matter background

For a quantitative treatment of the above ideas, we first deal with the simple case when the background matter has uniform density. We will

also stick to the simplest case of two flavors ν_e and ν_μ . We first recapitulate the case of vacuum oscillation in a way that is most suitable for the generalization to matter oscillation. If the mass eigenstates are ν_1 and ν_2 , the evolution equation for these states can be written as

$$i \frac{d}{dt} \begin{pmatrix} \nu_1(t) \\ \nu_2(t) \end{pmatrix} = H \begin{pmatrix} \nu_1(t) \\ \nu_2(t) \end{pmatrix}, \quad (5.51)$$

where H is diagonal in this basis:

$$H = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix} \simeq E + \begin{pmatrix} m_1^2/2E & 0 \\ 0 & m_2^2/2E \end{pmatrix}, \quad (5.52)$$

using the approximate expressions of Eq. (5.8). Recalling the form for the matrix U that connects mass eigenstates to flavor states, we can rewrite Eq. (5.51) as

$$i \frac{d}{dt} \begin{pmatrix} \nu_e(t) \\ \nu_\mu(t) \end{pmatrix} = H' \begin{pmatrix} \nu_e(t) \\ \nu_\mu(t) \end{pmatrix}, \quad (5.53)$$

where

$$\begin{aligned} H' &= U H U^\dagger \\ &= E + \frac{m_1^2 + m_2^2}{4E} + \frac{\Delta m^2}{4E} \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix} \end{aligned} \quad (5.54)$$

From this, we can see that the diagonalizing angle θ is given by

$$\tan 2\theta = \frac{2H'_{12}}{H'_{22} - H'_{11}} \quad (5.55)$$

in terms of the elements of the matrix H' , as is usual for a 2×2 matrix.

We now consider the same problem when the neutrinos are traveling through matter. For simplicity, we assume that the density of the background matter is uniform, with n_e , n_p and n_n denoting the number of electrons, protons and neutrons per unit volume. Elastic scattering off these particles change the effective masses of the neutrinos, whose magnitude we now estimate.

Consider first elastic scattering through charged current interactions. The only interaction of this sort is the $\nu_e e$ scattering, since there are no

μ 's or τ 's in the background at normal temperatures. The diagram for this process has been given in Fig. 2.1b on page 29. The contribution of this diagram to the effective Lagrangian is given by

$$i\mathcal{L}_{\text{eff}} = \left(\frac{ig}{\sqrt{2}} \right)^2 \left\{ \bar{e}(p_1) \gamma^\lambda \mathbb{L} \nu_e(p_2) \right\} \frac{ig_{\lambda\rho}}{M_W^2} \left\{ \bar{\nu}_e(p_3) \gamma^\rho \mathbb{L} e(p_4) \right\}, \quad (5.56)$$

where we have neglected the momentum dependence in the W -propagator since we are interested in the leading order contribution in the Fermi constant. Using the relation $g^2/8M_W^2 = G_F/\sqrt{2}$, we thus obtain

$$\begin{aligned} \mathcal{L}_{\text{eff}} &= -\frac{4G_F}{\sqrt{2}} \left\{ \bar{e}(p_1) \gamma_\lambda \mathbb{L} \nu_e(p_2) \right\} \left\{ \bar{\nu}_e(p_3) \gamma^\lambda \mathbb{L} e(p_4) \right\} \\ &= -\frac{4G_F}{\sqrt{2}} \left\{ \bar{\nu}_e(p_3) \gamma_\lambda \mathbb{L} \nu_e(p_2) \right\} \left\{ \bar{e}(p_1) \gamma^\lambda \mathbb{L} e(p_4) \right\}, \end{aligned} \quad (5.57)$$

where the second form is obtained via Fierz transformation. For forward scattering where $p_2 = p_3 = p$, this gives the following contribution that affects the propagation of the ν_e :

$$-\sqrt{2}G_F \bar{\nu}_{eL}(p) \gamma_\lambda \nu_{eL}(p) \left\langle \bar{e} \gamma^\lambda (1 - \gamma_5) e \right\rangle, \quad (5.58)$$

averaging the electron field bilinear over the background. To calculate that average, note that the axial current reduces to spin in the non-relativistic approximation, which is negligible for a non-relativistic collection of electrons. The spatial components of the vector current give the average velocity, which is negligible as well. So the only non-trivial average is

$$\left\langle \bar{e} \gamma^0 e \right\rangle = \left\langle e^\dagger e \right\rangle = n_e, \quad (5.59)$$

which gives a contribution to the effective Lagrangian

$$-\sqrt{2}G_F n_e \bar{\nu}_{eL} \gamma_0 \nu_{eL}. \quad (5.60)$$

- **Exercise 5.5** Write down the equation of motion of a neutrino, using the free Lagrangian augmented by the term in Eq. (5.60). Show that the effect of this term is to change the effective energy of the neutrino, viz.,

$$E = \sqrt{|\mathbf{p}|^2 + m^2} + \sqrt{2}G_F n_e. \quad (5.61)$$

One can interpret the density dependent term as a contribution to the potential energy:

$$V_{cc} = \sqrt{2}G_F n_e. \quad (5.62)$$

Turning now to neutral current contributions, we find in an exactly similar way the following contributions to effective energies of both ν_e and ν_μ :

$$V_{nc} = \sqrt{2}G_F \sum_f n_f \left[I_{3L}^{(f)} - 2 \sin^2 \theta_W Q^{(f)} \right] \quad (5.63)$$

where f stands for the electron, the proton or the neutron, $Q^{(f)}$ is the charge of f and $I_{3L}^{(f)}$ is the third component of weak isospin of the left-chiral projection of f . Thus, for the proton, $Q = 1$ and $I_{3L} = \frac{1}{2}$, whereas for the electron, $Q = -1$ and $I_{3L} = -\frac{1}{2}$. Also, for normal neutral matter, $n_e = n_p$ to guarantee charge neutrality. Therefore the contributions of the electron and the proton cancel each other and we are left with the neutron contribution, which is

$$V_{nc} = -\sqrt{2}G_F n_n / 2. \quad (5.64)$$

We emphasize that this neutral current contribution is the same for all flavors of neutrinos whereas the charged current contribution affects ν_e only. Thus, the evolution equation of neutrino beams should now be given by Eq. (5.53) with H' replaced by

$$\begin{aligned} \tilde{H} = E + \frac{m_1^2 + m_2^2}{4E} - \frac{1}{\sqrt{2}}G_F n_n \\ + \begin{pmatrix} -\frac{\Delta m^2}{4E} \cos 2\theta + \sqrt{2}G_F n_e & \frac{\Delta m^2}{4E} \sin 2\theta \\ \frac{\Delta m^2}{4E} \sin 2\theta & \frac{\Delta m^2}{4E} \cos 2\theta \end{pmatrix}. \end{aligned} \quad (5.65)$$

The effective mixing angle in matter, $\tilde{\theta}$, would accordingly be given by

$$\tan 2\tilde{\theta} = \frac{2\tilde{H}_{12}}{\tilde{H}_{22} - \tilde{H}_{11}} = \frac{\Delta m^2 \sin 2\theta}{\Delta m^2 \cos 2\theta - A}, \quad (5.66)$$

where for the sake of convenience, we defined

$$A = 2\sqrt{2}G_F n_e E. \quad (5.67)$$

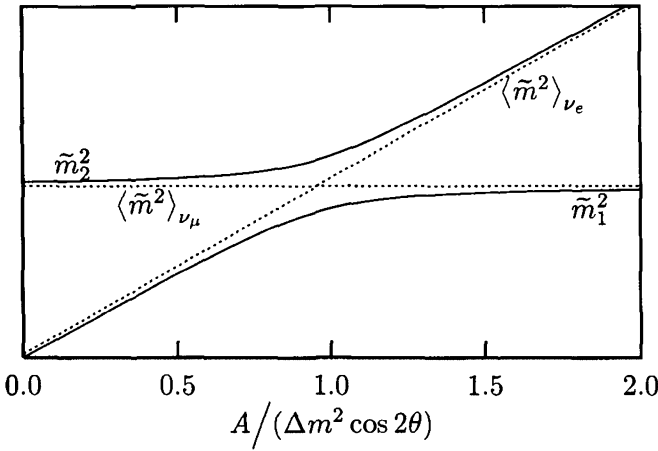


Figure 5.3: The effective masses of neutrinos in matter. The quantity A is proportional to matter density. The solid lines are mass squared values of the physical eigenstates, the dashed lines are the expectation values of squared mass for the states ν_e and ν_μ . We have taken $\theta = 0.15$ for the plot. The vertical scale is arbitrary.

The effective mixing angle thus changes inside matter. The change is most dramatic if $A = \Delta m^2 \cos 2\theta$, i.e., if the electron number density is given by

$$n_e = \frac{\Delta m^2 \cos 2\theta}{2\sqrt{2}G_F E}. \quad (5.68)$$

Then, even if the vacuum mixing angle θ is small, we obtain $\tilde{\theta} = \pi/4$, which is to say that ν_e and ν_μ mix maximally.

To understand the exact nature of this maximal mixing, let us write down the eigenvalues of $\tilde{H}$. The results are

$$E_\alpha = E - \frac{1}{\sqrt{2}}G_F n_n + \frac{\tilde{m}_\alpha^2}{2E}, \quad (5.69)$$

where

$$\tilde{m}_{1,2}^2 = \frac{1}{2} \left[(m_1^2 + m_2^2 + A) \mp \sqrt{(\Delta m^2 \cos 2\theta - A)^2 + (\Delta m^2 \sin 2\theta)^2} \right]. \quad (5.70)$$

The behavior of $\tilde{m}_1^2$ and $\tilde{m}_2^2$ as functions of A are shown schematically in Fig. 5.3. These values for effective mixing and masses, shown in Eqs.

(5.66) and (5.70), should be used if one wants to discuss neutrino oscillation in a background medium of uniform density.

5.5.2 Non-uniform matter background and resonant oscillation

It was realized by Mikheyev and Smirnov [26] that such density dependence of effective mass can produce interesting effect if the background matter density is non-uniform. To get a feeling for the kind of effects, consider a ν_e produced in a region of high density, like in the core of the sun. Assume, for the moment, that θ is very small. For $A = 0$ which corresponds to the vacuum, the lighter eigenstate ν_1 is almost purely ν_e whereas the heavier one, ν_2 , is almost purely ν_μ . But in the high density region where $A \gg \Delta m^2 \cos 2\theta$, $\tilde{\theta} \rightarrow \pi/2$, which means that the lighter eigenstate is almost purely ν_μ whereas the heavier one is ν_e to a good approximation. Thus, if we follow the effective mass squared values of ν_e and ν_μ , as shown by the dotted lines in Fig. 5.3, we see that the effective mass squared value of ν_e starts lower than that of ν_μ but with the increase of A , the difference diminishes until finally ν_e overtakes ν_μ . This phenomenon is known as *level-crossing*.

To analyze this phenomenon in detail, we continue using oscillations between two flavors of neutrinos only, which we call ν_e and ν_μ . Recall Eqs. (5.53) and (5.54), which give the equation governing neutrino propagation in matter. Ignoring the terms proportional to the unit matrix which are unimportant for the discussion about mixing, we can write the equation in the form

$$i \frac{d}{dx} \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = \frac{1}{2E} M^2 \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix}, \quad (5.71)$$

where [25, 26]

$$M^2 = \frac{1}{2} \begin{pmatrix} -\Delta m^2 \cos 2\theta + 2A & \Delta m^2 \sin 2\theta \\ \Delta m^2 \sin 2\theta & \Delta m^2 \cos 2\theta \end{pmatrix}. \quad (5.72)$$

In writing this equation, we have put $x \approx t$ since the neutrinos are relativistic.

The effective mixing angle, which controls the oscillation probability,

is given by

$$\sin^2 2\tilde{\theta} = \frac{(\Delta m^2 \sin 2\theta)^2}{(\Delta m^2 \cos 2\theta - A)^2 + (\Delta m^2 \sin 2\theta)^2}. \quad (5.73)$$

We see that when $A = \Delta m^2 \cos 2\theta$, the mixing angle is maximal and transition from ν_e to ν_μ occurs. This relation is equivalent to Eq. (5.66), but shows more readily that it is a resonance condition, with width $\Gamma \approx \Delta m^2 \sin 2\theta$.

So far, the discussion is no different from that of §5.5.1, where a uniform density of matter was assumed. Let us now imagine that the density is non-uniform, e.g., like that inside the sun decreases on the way out from the core. At any given point, however, the eigenvalues of the matrix M^2 in Eq. (5.72) are given by

$$\tilde{m}_{1,2}^2 = \frac{1}{2} \left[A \mp \sqrt{(\Delta m^2 \cos 2\theta - A)^2 + (\Delta m^2 \sin 2\theta)^2} \right], \quad (5.74)$$

which is essentially the same as Eq. (5.70) apart from the terms which contribute equally to the two eigenvalues. Propagation through matter is determined by these two eigenvalues, with the corresponding eigenstates $\tilde{\nu}_\alpha$ related to the weak eigenstates as follows:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = \begin{pmatrix} \cos \tilde{\theta} & \sin \tilde{\theta} \\ -\sin \tilde{\theta} & \cos \tilde{\theta} \end{pmatrix} \begin{pmatrix} \tilde{\nu}_1 \\ \tilde{\nu}_2 \end{pmatrix}. \quad (5.75)$$

The angle $\tilde{\theta}$ is given in Eq. (5.73). The equation of propagation for the $\tilde{\nu}_\alpha$'s can be obtained [27] by using Eqs. (5.71), (5.72) and (5.75), remembering that $\tilde{\theta}$ depends on the matter density which varies with distance:

$$i \frac{d}{dx} \begin{pmatrix} \tilde{\nu}_1 \\ \tilde{\nu}_2 \end{pmatrix} = \begin{pmatrix} \frac{\tilde{m}_1^2}{2E} & i \frac{d\tilde{\theta}}{dx} \\ -i \frac{d\tilde{\theta}}{dx} & \frac{\tilde{m}_2^2}{2E} \end{pmatrix} \begin{pmatrix} \tilde{\nu}_1 \\ \tilde{\nu}_2 \end{pmatrix}. \quad (5.76)$$

Notice that if the density is constant, the off-diagonal terms in this matrix vanish, so that the states denoted by $\tilde{\nu}_\alpha$ indeed become the stationary eigenstates.

For varying density, one can define the stationary eigenstates only for the Hamiltonian at a given point. To proceed, let us first assume the adiabaticity condition to hold. This means that the density varies

slowly, so that the off-diagonal elements in the matrix in Eq. (5.76) are small. As long as this condition is satisfied, any matter eigenstate $\tilde{\nu}_\alpha$ traverses unchanged with their relative admixture of ν_e and ν_μ changing adiabatically according to the value of electron density at a given point. We argued before that terms proportional to the unit matrix do not affect oscillation probabilities, so the only physical parameter in the diagonal elements must be their difference. Thus the *adiabaticity condition* is

$$\left| \frac{d\tilde{\theta}}{dx} \right| \ll \left| \frac{\tilde{m}_2^2 - \tilde{m}_1^2}{2E} \right|. \quad (5.77)$$

The condition can be rewritten by defining the *adiabaticity parameter*

$$\gamma \equiv \left| \frac{(\tilde{m}_2^2 - \tilde{m}_1^2)/2E}{d\tilde{\theta}/dx} \right|, \quad (5.78)$$

which must be large to ensure adiabatic conditions. Obviously, the value of this parameter depends on the local density, and is smallest at the resonance point, where the mass eigenvalues come closest. Adiabaticity condition is therefore guaranteed if the value of γ is large at the resonance point, i.e., if

$$\gamma_{\text{res}} \gg 1. \quad (5.79)$$

Using the resonant value of $\tilde{\theta}$ from Eq. (5.73) as well as the resonant density, we find

$$\gamma_{\text{res}} = \frac{\Delta m^2}{E} \cdot \frac{\sin^2 2\theta}{\cos 2\theta} \left| \frac{d}{dx} \ln n_e \right|_{\text{res}}^{-1}, \quad (5.80)$$

The physical meaning of the condition in Eq. (5.79) is the following. The resonance condition $A = \Delta m^2 \cos 2\theta$ is approximately satisfied within the half-width Γ , i.e., for a range of electron density

$$\delta n_e \approx \frac{\Gamma}{2\sqrt{2}G_F E}. \quad (5.81)$$

Denoting $\delta n_e = |dn_e/dx|_{\text{res}} \delta x$ and using $\Gamma = \Delta m^2 \sin 2\theta$, we see that the resonance condition is approximately valid in a distance

$$\delta x \approx \frac{\Delta m^2 \sin 2\theta}{|dn_e/dx|_{\text{res}} 2\sqrt{2}G_F E}. \quad (5.82)$$

The oscillation length at resonance, on the other hand, is given by

$$L_{\text{res}} = \frac{4\pi E}{\Delta m^2 \sin 2\theta}, \quad (5.83)$$

following the definition in Eq. (5.10). The ν_μ -fraction of the wave in ν_e has enough time to build if $\delta x \gg L_{\text{res}}$, which is precisely the resonance condition of Eq. (5.80).

To see the essence of resonant matter oscillation (or MSW effect, as it is called after Mikheyev, Smirnov [26] and Wolfenstein [25] who played pivotal roles in discovering this effect) we recall that [28] when the adiabaticity condition is satisfied, $\tilde{\nu}_\alpha$ remains unchanged. Let us assume ν_e 's are produced at a point where the electron density is infinite. Then from Eq. (5.66), we find $\tilde{\theta}_0 = \pi/2$, where the subscript '0' denotes the production point of the neutrino for the rest of this section. Eq. (5.75) then implies that $\tilde{\nu}_2$ corresponds to ν_e . As the beam travels to regions of gradually smaller density, the ν_e component in the beam, however, decreases continuously, since at a general point, $\tilde{\nu}_2 = \nu_e \sin \tilde{\theta} + \nu_\mu \cos \tilde{\theta}$, and $\tilde{\theta}$ decreases with decreasing density. Suppose finally the neutrino beam comes out of matter so that $n_e = 0$, and $\tilde{\theta}$ equals the vacuum mixing angle θ . Thus, the probability of finding the ν_e component of the beam decreases from almost unity to the final value of $\sin^2 \theta$. This is the *resonant conversion* phenomenon, which is most dramatic if θ is very small, so that the ν_e beam is almost totally depleted.

Let us now generalize to the more realistic case where the adiabaticity condition still holds, but the density at the production point is not infinite. Eq. (5.75) now shows that the produced neutrinos are either $\tilde{\nu}_1$ or $\tilde{\nu}_2$, with probabilities $\cos^2 \theta_0$ and $\sin^2 \theta_0$ respectively. If the neutrino is produced as $\tilde{\nu}_1$, it will travel as $\tilde{\nu}_1$, and will be detected as $\tilde{\nu}_1$. At the detection point, the effective mixing angle by θ , which means the probability of detecting ν_e is $\cos^2 \theta$. In the other case when the neutrino is produced as $\tilde{\nu}_2$, a ν_e is detected with probability $\sin^2 \theta$. Adding these two cases, we obtain that the survival probability in the adiabatic case is given by

$$\begin{aligned} P_{\nu_e \nu_e}^{(\text{ad})} &= \cos^2 \theta_0 \cos^2 \theta + \sin^2 \theta_0 \sin^2 \theta \\ &= \frac{1}{2} [1 + \cos 2\tilde{\theta}_0 \cos 2\theta]. \end{aligned} \quad (5.84)$$

There is of course an interference term as well, since we should have

added the amplitudes, not the probabilities, but this interference term turns out to be negligible in matter of practical interest.

- **Exercise 5.6** Assuming that the propagation of neutrinos is adiabatic, show that the survival probability of ν_e 's is given by

$$P_{\nu_e \nu_e}^{(\text{ad})} = \frac{1}{2} \left[1 + \cos 2\tilde{\theta}_0 \cos 2\theta + \sin 2\tilde{\theta}_0 \sin 2\theta \cos \left(\int dx' \frac{\tilde{m}_2^2 - \tilde{m}_1^2}{2E} \right) \right]. \quad (5.85)$$

In the more general case when the adiabaticity condition is not necessarily satisfied, there is a probability that the eigenstate $\tilde{\nu}_1$ will “jump” to the other eigenstate $\tilde{\nu}_2$, and vice versa. This probability of jumping from one eigenstate to another is called the Landau-Zenner-Stuckelberg probability or jumping probability and will be denoted by P_{LZS} . In case no jump occurs, which has a probability $1 - P_{\text{LZS}}$, the survival probability will be the same as the adiabatic survival probability given in Eq. (5.84). On the other hand, a neutrino which would have ended as a ν_μ in the adiabatic case would end as a ν_e if a jump occurs. Combining the two possibilities, we find the survival probability to be [29]

$$\begin{aligned} P_{\nu_e \nu_e} &= (1 - P_{\text{LZS}})P_{\nu_e \nu_e}^{(\text{ad})} + P_{\text{LZS}}P_{\nu_e \nu_\mu}^{(\text{ad})} \\ &= \frac{1}{2} \left[1 + (1 - 2P_{\text{LZS}}) \cos 2\tilde{\theta}_0 \cos 2\theta \right], \end{aligned} \quad (5.86)$$

- **Exercise 5.7** If a neutrino is produced in the far half of the sun so that it undergoes resonance twice before it reaches the earth, once on the way towards the core and once on its way out, show that

$$P_{\nu_e \nu_e} = \frac{1}{2} \left[1 + (1 - 2P_{\text{LZS}})^2 \cos 2\tilde{\theta}_0 \cos 2\theta \right]. \quad (5.87)$$

It now remains to calculate the jump probability P_{LZS} . In the semi-classical approximation which is valid if γ is not very small, this is done by using Landau's method of complex trajectory [30], which gives

$$\ln P_{\text{LZS}} = -\frac{1}{E} \text{Im} \int_{A_R}^{A_*} \frac{dA}{(dA/dx)} (\tilde{m}_2^2 - \tilde{m}_1^2), \quad (5.88)$$

where A_R is the value of A at resonance, and A_* is the value of A for which $\tilde{m}_2 = \tilde{m}_1$. This condition, of course, is satisfied for complex values of A , viz $A_* = e^{\pm 2i\theta} \Delta m^2$, and that is why the method is called the method of complex trajectory.

The final result depends on the nature of the density variation. If the variation is assumed to be linear, i.e., dA/dx is constant, one obtains

$$P_{\text{LZS}} = \exp\left(-\frac{\pi}{4}\gamma_{\text{res}}\right). \quad (5.89)$$

Different expressions, valid for exponentially varying density [31] as well as in regions where $\gamma \ll 1$, [32] have been worked out in the literature [30].

□ **Exercise 5.8** Supply the missing steps between Eqs. (5.88) and (5.89).

This phenomenon of level crossing proved crucial in understanding the neutrino flux from the Sun, a detailed account of which appears in Ch. 6. In view of this importance, there has been a great deal of work in this field [33]. Several authors have analyzed matter effects with three generations of neutrinos instead of two [34]. A rigorous field theoretic derivation of the effective mass matrix has been performed as well, the details of which will be presented in Ch. 16.

Part II

Models of neutrino mass

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Chapter 7

Neutrino mass in $SU(2)_L \times U(1)_Y$ models

7.1 Introduction

The standard model itself is based on the gauge group $SU(2)_L \times U(1)_Y$. But this fixes only the gauge bosons of the model. The fermions and Higgs contents have to be chosen somewhat arbitrarily. In the standard model, these choices are made in such a way that the neutrinos are massless, as discussed in Ch. 2. The masslessness is maintained to all orders in perturbation theory, and even after non-perturbative effects are included, as pointed out in §2.5.

But neutrinos are massive, and so one must have to incorporate neutrino mass in a realistic model of particle interactions. First, one might ask whether the inclusion of quantum gravity effects into the standard model might explain neutrino masses.

Clearly, as long as we treat gravity in perturbation theory, the symmetry arguments presented in §2.5 hold since all gravity coupling respect $B - L$ symmetry. However, once non-perturbative gravitational effects like black holes and worm holes are included, there is no guarantee that global symmetries will be respected in the low energy theory. The intuitive way to appreciate the argument is to note that throwing baryons into a black hole does not lead to any detectable consequence except through a net change in the baryon number of the universe. Since one can throw in an arbitrary number of baryons into the black hole, an arbitrary information loss about the net number of missing baryons would prevent us from defining a baryon number of the visible universe. Thus baryon number cannot be an exact symmetry in the presence of a black

hole. Similar arguments can be made for any global charge such as lepton number in the standard model. A field theoretic parameterization of this statement is that the effective low energy Lagrangian for the standard model in the presence of black holes and worm holes etc must contain baryon and lepton number violating terms. In the context of the standard model, the only such terms that one can construct are non-renormalizable terms of the form $\psi_L \phi \psi_L \phi / M_{\text{Pl}}$, where ψ_L is the lepton doublet, ϕ is the Higgs doublet, and M_{Pl} is the Planck mass, defined through Newton's gravitational constant by $M_{\text{Pl}} = 1/\sqrt{G_N}$. After gauge symmetry breaking, they lead to neutrino masses. However these masses are at most of order $v_{\text{wk}}^2 / M_{\text{Pl}} \simeq 10^{-5}$ eV [1]. But the solution of the atmospheric neutrino problem requires masses at least three orders of magnitude higher, as discussed in Ch. 5.

Thus one must seek physics beyond the standard model to explain observed evidences for neutrino masses. This does not necessarily mean going beyond the gauge group of the standard model, which fixes only the gauge bosons of the model. The fermions and Higgs contents have to be chosen somewhat arbitrarily. Even with the same gauge group as the standard model, one can conjecture extra fermions or Higgs bosons in the model so that the model predicts massive neutrinos. In this chapter, we will discuss some such simple modifications of the standard model.

7.2 Models with enlarged fermion sector

One of the peculiarities of the standard model, as commented before, is that it contains left and right chiral projections of all fermions except the neutrinos. This looks almost contrived.

To remedy this, let us add right-handed neutral fields $N_{\ell R}$ corresponding to each charged lepton ℓ . Like the other right-handed fields, they are assumed to be $SU(2)_L$ singlets. The definition of electric charge in Eq. (2.8) then implies that they also have $Y = 0$. We can summarize these statements as:

$$N_{\ell R} \quad : \quad (1, 0). \quad (7.1)$$

Thus, the $N_{\ell R}$ fields are singlets of the entire gauge group. In other words, they have no interaction with the gauge bosons. However, they affect the model non-trivially because of their other properties.

7.2.1 A simple model with Dirac neutrinos

The first non-trivial thing to notice is that the presence of these right-handed fields imply new gauge-invariant interactions in the Yukawa sector:

$$-\mathcal{L}'_Y = \sum_{\ell, \ell'} f_{\ell \ell'} \bar{\psi}_{\ell L} \hat{\phi} N_{\ell' R} + \text{h.c.}, \quad (7.2)$$

where $f_{\ell \ell'}$ are new coupling constants, and $\psi_{\ell L}$ is the lepton doublet of Eq. (2.10). The Higgs multiplet, ϕ , is the same as that appearing in the standard model. With a vev $v/\sqrt{2}$ as in Eq. (2.3), this gives rise to the following mass terms:

$$-\mathcal{L}_{\text{mass}} = \sum_{\ell, \ell'} f_{\ell \ell'} \frac{v}{\sqrt{2}} \bar{\nu}_{\ell L} N_{\ell' R} + \text{h.c.}. \quad (7.3)$$

This is a mass term for neutrinos if we identify the fields $N_{\ell R}$ as the right-handed components of the neutrinos, i.e., $N_{\ell R} = \nu_{\ell R}$. In the flavor space, it corresponds to a matrix with elements

$$M_{\ell \ell'} = \frac{v}{\sqrt{2}} f_{\ell \ell'}. \quad (7.4)$$

Such a matrix is called a *mass matrix*. In general, this matrix is not diagonal so that the fields $\nu_{\ell L}$, $N_{\ell R}$ do not correspond to the chiral projections of physical fermion fields.

To obtain the physical fields, one has to find the eigenvectors of the matrix M . In general, this can be done by diagonalizing M with a biunitary transformation:

$$U^\dagger M V = m, \quad (7.5)$$

where m is diagonal matrix. Thus, defining new states by the relations

$$\begin{aligned} \nu_{\ell L} &\equiv \sum_{\alpha} U_{\ell \alpha} \nu_{\alpha L}, \\ N_{\ell R} &\equiv \sum_{\alpha} V_{\ell \alpha} \nu_{\alpha R}, \end{aligned} \quad (7.6)$$

the terms in Eq. (7.3) can be rewritten as

$$-\mathcal{L}_{\text{mass}} = \sum_{\alpha} \bar{\nu}_{\alpha L} m_{\alpha} \nu_{\alpha R} + \text{h.c.}, \quad (7.7)$$

where m_{α} is the α^{th} diagonal element of the matrix m . This equation shows that the fields ν_{α} are fields with definite masses m_{α} and are therefore physical particles.

7.2.2 Neutrino mixing

The simple model above shows one of the major consequences of generic neutrino mass terms. The charged current interaction between the W bosons and the leptons is given by

$$\frac{g}{\sqrt{2}} \sum_{\ell} \bar{\ell}_L \gamma^{\mu} \nu_{\ell L} W_{\mu}^{-} + \text{h.c.}, \quad (7.8)$$

where $\nu_{\ell L}$ is the field which appears in the same $SU(2)_L$ doublet as the charged lepton ℓ .

Using Eq. (7.6), we can rewrite this in terms of the physical fields ν_{α} as

$$\frac{g}{\sqrt{2}} \sum_{\ell} \sum_{\alpha} \bar{\ell}_L \gamma^{\mu} U_{\ell\alpha} \nu_{\alpha L} W_{\mu}^{-} + \text{h.c.}. \quad (7.9)$$

This shows that in general all neutrinos can have charged current interaction with a given lepton ℓ (unless, of course, some particular element of U vanishes). This is in parallel to mixing in quark sector where mixing arises due to the matrix of Eq. (1.4). In the present case, this is appropriately called *neutrino mixing*.

An immediate consequence of neutrino mixing is that the generational lepton numbers like L_e , L_{μ} and L_{τ} are no more good global symmetries even at the classical level. This is because any mass eigenstate is a mixture of ν_e , ν_{μ} , ν_{τ} etc. At the classical level, the only global symmetry remaining in the leptonic sector is the total lepton number $L_e + L_{\mu} + L_{\tau}$. This feature would give rise to various flavor violating processes, e.g., $\mu \rightarrow e\gamma$, $\mu + e \rightarrow \tau + e$ and so on. These comments are true not just specifically for the present model, but for any model with neutrino mixing.

7.2.3 Shortcomings of the model

The model described above is very simple in the sense that it treats neutrino mass on exactly the same footing as the masses of other known fermions. The neutrinos are Dirac particles just like all other known fermions, which makes life easier because calculations with Dirac fermions are quite familiar to us. However, the model does not answer some questions.

First, the matrix $f_{\ell\ell'}$, defined in Eq. (7.2), is completely arbitrary. Without any knowledge of its elements, we have neither any idea of its eigenvalues, i.e., the neutrino masses, nor about the magnitudes of neutrino mixings. In the context of this model, these are then completely arbitrary parameters which should be determined by experiments, to which the model does not provide any guideline whatsoever.

Secondly, it also provides no answer to the question about the lightness of neutrinos. Of course, if the coupling constants $f_{\ell\ell'}$ are very small compared to the corresponding coupling constants which generate lepton or quark masses, the neutrinos will turn out to be lighter, in agreement with experimental bounds. But there is no good reason why $f_{\ell\ell'}$ must be small in this model.

In addition, the model is incomplete in a sense, which is discussed next. As we will see, this brings in new issues, absent in the simple model.

7.2.4 The complete model with right handed neutrinos

So far, we have considered only one type of term involving the right-handed neutrinos, viz., the Yukawa interaction of Eq. (7.2). But there can be more. Since the $N_{\ell R}$ fields are invariant under $SU(2)_L \times U(1)_Y$, so must be their conjugate fields $\widehat{N}_{\ell L}$. Thus, one can form gauge invariant bare mass terms

$$-\mathcal{L}_{\text{bare}} = \frac{1}{2} \sum_{\ell, \ell'} B_{\ell\ell'} \widehat{N}_{\ell L} N_{\ell' R} + \text{h.c.} . \quad (7.10)$$

These terms must be present if we write down the most general gauge-invariant Lagrangian involving the particles in the model. Of course, such terms violate $B - L$, but $B - L$ is a global symmetry that appears automatically with the particle content of the standard model and we are not obliged to impose it in the extended models for any fundamental reason. If one chooses to impose $B - L$ anyway, these terms are untenable and we go back to the model with Dirac neutrinos discussed earlier.

In the general case, the bare mass terms in Eq. (7.10) are allowed and add to the contributions of Eq. (7.3). Using the identity

$$\bar{\nu}_{\ell L} N_{\ell' R} = \widehat{N}_{\ell' L} \widehat{\nu}_{\ell R} , \quad (7.11)$$

we can write all the mass terms in the Majorana basis introduced in §4.5. We obtain

$$-\mathcal{L}_{\text{mass}} = \frac{1}{2} \left(\bar{\nu}_L \quad \overline{\widehat{N}_L} \right) \begin{pmatrix} 0 & M \\ M^\top & B \end{pmatrix} \begin{pmatrix} \widehat{\nu}_R \\ N_R \end{pmatrix} + \text{h.c.}, \quad (7.12)$$

where M and B are $\mathcal{N} \times \mathcal{N}$ matrices for $\mathcal{N}$ generations of fermions, and N_R , ν_L etc are $\mathcal{N}$ element column-vectors containing fields from $\mathcal{N}$ generations. Notice that the mass matrix is symmetric, as it should be according to the discussion of §4.5.

Upon diagonalization of the mass terms, one now obtains $2\mathcal{N}$ Majorana neutrinos in general. To see this, consider the simple case of a single generation, i.e., $\mathcal{N} = 1$. Then the mass matrix in Eq. (7.12) can be written as

$$\mathcal{M} = \begin{pmatrix} 0 & M \\ M & B \end{pmatrix} \quad (7.13)$$

where now M and B are simply numbers. Moreover, for the sake of simplicity, let us assume that both M and B are real and $B > 0$. Now choose an orthogonal matrix

$$O = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad (7.14)$$

with

$$\tan 2\theta = 2M/B. \quad (7.15)$$

Then

$$O\mathcal{M}O^\top = \begin{pmatrix} -m_1 & 0 \\ 0 & m_2 \end{pmatrix}, \quad (7.16)$$

a diagonal matrix, where

$$m_{1,2} = \frac{1}{2} \left(\sqrt{B^2 + 4M^2} \mp B \right). \quad (7.17)$$

Since $m_{1,2} \geq 0$, we still have a little problem, viz, the elements of the diagonal matrix in Eq. (7.16) are not all non-negative and therefore cannot be interpreted as the masses of physical fields. So let us write

$$\begin{pmatrix} -m_1 & 0 \\ 0 & m_2 \end{pmatrix} = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \equiv mK^2, \quad (7.18)$$

where m contains the positive mass eigenvalues and K^2 is a diagonal matrix of positive and negative signs. Then

$$\mathcal{M} = O^T m K^2 O. \quad (7.19)$$

Now, if we define the column vectors

$$\begin{pmatrix} n_{1L} \\ n_{2L} \end{pmatrix} \equiv O \begin{pmatrix} \nu_L \\ \hat{N}_L \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \nu_L \\ \hat{N}_L \end{pmatrix} \quad (7.20)$$

and

$$\begin{pmatrix} n_{1R} \\ n_{2R} \end{pmatrix} \equiv K^2 O \begin{pmatrix} \hat{\nu}_R \\ N_R \end{pmatrix} = \begin{pmatrix} -\cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \hat{\nu}_R \\ N_R \end{pmatrix}, \quad (7.21)$$

the mass terms reduce to

$$-\mathcal{L}_{\text{mass}} = m_1 \bar{n}_{1L} n_{1R} + m_2 \bar{n}_{2L} n_{2R} + \text{h.c.} \quad (7.22)$$

Thus, for a single generation, we obtain two eigenstates. Furthermore, from Eqs. (7.20) and (7.21), we see that

$$\begin{aligned} n_1 &= n_{1L} + n_{1R} = \cos \theta (\nu_L - \hat{\nu}_R) - \sin \theta (\hat{N}_L - N_R) \\ n_2 &= n_{2L} + n_{2R} = \sin \theta (\nu_L + \hat{\nu}_R) + \cos \theta (\hat{N}_L + N_R). \end{aligned} \quad (7.23)$$

Since $\hat{\nu}_R$ is the conjugate of ν_L and $\hat{N}_L$ of N_R , this immediately proves

$$n_1 = -\hat{n}_1, \quad n_2 = \hat{n}_2 \quad (7.24)$$

so that both n_1 and n_2 are Majorana particles. For $\mathcal{N}$ generations, one obtains $2\mathcal{N}$ Majorana particles in general.

Alternatively, one can also diagonalize by using the unitary matrix $U = KO$, with $K = \text{diag}(i \ 1)$. In this case, we can write $\mathcal{M} = U^T m U$ instead of Eq. (7.19). The definition of mass eigenstates in Eq. (7.20) and Eq. (7.21) change in this case, but the physical implications are the same.

- **Exercise 7.1** Will Eq. (7.24) be valid if we take $U = KO$ as indicated above? If not, show that the CP eigenvalues of the neutrino states come out to be the same in either formulation.

One redeeming aspect of this model is that it can provide a reason for the smallness of neutrino masses. To see this, refer to the one-generation model once again. In the mass matrix, the quantity M arises from ϕ -coupling and therefore it is natural to assume that it is of the same order as the masses of other fermions in the same generation. Let us suppose $B \gg M$. Then the eigenvalues in Eq. (7.17) reduce to

$$m_1 \simeq \frac{M^2}{B} \quad , \quad m_2 \simeq B. \quad (7.25)$$

Since $B \gg M$, it follows that $m_1 \ll M$, which means that the neutrino is much lighter than the charged fermions. Of course, there is another neutrino whose mass, B , is much larger than the charged fermion masses. This mechanism of making one particle light at the expense of making another one heavy is called the *see-saw mechanism* [2].

But one point must be noticed here. Cosmological arguments restrict the mass of any stable neutrino to be less than about 1 eV [see Ch. 18 for details]. Thus, if even ν_τ has to be lighter than this bound, Eq. (7.25) demands, with $M \sim m_\tau$,

$$B \gtrsim 5 \times 10^9 \text{ GeV}. \quad (7.26)$$

Thus, successful implementation of this model requires a mass scale much larger than the weak scale, which is of order 10^2 GeV . In general, models with such widely disparate scales suffer from the theoretical problem of maintaining the mass scales. The point is that even if the classical solution of the vacuum has two widely different scales, the quantum corrections tend to merge them. This problem, called the *hierarchy problem*, plagues grand unified theories necessarily, but there one achieves unification at the cost of hierarchy problem. In the present context, the hierarchy problem makes the model unattractive unless we embed it into a grand unified theory or there is an extra gauge symmetry such as for example $B - L$ whose scale can be associated with B .

7.3 Models with expanded Higgs sector

If we do not add any new fermion to the particle content of the standard model, we have, in each generation, only two degrees of freedom

corresponding to uncharged fermions, viz, $\nu_{\ell\text{L}}$ and $\widehat{\nu}_{\ell\text{R}}$. Thus, in whatever way they pick up mass, the mass must always be Majorana type, i.e., the mass terms must violate $B - L$. Thus, we are motivated to introduce new Higgs bosons which can violate $B - L$ symmetry in their interactions. Also, the neutrino masses must somehow be induced by the Yukawa couplings. With these two considerations in mind, we examine what are the fermionic bilinears that have a net $B - L$ quantum number.

For this, we can combine some lepton field with some antilepton field. The leptonic fields are given in Eq. (2.10). Their antiparticles are given by

$$\begin{aligned}\widehat{\psi}_{\text{R}} &= \gamma_0 C \epsilon \psi_{\text{L}}^* & : & (2, 1), \\ \widehat{\ell}_{\text{L}} &= \gamma_0 C \ell_{\text{R}}^* & : & (1, 2),\end{aligned}\tag{7.27}$$

where $\epsilon = i\tau_2$ is the 2×2 antisymmetric matrix. Note that in the first equation here, we have to take the conjugation with respect to two sets of indices. For the spinor index, this is done as in Eq. (4.10), and for the $SU(2)_L$ index as in Eq. (2.15). It is now easy to see that one can obtain two fermion bilinears which have a net $B - L$ quantum number:

$$\begin{aligned}\overline{\psi}_{\text{L}} \widehat{\psi}_{\text{R}} &\sim (2, 1) \times (2, 1) = (1, 2) + (3, 2), \\ \widehat{\ell}_{\text{L}} \ell_{\text{R}} &\sim (1, -2) \times (1, -2) = (1, -4),\end{aligned}\tag{7.28}$$

where in parentheses we have put the transformations under $SU(2)_L \times U(1)_Y$. The Higgs multiplets that can directly couple to these bilinears to form gauge invariant Yukawa couplings are as follows [3]:

- a triplet $\Delta : (3, -2)$,
- a singly charged singlet : $(1, -2)$,
- a doubly charged singlet : $(1, 4)$.

We now see how the introduction of these scalars, one at a time or in combinations, can give rise to neutrino masses.

7.3.1 Adding a triplet Δ

With the value of Y equal to -2 , the electric charges of the components of the triplet Δ are as follows

$$\Delta = \begin{pmatrix} \Delta_0 \\ \Delta_- \\ \Delta_{--} \end{pmatrix}. \quad (7.29)$$

and we have an additional Yukawa coupling outside the ones in the standard model:

$$-\mathcal{L}'_Y = \sum_{\ell, \ell'} f_{\ell \ell'} \bar{\psi}_{\ell L} \frac{1}{\sqrt{2}} \boldsymbol{\tau} \cdot \Delta \hat{\psi}_{\ell' R} + \text{h.c.} \quad (7.30)$$

where $\boldsymbol{\tau}$ denotes the Pauli matrices. Note that the fermion bilinear combination involving the Pauli matrices is just the symmetric (i.e., triplet) combination of the two doublets $\psi_{\ell L}$ and $\hat{\psi}_{\ell' R}$, whose dot product with the triplet Δ gives an $SU(2)_L$ singlet interaction in Eq. (7.30).

The Higgs potential now involves both Δ and the usual doublet ϕ . Let us assume that the parameters in this potential are such that, at the minimum,

$$\langle \phi_0 \rangle = \frac{v_2}{\sqrt{2}} \quad , \quad \langle \Delta_0 \rangle = \frac{v_3}{\sqrt{2}}. \quad (7.31)$$

In that case, after symmetry breaking, Eq. (7.30) produces Majorana mass terms for neutrinos

$$-\mathcal{L}_{\text{mass}} = \sum_{\ell, \ell'} \bar{\nu}_{\ell L} M_{\ell \ell'} \hat{\nu}_{\ell' R} + \text{h.c.} \quad (7.32)$$

where

$$M_{\ell \ell'} = \frac{v_3}{\sqrt{2}} f_{\ell \ell'}. \quad (7.33)$$

In Ch. 4, we proved that any mass matrix in the Majorana basis must be symmetric. To see that the statement is true here, we will have to show that $f_{\ell \ell'}$ is symmetric under the generation indices. To this end, we rewrite Eq. (7.30) in the following way, using the definition of the conjugate field from Eq. (4.27):

$$-\mathcal{L}'_Y = \sum_{\ell, \ell'} f_{\ell \ell'} \hat{\psi}_{\ell R}^\dagger C^{-1} \epsilon \frac{1}{\sqrt{2}} \boldsymbol{\tau} \cdot \Delta \hat{\psi}_{\ell' R} + \text{h.c.} \quad (7.34)$$

Written this way, it is clear that it is a bilinear involving $\hat{\psi}_R$ twice, and therefore it must obey Fermi statistics. The $SU(2)_L$ indices are contracted by the $\epsilon\tau$ matrix, which is symmetric. The spinor indices are contracted by the matrix C^{-1} , which is antisymmetric because C is antisymmetric, as shown in Eq. (4.26). The only other indices are the generation indices, and they must be symmetric in order that the entire combination is antisymmetric. Thus, $f_{\ell\ell'} = f_{\ell'\ell}$, which shows that the mass matrix is symmetric.

The diagonalization gives the mass eigenvalues and eigenstates. However, since the couplings $f_{\ell\ell'}$ are unknown, one cannot predict any pattern of neutrino mixing, just as in the model of §7.2.1. But, unlike that model, a different vev (viz, that of the triplet) is responsible for neutrino masses than for other fermion masses. Thus, one explanation for the lightness of the neutrinos could be forwarded here, viz, that the neutrinos could be light because $v_3 \ll v_2$.

Indeed, v_3 has to be somewhat smaller than v_2 from another consideration. Since Δ_0 couples to the W and the Z , its vev contributes to the masses of these gauge bosons. Instead of the standard model formulas, we now obtain

$$\begin{aligned} M_W^2 &= \frac{1}{4}g^2(v_2^2 + 2v_3^2) \\ M_Z^2 &= \frac{1}{4}(g^2 + g'^2)(v_2^2 + 4v_3^2), \end{aligned} \quad (7.35)$$

so that

$$\rho \equiv \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = \frac{1 + 2v_3^2/v_2^2}{1 + 4v_3^2/v_2^2}. \quad (7.36)$$

Using the experimental bounds on ρ from Eq. (2.34), we obtain

$$\frac{v_3}{v_2} < 0.07. \quad (7.37)$$

To provide an explanation of why neutrino masses are at least six orders of magnitude smaller than charged fermion masses in the same generation, v_3 in fact must be quite a bit smaller than this upper limit. We will discuss theories where such smallness can arise naturally.

□ **Exercise 7.2** Verify Eq. (7.35).

- **Exercise 7.3** In this model, identify the quadratic terms of the form $W_\mu^+ \partial^\mu H_-$ coming from the covariant derivative terms after symmetry breaking. This H_- , properly normalized, would then represent the unphysical Higgs boson eaten up by W_- . Show that

$$H_- = \frac{v_2 \phi_- + \sqrt{2} v_3 \Delta_-}{\sqrt{v_2^2 + 2v_3^2}}. \quad (7.38)$$

An important question that arises is that of the global quantum number $B - L$. It must be somehow broken in the model since Majorana neutrinos have resulted. The Yukawa coupling of Eq. (7.30) as such does not imply $B - L$ violation. Although the Higgs boson in this case couples to a fermion bilinear whose $B - L$ number equals 2, we can assign a $B - L$ value of -2 to Δ and then the Yukawa coupling is $B - L$ invariant. However, once this quantum number is carried by Δ , it is broken when Δ_0 acquires a vev. In fact, it breaks $B - L$ by 2 units, which is exactly what is necessary for the generation of Majorana mass terms.

7.3.2 Model with a singly charged singlet

This model was proposed by Zee [4]. He introduced an $SU(2)_L$ singlet particle h_- in the Higgs sector. The Yukawa coupling of this particle is

$$-\mathcal{L}_Y = \sum_{\ell, \ell'} f_{\ell \ell'} \bar{\psi}_{\ell L} \hat{\psi}_{\ell' R} h_- + \text{h.c.} \quad (7.39)$$

The coupling constants $f_{\ell \ell'}$ must be antisymmetric because of Fermi statistics.

As in the previous model, we can assign a $B - L$ quantum number of -2 to the field h_- . Since h has electric charge, its vev must vanish in a physically acceptable ground state, because otherwise electromagnetic gauge symmetry would be spontaneously broken. So, in order to generate Majorana masses for the neutrinos, one must have some alternative sources of $B - L$ violation.

This can now only come from the Higgs potential, since kinetic energy terms and gauge interactions do not violate any $U(1)$ quantum number. In the Higgs potential, if apart from h_- we have just one doublet ϕ as in the standard model, it is impossible to write down any $B - L$ violating term. Consequently, neutrinos are massless just as in the standard model.

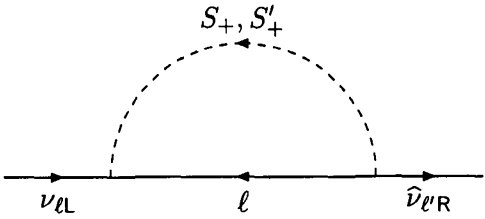


Figure 7.1: Self-energy diagrams giving neutrino masses in Zee's model.

The situation changes if there are more than one doublets in the theory. For example, let us say we have a doublet ϕ' in addition to the standard model doublet ϕ . If ϕ' couples to quarks and leptons in the same way as ϕ does, it must have a vanishing $B - L$ quantum number. In that case, the trilinear coupling

$$\mu\phi^T\epsilon\phi'h_- + \text{h.c.} \tag{7.40}$$

violates $B - L$ by 2 units. Such a coupling cannot exist with just one doublet because $\phi^T\epsilon\phi$, being the antisymmetric combination, vanishes.

There are now three different singly charged scalars in the theory: ϕ_+ , ϕ'_+ and h_+ . At the symmetry breaking, some combination of the first two of these will be eaten up by the W^+ gauge boson. So the physical spectrum will contain two charged scalars S and S' . Each of these would be mixtures of h_+ and the ϕ_+ , ϕ'_+ . Therefore, these physical scalars are not eigenstates of $B - L$. Thus, neutrinos will obtain Majorana masses at the one-loop level from the self-energy diagrams, shown in Fig. 7.1, involving these physical scalars S and S' .

By a naive power counting, one might be tempted to say that the diagram in Fig. 7.1 is divergent, so that it gives an infinite contribution to neutrino masses. This is not true because of the renormalizability of the theory, which requires that all infinities arising in the calculation can be absorbed into the definitions of the parameters in the classical Lagrangian. In this case, the tree-level Lagrangian does not have any neutrino mass term. So, if Fig. 7.1 gives an infinite mass term, we cannot absorb it in any tree-level term. However, the renormalizability of the model is guaranteed by the general proof of 't Hooft. Therefore, Fig. 7.1 must give finite masses.

The mass terms are particularly simple if we assume that only one

of the two Higgs doublets couples to leptons [5]. A straightforward calculation shows [see also §7.4]

$$M_{\ell\ell'} = A f_{\ell\ell'} (m_\ell^2 - m_{\ell'}^2) \quad (7.41)$$

where A is a constant. Since $f_{\ell\ell'} = -f_{\ell'\ell}$, as commented before, $M_{\ell\ell'}$ is symmetric like any mass matrix in the Majorana basis.

Unlike the models described before, this model gives a pattern of neutrino masses and mixings. To see that, let us define [5]

$$\begin{aligned} \tan \alpha &= \frac{f_{\mu\tau}}{f_{e\tau}} \left(1 - \frac{m_\mu^2}{m_\tau^2} \right) \\ \sigma &= \frac{f_{e\mu}}{f_{e\tau}} \frac{m_\mu^2}{m_\tau^2} \cos \alpha. \end{aligned} \quad (7.42)$$

Neglecting the electron mass which is tiny compared to the masses of the muon and the tau, the mass matrix of Eq. (7.41) can be rewritten as

$$M = m_0 \begin{pmatrix} 0 & \sigma & \cos \alpha \\ \sigma & 0 & \sin \alpha \\ \cos \alpha & \sin \alpha & 0 \end{pmatrix}. \quad (7.43)$$

where $m_0 = A m_\tau^2 f_{\tau e} / \cos \alpha$. If we assume that there is no large hierarchy among the couplings $f_{\ell\ell'}$, it follows that $\sigma \ll 1$ because of the factor m_μ^2/m_τ^2 . Then the diagonalization of the matrix M gives the eigenvalues

$$\begin{aligned} m_1 K_1^2 &= -m_0 \left(1 - \frac{1}{2} \sigma \sin 2\alpha \right), \\ m_2 K_2^2 &= m_0 \left(1 + \frac{1}{2} \sigma \sin 2\alpha \right), \\ m_3 K_3^2 &= -m_0 \sigma \sin 2\alpha, \end{aligned} \quad (7.44)$$

correct upto first order terms in σ . Here m_α are positive, and K_α^2 is $+1$ or -1 , depending on the signs of $\sigma \sin 2\alpha$ and m_0 . To eliminate the negative eigenvalues, we can again choose a matrix K described in §7.2.4 appropriately, so that m_α is the mass of the physical eigenstate. Eq. (7.44) now shows that there must be two physical neutrinos whose masses, m_1 and m_2 , are very close, whereas the other mass m_3 is much smaller. However, the combination of masses and mixings obtained in this model is not consistent with present observations from the oscillation data. We discuss this in detail in Ch. 11.

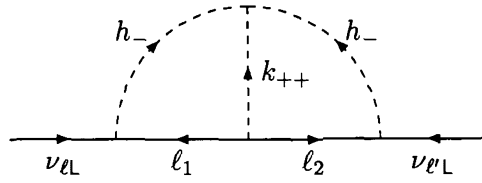


Figure 7.2: 2-loop diagram in Babu's model, giving rise to Majorana masses of the neutrinos.

7.3.3 Model with doubly charged singlet

The doubly charged singlet can have Yukawa couplings like

$$-\mathcal{L}'_Y = \sum_{\ell, \ell'} F_{\ell\ell'} \bar{\ell}'_L \ell'_R k_{++} + \text{h.c.} \quad (7.45)$$

This assigns a $B - L$ quantum number of 2 to the field k_{++} . In order to generate Majorana masses for the neutrinos, we need $B - L$ violation. This is not possible with any number of doublet Higgs bosons, since there can be no trilinear coupling like the one involving the singly charged Higgs bosons, as shown in Eq. (7.40). Thus, neutrinos are massless in all orders of perturbation theory [3].

The situation changes, as was pointed out by Babu [6], if the model contains the singly charged scalar h_- of §7.3.2 alongwith the k_{++} . Due to their Yukawa couplings, both h_+ and k_{++} carry 2 units of $B - L$ quantum number. With just one doublet ϕ , the trilinear $\phi\phi h$ coupling vanishes, as discussed in the previous section. However, there is one trilinear coupling, viz,

$$\mu h_- h_- k_{++} + \text{h.c.} \quad (7.46)$$

which breaks $B - L$. Thus, neutrino masses can be generated from 2-loop diagrams as shown in Fig. 7.2. It gives a mass matrix of the form

$$m_{\ell\ell'} = 8\mu \sum_{\ell_1, \ell_2} m_{\ell_1} m_{\ell_2} f_{\ell\ell_1} F_{\ell_1\ell_2} f'_{\ell_2\ell'} I_{\ell_1\ell_2} \quad (7.47)$$

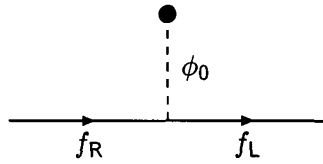


Figure 7.3: A graphic representation of how fermion masses arise.

where

$$I_{\ell_1 \ell_2} = \int \frac{d^4 p}{(2\pi)^4} \int \frac{d^4 q}{(2\pi)^4} \frac{1}{p^2 - m_{\ell_1}^2} \frac{1}{q^2 - m_{\ell_2}^2} \frac{1}{p^2 - m_h^2} \frac{1}{q^2 - m_h^2} \frac{1}{(p - q)^2 - m_k^2}. \quad (7.48)$$

Notice that the eigenvalues of m are naturally small here because of the suppression factors that appear in the 2-loop integration. This provides a rationale for the smallness of neutrino masses.

Another interesting prediction of this model is that, since the coupling matrix f is antisymmetric, the determinant of the matrix m vanishes if there are three generations. Thus, there must be one zero eigenvalue.

This does not exactly mean that there will be a massless neutrino, because the expression for $m_{\ell\ell'}$ has corrections coming from higher loops. Such corrections should be smaller than the 2-loop contribution in the perturbative regime. Nevertheless, they will modify the eigenvalues by small amounts. But in any case, we expect one eigenvalue to be much smaller than the other two.

7.4 The method of flavor diagrams

In calculations for obtaining physical quantities like the S -matrix elements, it is most convenient to use Feynman diagrams, where the external lines correspond to physical particles. However, in order to understand questions regarding mass and symmetry breaking, it is often beneficial to use another basis, which we call the *flavor basis*. The method and its advantages will be clear from the examples below.

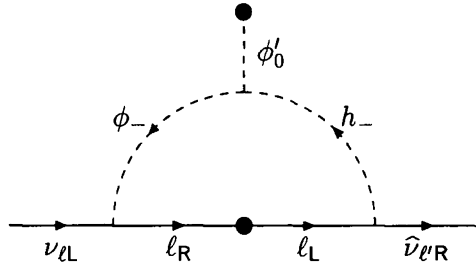


Figure 7.4: The flavor diagram for neutrino masses in Zee's model.

In this method, we draw Feynman-like diagrams using the fields that appear in the gauge-invariant Lagrangian. For the fermions, it means that we talk of the left- and right-chiral fields separately. For the Higgs field ϕ , we use the original classical field which appears, say, in Eq. (2.2), and not the quantum field that is obtained by the fluctuation of this classical field from its vev. Since the charged fermions are massless before symmetry breaking, the generation of mass after symmetry breaking can be represented as in Fig. 7.3. The blob at the end of the ϕ_0 line represents its annihilation into the vacuum, which manifests the fact that ϕ_0 has a vev. The figure now is a graphical illustration of the fact that a charged fermion mass is given by its Yukawa coupling with ϕ , multiplied by the vev of ϕ_0 , as expressed algebraically in Eq. (2.17).

If neutrinos are Dirac particles and obtain their mass through their coupling with ϕ as in the model of §7.2.1, then neutrino masses are also represented by Fig. 7.3, and we get no new insight. However, if the neutrinos get Majorana masses as in Zee's model described in §7.3.2, we can see the power of the method of flavor diagrams by trying to derive the neutrino mass matrix. In this case, we should draw, instead of Fig. 7.1, the neutrino mass diagram as in Fig. 7.4. Notice that here we use the neutrino states $\nu_{\ell L}$ and $\hat{\nu}_{\ell' R}$ which couple to one flavor of charged lepton via charged current interactions. The blob on the internal fermion line is a shorthand for Fig. 7.3 representing the mass of the charged lepton. The introduction of the mass is necessary on the internal fermion line to flip the helicity of $\hat{\ell}'$, without which we cannot obtain a right-handed particle on the right end of the diagram.

For the sake of simplicity, let us assume that the Yukawa couplings of

ϕ' are much smaller than those of ϕ , so that charged lepton masses come mainly through their coupling with ϕ . In this case, just a cursory look at Fig. 7.4 tells us that its contribution to $m_{\ell\ell'}$ must have the following features:

1. it has a factor of $f_{\ell\ell'}$ from the vertex with h_- ;
2. it has a factor of m_ℓ from the mass insertion;
3. it has a factor of v' from the vev of ϕ'_0 ;
4. it has a factor of the Yukawa coupling of ℓ with ϕ , which is m_ℓ/v .

Taking all these factors together, we obtain a contribution of $f_{\ell\ell'} m_\ell^2$, apart from factors which do not depend on fermion generations. Similarly, there would be one flavor diagram where the ϕ_- attaches to the right end of the fermion line, and the h_- line attaches to the left end. This gives a contribution proportional to $f_{\ell'\ell} m_{\ell'}^2$. Using the antisymmetry of f , we now obtain the form of the mass matrix given in Eq. (7.41), which shows the power of the method.

The flavor diagrams are also useful for examining the breaking of global quantum numbers. For example, if we compare Fig. 7.1 and Fig. 7.4, it is much easier to tell from the latter that $B - L$ is violated at the $h\phi\phi'$ vertex. Because of this, we use this method extensively in the next section and later in the book.

- **Exercise 7.4** By drawing flavor diagrams, show that the neutrino mass matrix in the model of §7.3.3 contains two factors of charged lepton masses, as shown in Eq. (7.47). Verify it by evaluating the Feynman diagrams for neutrino masses.

7.5 Models with spontaneous $B - L$ violation

In all but one models of neutrino mass discussed in this chapter, the mass is of Majorana type, arising out of $B - L$ violation. One might therefore wonder as to how exactly these models violate $B - L$. There are two ways to break global $B - L$ symmetry: *i*) by including explicit $B - L$ breaking terms in the Lagrangian; *ii*) by making the vacuum non-invariant under $B - L$ symmetry. The latter case, which was proposed by Chikashige, Mohapatra and Peccei (CMP) in 1980, implies the existence of a massless pseudo-scalar particle called the Majoron which we will denote by the symbol J . Even though a priori the existence of

the Majoron would imply long range forces between particles to which it couples, it was pointed out [7] that only spin dependent forces between matter arise in the non-relativistic limit from Majoron exchange. The laboratory limits on such forces are very weak, implying that the scale of $B - L$ breaking could be low. This is very interesting because, new physics associated with $B - L$ breaking could then be accessible to on-going experiments. Low energy manifestations of the Majoron of course depends on its transformation property under the weak $SU(2)_L$ group. Three classes of Majoron models have been discussed in the literature. The original Majoron of CMP transforms like an isosinglet and is dubbed the singlet Majoron. Subsequently, a Majoron model was proposed by Gelmini and Roncadelli [8] using the triplet Higgs field. This is known as the triplet Majoron. Several models also have been constructed where the Majoron has doublet transformation property [9, 10]. Below we discuss the present status of these models.

7.5.1 Constraints on Majoron models

The phenomenological constraints on Majoron models derive from known upper bounds of flavor changing currents in the leptonic sector. For example, in a model with Majoron, one expects the muon decay mode

$$\mu \rightarrow e + J \tag{7.49}$$

at some level. Experimental results on this mode is [11]

$$\Gamma(\mu \rightarrow e + J) < 3 \times 10^{-4}. \tag{7.50}$$

There are similar bounds from τ -decays but they are in general much weaker. Another class of bounds come from the absence of Majoron bremsstrahlung from final states particles in physical processes. For example, analysis of leptonic decays of kaons and pions [12] imply that the Majoron coupling to neutrinos must be smaller than about 10^{-2} or so.

In practice, the strongest constraints on Majoron models come from astrophysical considerations. The point is that Majorons can be produced in stars through the process

$$\gamma + e \rightarrow e + J, \tag{7.51}$$

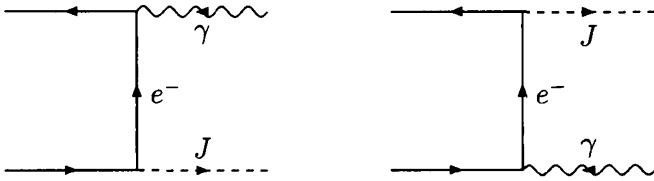


Figure 7.5: Photoproduction of Majoron.

and they can come out of the star, carrying some energy out. This thus contributes to the energy loss mechanism of stars and one should be able to put bounds on the process from known values of stellar luminosities.

Here we give a very simple derivation of the quantitative bound. The process in Eq. (7.51) occurs, at the tree level, through the diagrams in Fig. 7.5. For stellar core temperatures satisfying $T \ll m_e$, the cross-section of this process is easily seen to be

$$\sigma \simeq \frac{(eg_{eJ})^2}{12\pi} \frac{\omega^2}{m_e^4}, \quad (7.52)$$

where ω is the incident photon energy and g_{eJ} is the coupling of the electron with the Majoron J . The number of occurrences of the photoproduction reaction per unit volume per unit time is given by $n_e n_\gamma \sigma$, where n_e and n_γ are the number densities of the electron and the photon respectively. As a rough order-of-magnitude estimate, we can put $\omega \sim T$ and $n_\gamma \sim T^3$. Since at each occurrence, we obtain a Majoron of energy ω' of order T , the amount of energy going into Majorons per unit time per unit mass, L_{majo} , is given by

$$L_{\text{majo}} \simeq \frac{n_e n_\gamma \sigma \omega'}{\rho} \simeq \frac{\alpha g_{eJ}^2 n_e T^6}{3\rho m_e^4}, \quad (7.53)$$

where ρ is the mass per unit volume. For a rough estimate, we can neglect the contribution of neutrons to the mass and write the proton contribution as $\rho \simeq m_p n_p$. For $T \ll m_e$ which is the case of present interest, no positrons are present. Therefore, the proton number density n_p must equal n_e since the stellar core is electrically neutral. Thus we finally obtain

$$L_{\text{majo}} \simeq \frac{\alpha g_{eJ}^2 T^6}{3m_e^4 m_p}. \quad (7.54)$$

Let us assume that all this energy is coming out of the star, the Majoron being so weakly interacting that its mean free path is larger than the stellar radius. Then the stellar energy loss per unit time per unit mass because of Majoron emission is given by the last expression. For the sun, for example, the core temperature is $T \simeq 15.7 \times 10^6$ K. In the standard model of the sun, the energy loss rate per unit mass at the center is calculated [13] to be $17.5 \text{ erg s}^{-1} \text{ g}^{-1}$. Since the standard model works quite well, we can quite conservatively demand that the energy loss due to any new mechanism like Majoron emission is smaller than the above amount. This gives the bound [14]

$$g_{eJ} \lesssim 10^{-10}. \quad (7.55)$$

For red-giant stars which have higher core temperature, one obtains a more stringent bound,

$$g_{eJ} \lesssim 10^{-12}, \quad (7.56)$$

although it is less certain than the solar bound since the dynamics of the red giants is not as well understood as that of the sun. Observation of neutrinos from the supernova SN1987A puts bounds on Majoron coupling to neutrinos as well [15]. These bounds severely constrain many Majoron models, some examples of which follow.

7.5.2 Majoron in the model with right-handed neutrinos

We start with the complete model with right-handed neutrinos, described in §7.2.4. In the model, the couplings of the doublet ϕ conserves $B - L$, but the bare mass terms of Eq. (7.10) break $B - L$ by two units and give rise to Majorana neutrinos. If $B - L$ breaks that way through the explicit occurrence of the bare mass term, we do not expect any Majoron. However, consider an alternative situation where the model contains a gauge singlet, spin-0 field S having the following Yukawa coupling:

$$-\mathcal{L}_Y' = \sum_{\ell, \ell'} b_{\ell\ell'} \bar{\hat{N}}_{\ell} N_{\ell'R} S + \text{h.c.} \quad (7.57)$$

If S develops vev, neutrinos obtain mass just as in §7.2.4, where

$$B_{\ell\ell'} = b_{\ell\ell'} \langle S \rangle. \quad (7.58)$$

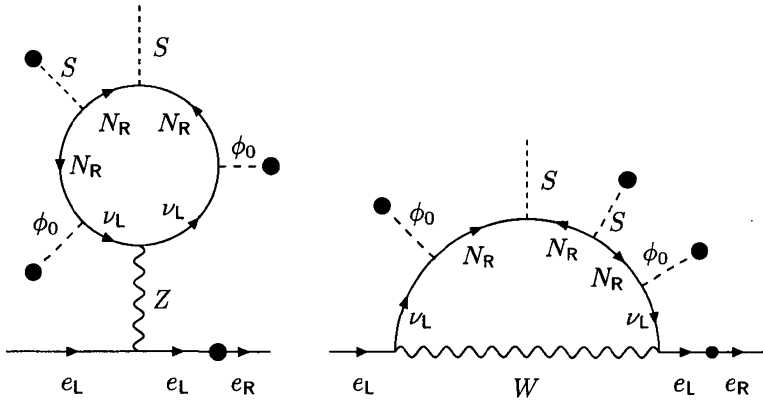


Figure 7.6: Majoron coupling with charged fermions in the singlet Majoron model.

The difference with the earlier model is that here all terms involving fermions conserve $B - L$ provided we assign 2 units of $B - L$ quantum number to S . In the scalar potential also, let us forbid any term that violates $B - L$. Then $B - L$ would be a global symmetry of the entire Lagrangian. But since the scalar S carries this quantum number, $B - L$ will be broken spontaneously if S develops a vev. As a result, we will obtain a Majoron [7] J . In the simple case when all vevs are real, J would simply be the imaginary part of the complex field S . At one-loop level, couplings are induced through the diagrams of Fig. 7.6. From these flavor diagrams, we can identify the following characteristics of the effective coupling g_{eJ} :

- it must involve a factor of G_F from the propagator and the couplings of the W or the Z ;
- it must be proportional to m_e which is necessary to get a helicity flip on the electron line;
- it should involve two powers of M from the neutrino coupling to ϕ and the associated ϕ vevs.

Since the heavy mass in the loop is of order B , it provides the damping scale, so that we can write

$$g_{eJ} \sim \frac{1}{16\pi^2} G_F m_e \frac{M^2}{B} \sim \frac{1}{16\pi^2} G_F m_e m_\nu, \quad (7.59)$$

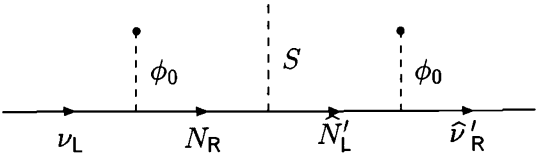


Figure 7.7: Flavor diagram for neutrino decay via a Majoron emission.

which easily satisfies the astrophysical bound of Eq. (7.56) for any acceptable value of neutrino masses. With such small effective couplings with charged fermions, rates for $\mu \rightarrow e + J$ etc are also very slow.

However, as we discussed before, the model is unattractive because of the presence of hierarchy of mass scales provided the neutrinos are stable. One can, however, question the stability of neutrinos in this model, since all but the lightest of the neutrinos can decay through a Majoron emission:

$$\nu \rightarrow \nu' + J. \tag{7.60}$$

Naïvely, from the flavor diagram of Fig. 7.7 one might expect that the amplitude for this process is proportional to two powers of the mixing between the light ν and the heavy N states, i.e., goes as $(M/B)^2$. A detailed analysis shows that there is an accidental cancellation of the $(M/B)^2$ term [16]. This cancellation persists even if one considers loop effects [17]. The leading non-vanishing term in the amplitude [18, 17] goes like $(M/B)^4$. This makes the process so slow that the lifetime, for all allowed values of ν_μ and ν_τ masses, exceeds the age of the universe by many orders of magnitude. Thus, in the time-scale of the age of the universe, the light neutrinos are indeed stable in this model and the model has the hierarchy problem built in it.

7.5.3 Majorons in models with extended Higgs sector

If neutrinos obtain mass through their coupling with a triplet Higgs Δ , as discussed in §7.3.1, the triplet must carry 2 units of $B - L$ quantum number and develop a vev at the minimum of the scalar potential. In that case, $B - L$ is spontaneously broken if it were a symmetry of the classical Lagrangian.

The latter condition is not satisfied if we insist on writing down the full gauge-invariant potential involving ϕ and Δ , which contains a term

$$A\phi^\top \epsilon \tau \cdot \Delta \phi + \text{h.c.} \quad (7.61)$$

that breaks $B-L$ explicitly. However, if we forbid this term by imposing $B-L$, then v_3 , the vev of Δ_0 , produces a Majoron J . This variant of the triplet model, proposed by Gelmini and Roncadelli [8], has interesting consequences, since the Majoron can mediate ν - ν scattering at tree level and therefore gives a large cross-section.

In the model with h_- of §7.3.2, the trilinear coupling $\phi^\top \epsilon \phi' h_-$ of Eq. (7.40) breaks $B-L$ only if we assume that ϕ' is neutral under $B-L$ just as ϕ is. However, one can also contemplate that, since h_- carries 2 units of $B-L$, ϕ' carries -2 units and the trilinear coupling conserves $B-L$. One then will have to forbid other possible $B-L$ violating terms like $(\phi'^\dagger \phi)(\phi'^\dagger \phi)$ in the scalar potential. If that is done, then the vev v' of ϕ'_0 breaks $B-L$ spontaneously and a Majoron results [10].

Both these models fall in a general class in the sense that in both models, the Majoron carries non-trivial quantum numbers of the gauge group $SU(2)_L \times U(1)_Y$. This fact has interesting consequences, some of which rule out such models with present experimental data.

To show that, let us specialize to the model of Gelmini and Roncadelli. In this model, the vev of the field Δ_0 breaks $B-L$ spontaneously. But the same vev also contributes to the mass of the Z , so the unphysical Higgs eaten up by Z is a linear superposition of Δ_0 and ϕ_0 , the latter being the component of the usual doublet. In fact, the combination eaten up by Z is

$$\frac{v_2 \operatorname{Im} \phi_0 + 2v_3 \operatorname{Im} \Delta_0}{\sqrt{v_2^2 + 4v_3^2}}, \quad (7.62)$$

assuming v_2 and v_3 to be real. The Majoron must be an orthogonal combination, and therefore is given by

$$J = \frac{2v_3 \operatorname{Im} \phi_0 - v_2 \operatorname{Im} \Delta_0}{\sqrt{v_2^2 + 4v_3^2}} \simeq -\operatorname{Im} \Delta_0 + \frac{2v_3}{v_2} \operatorname{Im} \phi_0, \quad (7.63)$$

where in the last step we have used the fact that $v_3/v_2 \ll 1$, as argued in Eq. (7.37). Thus J contains a small part of ϕ_0 and therefore couples

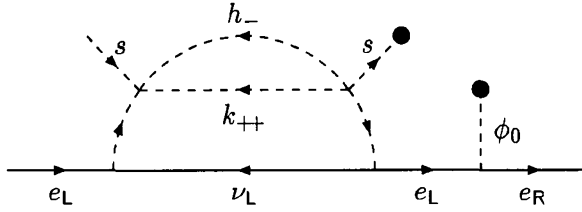


Figure 7.8: Majoron coupling with electron in no-hierarchy Majoron model.

directly to charged fermions. Since the ϕ -coupling to the electron is $\sqrt{2}m_e/v_2$, the coupling of J is

$$g_{eJ} = \frac{\sqrt{2}m_e}{v_2} \cdot \frac{2v_3}{v_2}. \quad (7.64)$$

The astrophysical bound of Eq. (7.56) then gives

$$v_3/v_2 \lesssim 10^{-6}, \quad (7.65)$$

which gives $v_3 \lesssim 20 \text{ keV}$.

With such a small vev, the mass of the field containing mostly $\text{Re } \Delta_0$ would be naturally of order v_3 . In that case, Z can easily decay into the Majoron and $\text{Re } \Delta_0$. This decay mode contributes twice as much to the total width of Z as a pair of neutrino and antineutrino does [19]. For the doublet Majoron model, the same argument can be used to show that $v'/v \lesssim 10^{-6}$. Here, Z can decay to the Majoron and the $\text{Re } \phi'_0$, and the rate is half of that of a neutrino antineutrino pair. Measured widths of the Z -boson does not have any room for these particles. So, both these models, although interesting at the time they were proposed, have been ruled out by now. One can, of course, make more complicated viable Majoron models where the Majoron is a mixture of, say, a doublet and a singlet.

- **Exercise 7.5** Calculate the decay width of the Z boson decaying to the Majoron and the real part of the same multiplet when the Majoron transforms like a n -dimensional representation of $SU(2)_L$. Compare this with the rate of $Z \rightarrow \nu\bar{\nu}$ derived in Exercise 2.4.

In contrast, if we want to break $B - L$ spontaneously in the model involving h_- and k_{++} , we can introduce a gauge singlet Higgs field s

whose important interaction is a quartic coupling

$$\lambda h_- h_- k_{++} s + \text{h.c.} , \quad (7.66)$$

which assigns a $B - L$ number 2 to s . The $\text{Im } s$ appears as the Majoron when $\text{Re } s$ develops a vev [20]. The implication for neutrino mass is the same as in Babu's model of §7.3.3 if we identify

$$\mu = \lambda \langle s \rangle , \quad (7.67)$$

where μ is the coupling constant defined in Eq. (7.46), associated with the trilinear scalar interaction of Babu's model. Thus, neutrino masses are small because they arise at 2-loop level. On the other hand, in absence of right-handed neutrinos in the model, s does not couple to any fermions at tree level. The coupling to electrons is generated at the 2-loop level through the diagram in Fig. 7.8 which can easily satisfy the astrophysical bound even if $\langle s \rangle$ is of the same order as the weak scale. Thus, this model does not suffer from any hierarchy problem.

Chapter 8

Neutrino mass in Left-Right symmetric models

An attractive extension of the standard model which may manifest itself in the multi-TeV (or perhaps higher) range of energies is the left-right symmetric model of weak interaction. As the name implies, in this model the left and right chiralities of fermions are assumed to play an identical role prior to symmetry breaking (or at high energies above all symmetry breaking scales). It therefore follows that in the symmetric phase of this model, weak interactions conserve parity [1], a property already shared by electromagnetic, strong and gravitational interactions. This is perhaps closer to the spirit of unified theories than the standard model.

Left-right symmetric treatment of weak interactions requires that all left handed fermions must have a right handed partner in the spectrum. Thus, an immediate consequence of these models is the existence of a new particle, the right handed neutrino, denoted by ν_R or sometimes by N_R . This model therefore obeys complete quark-lepton symmetry in its spectrum and automatically leads to massive neutrinos.

The smallest gauge group that implements the hypothesis of left-right symmetry of weak interactions is $SU(2)_L \times SU(2)_R \times U(1)_{B-L}$. In the original versions of the model [1], it was not realized that the generator of the $U(1)$ symmetry is indeed $B - L$. It was appreciated subsequently [2], when it was pointed out that the electric charge formula for these models acquires a rather attractive form:

$$Q = I_{3L} + I_{3R} + \frac{B - L}{2}, \quad (8.1)$$

which looks quite similar to the Gell-Mann Nishijima relation for electric charge in the case of strong interactions. A number of very important

implications of parity symmetry breaking can be deduced from this equation [3]. For instance, at an energy scale where $SU(2)_L$ is unbroken but $SU(2)_R$ is broken, one finds from Eq. (8.1) that

$$\Delta I_{3R} = -\frac{1}{2}\Delta(B-L). \quad (8.2)$$

This equation, for purely baryon number conserving interactions (i.e., $\Delta B = 0$), implies that $|\Delta L| = 2|\Delta I_{3R}|$. If symmetry breaking is chosen so as to give $\Delta I_{3R} = 1$, we find $\Delta L = 2$, which implies Majorana neutrinos and neutrinoless double beta decays. Even if $\Delta I_{3R} = \frac{1}{2}$, Majorana neutrinos can arise in higher orders. Similarly, for purely lepton number conserving interactions, this implies violation of baryon number by two units, leading to processes such as neutron-antineutron oscillation [3, 4]. Our concern in this chapter will be neutrino masses in left-right symmetric models.

8.1 The gauge sector

8.1.1 Symmetry breaking

In the left-right symmetric models, the quarks and leptons are assigned to the following irreducible representations of the gauge group $SU(2)_L \times SU(2)_R \times U(1)_{B-L}$:

$$\begin{aligned} q_L &= \begin{pmatrix} u_L \\ d_L \end{pmatrix} : (2, 1, 1/3), & q_R &= \begin{pmatrix} u_R \\ d_R \end{pmatrix} : (1, 2, 1/3) \\ \psi_L &= \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix} : (2, 1, -1), & \psi_R &= \begin{pmatrix} N_{eR} \\ e_R \end{pmatrix} : (1, 2, -1). \end{aligned} \quad (8.3)$$

As in §7.2, we write the right handed neutrino with a different symbol than the left handed one to keep open the possibility that they might end up being parts of two different Majorana particles.

The gauge invariant Lagrangian for the quarks and leptons leads to

the following gauge boson interactions with fermions:

$$\begin{aligned}
 \mathcal{L}_{\text{gauge}} = & g_L \left[\bar{q}_L \gamma_\mu \frac{\tau}{2} q_L + \bar{\psi}_L \gamma_\mu \frac{\tau}{2} \psi_L \right] \cdot \mathbf{W}_L^\mu \\
 & + g_R \left[\bar{q}_R \gamma_\mu \frac{\tau}{2} q_R + \bar{\psi}_R \gamma_\mu \frac{\tau}{2} \psi_R \right] \cdot \mathbf{W}_R^\mu \\
 & + g' \left[\frac{1}{6} \bar{q} \gamma_\mu q - \frac{1}{2} \bar{\psi} \gamma_\mu \psi \right] B^\mu
 \end{aligned} \tag{8.4}$$

where $\mathbf{W}_L^\mu$, $\mathbf{W}_R^\mu$ and B^μ are the gauge bosons corresponding to the groups $SU(2)_L$, $SU(2)_R$ and $U(1)_{B-L}$ respectively, whereas g_L , g_R and g' are the corresponding gauge coupling constants. We now require that the theory be invariant under parity operation $\mathcal{P}$ under which various fields transform as follows:

$$\psi_L \leftrightarrow \psi_R, \quad q_L \leftrightarrow q_R, \quad \mathbf{W}_L \leftrightarrow \mathbf{W}_R. \tag{8.5}$$

This requires $g_L = g_R = g$, reducing the number of arbitrary gauge coupling constants to two as in the standard model. As in the standard model, we can parameterize g and g' in terms of two parameters: the electric charge of the positron, e and the Weinberg angle, θ_W , defined by

$$g = e / \sin \theta_W, \tag{8.6}$$

which then leads to

$$g' = \frac{e}{\sqrt{\cos 2\theta_W}}. \tag{8.7}$$

□ **Exercise 8.1** For any gauge theory, if the electric charge operator is given by the combination $Q = \sum_n a_n T_n$, where T_n are generators and a_n are numerical coefficients, show that the electric charge is given by [5]

$$\frac{1}{e^2} = \sum_n \frac{a_n^2}{g_n^2}, \tag{8.8}$$

where g_n is the coupling constant accompanying T_n . Use this equation to derive Eq. (8.7) from Eqs. (8.1) and (8.6).

Let us now consider the breaking of this gauge symmetry. In order to maintain left-right symmetry, we must choose Higgs multiplets which

are left-right symmetric. The first candidate for this purpose seems to be

$$\Phi = \begin{pmatrix} \phi^0 & \phi'^+ \\ \phi^- & \phi'^0 \end{pmatrix} \quad : \quad (2, 2, 0), \quad (8.9)$$

which can couple to the fermion bilinears $\bar{q}_L q_R$ and $\bar{\psi}_L \psi_R$. Thus, after symmetry breaking, non-zero vevs of the electrically neutral components of Φ ,

$$\langle \Phi \rangle = \begin{pmatrix} \kappa & 0 \\ 0 & \kappa' \end{pmatrix}, \quad (8.10)$$

can give masses to quarks and leptons. However, these vevs are not sufficient to break the gauge symmetry. Since Φ is neutral under $B-L$, $U(1)_{B-L}$ is not broken by the vevs of Φ . Moreover, from Eq. (8.1), we see that the electrically neutral components of Φ have $I_{3L} + I_{3R} = 0$. Thus, the gauge symmetry is broken to $U(1)_{I_{3L}+I_{3R}} \times U(1)_{B-L}$, and not to $U(1)_Q$ as observed in the real world. One must then introduce more Higgs multiplets to obtain the desired symmetry breaking pattern.

There are various ways to achieve this goal. In the early days of the development of the left-right symmetry, the breaking of the gauge symmetry [1, 6] was implemented by choosing the Higgs multiplets

$$\chi_L : (2, 1, 1) \quad , \quad \chi_R : (1, 2, 1). \quad (8.11)$$

Later, it was shown that in order to understand the smallness of neutrino masses, it is preferable to introduce the following set of fields [7]:

$$\Delta_L : (3, 1, 2) \quad , \quad \Delta_R : (1, 3, 2). \quad (8.12)$$

The choice in Eq. (8.11) implies Dirac neutrinos whereas the choice in Eq. (8.12) leads to Majorana neutrinos. We first discuss the Majorana neutrino alternative and comment later on the Dirac neutrino case.

The gauge symmetry breaking proceeds in two stages. In the first stage, the electrically neutral component of Δ_R , denoted by Δ_R^0 , acquires a vev v_R and breaks the gauge symmetry down to $SU(2)_L \times U(1)_Y$ where

$$\frac{Y}{2} = I_{3R} + \frac{B-L}{2}. \quad (8.13)$$

The parity symmetry breaks down at this stage. In the second stage, the vevs of the electrically neutral components of Φ break the symmetry down to $U(1)_Q$. Experimental constraints force the relation that $\kappa, \kappa' \ll v_R$, as we will see later. Here we take κ and κ' real for simplicity. Making them complex leads to an interesting model of CP-violation.

At the first stage, the charged right handed gauge bosons denoted by $W_R^\pm$ and a neutral gauge boson called Z' acquire masses proportional to v_R and become much heavier than the usual left handed $W_L^\pm$ and the Z bosons which pick up masses proportional to κ and κ' only at the second stage. In general the different gauge bosons mix and lead to a 2×2 mass matrix describing the W_L - W_R system and a 3×3 mass matrix describing the neutral gauge bosons W_{3L} , W_{3R} and B . The charged gauge boson mass matrix turns out to be

$$\begin{pmatrix} \frac{1}{2}g^2(\kappa^2 + \kappa'^2 + 2v_L^2) & g^2\kappa\kappa' \\ g^2\kappa\kappa' & \frac{1}{2}g^2(\kappa^2 + \kappa'^2 + 2v_R^2) \end{pmatrix}. \quad (8.14)$$

Here v_L denotes the vev of Δ_L^0 which is assumed to be much smaller than κ, κ' . The eigenstates of this matrix are

$$\begin{aligned} W_1 &= W_L \cos \zeta + W_R \sin \zeta \\ W_2 &= -W_L \sin \zeta + W_R \cos \zeta, \end{aligned} \quad (8.15)$$

where

$$\tan 2\zeta = \frac{2\kappa\kappa'}{v_R^2 - v_L^2}. \quad (8.16)$$

In what follows, we will assume that $\kappa' \ll \kappa$, although strictly speaking it is not required from phenomenological considerations. With this assumption, ζ is small, i.e., the physical charged gauge bosons W_1 and W_2 are the same as W_L and W_R to a good approximation. The masses of these gauge bosons can be obtained from the relations

$$\begin{aligned} M_{W_L}^2 &\simeq M_{W_1}^2 \cos^2 \zeta + M_{W_2}^2 \sin^2 \zeta \\ M_{W_R}^2 &\simeq M_{W_1}^2 \sin^2 \zeta + M_{W_2}^2 \cos^2 \zeta, \end{aligned} \quad (8.17)$$

where $M_{W_L}^2$ and $M_{W_R}^2$ are the diagonal elements of the mass matrix in Eq. (8.14). The charged current weak interactions can be written,

suppressing generation indices for the sake of clarity, as

$$\begin{aligned} \mathcal{L}_{\text{cc}} = & \frac{g}{\sqrt{2}} [(\bar{u}_L \gamma_\mu d_L + \bar{\nu}_L \gamma_\mu e_L) W_L^\mu \\ & + (\bar{u}_R \gamma_\mu d_R + \bar{\nu}_R \gamma_\mu e_R) W_R^\mu] + \text{h.c.} . \end{aligned} \quad (8.18)$$

It is clear that for $M_{W_L} \ll M_{W_R}$, the charged current weak interactions will appear nearly maximally parity violating at low energies. Any deviation from the pure left-handed (or $V - A$) structure of charged weak current will constitute evidence for the right-handed currents and therefore for a left-right symmetric structure of weak interactions.

For the discussion of neutral currents and gauge bosons, it is convenient to define the following basis:

$$\begin{aligned} A &= \sin \theta_W (W_{3L} + W_{3R}) + \sqrt{\cos 2\theta_W} B \\ Z &= \cos \theta_W W_{3L} - \sin \theta_W \tan \theta_W W_{3R} - \tan \theta_W \sqrt{\cos 2\theta_W} B \\ Z' &= \frac{\sqrt{\cos 2\theta_W}}{\cos \theta_W} W_{3R} - \tan \theta_W B . \end{aligned} \quad (8.19)$$

Here, A is the photon, which remains massless after symmetry breaking.

- **Exercise 8.2** Write the mass matrix for the neutral vector bosons when the symmetry is broken by the vevs of Φ , Δ_L and Δ_R . Show that the photon eigenstate given in the text is indeed a zero-mass eigenstate of this mass matrix.
- **Exercise 8.3** Show that when χ_L and χ_R are used instead of Δ_L and Δ_R for symmetry breaking, the photon is still given by the same combination.

The two massive neutral gauge bosons can be called Z_1 and Z_2 , which are mixtures of Z and Z' :

$$\begin{aligned} Z_1 &= Z \cos \xi + Z' \sin \xi \\ Z_2 &= -Z \sin \xi + Z' \cos \xi . \end{aligned} \quad (8.20)$$

so that the masses of these gauge bosons can be calculated from the relations

$$\begin{aligned} M_{Z_1}^2 &= M_Z^2 \cos^2 \xi + M_{Z_2}^2 \sin^2 \xi \\ M_{Z_2}^2 &= M_{Z_1}^2 \sin^2 \xi + M_{Z_2}^2 \cos^2 \xi . \end{aligned} \quad (8.21)$$

Here,

$$\begin{aligned}
 M_Z^2 &= \frac{g^2}{2 \cos^2 \theta_W} (\kappa^2 + \kappa'^2 + 4v_L^2) \\
 M_{Z'}^2 &= \frac{g^2}{2 \cos^2 \theta_W \cos 2\theta_W} (4v_R^2 \cos^4 \theta_W \\
 &\quad + (\kappa^2 + \kappa'^2) \cos^2 2\theta_W + 4v_L^2 \sin^4 \theta_W), \quad (8.22)
 \end{aligned}$$

and the mixing angle ξ , neglecting v_L and using $\kappa^2 + \kappa'^2 \ll v_R^2$, is given by

$$\tan 2\xi \simeq \frac{(\cos 2\theta_W)^{3/2}}{2 \cos^4 \theta_W} \frac{\kappa^2 + \kappa'^2}{v_R^2} \simeq 2\sqrt{\cos 2\theta_W} M_Z^2 / M_{Z'}^2. \quad (8.23)$$

Obviously, as $v_R \rightarrow \infty$, $\xi \rightarrow 0$, so that $M_{Z_1}^2 \rightarrow M_Z^2$. In this limit, neglecting v_L , the masses of W_1 and Z_1 satisfy the standard model relation $M_{W_1} = M_{Z_1} \cos \theta_W$. The interaction of W_1 and Z_1 also reduce to those of the standard model in this limit. The $1/v_R^2$ corrections to the four-Fermi interactions can be obtained by considering the neutral current Lagrangian:

$$\begin{aligned}
 \mathcal{L}_{\text{nc}} &= \frac{g}{\cos \theta_W} \left[K_L^\mu Z_\mu + \frac{1}{\sqrt{\cos 2\theta_W}} \left\{ \sin^2 \theta_W K_L^\mu + \cos^2 \theta_W K_R^\mu \right\} Z'_\mu \right] \\
 &\simeq \frac{g}{\cos \theta_W} \left[\left\{ K_L^\mu - \frac{\xi}{\sqrt{\cos 2\theta_W}} \left(\sin^2 \theta_W K_L^\mu + \cos^2 \theta_W K_R^\mu \right) \right\} Z_{1\mu} \right. \\
 &\quad \left. + \frac{1}{\sqrt{\cos 2\theta_W}} \left\{ \sin^2 \theta_W K_L^\mu + \cos^2 \theta_W K_R^\mu \right\} Z_{2\mu} \right], \quad (8.24)
 \end{aligned}$$

where

$$K_L^\mu = \sum_f \bar{f} \gamma^\mu \left[I_{3L} \mathbb{1} - Q \sin^2 \theta_W \right] f, \quad (8.25)$$

and K_R^μ can be obtained by replacing I_{3L} by I_{3R} and changing the sign of γ_5 .

□ **Exercise 8.4** In the four-Fermi limit, find the leading-order corrections to the neutral current parameters $g_V^{(e)}$ and $g_A^{(e)}$ defined in Eq. (2.28).

8.1.2 Constraints on the masses of the gauge bosons

We are now ready to discuss the bounds on the masses of the gauge bosons W_2 and Z_2 as well as the left-right mixing parameter ζ . The most model-independent limit is on the mass of the Z_2 boson. This is obtained by analyzing neutrino neutral current data, where one searches for deviations from the predictions of the standard model. The present experimental accuracy in the neutral current data implies that [8, 9]

$$M_{Z_2} \geq 389 \text{ GeV} . \quad (8.26)$$

A Z_2 with leptonic couplings given in Eq. (8.24) and mass in the hundreds of GeV range should also be detectable in $p\bar{p}$ collider experiments. Analysis of these data leads to a somewhat more stringent bound [10] of 445 GeV on M_{Z_2} . The implication for future e^+e^- machines have also been studied extensively [11].

To obtain limits on M_{W_2} and ζ , an obvious thing to do is to look for deviations from the predictions of the $V - A$ theory for muon decay. However, since right-handed leptonic charged currents involve the right-handed neutrino field, one needs the mass of the right-handed neutrino in order to carry out the analysis. Since, as discussed in Ch. 4, the neutrinos can be Majorana particles, the N_R can have a different mass than the ν_L . For right-handed neutrinos to contribute to muon decay, N_R mass must satisfy $m_N \ll m_\mu$. Assuming this to be true, the muon decay parameters were analyzed [12] using existing data to obtain bounds on M_{W_2} and ζ . Subsequently, more refined experiments have been carried out. The most stringent limits at this moment come from the measurement of the ξ -parameter in μ -decay [13] using 100% stopped polarized muons. The limits are

$$\begin{aligned} M_{W_2} &\geq 432 \text{ GeV} && \text{for arbitrary } \zeta \\ \zeta &\leq 0.035 && \text{for } M_{W_2} \rightarrow \infty . \end{aligned} \quad (8.27)$$

In general these two bounds are correlated and one gets elliptical regions in the M_{W_2} - ζ plane which give the allowed and forbidden values for the above parameters. In Fig.8.1, we have summarized the various constraints on M_{W_2} and ζ .

For values of right-handed neutrino mass close to or bigger than the mass of the muon, the above analysis does not shed light on the

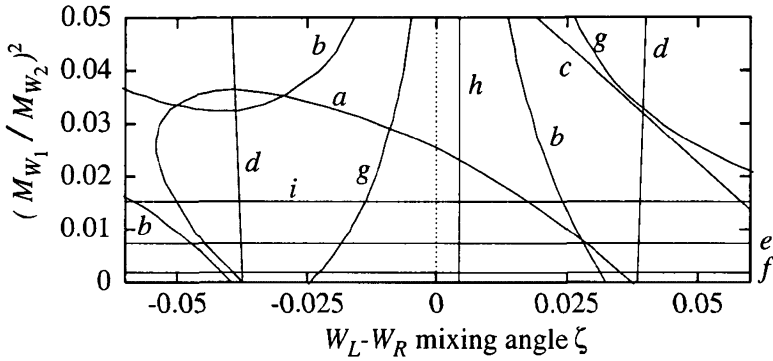


Figure 8.1: Various constraints between the heavy charged gauge boson mass M_{W_2} and the gauge boson mixing angle ζ . For each line, only the region towards the origin is allowed by a particular experiment. The constraints come from the following sources (see Ref. [14]): (a) endpoint spectrum in muon decay; (b) asymmetry parameter in the β -decay of ^{19}Ne ; (c) electron polarization in Gamow-Teller type β decay; (d) Michel parameter ρ in muon decay; (e) neutrinoless double beta decay and vacuum stability; (f) K_L - K_S mass difference; (g) relative lepton polarization in Fermi and Gamow-Teller decays; (h) unitarity; (i) direct collider search.

strength of the right-handed interactions and one must look at weak processes involving only hadrons. The most stringent bound arises from consideration of K_L - K_S mass difference, where one finds [15]

$$\Delta m_{K_1^0 \rightarrow K_2^0} \simeq \left(\Delta m_{K_1^0 \rightarrow K_2^0} \right)_{\text{SM}} \left[1 - 430 \left(\frac{M_{W_L}^2}{M_{W_R}^2} \right) \right], \quad (8.28)$$

where the short distance contribution calculated in the standard model has been denoted by the symbol $\left(\Delta m_{K_1^0 \rightarrow K_2^0} \right)_{\text{SM}}$. The standard model contribution can account for the bulk of the observed mass difference and gives the right sign as well. Thus, Eq. (8.28) implies

$$\frac{M_{W_L}^2}{M_{W_R}^2} < \frac{1}{430}, \quad (8.29)$$

or $M_{W_R} \geq 1.6 \text{ TeV}$, which is the most stringent bound on M_{W_R} . The original derivation of this result, though important, was incomplete in three ways: (a) the effect the intermediate t -quark was not taken into

account; (b) the Higgs boson effects, which are part of the complete left-right models, were not included; (c) the diagrams on which the analysis was based did not form a gauge invariant set. These issues were examined in subsequent papers [16, 17]. For the case of equal mixing angles in the left and right handed fermionic sectors, it is by now accepted that $M_{WR} \geq 1.6 \text{ TeV}$. These bounds have been further strengthened by nearly a factor of 2 by the inclusion of higher order QCD effects [18]. It has been noted that if the neutrinos are Dirac particles or if they are Majorana particles with $m_{N_R} \lesssim 10 \text{ MeV}$, observation of the neutrino signal from the supernova SN1987A provides a lower limit [19] on M_{WR} and ζ : $M_{WR} \geq 22 \text{ TeV}$ and $\zeta \leq 10^{-5}$.

8.2 Majorana neutrinos

8.2.1 The see-saw mechanism

It has been pointed out in §7.2.4 that a simple way to understand the smallness of neutrino mass is to assume it to be a Majorana particle and use the see-saw mechanism [7, 21].

In the left-right model, the see-saw mechanism appears as follows. We work with the Higgs multiplets [7] Φ along-with the triplets introduced in Eq. (8.12). The most general Yukawa couplings involving the leptons are given by

$$\begin{aligned}
 -\mathcal{L}_Y = & \sum_{a,b} h_{ab}^{(\ell)} \bar{\psi}_{aL} \Phi \psi_{bR} + \tilde{h}_{ab}^{(\ell)} \bar{\psi}_{aL} \tilde{\Phi} \psi_{bR} \\
 & + f_{ab} \left[\psi_{aL}^T C^{-1} \epsilon \tau \cdot \Delta_L \psi_{bL} + (L \rightarrow R) \right] + \text{h.c.}, \quad (8.30)
 \end{aligned}$$

where $\tilde{\Phi} = \tau_2 \Phi^* \tau_2$, and a, b now label different generations. It is then clear that at the first stage of symmetry breaking, we have $\langle \Delta_R^0 \rangle = v_R \neq 0$, leading to a heavy Majorana mass for the right-handed neutrinos, given by the matrix $f_{ab} v_R$. At the second stage, once the neutral components in Φ develop non-zero vevs as in Eq. (8.10), we obtain the following form for the mass matrix of the neutrinos:

$$\begin{pmatrix} 0 & m_D \\ m_D^T & f v_R \end{pmatrix}, \quad (8.31)$$

where

$$m_D = h^{(\ell)} \kappa + \tilde{h}^{(\ell)} \kappa'. \quad (8.32)$$

For $\mathcal{N}$ generations of fermions, Eq. (8.31) gives a $2\mathcal{N} \times 2\mathcal{N}$ matrix, where all the elements shown are $\mathcal{N} \times \mathcal{N}$ blocks. One can block diagonalize this matrix by a similarity transformation using the orthogonal matrix [20]

$$\begin{pmatrix} 1 - \frac{1}{2} \rho \rho^\top & \rho \\ -\rho^\top & 1 - \frac{1}{2} \rho^\top \rho \end{pmatrix}. \quad (8.33)$$

where $\rho = m_D f^{-1} / v_R$. This diagonalization is correct up to terms smaller than of order ρ^2 and one obtains the mass matrix for the light neutrinos to be

$$m^{\text{light}} = -\frac{1}{v_R} m_D f^{-1} m_D^\top. \quad (8.34)$$

Of course, this matrix needs further diagonalization to obtain the light neutrino eigenvalues and eigenstates. If we ignore mixing between generations in the first approximation and assume that $m_D \simeq m_\ell$, where m_ℓ is the mass of the charged lepton, we obtain from Eq. (8.34) the relation

$$m_{\nu_\ell} \simeq \frac{m_\ell^2}{M_{N_\ell}} \quad (8.35)$$

where M_{N_ℓ} is the mass of the heavy right-handed neutrino. If M_{N_ℓ} is assumed to be generation independent, one gets the following quadratic mass formula for neutrino masses:

$$m_{\nu_e} : m_{\nu_\mu} : m_{\nu_\tau} = m_e^2 : m_\mu^2 : m_\tau^2. \quad (8.36)$$

Here and elsewhere, when we talk about the “mass” of ν_e , for example, we imply the mass eigenvalue of the physical state which contains mostly the ν_e component, assuming mixings are small. It is clear from Eq. (8.34) that as $v_R \rightarrow \infty$, $m_\nu \rightarrow 0$. Therefore the smallness of the neutrino mass is connected to the suppression of $V + A$ currents in this scenario [7, 21].

While the above expression in Eq. (8.34) provides a natural explanation of why neutrino masses are small, without further assumption, it does not give us any idea about the real magnitudes of the masses

nor about the value of the right handed scale given certain values for them. The reason is that without further assumptions, we do not know the value of the Dirac mass term in Eq. (8.34).

For instance, if we take the see-saw expression in Eq. (8.34) literally (i.e. $m_D \sim m_\ell$), then a TeV scale W_R will lead to a spectrum an eV-keV-MeV type spectrum for neutrinos. In other words, if $m_{\nu_e} \approx 1$ eV, one gets $m_{\nu_\mu} \approx 40$ keV and $m_{\nu_\tau} \approx 12$ MeV. This kind of a spectrum is now definitively ruled out by data. In fact, present neutrino data seems to have clearly established that all neutrino masses are in the eV to sub-eV range. The question then is: is a TeV scale right handed symmetry ruled out by neutrino data? The answer is of course ‘no’ since it could very easily be that the Dirac mass of the neutrino could arise as a loop effect and therefore be of order $m_D \simeq \alpha m_{q,\ell}/4\pi$. If this is the case, then with a TeV scale v_R , the neutrino spectrum becomes roughly $10^{-6} \times (\text{eV} : \text{keV} : \text{MeV})$. This is in fact completely consistent with observations.

The usual assumption is however that the Dirac mass terms for the neutrinos arise at the tree level and are proportional to either the quark masses or the charged lepton masses. The first possibility occurs in grand unification models based on the $SO(10)$ symmetry group, and the second in some models based on the group E_6 , which are discussed in Ch. 9. Such scenarios for Dirac masses would then imply that the scale of right handed symmetry is in the range of 10^{12} to 10^{15} GeV. This would push most aspects of the left-right models beyond the range of observability.

Below we discuss the implications of both the TeV scale v_R models as well as those with super high values for the v_R .

8.2.2 Implications of TeV scale W_R models for leptons

If it turns out that neutrino masses do allow one to maintain a TeV scale for the W_R , then there are several interesting implications for leptonic physics which are independent of the model details used to make the Dirac masses of neutrinos small. We give a sample of those results below.

A low W_R scale allows new decay modes for the heaviest neutrino in addition to the conventional W_L induced decay $\nu_h \rightarrow \nu_{\text{light}} + \gamma$. The decay arises from the exchange of the neutral member of the Higgs triplet

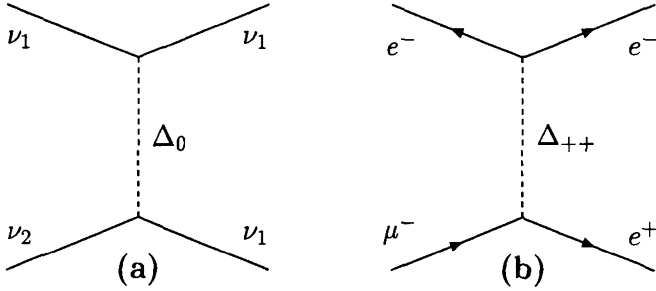


Figure 8.2: Tree-level diagrams for (a) $\nu_2 \rightarrow 3\nu_1$ and (b) $\mu \rightarrow 3e$ in the left-right model.

$\Delta_{L,0}$ (see Fig. 8.2a). The decay rate via this mode can be written as

$$\Gamma_{\nu_2 \rightarrow 3\nu_e} = \left(\frac{f_{11}f_{12}}{M_{\Delta_L^0}^2} \right)^2 \frac{m_2^5}{192\pi^3}. \quad (8.37)$$

In order to estimate the lifetime for ν_2 , we need some idea about the couplings f_{ij} and the mass of the Δ triplet. The following discussion gives some idea about them. The same set of couplings f that give rise to neutrino decay also give rise to $\mu^- \rightarrow e^-e^-e^+$ (commonly called $\mu \rightarrow 3e$) decay [26] via the diagrams shown in Fig. 8.2b. The upper limit on this decay rate is very stringent:

$$B(\mu \rightarrow 3e) \leq 1.0 \times 10^{-12}. \quad (8.38)$$

The rate for $\mu \rightarrow 3e$ derived from Fig. 8.2b is

$$\Gamma_{\mu \rightarrow 3e} = \left(\frac{f_{ee}f_{e\mu}}{M_{\Delta^{++}}^2} \right)^2 \frac{m_\mu^5}{192\pi^3}, \quad (8.39)$$

where $M_{\Delta^{++}}$ is the mass of the lighter doubly charged particle. The bound of Eq. (8.38) now implies

$$\frac{f_{ee}f_{e\mu}}{M_{\Delta^{++}}^2} \leq 10^{-6} G_F. \quad (8.40)$$

Since the Z -decay modes do not reveal the existence of any neutral scalar triplet, Δ_L^0 must be heavier than 45 GeV. On the other hand,

since Δ_L^0 and Δ_L^{++} belong to the same $SU(2)_L$ multiplet, their mass difference should be of order the $SU(2)_L$ breaking, i.e., a few hundred GeV at most. With these constraints, the masses of Δ_L^0 and Δ_L^{++} must be within a factor of about 10 of each other. Thus, for example, the $\mu \rightarrow 3e$ rate might be suppressed if, say, $f_{ee} < 10^{-6}$. In any case, using Eq. (8.39) for $\mu \rightarrow 3e$ decay, we estimate $\tau_{\nu_2} \simeq 10^{39}$ years for $m_2 \simeq 1$ eV.

There are two other interesting physical processes mediated by the exchange of Δ_L^+ and Δ_L^{++} . The Δ_L^+ exchange leads to an anomalous muon decay of the following type:

$$\mu^- \rightarrow e^- + \nu_e + \widehat{\nu}_\mu. \quad (8.41)$$

Note that the conventional muon decay is $\mu^- \rightarrow e^- + \widehat{\nu}_e + \nu_\mu$. The Δ_L^{++} exchange, on the other hand, leads to muonium to antimuonium transition:

$$\mu^+ e^- \rightarrow \mu^- e^+. \quad (8.42)$$

The rates of these processes are given in terms of the following effective couplings:

$$\begin{aligned} G_{\mu \rightarrow e \nu_e \widehat{\nu}_\mu} &\simeq \frac{f_{ee} f_{\mu\mu}}{M_{\Delta_L^+}^2} \\ G_{\mu^+ e^- \rightarrow \mu^- e^+} &\simeq \frac{f_{ee} f_{\mu\mu}}{M_{\Delta_L^{++}}^2}. \end{aligned} \quad (8.43)$$

The present experimental upper limits for these processes are at the level of $10^{-2} G_F$ and $10^{-3} G_F$ respectively [28, 29].

8.3 Physics involving right-handed neutrinos

The left-right symmetric model with Majorana neutrinos has several interesting implications on the properties of neutrinos, which we discuss below. To understand these properties, it is useful to restrict our attention only to one generation. In this case the quantities m_D and $M \equiv f v_R$, appearing in the matrix of Eq. (8.31), become just numbers.

8.3.1 Flavor changing neutral currents

The diagonalization of the mass matrix was discussed in §7.2.4. The eigenvalues of the matrix are found to be

$$m_\nu \simeq m_D^2/M \quad \text{and} \quad m_N \simeq M, \quad (8.44)$$

where ν and N denote the light and heavy Majorana neutrinos corresponding to a given generation.

The coupling with the Z -boson can be written in the flavor basis as

$$\frac{g}{2 \cos \theta_W} Z^\mu \bar{\nu}_{eL} \gamma_\mu \nu_{eL}, \quad (8.45)$$

since the N_e -field does not couple to the Z -boson. Recalling Eq. (7.20), we can write this coupling in terms of the mass eigenstates as follows:

$$\begin{aligned} \frac{g}{2 \cos \theta_W} Z^\mu \Big[& \cos^2 \theta \bar{\nu} \gamma_\mu L \nu + \sin^2 \theta \bar{N} \gamma_\mu L N \\ & + \cos \theta \sin \theta (\bar{N} \gamma_\mu L \nu + \bar{\nu} \gamma_\mu L N) \Big] \end{aligned} \quad (8.46)$$

where θ is the mixing angle, given in Eq. (7.15). Since $m_N \gg m_D$, we can approximately write

$$\theta \simeq \frac{m_D}{M} \simeq \left(\frac{m_\nu}{m_N} \right)^{1/2}. \quad (8.47)$$

For Z coupling to a single Majorana fermion, the vector part of the coupling vanishes and only the axial vector part contributes. However, Eq. (8.46) shows that Z can also couple to two different fermions. In other words, Z has flavor changing neutral currents (FCNC) so far as its coupling to neutral fermions are concerned. This is unlike the case in the standard model. For $m_N < M_Z$, this will lead to the following decay width for $Z \rightarrow \nu N$:

$$\Gamma(Z \rightarrow \nu N) \approx \theta^2 \cdot 165 \text{ MeV}. \quad (8.48)$$

If we assume $m_D \simeq m_e$ as in the previous section, we get $m_{N_1} \geq 800 \text{ GeV}$ if we take the $m_{\nu_1} \leq 0.2 \text{ eV}$. From Eq. (8.47), this implies $\theta \leq 5 \times 10^{-7}$. Thus the ν - N mixing is small in the see-saw limit. If, however, the see-saw limit is relaxed, one could get bigger values for θ , which would then lead to an appreciable decay width of the Z through Eq. (8.48). This decay mode has a distinctive signature since, as we will see below, decay modes of N_1 can involve charged leptons or hadrons which can be easily detected.

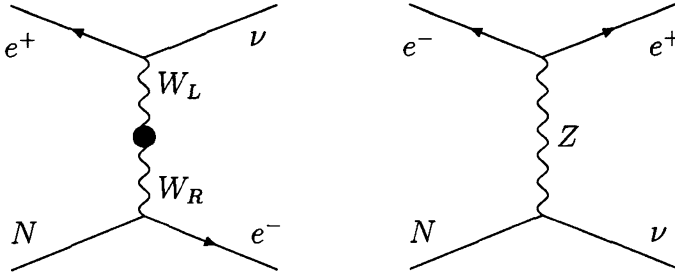


Figure 8.3: Diagrams for the decay $N \rightarrow \nu + e^+ + e^-$.

8.3.2 Decay of the right-handed neutrinos

In the see-saw picture, the right-handed neutrino mass is in the tens of GeV range. If we give up the see-saw picture, its mass could be arbitrary. It is therefore important to know its decay properties so that one could restrict its mass and coupling from observations discussed in this section. If we consider only one generation, we find that N has the following possible decay modes if we restrict the discussion to modes having not more than three particles in the final state:

$$N \rightarrow \begin{cases} \nu + e^+ + e^- \\ \nu + \gamma \\ \nu + \gamma + \gamma \\ 3\nu \\ e^\pm + \text{hadrons} . \end{cases} \quad (8.49)$$

The hadronic decay becomes predominant above the pion threshold, i.e., for $m_N \gtrsim 140 \text{ MeV}$. For masses below that threshold, the only decay modes are leptonic and radiative. Let us first discuss these non-hadronic modes.

• $N \rightarrow \nu + e^+ + e^-$

This decay proceeds via the W_L - W_R mixing graph shown in Fig. 8.3a and the flavor changing neutral current graph in Fig. 8.3b. The partial decay width of N into this channel is given by

$$\Gamma(N \rightarrow \nu + e^+ + e^-) \simeq \frac{G_F^2 m_N^5}{192\pi^3} (\zeta^2 + \theta^2 \Delta) , \quad (8.50)$$

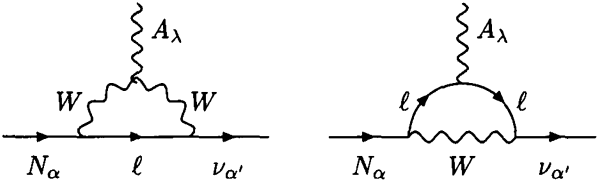


Figure 8.4: Diagrams for the decay $N \rightarrow \nu + \gamma$.

where $\Delta = \frac{1}{4} - \sin^2 \theta_W + 2 \sin^4 \theta_W$.

• $N \rightarrow \nu + \gamma$

This decay is absent at the tree level and is induced at the one loop level via the W_L - W_R mixing graph of Fig. 8.4. The decay amplitude for this process can be written as

$$\bar{\nu} \sigma^{\mu\rho} (a + b \gamma_5) N q_\rho \epsilon_\mu^*, \quad (8.51)$$

where ϵ_μ is the polarization and q is the momentum of the photon. Consider the simple case of CP-conserving theories. The Majorana neutrinos have specific CP eigenvalues in this case. If, for example, the CP eigenvalue of N and ν are opposite, one obtains $a = 0$ in the above equation. Detailed calculation, described in §13.3.3, shows that

$$b = - \frac{e G_F}{\sqrt{2} \pi^2} \sum_\ell m_\ell U_{\ell\nu} V_{\ell N}, \quad (8.52)$$

where U and V are the mixing matrices in the left and right handed sectors respectively. Due to the presence of the factor m_ℓ in Eq. (8.52), decay into the ν_τ mode is likely to dominate and we find [32] the width to be

$$\Gamma(N \rightarrow \nu_\tau + \gamma) \simeq 8 \times 10^{-22} \zeta^2 \theta_{e\tau}^2 \left(\frac{m_N}{1 \text{ MeV}} \right)^3 \text{ MeV}, \quad (8.53)$$

where $\theta_{e\tau}$ stands for the mixing.

- $N \rightarrow 3\nu$

This decay proceeds through the flavor changing Z -exchange graph similar to that in Fig. 8.3b and leads to the following decay width:

$$\Gamma(N \rightarrow 3\nu) \simeq \frac{G_F^2}{192\pi^3} \theta^2 m_N^5. \quad (8.54)$$

Using Eq. (8.53), we find the relative branching ratios

$$\frac{\Gamma(N \rightarrow 3\nu)}{\Gamma(N \rightarrow \nu\gamma)} \simeq 2 \times 10^{-3} \left(\frac{\theta}{\zeta} \right)^2 \left(\frac{m_N}{1 \text{ MeV}} \right)^2. \quad (8.55)$$

- $N \rightarrow e^\pm + \text{hadrons}$

This decay mode occurs for $m_N \geq 150 \text{ MeV}$ or so [33]. Due to the Majorana nature of N , one obtains

$$\Gamma(N \rightarrow e^+ + \text{hadrons}) = \Gamma(N \rightarrow e^- + \text{hadrons}) \quad (8.56)$$

Depending on the value of m_N , one or two generation of quarks will contribute. If only the first generation contributes, we get

$$\Gamma(N \rightarrow e^+ + \text{hadrons}) = \frac{G_F^2}{192\pi^3} \cdot 9m_N^5 \left(\frac{M_{W_L}}{M_{W_R}} \right)^4. \quad (8.57)$$

For $M_{W_R} \simeq 2 \text{ TeV}$, if the charm threshold is open, we get

$$\Gamma(N \rightarrow e^+ + \text{hadrons}) = 1.3 \times 10^{-15} \left(\frac{m_N}{5 \text{ GeV}} \right)^5 \text{ GeV}. \quad (8.58)$$

8.4 Naturalness of the see-saw formula

The see-saw picture discussed in §8.2.1 has a problem of naturalness associated with it. The neutrino mass matrix written in Eq. (8.31) is derived from the Eq. (8.30) on the assumption that the Higgs field Δ_L has got a zero vev. However, an actual analysis of the Higgs potential does not bear out this result. To see this, let us write the full Higgs potential involving Φ , Δ_L and Δ_R . For this purpose, it is useful to introduce the matrices $\Delta_L \equiv \epsilon \tau \cdot \mathbf{\Delta}_L$ and a similar matrix form of Δ_R .

Along-with the matrix Φ defined in Eq. (8.9), we can write the potential now:

$$\begin{aligned}
 V(\Phi, \Delta_L, \Delta_R) = & - \sum_{i,j} \mu_{ij}^2 \text{Tr } \Phi_i^\dagger \Phi_j - \mu^2 (\text{Tr } \Delta_L^\dagger \Delta_L + \text{Tr } \Delta_R^\dagger \Delta_R) \\
 & + \sum_{ijkl} \lambda_{ijkl} \text{Tr } (\Phi_i^\dagger \Phi_j) \text{Tr } (\Phi_k^\dagger \Phi_l) + \sum_{ijkl} \lambda'_{ijkl} \text{Tr } (\Phi_i^\dagger \Phi_j \Phi_k^\dagger \Phi_l) \\
 & + \rho_1 [(\text{Tr } \Delta_L^\dagger \Delta_L)^2 + (\text{Tr } \Delta_R^\dagger \Delta_R)^2] \\
 & + \rho_2 [(\text{Tr } \Delta_L^\dagger \Delta_L \Delta_L^\dagger \Delta_L) + (\text{Tr } \Delta_R^\dagger \Delta_R \Delta_R^\dagger \Delta_R)] \\
 & + \rho_3 (\text{Tr } \Delta_L^\dagger \Delta_L) (\text{Tr } \Delta_R^\dagger \Delta_R) \\
 & + \rho_4 [(\text{Tr } \Delta_L^\dagger \Delta_L) (\text{Tr } \Delta_L \Delta_L) + (\text{Tr } \Delta_R^\dagger \Delta_R) (\text{Tr } \Delta_R \Delta_R)] \\
 & + \sum_{ij} \alpha_{ij} (\text{Tr } \Phi_i^\dagger \Phi_j) (\text{Tr } \Delta_L^\dagger \Delta_L + \text{Tr } \Delta_R^\dagger \Delta_R) \\
 & + \sum_{ij} \beta_{ij} \text{Tr } [\Phi_i^\dagger \Phi_j (\Delta_L^\dagger \Delta_L + \Delta_R^\dagger \Delta_R)] \\
 & + \sum_{ij} [\gamma_{ij} \text{Tr } \Delta_L^\dagger \Phi_i \Delta_R \Phi_j^\dagger + \text{h.c.}], \tag{8.59}
 \end{aligned}$$

where the sums over i, j, k and l always runs from 1 to 2, with $\Phi_1 = \Phi$ and $\Phi_2 \equiv \epsilon \Phi^* \epsilon$. In the presence of the γ_{ij} terms, minimization of the Higgs potential leads to an equation [7]

$$v_L = \left(\frac{\gamma_{12}}{2(\rho_1 + \rho_2) - \rho_3} \right) \frac{\kappa^2}{v_R}, \tag{8.60}$$

where $v_L = \langle \Delta_L^0 \rangle$ and we have assumed $\kappa' \ll \kappa$ for simplicity. We thus see that $v_L \simeq \gamma \kappa^2 / v_R$, leading to non-vanishing entries in the upper left corner of the mass matrix of Eq. (8.31). To see the implications of this term, let us restrict our attention to the case of one generation. We then see that the light neutrino mass is given by

$$m_\nu \simeq \gamma \frac{f \kappa^2}{v_R} - \frac{m_D^2}{f v_R}. \tag{8.61}$$

This formula, which is different from the conventional see-saw formula in Eqs. (7.25) and (8.34), will be called the type II see-saw formula.

First point to realize is that, as $v_R \rightarrow \infty$, the light neutrino mass goes to zero. Thus, the analytical connection between vanishing of neutrino mass and the absence of $V + A$ interactions is preserved. However,

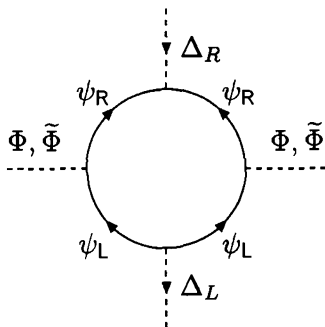


Figure 8.5: One-loop diagram giving rise to a non-trivial correction to the tree level value of γ_{ij} .

if we look at the magnitude of the neutrino masses, we see that, for $v_R \simeq 10 \text{ TeV}$ and $f \simeq 10^{-2}$ or so, we require $\gamma \lesssim 10^{-7}$ to get m_{ν_e} in the electron-volt range. This clearly requires a fine tuning of parameters and we ought to look for ways to avoid this unpleasant situation.

Let us ask ourselves what happens if we set $\gamma_{ij} = 0$ in the tree level potential. One finds that it is induced by the diagram of Fig. 8.5 which is logarithmically divergent and therefore one must introduce tree level counterterms to absorb this infinity. Thus, setting it to zero at the tree level does not solve our problem. It is however interesting that the diagram of Fig. 8.5 involves only Yukawa couplings and if we assume that the theory has a natural cutoff at $\Lambda \simeq M_{\text{Pl}}$, we estimate the magnitude of the one-loop corrections to be $\gamma_{1\text{-loop}} \simeq (f^2 h^2 / 16\pi^2) \ln(\Lambda / M_{W_R})$. Assuming the third generation to make the dominant contribution, we get $\gamma \simeq 10^{-7}$ for our choice of parameters, as required. Thus, choosing such a small value of γ is not as bad as it might appear.

There is however a more natural way to solve this problem [34] if we introduce a parity odd real scalar field σ into the theory [35]. In the presence of this particle, the Higgs potential acquires the following terms in addition to those already presented in Eq. (8.59):

$$V_\sigma = -\mu_\sigma^2 \sigma^2 + \lambda \sigma^4 + m_0 \sigma (\text{Tr } \Delta_L^\dagger \Delta_L - \text{Tr } \Delta_R^\dagger \Delta_R) + \dots \quad (8.62)$$

Minimization of the full potential $V + V_\sigma$ gives

$$\langle \sigma \rangle = M_p \neq 0. \quad (8.63)$$

Since σ is parity odd, its vev breaks only the discrete parity symmetry but keeps the gauge symmetry intact. If $M_p \gg M_{W_R}$, then one finds

$$v_L \simeq \gamma \frac{\kappa^2 v_R}{M_p^2} \quad (8.64)$$

instead of Eq. (8.60). Thus, for example, if $M_p \geq 10^4 v_R$, this makes the $\nu_L \nu_L$ mass term in the see-saw matrix very small, helping to restore naturalness of the see-saw picture. In Ch. 9, we will see that this mechanism of decoupling parity and $SU(2)_R$ breaking scales has a smooth realization in the $SO(10)$ model of grand-unification. An important phenomenological consequence of this idea is that the Δ_L -triplet acquires masses of order M_p and therefore cannot play any role in neutrino decay. Thus, some new mechanism for neutrino decay must be introduced into the theory in order to avoid cosmological problems [36].

8.5 Dirac neutrinos

In the original version of the left-right symmetric models [1, 6], though the neutrinos were considered to be Dirac particles, the smallness of their mass was not understood. More recently, a version of these models have been developed [37, 38] wherein the neutrino mass vanishes at the tree level but arises at one or two loop level, thus making it naturally much smaller than the charged fermion masses. An interesting feature of these models is that the neutrino masses scale linearly with the charged fermion mass and one has a mass formula of the type

$$m_{\nu_\ell} \simeq 10^{-7} m_\ell, \quad (8.65)$$

ignoring generational mixings. These models are inherently different from the class of left-right models that lead to small Majorana masses for neutrinos. We provide a brief sketch of these models in this section.

These models use heavy, vectorlike singlet quark and leptons in their construction and use the see-saw mechanism for quarks and charged leptons instead of the neutrino. The idea of using partial see-saw mechanism for down-type quarks and charged leptons was already discussed (e.g. in [39]). But later it was proposed [40] that the entire fermion spectrum be generated via the see-saw mechanism by postulating the

following heavy fermions in the left-right symmetric models in addition to the already existing light fermions of Eq. (8.3):

$$\begin{aligned} \mathcal{U}_{L,R} \left(1, 1, \frac{4}{3}\right), \quad \mathcal{D}_{L,R} \left(1, 1, -\frac{2}{3}\right), \\ E_{L,R}^- \left(1, 1, -2\right), \quad E_{L,R}^0 \left(1, 1, 0\right). \end{aligned} \quad (8.66)$$

An advantage of these models is their simple Higgs structure. Only one pair of Higgs doublets $\chi_{L,R}$ and a parity odd real singlet scalar (needed to generate the left-right asymmetry) are sufficient:

$$\chi_L (2, 1, 1), \quad \chi_R (1, 2, 1), \quad \sigma(1, 1, 0). \quad (8.67)$$

The most general gauge invariant coupling is given by

$$\begin{aligned} -\mathcal{L}_Y = & \left[h^d \bar{q}_L \chi_L \mathcal{D}_R + h^u \bar{q}_L \tilde{\chi}_L \mathcal{U}_R + h^\ell \bar{\psi}_L \chi_L E_R^- \right. \\ & \left. + h^\nu \bar{\psi}_L \tilde{\chi}_L E_R^0 + (L \leftrightarrow R) \right] + \text{h.c.} \\ & + \bar{\mathcal{U}} (f_{\mathcal{U}} + i\gamma_5 \sigma f'_{\mathcal{U}}) \mathcal{U} + \bar{\mathcal{D}} (f_{\mathcal{D}} + i\gamma_5 \sigma f'_{\mathcal{D}}) \mathcal{D} \\ & + \bar{E}^- (f_{E^-} + i\gamma_5 \sigma f'_{E^-}) E^- + \bar{E}^0 (f_{E^0} + i\gamma_5 \sigma f'_{E^0}) E^0 \\ & + (E_L^0{}^\top C^{-1} \mathcal{M}^L E_L^0 + E_R^0{}^\top C^{-1} \mathcal{M}^R E_R^0 + \text{h.c.}). \end{aligned} \quad (8.68)$$

where we have dropped the generation indices for the sake of clarity. At the minimum of the Higgs potential, one has $\langle \chi_{L,R} \rangle = v_{L,R}$, leading to the charged lepton and quark mass matrices of the form

$$\begin{pmatrix} 0 & h v_L \\ h^\top v_R & M \end{pmatrix}. \quad (8.69)$$

It is clear from this that the light quark masses are given by

$$m_{\text{light}} \simeq h^2 v_L v_R / M. \quad (8.70)$$

The interesting aspect of the above quark and charged lepton mass formula is that, if we chose $v_R/M \simeq 10^{-1}$ and $h \simeq 10^{-2}$, one can understand small masses of the first generation fermions. This ameliorates somewhat the severe finetuning problem of the standard model. A very exciting development is that within this scenario, a mechanism has been proposed [41] wherein the mass and mixing patterns among quarks and leptons receive a very plausible explanation.

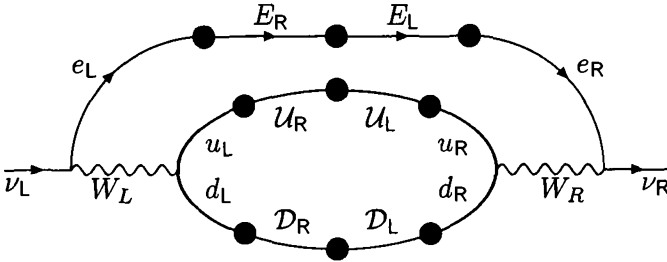


Figure 8.6: 2-loop diagram giving rise to Dirac neutrino masses.

Turning now to the neutrino sector, one can use [40] a generalized see-saw mechanism for neutrino masses leading to Majorana neutrinos. But to understand the smallness of neutrino masses, one needs the scale of new physics (i.e., the singlet quark masses, the right handed scale etc) to be in the range of 10^{10} GeV or so. However, if the E^0 is omitted from the heavy fermion sector, the neutrino mass vanishes at the tree level and arise only at the two loop level [37]. A typical relevant two loop graph is shown in Fig. 8.6. This graph leads to small and finite Dirac neutrino masses and leads to a linear mass formula given in Eq. (8.65) for $M_{W_R} \simeq 10$ TeV. In this picture, therefore, by increasing the W_R scale by one order of magnitude, one may, if one wishes, accommodate the matter oscillation solution of the solar neutrino problem [see Ch. 6 for details].

In this class of models, one does not expect any new charged Higgs bosons that couple to quarks although there may be charged Higgs bosons coupling to leptons in some versions [37]. Furthermore, in this model one expects tree level flavor changing neutral current signals such as $Z \rightarrow b\hat{s}$ etc.

Chapter 9

Neutrino mass in Grand unified models

At the present time, neutrino masses and mixings constitute the only solid experimental evidence for physics beyond the standard model. Cosmological observations strongly suggest the existence of dark matter in the Universe, which could also be taken as evidence for new physics as a possible explanation of the matter anti-matter asymmetry in Nature.

There are however plenty of theoretical arguments for new physics beyond the standard model such as understanding the origin of fermion masses and mixings or origin of weak symmetry breaking, solution to strong CP-problem. etc. Secondly, one can imagine scenarios beyond the standard model arising from aesthetic requirements such as the unification of all forces and matter at short distances. The latter have been a major cornerstone of many theoretical discussions since the 1970s. In this chapter, we consider these so called grand unified theories which unify strong, electromagnetic and weak interactions, and study the constraints imposed on them by various neutrino observations. It would of course be most desirable to have this scenario not only explain the smallness of neutrino masses (e.g. via the see-saw mechanism) but also the large neutrino mixings which are so different from the mixings in the quark sector.

While these requirements might sound severe, they still allow many possible choices. We illustrate them using three popular grand-unification groups: $SU(5)$, $SO(10)$ and E_6 , and see how detailed consideration of neutrino masses restrict the choice of Higgs structure, possible existence of intermediate scale, $\sin^2 \theta_W$ etc. For other details of the grand-unified theories, we refer the reader to existing books on the subject [1].

9.1 $SU(5)$

In the $SU(5)$ model [2], the fermions of each generation are assigned to $\{10\}$ and $\{\bar{5}\}$ representations, denoted respectively by T and F :

$$F = \begin{pmatrix} \hat{d}_1 \\ \hat{d}_2 \\ \hat{d}_3 \\ e \\ \nu \end{pmatrix}_L, \quad T = \begin{pmatrix} 0 & \hat{u}_3 & -\hat{u}_2 & u_1 & d_1 \\ & 0 & \hat{u}_1 & u_2 & d_2 \\ & & 0 & u_3 & d_3 \\ & & & 0 & e^+ \\ & & & & 0 \end{pmatrix}_L. \quad (9.1)$$

Here $\hat{u}$ denotes the conjugate state so that $\hat{u}_L$ for instance is the CPT -conjugate state of the right-handed helicity states of the up quark. The indices 1,2,3 stand for three different colors. The matrix T is antisymmetric, so we did not write down the elements below the diagonal.

The breaking down of the gauge symmetry to $SU(3)_c \times U(1)_Q$ is achieved by a choice of only two Higgs multiplets:

$$H \equiv \{\bar{5}\} \quad , \quad \Phi \equiv \{24\}. \quad (9.2)$$

The vacuum expectation value of Φ is chosen such that it breaks $SU(5)$ down to $SU(3)_c \times SU(2)_L \times U(1)_Y$ as follows

$$\langle \Phi \rangle = \text{diag} \left(V, V, V, -\frac{3}{2}V, -\frac{3}{2}V \right). \quad (9.3)$$

Denoting the gauge bosons of the $SU(5)$ model by

$$\left(\begin{array}{c|cc} \frac{1}{\sqrt{2}} \sum_{a=1}^8 \lambda^a G^a + \sqrt{\frac{2}{15}} B_{24} & X_1^{4/3} & Y_1^{1/3} \\ & X_2^{4/3} & Y_2^{1/3} \\ & X_3^{4/3} & Y_3^{1/3} \\ \hline X_1^{-4/3} & X_2^{-4/3} & X_3^{-4/3} \\ Y_1^{-1/3} & Y_2^{-1/3} & Y_3^{-1/3} \end{array} \middle| \begin{array}{c} \\ \\ \\ \frac{1}{\sqrt{2}} \boldsymbol{\tau} \cdot \mathbf{W} - \sqrt{\frac{3}{10}} B_{24} \end{array} \right) \quad (9.4)$$

we find that

$$M_X^2 = M_Y^2 = \frac{25}{8} g^2 V^2. \quad (9.5)$$

The final stage of symmetry breaking down to $U(1)_Q$ occurs via the non-zero vev of the $\{5\}$ -dimensional Higgs field H :

$$\langle H \rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ v/\sqrt{2} \end{pmatrix}. \tag{9.6}$$

This gives masses:

$$M_W = \frac{1}{2} g v \quad , \quad M_Z = \frac{g v}{2 \cos \theta_W} \tag{9.7}$$

where θ_W is the Weinberg angle. In this case, since there is only one gauge coupling constant, Weinberg angle is predicted at the unification scale to satisfy

$$\tan \theta_W = \sqrt{3/5}. \tag{9.8}$$

□ **Exercise 9.1** Using any one of the representations in Eq. (9.1), show that

$$\text{Tr} [(Y/2)^2] = \frac{5}{3} \text{Tr} [(I_{3L})^2]. \tag{9.9}$$

From this, deduce Eq. (9.8).

The unification scale V is obtained from the constraint that the three different gauge couplings g_3 , g_2 and g_1 at low energies unify to a single gauge coupling g_U above the unification scale. The evolution of the gauge couplings with the mass scale μ is given via the renormalization group equations for each gauge group which in the one-loop approximation looks as follows

$$\frac{dg_n}{d \ln \mu} = b_n \frac{g_n^3}{16 \pi^2} \quad \text{for } n = 1, 2, 3. \tag{9.10}$$

The coefficient b_n is given by $\frac{4}{3} \mathcal{N} - \frac{11}{3} n$ for $n = 2, 3$ if we neglect the Higgs contribution and assume that the only fermions are $\mathcal{N}$ generations of the fields shown in Eq. (9.1). Under the same assumptions, $b_1 = \frac{4}{3} \mathcal{N}$.

The values of the three gauge couplings at the scale M_Z have been experimentally determined from the LEP and SLC experiments to be:

$$\begin{aligned}\alpha_1^{-1}(M_Z) &= 58.89 \pm 0.11; \\ \alpha_2^{-1}(M_Z) &= 29.75 \pm 0.11 \\ \alpha_3(M_Z) &= 0.121 \pm .004 \pm .001.\end{aligned}\tag{9.11}$$

It is then easy to show that the three gauge couplings of the standard model do not unify as they are extrapolated to high scales. To see this, let us write down explicitly the solutions to the 1-loop evolution equations for the gauge couplings:

$$\begin{aligned}\alpha_3^{-1}(M_Z) &= \alpha_3^{-1}(M_U) - \frac{7}{2\pi} \ln \left(\frac{M_U}{M_Z} \right); \\ \alpha_2^{-1}(M_Z) &= \alpha_2^{-1}(M_U) - \frac{19}{12\pi} \ln \left(\frac{M_U}{M_Z} \right); \\ \alpha_1^{-1}(M_Z) &= \alpha_1^{-1}(M_U) + \frac{41}{20\pi} \ln \left(\frac{M_U}{M_Z} \right).\end{aligned}\tag{9.12}$$

Unification condition is that $\alpha_i(M_U) = \alpha_U$ for all i . One can then eliminate the two unknowns α_U and $\ln(M_U/M_Z)$ to obtain the following constraint on $\alpha_i(M_Z)$:

$$1.9\alpha_3^{-1}(M_Z) - 2.9\alpha_2^{-1}(M_Z) + \alpha_1^{-1}(M_Z) = 0\tag{9.13}$$

which is not satisfied by the LEP and SLC data for the gauge couplings. Therefore, $SU(5)$ is not acceptable as a viable model of grand unification unless it is “polluted” with extra Higgs bosons, in which case, the beauty and predictivity of the model is lost. Regardless of this unfortunate lack of unification however, the $SU(5)$ is an excellent tool for studying many generic properties of grand unified theories. Furthermore, the lack of unification in the non-supersymmetric $SU(5)$ model does not mean that the idea of grand unification is not right. As we will discuss subsequently, the $SO(10)$ model as well as the supersymmetric version of the $SU(5)$ model do lead to unification of coupling constants in an interesting manner. We therefore turn to the discussion of fermion masses in this model and more specifically the neutrino masses.

- **Exercise 9.2** Write down the evolution equations for $\alpha_n = g_n^2/4\pi$, beginning with Eq. (9.10). Solve these to obtain Eq. (9.12).

9.2 Neutrino masses in $SU(5)$ model

The most general gauge invariant Yukawa couplings in the $SU(5)$ model are given by

$$\mathcal{L}_Y = h_1 T_{ij}^T C^{-1} F^i H^j + h_2 \epsilon^{ijklm} T_{ij}^T C^{-1} T_{kl} H_m + \text{h.c.}, \quad (9.14)$$

where the displayed indices are $SU(5)$ indices. We have suppressed generation indices for the sake of clarity. Also, complex conjugation is implied by lowering and raising of indices. Thus, for example, F^i denotes the complex conjugate of F_i .

Once the $\{5\}$ -dimensional Higgs boson H acquires a vev, Eq. (9.14) leads to

$$\begin{aligned} M^{(d)} &= M^{(\ell)} = h_1 v / \sqrt{2} \\ M^{(u)} &= h_2 v / \sqrt{2}, \end{aligned} \quad (9.15)$$

where, as in Ch. 2, $M^{(u)}$ and $M^{(d)}$ denote the mass matrices of the positively and negatively charged quarks respectively, whereas $M^{(\ell)}$ is the mass matrix for charged leptons. The neutrinos remain massless. To see that it never acquires a mass through radiative corrections, we note that $\mathcal{L}_Y$ is invariant under the global $U(1)_{\mathcal{F}}$ symmetry, if we assign $\mathcal{F}(H_i) = -2/3$, $\mathcal{F}(T_{ij}) = +1/3$ and $\mathcal{F}(F_i) = 1$. This combines with the $U(1)_Y$ to generate the $B - L$ symmetry:

$$B - L = \frac{3}{5} \mathcal{F} - \frac{2}{\sqrt{15}} \lambda_{24}, \quad (9.16)$$

where λ_{24} is the generator which couples to the gauge boson B_{24} introduced in Eq. (9.4). It is easy to see that the Higgs fields developing vevs do not have any $B - L$ quantum number. Thus $B - L$ remains unbroken even after symmetry breaking. As a result, neutrinos cannot acquire any Majorana mass, which requires $B - L$ violation. It cannot acquire Dirac mass since there is no right-handed neutrino in the model. Thus, $m_\nu = 0$ in the simple $SU(5)$ model.

The simplest way to modify the $SU(5)$ model to generate a massive neutrino is to include a $\{15\}$ -dimensional Higgs boson $S_{ij} = S_{ji}$ (symmetric in the $SU(5)$ indices). In the presence of this multiplet, there is a Yukawa coupling of the form:

$$\mathcal{L}_Y' = f F_i^T C^{-1} F_j S^{ij} + \text{h.c.} \quad (9.17)$$

The multiplet S under $SU(3)_c \times SU(2)_L \times U(1)_Y$ contains an isotriplet Higgs boson Δ_L used in §7.3.1 for extending the standard model to generate neutrino mass. If we assign $\mathcal{F}(S) = +2$, then the model has exact $B - L$ number symmetry. By assigning a non-zero vev to S_{55} , i.e.

$$\langle S_{55} \rangle = \frac{u}{\sqrt{2}}, \quad (9.18)$$

the global $B - L$ symmetry is spontaneously broken, leading to a triplet Majoron [3]. Measurements of the Z -width [4] have ruled out this model. Let us therefore consider a variation of this model where $B - L$ is explicitly broken in the Higgs potential by the following term:

$$V = \mu_S H_i H_j S^{ij} + \text{h.c.} . \quad (9.19)$$

This breaks $U(1)_{\mathcal{F}}$ symmetry and therefore the $B - L$ symmetry explicitly. The vev of S in this case is determined to be

$$\langle S_{55} \rangle = \frac{u}{\sqrt{2}} \simeq \frac{\mu_S v^2}{V^2} . \quad (9.20)$$

Even if μ_S is chosen to be of order V , we get $u \sim 10^{-12} v \simeq 10^{-1} \text{ eV}$. Thus, in this model, smallness of neutrino mass arises naturally, and the neutrino is a Majorana particle.

This model of course has the same problem as the minimal $SU(5)$ model as far as unification is concerned. It is, therefore, phenomenologically not viable. Yet we presented it to illustrate the techniques and ideas connected with generating neutrino mass in grand-unified theories.

9.3 $SO(10)$

The next higher symmetry useful for grand-unification of particle interactions is $SO(10)$. An interesting aspect of $SO(10)$ is that it contains the left-right symmetric gauge group $SU(2)_L \times SU(2)_R \times SU(4)_C$ and, therefore, it automatically contains the right-handed neutrino. Since it has quarks and leptons in the same irreducible representation, the mechanism responsible for generating quark lepton masses automatically makes the neutrino massive. Thus, unlike the $SU(5)$ model, in $SO(10)$ model massive neutrinos arise naturally. Below we will describe

ways to understand their smallness and study the kind of constraints they impose on the details of $SO(10)$ grand-unification.

- **Exercise 9.3** Show that the generators of an orthogonal group are antisymmetric matrices. For the $SO(2n)$ groups, the generators in the fundamental (i.e., $2n$ -dimensional) representation can be chosen to be the matrices T_{ab} , whose elements are given by

$$(T_{ab})_{pq} \equiv -i(\delta_{ap}\delta_{bq} - \delta_{bp}\delta_{aq}). \quad (9.21)$$

Deduce the algebra of $SO(2n)$ groups from this. Show that the mutually commuting generators are $T_{12}, T_{34}, T_{56} \dots$

To begin with, the two maximal continuous subgroups of $SO(10)$ are:

- $G_{224} = SU(2)_L \times SU(2)_R \times SU(4)_C$;
- $G_5 = SU(5) \times U(1)$.

The spinor representation [5] of $SO(10)$ is $\{16\}$ -dimensional and decomposes under G_{224} and G_5 as follows:

$$\begin{aligned} G_{224} &: \{16\} \supset (2, 1, 4) + (1, 2, \bar{4}) \\ G_5 &: \{16\} \supset \{10\}_1 + \{\bar{5}\}_{-3} + \{1\}_5. \end{aligned} \quad (9.22)$$

From the $SU(5)$ content of the spinor representation, it is clear that all the known fermions of a single generation can be fitted into the spinor. The $SU(5)$ singlet piece of the spinor can be identified with the right-handed neutrino.

Let us first discuss the symmetry breaking of $SO(10)$ down to the standard model. Since $SO(10)$ has rank 5, there are many possible chains of symmetry breaking including one in which $SU(5)$ appears at an intermediate scale. Since $SU(5)$ is ruled out as a grand-unification symmetry, we will focus on the more interesting chains which contain the left-right symmetric group $SU(4)_C \times SU(2)_L \times SU(2)_R$ as an intermediate symmetry. It is important to realize that the actual maximal subgroup of $SO(10)$ is [6]:

$$SO(10) \supset SU(4)_C \times SU(2)_L \times SU(2)_R \times D, \quad (9.23)$$

where D is a discrete symmetry under which $f_L \rightarrow \hat{f}_L$, i.e., it interchanges the $(2, 1, 4)$ and the $(1, 2, \bar{4})$ sub-multiplets of the $SO(10)$ spinor.

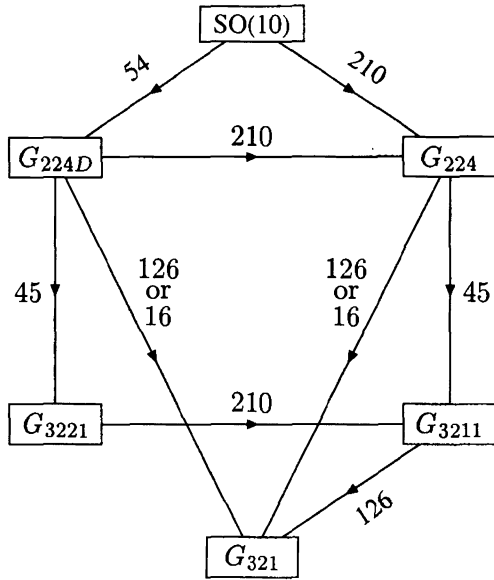


Figure 9.1: Some interesting chains of symmetry breaking from $SO(10)$ down to the standard model group which includes the left-right symmetric group as an intermediate stage. The representation whose vev can induce a certain breaking has also been shown.

It essentially plays the role of charge conjugation, at least on the fermion fields. This symmetry was called D -parity [6] and it plays an important role in understanding neutrino mass, as already alluded to in Ch. 8. Besides, it also has important bearings on the question of baryon asymmetry and domain wall problem in $SO(10)$ models [7]. Two of the important implications of D -parity are: a) it implies $g_L = g_R$ and b) it requires $\eta_B = \eta_{\bar{B}}$ where η_B is the number of baryons in the universe.

- **Exercise 9.4** Show that the following identification of the generators of the subgroups of the $SO(10)$ grand-unification group satisfies the algebras of the subgroups:

$$\begin{aligned}
 SU(2)_L &: \frac{1}{2}(T_{89} + T_{70}), \frac{1}{2}(T_{97} + T_{80}), \frac{1}{2}(T_{78} + T_{90}) \\
 SU(2)_R &: \frac{1}{2}(T_{89} - T_{70}), \frac{1}{2}(T_{97} - T_{80}), \frac{1}{2}(T_{78} - T_{90})
 \end{aligned}$$

$$\begin{aligned}
SU(3)_c : & \frac{1}{2}(T_{23} - T_{14}), \frac{1}{2}(T_{31} - T_{24}), \frac{1}{2}(T_{12} - T_{34}), \\
& \frac{1}{2}(T_{25} - T_{16}), \frac{1}{2}(T_{51} - T_{26}), \frac{1}{2}(T_{45} - T_{36}), \\
& \frac{1}{2}(T_{53} - T_{46}), \frac{1}{2\sqrt{3}}(T_{12} + T_{34} - 2T_{56}) \\
B - L : & \propto T_{12} + T_{34} + T_{56}.
\end{aligned} \tag{9.24}$$

If we demand the intermediate symmetry to contain an $SU(2)_L \times SU(2)_R$ group, then some of the possible symmetry breaking chains are listed in Fig. 9.1, where we have also listed the dimensionality of Higgs bosons responsible for each stage of symmetry breaking. To appreciate the effect of a particular kind of Higgs multiplet on symmetry breaking from G to G' , we give the decomposition of the multiplets used in Fig. 9.1 under the group G' . For instance, for the case of $\{45\}$, $G = G_{224}$ or G_{224D} and $G' = G_{3221}$. So we give the decomposition of the $\{45\}$ -dimensional representation of $SO(10)$ under $SU(3)_c \times SU(2)_L \times SU(2)_R \times U(1)_{B-L}$:

$$\begin{aligned}
\{45\} \xrightarrow{G_{3221}} & (1, 3, 1, 0) + (1, 1, 3, 0) + (8, 1, 1, 0) + (1, 1, 1, 0) \\
& + (3, 2, 2, \frac{2}{3}) + (\bar{3}, 2, 2, -\frac{2}{3}) + (3, 1, 1, \frac{1}{3}) + (\bar{3}, 1, 1, -\frac{1}{3}).
\end{aligned} \tag{9.25}$$

We see that $\{45\}$ contains a singlet of the group G_{3221} . If the multiplet develops a vev in that direction, then we obtain G_{3221} as the unbroken subgroup.

Let us now examine the multiplets which can break the $SO(10)$ symmetry down to G_{224} or G_{224D} . For this, we need the multiplets which contains a singlets under the broken symmetries. Note that

$$\{54\} \xrightarrow{G_{224}} (2, 2, 6) + (1, 1, 20) + (3, 3, 1) + (1, 1, 1) \tag{9.26}$$

$$\begin{aligned}
\{210\} \xrightarrow{G_{224}} & (1, 1, 15) + (2, 2, 20) + (3, 1, 15) + (1, 3, 15) \\
& + (2, 2, 6) + (1, 1, 1)
\end{aligned} \tag{9.27}$$

$$\begin{aligned}
\{16\} \xrightarrow{G_{321}} & (3, 2, \frac{1}{3}) + (1, 2, -1) + (\bar{3}, 1, -\frac{4}{3}) + (\bar{3}, 1, \frac{2}{3}) \\
& + (1, 1, 2) + (1, 1, 0).
\end{aligned} \tag{9.28}$$

For the $\{126\}$ -dimensional representation, we first give its reduction under G_{224} :

$$\{126\} \xrightarrow{G_{224}} (1, 1, 6) + (2, 2, 15) + (3, 1, 10) + (1, 3, \bar{10}). \tag{9.29}$$

The multiplet responsible for breakdown of G_{3221} or G_{224} down to the standard model is clearly contained in $(1, 3, \overline{10})$.

Again, as in the case of $SU(5)$, the gauge bosons contained in the quotient group $SO(10)/SU(4)_C \times SU(2)_L \times SU(2)_R$ become massive at the first stage of symmetry breaking and contribute to baryon non-conserving processes such as proton decay. There also exist detailed renormalization group analysis including two loop contributions [7, 8], which gives the predictions for $\sin^2 \theta_W$ for different symmetry breaking chains. In contrast with the minimal non-supersymmetric $SU(5)$ model, in this model the coupling constants truly unify [8].

Two chains important for the discussion of neutrino masses are summarized here.

- The first chain is

$$SO(10) \xrightarrow[\{54\}]{M_U} G_{224D} \xrightarrow[\{210\}]{M_p} G_{224} \xrightarrow[\{210\}]{M_C=M_{W_R^+}} G_{2113} \xrightarrow[\{126\}]{M_{Z'}} G_{321}. \quad (9.30)$$

In this case, the grand-unification scale $M_U \simeq 10^{16.6}$ GeV, $M_p \simeq 10^{14}$ GeV, $M_{W_R} = M_C \simeq 10^5$ - 10^7 GeV with $M_{Z'} \leq 1$ TeV, leading to a prediction of $\sin^2 \theta_W \simeq .227$. The important aspect of this symmetry breaking chain relevant to neutrino mass is the value of $M_{Z'}$ (or the scale of $B - L$ breaking) near the TeV scale.

- The second chain of symmetry breaking is given by

$$SO(10) \xrightarrow[\{45\}]{M_U=M_C=M_p} G_{3221} \xrightarrow[\{126\}]{M_{W_R}, M_{Z'}} G_{321}. \quad (9.31)$$

This chain is highly predictive and implies $M_U \simeq 10^{15.4}$ GeV for $\sin^2 \theta_W \simeq 0.23$ and $M_{W_R} \simeq M_{B-L} \simeq 10^{12}$ GeV.

9.4 Neutrino mass in $SO(10)$ models

Let us now discuss fermion masses in $SO(10)$ theories. To see the Higgs bosons which lead to gauge invariant Yukawa couplings, we note that

$$\{16\} \otimes \{16\} = \{10\} \oplus \{120\} \oplus \{126\}. \quad (9.32)$$

The Higgs bosons that give mass to quarks and leptons must belong to $\{10\}$, $\{120\}$ and $\{\overline{126}\}$ dimensional representations of $SO(10)$. We can write the Yukawa couplings as:

$$\begin{aligned} \mathcal{L}_Y = & \sum_{a,b} f_{\{10\}}^{ab} \psi_a^T \mathcal{B} C^{-1} \Gamma^i \psi_b \{10\}_i + f_{\{120\}}^{ab} \psi_a^T \mathcal{B} C^{-1} \Gamma^i \Gamma^j \Gamma^k \psi_b \{120\}_{ijk} \\ & + f_{\{\overline{126}\}}^{ab} \psi_a^T \mathcal{B} C^{-1} \Gamma^i \Gamma^j \Gamma^k \Gamma^l \Gamma^m \psi_b \{\overline{126}\}_{ijklm} + \text{h.c.} \end{aligned} \quad (9.33)$$

where Γ_i are analogs of Dirac gamma matrices for $SO(10)$ and $\mathcal{B}$ is the analog of the conjugation matrix for the spinor of $SO(10)$; a and b are generation indices.

It turns out that $\{120\}$ couplings are antisymmetric under the interchange of the $\{16\}$ fermions, whereas $\{10\}$ and $\{126\}$ are symmetric under same interchange. Hence, the Yukawa coupling matrices f have the following properties:

$$\begin{aligned} f_{\{10\}}^{ab} &= f_{\{10\}}^{ba} \\ f_{\{120\}}^{ab} &= -f_{\{120\}}^{ba} \\ f_{\{\overline{126}\}}^{ab} &= f_{\{\overline{126}\}}^{ba} . \end{aligned} \quad (9.34)$$

The fermion masses arise when any of $\{10\}_H$, $\{120\}_H$ or $\{126\}_H$ acquire nonzero vev's, with appropriate symmetry conditions satisfied by the mass matrices. Let us consider first the case where only one $\{10\}$ -dimensional Higgs boson is used to generate fermion masses. In this case, we have at grand-unification scale:

$$M_{ab}^{(u)} = M_{ab}^{(d)} = M_{ab}^{(\ell)} = M_{ab}^{(0)} . \quad (9.35)$$

When these relations are extrapolated to the weak scale, different corrections apply to quark and leptons and we get,

$$M^{(u)}(M_W) \simeq M^{(d)}(M_W) \simeq 3M^{(\ell)}(M_W) \simeq 3M^{(0)}(M_W) . \quad (9.36)$$

The degeneracy between up and down quark masses is clearly unrealistic. However, if we employ two $\{10\}$ -dimensional Higgs bosons, the up-down mass degeneracy is split. However, we still have the unphysical relation

$$M^{(u)}(M_W) \simeq 3M^{(0)}(M_W) , \quad (9.37)$$

which is clearly unacceptable since it implies Dirac neutrinos with very large masses.

This problem is solved by adding a $\{126\}$ -dimensional Higgs boson and assigning vev to the $SU(5)$ singlet component of it. It gives Majorana mass to the right-handed neutrino exactly as in the case of the left-right symmetric model, leading to a Majorana mass for the right handed neutrinos:

$$\mathcal{L}_{\text{Maj-mass}}^{N_R} = f_{\{126\}} \cdot \left(\frac{M_{BL}}{g} \right) N_R^T C^{-1} N_R, \quad (9.38)$$

where we have denoted the vev of the $SU(5)$ singlet component of $\{126\}$ by M_{BL}/g . Again as the case of left-right symmetric theories, an $SO(10)$ invariant coupling of the form $\{126\} \cdot \{126\} \cdot \{10\} \cdot \{10\}$ appears in the Higgs potential, which leads to a non-zero vev for the $\nu_L \nu_L$ component of $\{126\}$ and a direct Majorana mass for ν_L is induced as follows:

$$\mathcal{L}_{\text{Maj-mass}}^{\nu_L} = \lambda f_{\{126\}} \frac{\kappa^2}{M_{BL}/g} \nu_L^T C^{-1} \nu_L, \quad (9.39)$$

where κ denotes the $SU(2)_L$ -breaking scale. This leads to a neutrino mass matrix of the form

$$\begin{pmatrix} \lambda f_{\{126\}} \frac{\kappa^2}{M_{BL}/g} & \frac{1}{3} f_{\{10\}} \kappa \\ \frac{1}{3} f_{\{10\}} \kappa & f_{\{126\}} M_{BL}/g \end{pmatrix}. \quad (9.40)$$

It gives the following expression for the light neutrino masses:

$$m_\nu^{\text{light}} \simeq \lambda f_{\{126\}} \frac{\kappa^2}{M_{BL}/g} - \frac{1}{9} f_{\{10\}} f_{\{126\}}^{-1} f_{\{10\}} \frac{\kappa^2}{M_{BL}/g}. \quad (9.41)$$

This formula is very similar to the type II see-saw formula of Eq. (8.61), as opposed to the type I seesaw formula of Eqs. (7.25) and (8.34). We see that as $M_{BL} \rightarrow \infty$, $m_\nu \rightarrow 0$. In models where $B-L$ and $SU(2)_R$ are broken together, we obtain the analytic connection of m_ν and suppression of $V+A$ interactions as advocated in the context of the left-right model [9]. However, the presence of the first term destroys the see-saw formula. But since in a minimal Higgs scenario, this term is bound to appear, we will first discuss its effect on neutrino masses.

Looking at the orders of magnitude, it is clear that the ratio of the second term to the first is $\delta \approx \frac{f_{\{10\}}^2}{9\lambda f_{\{126\}}^2}$. But $f_{\{10\}}$ is proportional

to $M^{(u)}/M_W$. Neglecting inter-generational mixings for the moment, we expect its value to be a small number of order 10^{-4} for the first generation, about $\frac{1}{2} \times 10^{-2}$ for second generation and $\frac{1}{4}$ or greater for the third generation. Therefore, if $f_{\{126\}} \simeq 10^{-1}$ and $\lambda \simeq 10^{-1}$ to 10^{-2} , we expect that

$$\delta \sim \begin{cases} 10^{-5} \text{ to } 10^{-6} & \text{for the 1st generation,} \\ 10^{-2} & \text{for the 2nd generation.} \end{cases} \quad (9.42)$$

Therefore the first term is likely to dominate in this “minimal” version of the $SO(10)$ model. In this case, we expect neutrinos of all three generations to be almost degenerate. Their masses however depend on M_{BL} . Existing upper limits [10] on m_{ν_e} of 2.2 eV therefore imply $M_{BL} \geq 10^{10}$ to 10^{11} GeV. If indeed the scale is near that limit, assuming the elements of $f_{\{126\}}$ to be all of the same order of magnitude, we would expect

$$m_{\nu_e} \approx m_{\nu_\mu} \approx m_{\nu_\tau} \approx \text{few eV.} \quad (9.43)$$

Let us next turn to the case where the direct ν_L Majorana mass vanishes (e.g. by assuming $\lambda = 0$ or by a D -parity breaking mechanism to be described below). In this case, if we ignore quark mixing, we have for neutrino masses the following formula:

$$\begin{aligned} m_{\nu_e} &\approx \frac{1}{9} g \frac{m_u^2}{f_{\{126\}} M_{BL}} \\ m_{\nu_\mu} &\approx \frac{1}{9} g \frac{m_c^2}{f_{\{126\}} M_{BL}} \\ m_{\nu_\tau} &\approx \frac{1}{9} g \frac{m_t^2}{f_{\{126\}} M_{BL}} \end{aligned} \quad (9.44)$$

We have of course assumed the $f_{\{126\}}$ coupling to be roughly generation independent. An important improvement of the $SO(10)$ model over the left-right symmetric model is that here the neutrino Dirac masses are predicted to be equal to the up-quark masses of each generation. The factor $1/9$ is a crude estimate of the effect of renormalization group extrapolation down to the weak scale. For more accurate estimates of the renormalization effects, see Ref.[11].

This first of all predicts the following quadratic mass formula for neutrinos:

$$m_{\nu_e} : m_{\nu_\mu} : m_{\nu_\tau} = m_u^2 : m_c^2 : m_t^2 . \quad (9.45)$$

Therefore, using [12] $m_t = 180 \text{ GeV}$ and the upper limit of 18 MeV on ν_τ mass given in Eq. (3.8), we predict

$$m_{\nu_\mu} \leq 1.5 \text{ keV} . \quad (9.46)$$

This is to be compared with the existing experimental upper limit of 170 keV .

At this stage, depending on the symmetry breaking chain one chooses, the neutrino mass spectrum can be different. Some interesting scenarios are described below.

Intermediate M_{BL} scenario [Eq. (9.31)]

An interesting scenario is the two step $SO(10)$ breaking scenario [13], where $SO(10)$ breaks to the standard model via a left-right intermediate symmetry. Here, $M_{BL} \simeq 10^{12} \text{ GeV}$ from constraints of $\sin^2 \theta_W$. In this case, using Eq. (9.44) we predict

$$\begin{aligned} m_{\nu_e} &\approx 10^{-7} \text{ eV} , \\ m_{\nu_\mu} &\approx 10^{-3} \text{ eV} , \\ m_{\nu_\tau} &\approx 10^{-1} \text{ eV} . \end{aligned} \quad (9.47)$$

Of course these numbers have to be taken as rough estimates since $f_{\{126\}}$ is unknown.

The interesting point about this scenario is that this kind of mass spectrum is required to explain the solar neutrino puzzle via an MSW type matter oscillation effect, which will be discussed in Ch.6. This model therefore deserves special attention. An additional theoretical argument in favor of this model is that $U(1)_{PQ}$ -symmetry needed to solve the strong CP-problem in the class of invisible axion models [14] also has a symmetry breaking scale around 10^{12} GeV . The models of Ref. [13] emphasize this point by identifying the $U(1)_{PQ}$ symmetry breaking scale. The proton lifetime in this model is predicted to be around 10^{34} yrs and may be within reach of dedicated next generation experimental searches.

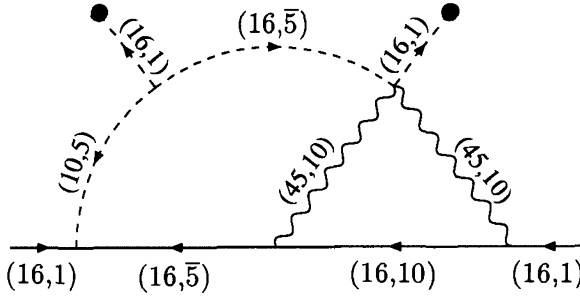


Figure 9.2: Two-loop diagram giving rise to the Majorana mass of right handed neutrinos in Witten's model. The notation $(16,1)$ means the $SU(5)$ singlet part of the 16 representation of $SO(10)$.

Loop-induced heavy mass for N_R

An interesting scenario which does not fall into either of the above cases was proposed by Witten [15] where the actual $B - L$ breaking scale is high (i.e. perhaps of order M_G) but the effective N_R mass that goes into the see-saw matrix is actually much lower. The way that happens is that, even though the $B - L$ symmetry is broken at the tree level, it can be broken by a properly chosen $SO(10)$ representation which has no Yukawa coupling to the fermions. As a result, no tree level mass for N_R will be present. However N_R mass can in general arise from radiative corrections and will therefore be much smaller. An explicit realization of this idea [15] is to break $B - L$ symmetry by a 16-dimensional Higgs multiplet. In this case the N_R mass arises at the two-loop level from the diagram of Fig. 9.2. We get

$$M_N = \frac{f_{\{10\}} g^4}{(16\pi^2)^2} \frac{v_{BL}^2}{M_G}. \quad (9.48)$$

This leads to $M_N \sim 10^{-6}$ to $10^{-7} M_G \approx 10^8 \text{ GeV}$, which in turn yields neutrino masses much larger than given in Eq. (9.47). Notice that, since the heavy neutrino mass is proportional to the Yukawa coupling $f_{\{10\}}$, the see-saw mechanism now gives light neutrino masses proportional to the single power of up quark masses.

In concluding this section, we mention how the D -parity breaking scenario of $SO(10)$ models helps to restore the see-saw picture for neutrino masses by suppressing the direct ν_L -Majorana mass term in Eq.

(9.40). The basic idea is the same as in Ch. 8 except here we break the $SO(10)$ symmetry either by a $\{210\}$ -dimensional representation or the $\{45\}$ -dimensional representation. Writing the $\{210\}$ dimensional representation as a 4-index antisymmetric tensor $\{210\}_{ijkl}$, the D -parity and $SO(10)$ breaking is achieved by giving the vev

$$\langle \{210\}_{7890} \rangle = M_U/g. \quad (9.49)$$

This lowers the $SO(10)$ symmetry to $SU(2)_L \times SU(2)_R \times SU(4)_C$. If instead we use the antisymmetric rank-2 tensor representation $\{45\}$, we can give vev to

$$\langle \{45\}_{12} \rangle = \langle \{45\}_{34} \rangle = \langle \{45\}_{56} \rangle = M_U/g. \quad (9.50)$$

This breaks D -parity and reduces the $SO(10)$ symmetry to $SU(3)_c \times SU(2)_L \times SU(2)_R \times U(1)_{B-L}$. The consequence of D -parity breaking is that it asymmetrizes the Higgs spectrum: for instance, if we denote M_p to be the D -parity breaking scale, then the masses of Δ_L and Δ_R in $\{126\}$ become different, i.e.,

$$M_{\Delta_L}^2 \simeq M_p^2, \quad (9.51)$$

whereas

$$M_{\Delta_R}^2 \simeq M_{BL}^2. \quad (9.52)$$

As a result, we have

$$\begin{aligned} \langle \Delta_L^0 \rangle &\simeq \lambda \frac{\kappa^2 M_{BL}}{M_p^2} \\ &= \lambda \left(\frac{M_{BL}}{M_p} \right)^2 \left(\frac{\kappa^2}{M_{BL}} \right). \end{aligned} \quad (9.53)$$

Eq. (9.42) then implies that if $M_{BL}/M_p \leq 10^{-3}$, the true see-saw picture emerges [16].

9.5 Predictive $SO(10)$ scenarios for neutrino masses

In the previous section, we noted that the see-saw mechanism for neutrino masses finds its natural embedding in the $SO(10)$ models. While

this provides a convincing way to understand the smallness of the neutrino masses compared to the corresponding charged fermion masses, it falls far short of predicting the detailed mass and mixing patterns for neutrinos in general. The reason for this is that, while in the $SO(10)$ model the Dirac masses for the neutrinos are related to the up quark mass matrix, the Majorana mass matrix for the right handed neutrinos involves unknown Yukawa couplings. There is however a class of minimal $SO(10)$ models where the Majorana mass for the heavy right handed neutrino gets related to the quark and lepton masses and therefore gets predicted. A model of this sort therefore leads to firm predictions for the mixings as well as masses for neutrinos [17]. The model is also consistent with all other phenomenological constraints and could therefore be the first $SO(10)$ model to be tested by the current generation of neutrino experiments. Let us briefly describe the model and explain why it has the predictive power in the neutrino sector. We will present only the non-supersymmetric version although a supersymmetric version has been constructed [18].

The Higgs sector of the model is assumed to consist of a $\{210\}$ -dimensional representation to break the $SO(10)$ symmetry down to $SU(2)_L \times SU(2)_R \times SU(4)_c$ model at the GUT scale. The only other Higgs fields of the theory are a complex $\{10\}$ and a $\{126\}$ dimensional (denoted by $\mathbf{H}$ and $\mathbf{\Delta}$ respectively). To start with we will assume that as usual the $\{126\}$ -dimensional multiplet breaks the $SU(2)_L \times SU(2)_R \times SU(4)_c$ group down to the standard model gauge group. It however turns out that the Higgs potential of this model consists of terms of the form $\Delta\bar{\Delta}\Delta H$. In the G_{224} decomposition, this coupling is of the form $(1, \mathbf{3}, \bar{10})(1, \mathbf{3}, 10)(2, \mathbf{2}, \mathbf{15})(2, \mathbf{2}, \mathbf{1})$. It is then clear that after electroweak symmetry breaking via the $\{10\}$ -dimensional multiplet, the $(2, \mathbf{2}, \mathbf{15})$ sub-multiplet of the $\{126\}$ -Higgs acquires a vev. This simple observation has profound implications for the fermion masses. To see this, let us further assume that the Yukawa couplings of the fermions have only the following terms:

$$\mathcal{L}_Y = h_{ab}\psi_a\psi_b H + f_{ab}\psi_a\psi_b\bar{\Delta} \quad (9.54)$$

Note that we have not included a coupling of the form $\psi\psi H^*$ which is forbidden if the model is supersymmetrized. The same can also happen if the model has a Peccei-Quinn symmetry. Now we see that the coupling f_{ab} plays a role in heavy neutrino Majorana masses after $B - L$

breaking and also contributes to quark and lepton masses due to the induced $(\mathbf{2}, \mathbf{2}, \mathbf{15})$ vev. The important point to note is that the Yukawa Lagrangian of the model has a total of twelve parameters which can be counted as follows. We can choose a basis so that h is diagonal, so that there are three parameters there. The coupling of Δ , being symmetric, has six parameters. As a result the Yukawa couplings give nine parameters. Three parameters come from the four Higgs vev's (two from H and two from Δ) after the W -mass constraint is used. Note that in the fermion sector there are nine masses (up, down and charged leptons) and three CKM mixing angles which are known. Thus all parameters in the Yukawa sector of the theory are predicted. Thus the f_{ab} which give the right handed neutrino masses are now predicted yielding a predictive scenario for neutrino masses and mixings. We do not reproduce the detailed predictions of the model [17], merely the formulae for the fermion masses that arises in this model:

$$\begin{aligned}
 M_u &= h\kappa_u + f v_u \\
 M_d &= h\kappa_d + f v_d \\
 M_\nu^D &= h\kappa_u - 3f v_u \\
 M_\ell &= h\kappa_d - 3f v_d \\
 M_{\nu_R} &= f v_R
 \end{aligned} \tag{9.55}$$

where $\kappa_{u,d}$ and $v_{u,d}$ are the vev's of the standard model Higgs doublets in $\{10\}$ and $\{126\}$ Higgs multiplets. Note that the same Yukawa couplings are involved in all mass matrices which is at the heart of the predictive power of this model.

An important consequence of this model is the sum-rule

$$kM_\ell = rM_d + U_{\text{CKM}}^\text{T} M_u U_{\text{CKM}}, \tag{9.56}$$

where $M_{u,d}$ are diagonal matrices. In order to predict neutrino masses and mixings in this model, there are two ways to go: (i) use either type I seesaw formula (Eq. (7.25)) or (ii) type II seesaw formula (Eq. (8.61)). Furthermore, in either case, there are several sets of predictions for the neutrino masses and mixings depending on the choice of the signs for the quark masses and one can then compare those numbers with data.

Case (i) : For the case of type I see-saw, if one ignores CP violating phases in the theory, then the predictions for neutrino masses are

found to be in disagreement with data [19]. However, if one includes CP phases, one can find solutions for neutrinos that are consistent with data [20].

Case (ii) : On the other hand, if one uses type II see-saw and works in the parameter range where the triplet term dominates, then one obtains a sum-rule relating neutrino masses to quark and charged lepton masses:

$$M_\nu = c(M_d - M_\ell). \quad (9.57)$$

This equation leads in a very natural way to maximal atmospheric as well as solar neutrino mixings [21] while giving the ratio $\Delta m_\odot^2 / \Delta m_{\text{atm}}^2$ in agreement with present data. The key reason for this is that the extrapolated values of the b -quark and τ -lepton masses at the GUT scale are nearly equal, thus enhancing all mixing angles.

Several other procedures for predicting the neutrino masses and mixings in $SO(10)$ models have also been considered in recent literature by postulating specific forms for the quark mass matrices and the right handed neutrino masses [22]. We do not discuss them here.

9.6 Neutrino masses in E_6

In this section, we turn to E_6 grand-unified theories and study the neutrino masses. Let us first discuss a few facts about the E_6 theories [23]. There are several maximal subgroups of E_6 :

$$E_6 \supset \begin{cases} SO(10) \times U(1) \\ SU(3) \times SU(3) \times SU(3) \\ SU(6) \times SU(2). \end{cases} \quad (9.58)$$

The fundamental representation of E_6 is $\{27\}$ -dimensional. Under the above subgroups $\{27\}$ decomposes as follows:

$$\begin{aligned} \{27\} &\xrightarrow{SO(10) \times U(1)} \{16\}_{+1} \oplus \{10\}_{-2} \oplus \{1\}_{+4} \\ &\xrightarrow{[SU(3)]^3} (3, \bar{3}, 1) \oplus (\bar{3}, 1, 3) \oplus (1, 3, \bar{3}) \\ &\xrightarrow{SU(6) \times SU(2)} (15, 1) \oplus (\bar{6}, 2). \end{aligned} \quad (9.59)$$

A look at the $SO(10) \times U(1)$ reduction of $\{27\}$ makes it clear that $\{27\}$ is suitable for assigning the fermion fields. Clearly it contains additional fermion fields than are present in the standard model. Their discovery, direct or indirect, will provide evidence for E_6 grand-unification. We will use the $[SU(3)]^3$ decomposition and assign the fermions as follows:

$$\{27\} \supset \underbrace{(3, \bar{3}, 1)}_{\begin{pmatrix} u \\ d \\ g \end{pmatrix}} \oplus \underbrace{(\bar{3}, 1, 3)}_{\begin{pmatrix} \hat{u} \\ \hat{d} \\ \hat{g} \end{pmatrix}} \oplus \underbrace{(1, 3, \bar{3})}_{\begin{pmatrix} E_u^0 & E_u^+ & e^+ \\ E_d^- & E_d^0 & \hat{\nu} \\ e^- & \nu & n \end{pmatrix}} \quad (9.60)$$

Among the new fermions required by E_6 are: a new color triplet isosinglet quark g and its antiparticle $\hat{g}$, and in addition five new leptons: E_u^0 , E_u^+ , E_d^- , E_d^0 and n , the last one being a singlet under $SO(10)$. The model has got five neutral leptons in each generation. Therefore its neutral lepton sector promises to be rich in physics. With so many neutral particles in a single generation, we choose to ignore generational mixing for the rest of this chapter and consider just the first generation.

Turning now to the gauge bosons, the adjoint representation of E_6 is $\{78\}$ -dimensional. Under $SO(10) \times U(1)$ and $[SU(3)]^3$ it decomposes as follows:

$$\begin{aligned} \{78\} &\xrightarrow{SO(10) \times U(1)} \{45\}_0 \oplus \{16\}_{+3} \oplus \{\overline{16}\}_{-3} \oplus \{1\}_0 \\ \{78\} &\xrightarrow{[SU(3)]^3} (8, 1, 1) \oplus (1, 8, 1) \oplus (1, 1, 8) \oplus (3, 3, 3) \oplus (\bar{3}, \bar{3}, \bar{3}). \end{aligned} \quad (9.61)$$

In order to study neutral lepton masses, let us briefly touch on the question of symmetry breaking.

The breaking of $E_6 \rightarrow SO(10)$ can be easily achieved by means of a $\{27\}$ -dimensional Higgs boson. On the other hand to break $E_6 \rightarrow [SU(3)]^3$, one needs to include $\{650\}$ dimensional representation. It turns out that in superstring inspired models, the flux breaking mechanism [24] breaks E_6 down to $[SU(3)]^3$ automatically.

Even in a single generation, the neutrino mass matrix in this model is a 5×5 matrix and is in general very involved. One simple possibility is to use $\{27\}$ -dimensional Higgs boson and $\{351'\}$ Higgs boson denoted by H_1 and Δ respectively. Denoting the $\{27\}$ -dimensional fermion by

ψ , we have the following Yukawa couplings:

$$\mathcal{L}_Y = h_1 \psi \psi H_1 + h_2 \psi \psi \Delta. \quad (9.62)$$

If we give a vev to $\langle n(H) \rangle$, then it gives a Dirac mass coupling $E_u^0 E_d^0$ in Eq. (9.60). If we choose $\langle n(H) \rangle$ heavy enough, two neutral leptons decouple from the low energy spectrum. We may then try to employ a see-saw type mechanism to get a light Majorana neutrino. The right-handed neutrino Majorana mass can be obtained from the Δ -coupling in Eq. (9.62) if we consider the $\{126\}$ dimensional representation of $SO(10)$ that is contained in the $SO(10) \times U(1)$ decomposition of E_6 :

$$\begin{aligned} \{351'\} \xrightarrow{SO(10) \times U(1)} & \{1\}_{-8} \oplus \{10\}_{-2} \oplus \{\overline{16}\}_{-5} \oplus \{54\}_4 \\ & \oplus \{\overline{126}\}_{-2} \oplus \{\overline{144}\}_1. \end{aligned} \quad (9.63)$$

The Dirac masses of the neutrinos can be obtained by using another $\{27\}$ -dimensional Higgs H_2 multiplet and giving vev to the following components:

$$\langle E_u^0(H_2) \rangle = \kappa_1, \quad \langle E_d^0(H_2) \rangle = \kappa_2. \quad (9.64)$$

In this case, one has in the symmetry limit the following mass relations:

$$\begin{aligned} m_u &= m_D = h_2 \kappa_1 \\ m_d &= m_e = h_2 \kappa_2. \end{aligned} \quad (9.65)$$

Then, a see-saw scenario similar to the case of $SO(10)$ emerges. In this case, the fifth neutral lepton n_0 remains fully decoupled from other matter.

On the other hand, if the $\{351'\}$ is not included in the Higgs spectrum, one can use an extra singlet fermion, S and use an extra $\{27\}$ -dimensional Higgs boson H_3 whose vev is $\langle \hat{\nu}(H_3) \rangle = V_{BL}$ and whose Yukawa coupling with fermions is h_3 . In this case, at low energies $\mu \ll \langle n(H_2) \rangle$, we obtain a 3×3 neutrino mass matrix [25]:

$$\begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & h_3 V_{BL} \\ 0 & h_3 V_{BL} & 0 \end{pmatrix}. \quad (9.66)$$

Since this matrix has zero determinant, one of the eigenvalues must be zero. This eigenstate is given by

$$\nu + \epsilon \hat{\nu} \quad \text{where } \epsilon \simeq \frac{m_D}{h_3 V_{BL}} . \tag{9.67}$$

The other two eigenvalues must be equal and opposite since the trace of the matrix in Eq. (9.66) is zero. In such a case, as was discussed in §4.5, they form a Dirac neutrino with mass $|h_3 V_{BL}|$. This is a rather interesting scenario that has been used in superstring models.

Chapter 10

Neutrino mass in supersymmetric models

10.1 Introduction

A symmetry between bosons and fermions, known as supersymmetry, is one of the fundamental new symmetries of nature that has been the subject of intense discussion in modern particle physics. This symmetry was introduced in the early 1970's by Golfand, Likhtman, Akulov, Volkov, Wess and Zumino. In addition to the obvious fact that it provides the hope of an unified understanding of the two known forms of matter, the bosons and fermions, it has also provided a mechanism to solve two conceptual problems of the standard model, viz. the possible origin of the weak scale as well as its stability under quantum corrections. The recent developments in strings, which embody supersymmetry in an essential way, also promise the fulfillment of the eternal dream of all physicists to find an ultimate theory of everything. It would thus appear that a large body of contemporary particle physicists have accepted that theory of particles and forces must incorporate supersymmetry, although no direct experimental evidence of such a symmetry has been found yet. This will of course have important implications for the physics of the neutrinos and the related phenomenon of lepton number violation, which we will consider in this chapter. Let us begin with a brief overview of the basic notions of supersymmetry.

There are many ways to introduce supersymmetric field theories and we refer to several texts [1, 2] and reviews [3] for more thorough discussion. Here we will content ourselves with a brief outline only.

Since supersymmetry transforms a boson to a fermion and vice versa,

an irreducible representation of supersymmetry will contain in it both fermions and bosons. Therefore in a supersymmetric theory, all known particles are accompanied by a superpartner which is a fermion if the known particle is a boson and vice versa. For instance, the electron (e) super-multiplet will contain its superpartner $\tilde{e}$, (called the selectron) which has spin zero. The photon (γ) super-multiplet will contain its superpartner, $\tilde{\gamma}$, known as the photino, which is a Majorana field of spin- $\frac{1}{2}$. We will adopt the notation that the superpartner of a particle will be denoted by the same symbol as the particle with a 'tilde' as above. The name of the superpartner of any known fermion will have an extra letter 's' at the beginning, and known bosons will have superpartners with names ending with the letters '-ino'. Furthermore, while supersymmetry does not commute with the Lorentz transformations, it commutes with all internal symmetries. As a result, all non-Lorentzian quantum numbers for both the fermion and boson in the same super-multiplet are the same.

A convenient mathematical way to discuss supersymmetry is to extend space-time to a superspace where one augments the normal space-time with the addition of anticommuting Grassmannian coordinates denoted by θ , where θ is a Majorana spinor. As elaborated in Ch. 4, such spinors have two independent complex components, or four independent real components. Equivalently, one can use the chiral projections θ_L and θ_R , each of which has two independent real components. Operator valued functions in the superspace will be called superfields and denoted by $\Phi(x, \theta_L, \theta_R)$. The Grassmannian property of the θ coordinates (i.e., $\theta_1\theta_2 = -\theta_2\theta_1$ and $(\theta_i)^2 = 0$) then implies that an expansion of the superfields Φ in the θ coordinate terminates after a few terms. The coefficients of such an expansion will be functions of x only, and are therefore fields of the ordinary field theory. Those fields will create and annihilate the known particles and their superpartners. The superfields can of course have any spin but for the description of the particles of the standard model, it is enough to consider only superfields with spin zero.

There exist three kinds of spin-zero superfields which form irreducible representations of supersymmetry: (i) the chiral and anti-chiral superfields, (ii) vector superfields and (iii) linear multiplets. Again for our purpose, we only need the first two kinds of multiplets. The chiral (and anti-chiral) multiplets contain a spin zero and a spin half particle and

will be used to describe matter as well as Higgs fields. On the other hand the vector multiplets contain a spin one and a spin half fields and will be used to describe gauge fields. It turns out that real vector fields allow transformations, which can be identified with the gauge transformations as is needed if they are to describe gauge fields.

10.2 The Lagrangian for supersymmetric field theories

In order to write down the action for a supersymmetric field theory, let us start by considering generic chiral fields denoted by $\Phi(x, \theta_L)$ with component fields given by (ϕ, ψ) and gauge fields denoted by $V(x, \theta_L, \theta_R)$ with component gauge and gaugino fields given by (A^μ, λ) . The action in the superfield notation is

$$S = \int d^4x \int d^2\theta_L d^2\theta_R \Phi^\dagger e^V \Phi + \int d^4x \int d^2\theta_L (P(\Phi) + W(V)W(V) + \text{h.c.}). \quad (10.1)$$

In the above equation, the first term gives the gauge invariant kinetic energy term for the matter fields Φ ; $P(\Phi)$ is a holomorphic function of Φ and is called the superpotential; it leads to the Higgs potential of the usual gauge field theories. Secondly, $W(V) \equiv \mathcal{D}^2 \mathcal{D}V$ where $\mathcal{D} \equiv \partial\theta - i\sigma \cdot \partial x$, and the term involving $W(V)$ leads to the gauge invariant kinetic energy term for the gauge fields as well as for the gaugino fields. In terms of the component fields the Lagrangian can be written as

$$\mathcal{L} = \mathcal{L}_g + \mathcal{L}_{\text{matter}} + \mathcal{L}_Y - V(\phi) \quad (10.2)$$

where

$$\begin{aligned} \mathcal{L}_g &= -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{1}{2} \bar{\lambda} \gamma^\mu i D_\mu \lambda \\ \mathcal{L}_{\text{matter}} &= |D_\mu \phi|^2 + \bar{\psi} \gamma^\mu i D_\mu \psi \\ \mathcal{L}_Y &= \sqrt{2} g \bar{\lambda} \psi \phi^\dagger + \psi_a \psi_b P_{ab} \\ V(\phi) &= |P_a|^2 + \frac{1}{2} \mathcal{D}_\alpha \mathcal{D}_\alpha \end{aligned} \quad (10.3)$$

where D_μ stands for the covariant derivative with respect to the gauge group and $\mathcal{D}_\alpha$ stands for the so-called $\mathcal{D}$ -term and is given by $\mathcal{D}_\alpha =$

$g\phi^\dagger T_\alpha \phi$ (g is the gauge coupling constant and T_α are the generators of the gauge group). P_a and P_{ab} are the first and second derivative of the superpotential P with respect to the superfield with respect to the field Φ_a , where the index a stands for different matter fields in the model.

Several new features of a supersymmetric field theories (as compared to a non-supersymmetric one) are evident from the Eq. (10.3). First, note that the Yukawa couplings and the first contribution to the Higgs potential arise from the same function $P(\Phi)$ and therefore their parameters are related. The second term in the Higgs potential involves only the gauge coupling as the free parameter. Thus the number of parameters in a SUSY field theory are expected to be fewer than the usual gauge theories. Secondly, there is a new kind of Yukawa coupling involving the gaugino, matter fermion ψ and the superpartner of the matter field with the Yukawa coupling given by the gauge coupling constant g . These generic features have many important implications for the phenomenology of these models.

To see one immediate phenomenological implication of supersymmetry, consider a superpotential $P(\Phi) = m\Phi^2$. Using Eq. (10.3), it is very easy to see that the scalar field ϕ and the fermion field ψ both have the same mass. This is again a generic prediction of the SUSY models, i.e. the particle and its superpartner have the same mass in the supersymmetric limit. This is clearly against observations for all known particles implying that in a realistic model, supersymmetry must be broken. As in the case of other global symmetries, supersymmetry breaking can be explicit or spontaneous. In the latter case, the analogs of Nambu-Goldstone theorem and the Higgs-Kibble mechanism imply that if supersymmetry is global and is spontaneously broken (i.e., $Q_{\text{susy}}|0\rangle \neq 0$), then there must exist a massless fermion in the particle spectrum (to be called the Goldstino) and it will obey low energy theorems — the analog of Adler's zeros for the pion. Since no particle with these properties is known to exist in nature, we have to assume that supersymmetry is either explicitly broken or more elegantly, supersymmetry is a local symmetry which is spontaneously broken. In the latter case, the analog of the Higgs-Kibble mechanism for this case leads to the conclusion that the Goldstino becomes the longitudinal mode of the gauge particle corresponding to the local supersymmetry. A very exciting aspect of local supersymmetry is that the corresponding gauge super-multiplet

contains the graviton (which has spin 2) and its superpartner called the gravitino (which has spin 3/2). The Higgs mechanism for local SUSY leads to the Goldstino being the longitudinal mode of the massive spin 3/2 gravitino. It turns out that in this process, the theory also generates SUSY breaking terms involving the matter and other fields [1, 2]. It is beyond the scope of this book to provide derivation of these results.

- **Exercise 10.1** Using the superpotential $m\Phi^2 + \lambda\Phi^3$ for a chiral superfield Φ , show that the component scalar field ϕ and the fermion field ψ both have the same mass. What is the relation between the Yukawa coupling and the quartic scalar coupling?

10.3 Soft breaking of supersymmetry

One other very important property of supersymmetric field theories is their ultraviolet behavior. They have the extremely important property that in the exact supersymmetric limit, the parameters of the superpotential $P(\Phi)$ do not receive any (finite or infinite) corrections from Feynman diagrams involving the loops. In other words, if the value of a superpotential parameter is fixed at the classical level, it remains unchanged to all orders in perturbation theory. This is known as the non-renormalization theorem [4].

This observation was realized as the key to solving the Higgs mass problem of the standard model as follows: the radiative corrections to the Higgs mass in the standard model are quadratically divergent and admits the Planck scale as a natural cutoff if there is no new physics upto that level. Since the Higgs mass is directly proportional to the mass of the W -boson, the loop corrections would push the W -boson mass to the Planck scale, destabilizing the standard model. On the other hand in the supersymmetric version of the standard model (to be called MSSM), in the limit of exact supersymmetry, there are no radiative corrections to any mass parameter and therefore to the Higgs boson mass which is a parameter of the superpotential. Thus if the world could be supersymmetric at all energy scales, the weak scale stability problem would be easily solved. However, since supersymmetry must be a broken symmetry, one has to ensure that the terms in the Hamiltonian that break supersymmetry do not spoil the non-renormalization theorem in a way that infinities creep into the self mass correction to the Higgs boson. This is precisely what happens if effective supersymmetry breaking terms

are “soft” which means they are of the following type:

1. $m_a^2 \phi_a^\dagger \phi_a$, where ϕ is the bosonic component of the chiral superfield Φ_a ;
2. $m \int d^2\theta_L \theta_L^2 \left(AP^{(3)}(\Phi) + BP^{(2)}(\Phi) \right)$, where $P^{(3)}(\Phi)$ and $P^{(2)}(\Phi)$ are the second and third order polynomials in the superpotential.
3. $\frac{1}{2} m_\lambda \lambda^\top C^{-1} \lambda$, where λ is the gaugino field.

It can be shown that the soft breaking terms only introduce finite loop corrections to the parameters of the superpotential. Since all the soft breaking terms require couplings with positive mass dimension, the loop corrections to the Higgs mass will depend on this mass and we must keep these masses less than a TeV so that the weak scale remains stabilized. This has the interesting implication that superpartners of the known particles are accessible to the ongoing and proposed collider experiments [5].

The mass dimensions associated with the soft breaking terms depend on the particular way that supersymmetry is broken. It is usually assumed that supersymmetry is broken in a sector that involves fields which do not have any quantum numbers under the standard model group. This is called the hidden sector. The supersymmetry breaking is transmitted to the visible sector either via the gravitational interactions [6] or via the gauge interactions of the standard model [7] or via anomalous $U(1)$ $\mathcal{D}$ -terms [8]. There is one particular scenario using gravity to transmit the supersymmetry breaking which has been very widely discussed and forms much of the basis for the discussion in current supersymmetry phenomenology. This was suggested by Polonyi and is based upon the following assumptions: first, the superpotential describing the model is a sum of two terms: $P = P_H + P_{\text{vis}}$. The P_H , the hidden sector potential consists of a gauge singlet field which we will denote by z and P_{vis} consists of the known fields of the standard model. Let us first give P_H :

$$P_H = \mu^2(z + \beta) \quad (10.4)$$

where μ and β are mass parameters to be fixed by various physical considerations. For instance, requiring the cosmological constant to vanish fixes $\beta = (2 - \sqrt{3})M_{\text{Pl}}$. Given this potential, supergravity calculus predicts that the generic soft breaking parameters m are given by

$m \sim \mu^2/M_{\text{Pl}}$. Requiring m to be in the TeV range implies that $\mu \sim 10^{11}$ GeV. This model also predicts A and B given above. It is also important to point out that the super-Higgs mechanism operates at the Planck scale. Therefore all parameters derived at the tree level of this model need to be extrapolated to the electroweak scale. So after the soft-breaking Lagrangian is extrapolated to the weak scale, it will look like:

$$\begin{aligned} \mathcal{L}^{SB} = & m \sum_{i,j,k} A_{ijk} \phi_i \phi_j \phi_k + \sum_{i,j} B_{ij} \phi_i \phi_j \\ & + \sum_i \mu_i^2 |\phi_i|^2 + \sum_a \frac{1}{2} M_a \lambda_a^\top C^{-1} \lambda_a + \text{h.c.} \end{aligned} \quad (10.5)$$

10.4 Supersymmetric standard model

We will now apply the discussions of the previous section to construct the supersymmetric extension of the standard model so that the goal of stabilizing the Higgs mass is indeed realized in practice. In Table 10.1, we give the particle content of the model.

First note an important difference between the standard model and its supersymmetric version apart from the presence of the superpartners: the latter has a second Higgs doublet. This is required both to give masses to quarks and leptons as well as to make the model anomaly-free. The gauge interaction part of the model is easily written down following the rules laid out in Ch. 2. In the weak eigenstate basis, weak interaction Lagrangian for the quarks and leptons is exactly the same as in the standard model. As far as the weak interactions of the squarks and the sleptons is concerned, the generation mixing angles are very different from those in the corresponding fermion sector due to supersymmetry breaking. This has the phenomenological implication that the gaugino-fermion-sfermion interaction changes generation leading to potentially large flavor changing neutral current effects such as $K^0-\bar{K}^0$ mixing, $\mu \rightarrow e\gamma$ decay etc unless the sfermion masses of different generations are chosen to be very close in mass.

Let us now proceed to a discussion of the superpotential of the model. It consists of two parts:

$$W = W_1 + W_2, \quad (10.6)$$

Table 10.1: The particle content of the supersymmetric standard model. For matter and Higgs fields, we have shown the left-chiral fields only. The right-chiral fields will have a conjugate representation under the gauge group.

Superfield	Particles	Superpartners	gauge transformation
Quarks Q	(u, d)	$(\tilde{u}, \tilde{d})$	$(3, 2, \frac{1}{3})$
Antiquarks $\hat{U}$	$\hat{u}$	$\tilde{\hat{u}}$	$(3^*, 1, -\frac{4}{3})$
Antiquarks $\hat{D}$	$\hat{d}$	$\tilde{\hat{d}}$	$(3^*, 1, \frac{2}{3})$
Leptons L	(ν, e)	$(\tilde{\nu}, \tilde{e})$	$(1, 2 - 1)$
Antileptons $\hat{E}$	$\hat{e}$	$\tilde{\hat{e}}$	$(1, 1, 2)$
Higgs Boson $\mathbf{H}_u$	(H_u^+, H_u^0)	$(\tilde{H}_u^+, \tilde{H}_u^0)$	$(1, 2, +1)$
Higgs Boson $\mathbf{H}_d$	(H_d^0, H_d^-)	$(\tilde{H}_d^0, \tilde{H}_d^-)$	$(1, 2, -1)$
Color Gauge Fields	G_a	$\tilde{G}_a$	$(8, 1, 0)$
Weak Gauge Fields	$W^\pm, Z$	$\tilde{W}^\pm, \tilde{Z}$	
Photon	γ	$\tilde{\gamma}$	

where

$$\begin{aligned}
 W_1 &= h_\ell^{ij} \hat{E}_i L_j \mathbf{H}_d + h_d^{ij} Q_i \hat{D}_j \mathbf{H}_d + h_u^{ij} Q_i \hat{U}_j \mathbf{H}_u + \mu \mathbf{H}_u \mathbf{H}_d, \\
 W_2 &= \lambda_{ijk} L_i L_j \hat{E}_k + \lambda'_{ijk} L_i Q_j \hat{D}_k + \lambda''_{ijk} \hat{U}_i \hat{D}_j \hat{D}_k,
 \end{aligned}
 \tag{10.7}$$

i, j, k being generation indices. We first note that the terms in W_1 conserve baryon and lepton number whereas those in W_2 do not. The latter are known as the R -parity breaking terms where R -parity is defined as

$$R = (-1)^{3(B-L)+2S}, \tag{10.8}$$

where S is the spin of the particle. It is interesting to note that the R -parity symmetry defined above assigns even R -parity to known particles of the standard model and odd R -parity to their superpartners. This has the important experimental implication that for theories that conserve R -parity, the super-partners of the particles of the standard model must always be produced in pairs. The lightest superpartner, generally called the LSP, must therefore be a stable particle. If the LSP turns out to be neutral, it is called neutralino, which can be thought of as the dark matter particle of the universe.

We now embed this model into the minimal $N = 1$ supergravity model with a Polonyi type hidden sector. As a result, we get the mass splitting for the squarks and sleptons from the quarks and the leptons. We also get trilinear scalar interactions among the sfermions as follows:

$$\begin{aligned} \mathcal{L}^{SB} = & m_{3/2}[A_{e,ab}\tilde{E}_a\tilde{L}_bH_d + A_{d,ab}\tilde{Q}_aH_d\tilde{d}_b + A_{u,ab}\tilde{Q}_aH_u\tilde{u}_b] \\ & + B\mu m_{3/2}H_uH_d + \sum_{i=\text{scalars}} \mu_i^2\phi_i^\dagger\phi_i + \sum_a \frac{1}{2}M_a\lambda^\top C^{-1}\lambda_a \quad (10.9) \end{aligned}$$

There will also be the corresponding terms involving the R -parity breaking terms, which we omit here for simplicity.

The subject of supersymmetry is now a vast topic which touches many areas of particle physics starting from physics just beyond the standard model to physics of grand unification and beyond to superstrings. In this chapter, we will discuss only those aspects that have a bearing on neutrino mass. This discussion can be neatly split into two parts: (i) the MSSM with R -parity breaking [2] and (ii) supersymmetric left-right model which leads to automatic R -parity conservation [9]. Both kinds of theories lead to non-vanishing neutrino masses in two very different ways. Related lepton number violating phenomena also have different complexions as we will discuss in some of the later chapters.

Before proceeding to the discussion of neutrino masses in the supersymmetric models, we wish to point out that one reason for the popularity of the supersymmetric models is the observation that unlike the non-supersymmetric $SU(5)$, in SUSY $SU(5)$ models, the three gauge couplings unify at a scale of about 2×10^{16} GeV. The difference arises from the fact that the beta functions receive additional contributions from the superpartners of various standard model particles. As a result the b_n 's defined in Eq. (9.10) are now given by $2N + N_H - 3n$ where N is the number of generations and n is the $SU(n)$ group in question and N_H denotes the Higgs contribution. It is easy to see that in this case the three gauge coupling starting at their values at M_Z nicely unify at the scale mentioned above.

10.5 Neutrino mass in MSSM

It is clear from Eq. (10.7) that in the absence of the R -parity violating superpotential W_2 , MSSM conserves lepton and baryon number to all or-

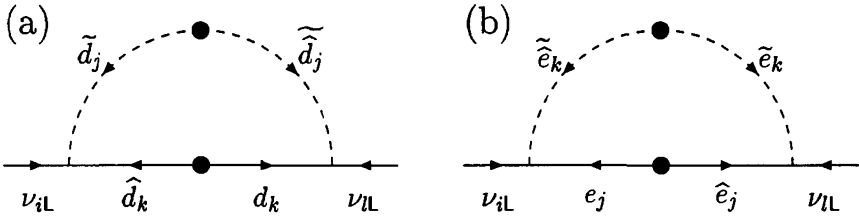


Figure 10.1: One-loop diagrams giving neutrino masses in R -parity violating MSSM.

ders in perturbation theory. Therefore as in the standard model, MSSM without R -parity violation implies $m_\nu = 0$. Thus any evidence for non-vanishing neutrino masses will imply either the presence of R -parity violating terms (either arising from explicit breaking or spontaneous breaking via the nonzero vacuum expectation values of the sneutrino field $\langle \tilde{\nu} \rangle \neq 0$ [10] or new physics beyond MSSM, such as SUSY left-right model or SUSY grand unification.

- **Exercise 10.2** Suppose the R -parity violating terms are absent in the superpotential. In this case, the Lagrangian would conserve $B - L$. If one of the sneutrino fields obtain a vev v_ν along with the vevs of H_u and H_d , $B - L$ will be broken spontaneously and a Majoron will result [10]. Identify the Majoron field.

Within the framework of supersymmetry, there is however no good theoretical reason to omit W_2 from the MSSM since all symmetries of the theory allow it. Once W_2 is included, it is clear that both lepton and baryon numbers are violated. Thus they lead to non-zero Majorana mass for the left-handed neutrinos without the need for a right handed neutrino [11]. Indeed such mass terms for neutrinos arise at the one loop level from the Feynman diagrams in Fig. 10.1. The contribution of Fig. 10.1a to the Majorana mass matrix of left-handed neutrinos can now be easily estimated from this diagram. First, notice that both the vertices are of the λ' -type. The blob on the internal fermion line represents just the fermion mass, which gives a factor of m_{d_k} . The blob on the internal scalar line, however, is somewhat more complicated because it can come from two sources. These are shown in Fig. 10.2. The contribution in Fig. 10.2a comes from the soft supersymmetry breaking term in Eq. (10.9), and this can be written as $m_{3/2} A_d v_d$, where $v_d = \langle H_d^0 \rangle$. We will denote this contribution by $A m_{d_k}$ with a suitably defined dimensionless

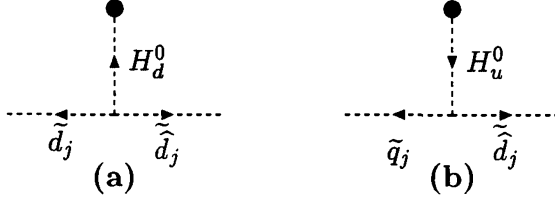


Figure 10.2: Two contributions to the blob on the sfermion line which appears in Fig. 10.1a.

constant A . The other contribution, represented in Fig. 10.2b, comes from the supersymmetric scalar potential whose general form was given in Eq. (10.3). The first term in that equation is $|P_a|^2$, where the subscript meant differentiation with respect to the superfield Φ_a . Applying it to Eq. (10.7), we find one term which is of the form

$$\left| h_d^{ij} Q_i \hat{D}_j + \mu \mathbf{H}_u \right|^2, \quad (10.10)$$

which is obtained by taking the derivative with respect to the superfield $\mathbf{H}_d$. The contribution shown in Fig. 10.2b is the cross term in this expression, obtained after putting the vev of H_u^0 . This contribution has a factor of μ , and is proportional to the down-type quark masses due to the first term of Eq. (10.10). However, the vev appearing in the down-type mass matrix is v_d , not v_u . Thus, denoting v_u/v_d by $\tan \beta$, the contribution of this diagram to the blob is $\mu m_{d_j} \tan \beta$. Collecting all the terms together, we can now write

$$m_{il} \simeq \frac{\lambda'_{ijk} \lambda'_{lkj}}{16\pi^2} \frac{m_{d_k} m_{d_j} (A + \mu \tan \beta)}{M^2} \quad (10.11)$$

where M denotes a typical mass of the squarks and the factor of $16\pi^2$ in the denominator comes from loop integration.

The contribution of the Fig. 10.1b is given by a similar formula with m_d in the above formula replaced by m_l of the corresponding generation. For a typical supersymmetric theory, we can expect the parameters A, μ, M to be in the 100 GeV range. Choosing them to be 100 GeV and keeping the dominant contribution from the b -quark mass, we estimate

$$m_{\nu_{ii}} \simeq 50 \text{ eV} \left(\frac{\lambda'_{ijk} \lambda'_{lkj}}{10^{-4}} \right). \quad (10.12)$$

Table 10.2: Examples of bounds on the R -parity breaking couplings.

Coupling	Upper limit	Source
λ_{12k}	$0.04 \times \left(M_{\tilde{e}_k} / 100 \text{ GeV} \right)$	Beta decay and muon decay universality
$\lambda_{13k, 23k}$	$0.1 \times \left(M_{\tilde{e}_k} / 100 \text{ GeV} \right)$	Leptonic tau decay
λ'_{11k}	$0.05 \times \left(M_{\tilde{d}_k} / 100 \text{ GeV} \right)$	$\pi \rightarrow e \nu_e$ decay

The presence of the couplings in W_2 leads to corrections to many standard model processes as well as processes that are forbidden in the standard model. In particular, the couplings λ_{ijk} and λ'_{ijk} , being lepton number violating, contributes not only to Majorana neutrino mass but also to other lepton number violating effects such as neutrinoless double beta decay, muonium-antimuonium transition etc., some of which will be discussed in the later chapters. Present low energy data therefore lead to upper bounds on various λ_{ijk} and λ'_{ijk} [12, 13] some typical examples of which are summarized in Table 10.2.

In this table, we have only given a small subset of possible constraints on the R -parity violating parameters, but they are enough to show that these couplings must be small. Thus we see that for present upper limits for the R -parity violating couplings, the typical contributions to the neutrino masses are in the eV range. One could therefore use the limits on neutrino masses to put constraints on the parameters of the supersymmetric models.

- **Exercise 10.3** The presence of the λ'' term also leads to baryon number violation. Show that if λ'' were nonzero, in combination with λ' , it would lead to rapid proton decay at the tree level. Estimate the decay rate from this, and use it to show that the present lower bounds on proton lifetime imply that $\lambda'' \lambda' \leq 10^{-24}$ for squark masses of order 100 GeV.

Another point worth emphasizing about the role of R -parity violating couplings in the discussion of neutrino masses is that one can generate interesting patterns for the neutrino masses by imposing symmetries on the couplings. As an example, if we impose $L_e - L_\mu$ invariance, all λ' 's

vanish and the only nonzero λ 's are: λ_{123} , λ_{131} and λ_{232} . Thus in this case only Fig. 10.1b contributes and it is easily checked that only $m_{\nu_{12}} \neq 0$ and all other contributions vanish. In particular, even though we would have expected on the basis of lepton number symmetry considerations that ν_τ could have a mass, the antisymmetry of the couplings λ makes it vanish.

10.6 Supersymmetric Left-Right model

It is well-known that one of the attractive features of the supersymmetric models is its ability to provide a candidate for the cold dark matter of the universe. This however relies heavily on the theory obeying R -parity conservation. In the MSSM, this is achieved by imposing global baryon and lepton number conservation on the theory as additional requirements. First of all, this takes us one step back from the non-supersymmetric standard model where the conservation B and L arise automatically from the gauge symmetry and the field content of the model. Secondly, there is a prevalent lore supported by some calculations that in the presence of non-perturbative gravitational effects such as black holes or worm holes, any externally imposed global symmetry must be violated by Planck suppressed operators [14]. In this case, the R -parity violating effects again become strong enough to cause rapid decay of the lightest R -odd neutralino so that again there is no dark matter particle in supersymmetric theories. It is therefore desirable to seek supersymmetric theories where, like the standard model, R -parity conservation (hence baryon and lepton number conservation) becomes automatic i.e. guaranteed by the field content and gauge symmetry. It was realized in mid-80's [9] that such is the case in the supersymmetric version of the left-right model that implements the see-saw mechanism. We briefly discuss this model in the section.

The gauge group for this model is $SU(2)_L \times SU(2)_R \times U(1)_{B-L} \times SU(3)_c$. The chiral superfields denoting left-handed and right-handed quark superfields are denoted by $Q \equiv (u, d)$ and $\hat{Q} \equiv (\hat{u}, \hat{d})$ respectively and similarly the lepton superfields are given by $L \equiv (\nu, e)$ and $\hat{L} \equiv (\hat{\nu}, \hat{e})$. The Q and L transform as left-handed doublets with the obvious values for the $B - L$ and the $\hat{Q}$ and $\hat{L}$ transform as the right-handed doublets with opposite $B - L$ values. The symmetry breaking is

achieved by the following set of Higgs superfields: $\phi_a(2, 2, 0, 1)$ ($a = 1, 2$); $\Delta(3, 1, +2, 1)$; $\overline{\Delta}(3, 1, -2, 1)$; $\Delta'(1, 3, -2, 1)$ and $\overline{\Delta}'(1, 3, +2, 1)$. Unlike in the MSSM, the allowed terms in the superpotential are very limited in this case:

$$W = h_a Q \phi_a \hat{Q} + h'_a L \phi_a \hat{L} + f(LL\Delta + \hat{L}\hat{L}\overline{\Delta}) + \mu_{ab} \text{Tr } \phi_a \phi_b + M(\Delta\overline{\Delta} + \Delta'\overline{\Delta}') \quad (10.13)$$

It is clear from the above equation that this theory has no baryon or lepton number violating terms and it allows for a dark matter particle. In the second paper [9] a subtle point about the model was noted. If we take the minimal version of this model, parity violation requires that R -parity must be spontaneously violated. However, one can add superfields such as $\delta(3, 1, 0, 1) + \bar{\delta}(1, 3, 0, 1)$ in which case spontaneous R -parity violation is not required to generate a parity violating global minimum. Such extra fields are always present if the model emerges from a grand unified theory or an underlying composite model.

The phenomenology of this model has been extensively studied [15] in recent papers. The neutrino masses in this model arise from the same see-saw mechanism as in the usual left-right models discussed in the Ch. 8. The associated phenomenology is quite similar and we do not go into it any further.

Chapter 11

Large neutrino mixings

11.1 Introduction

A puzzling aspect of neutrino mass physics is the mixing pattern among neutrinos. It appears to be very different from that among the quarks. We see from Eq. (2.20) and Eq. (1.7) that the the mixings among quarks are small and are roughly of nearest neighbor type which means that the mixing among two adjacent families though small, is still larger than among two not adjacent. A convenient approximate parameterization for this is the Wolfenstein parameterization [1]:

$$V_{\text{CKM}} = \begin{pmatrix} 1 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix}, \quad (11.1)$$

where λ is roughly equal to $\sin\theta_c$, θ_c being the Cabibbo angle. The parameters A , ρ and η appearing in this matrix are of order 1.

In order to understand this mixing from a fundamental theory, which gives the mass matrices for up-type and down-type quarks, this implies a certain pattern for the mass matrices of the quarks. Since a mass matrix is in general a basis dependent object, we will choose a particular basis by making a unitary transformation. A convenient basis is to choose is one in which the mass matrix of the up-type quarks is diagonal. Then the above mixing matrix implies that the down quark mass matrix roughly has the form

$$M_d = m_b \begin{pmatrix} \lambda^4 & \lambda^5 & \lambda^3 \\ \lambda^5 & \lambda^2 & \lambda^2 \\ \lambda^3 & \lambda^2 & 1 \end{pmatrix} \quad (11.2)$$

Attempts to gain a theoretical understanding of quark mixings often start with ways to generate mass textures of the above type. Although as yet a satisfactory understanding of quark masses and mixings does not exist, many useful hints that may go into the final understanding have emerged e.g. $\lambda \approx \sqrt{m_d/m_s}$ and $V_{cb} \sim m_s/m_b$ etc. and useful strategy is to look for symmetries that connect different families.

11.2 Hints for understanding large mixings

Let us briefly discuss the case of quarks which seem to have very small mixings. In a sense, in the context of $SU(2)_L$ gauge theories, the intuitive expectation would be to have small mixings. The argument goes as follows: the gauge boson masses seem to display a custodial symmetry in their mass relation in the limit of $g' \rightarrow 0$ i.e. we have $M_W = M_Z$ as the $U(1)_Y$ couplings g' goes to zero. This relation changes to $M_W = 0.88M_Z$ in the presence of nonzero $U(1)_Y$ coupling (a 12% correction). Since up and the down quarks are part of the $SU(2)_L$ representation, in the limit of the custodial symmetry one should expect the mass matrices of the up and down quarks to be equal and therefore the quark mixings ought to vanish. The custodial symmetry is of course broken not only by the $U(1)_Y$ gauge coupling but also by Yukawa couplings, which would therefore generate the small mixing angles. Thus one could argue that small quark mixings are more natural since one would expect $M_u \approx \kappa M_d$. A simple realization of this idea can be given using the left-right symmetric models [2].

Now turning to the neutrino mixings, one could extend the above intuitive logic to say that if neutrinos were Dirac fermions they would perhaps behave very much like the quark sector and would be accompanied by small mixing angles contrary to observations. However when neutrinos are Majorana particles, they could behave very differently. For instance the see-saw mechanism itself is a new contribution to neutrino masses which is absent for the quarks and could therefore be responsible for making the neutrino mass matrices very different from the charged leptons.

To elaborate on this a bit, let us write down the see-saw formula:

$$\mathcal{M}_\nu = f v_L - M_D (f v_R)^{-1} M_D^T \quad (11.3)$$

where M_D stands for the Dirac mass matrix for the neutrinos. The approximate custodial symmetry argument above would imply that $M_D \simeq M_\ell$. However f is a completely new matrix whose presence in the above equation is due to the fact that the neutrinos are Majorana particles. Thus even if the M_D and M_ℓ are very similar in form, the flavor structure in f could be responsible for the very different mixing pattern among neutrinos.

One moral of this discussion therefore is that one should seek an understanding of the large mixings in the see-saw mechanism and in the coupling matrix f . Furthermore since see-saw formula is a high scale phenomenon, one could expect more symmetries to manifest themselves in the matrix f than in the low scale couplings like the fermion Yukawa couplings. A large body of current attempts to understand large neutrino mixings are carried out under assumptions which are some variations of this general philosophy [3]. Below we illustrate this with some examples.

11.3 Mass matrices for neutrinos

11.3.1 Diagonalization of the neutrino mass matrix

To discuss neutrino mixings, let us denote the the flavor eigenstates by the symbol ν_ℓ as before, where $\ell = e, \mu, \tau$ etc. The mass eigenstates, on the other hand, will be denoted by the symbols ν_α , where $\alpha = 1, 2, 3$ etc. As pointed out in the quark case, it is convenient to start with a basis in which the charged lepton masses are diagonal. In this basis, the neutrino mass terms can be written as

$$-\mathcal{L}_{\text{mass}} = \sum_{\ell, \ell'} \nu_{\ell\text{L}}^\dagger C^{-1} \mathcal{M}_{\ell\ell'} \nu_{\ell'\text{L}} + \text{h.c.} . \quad (11.4)$$

As discussed in Ch. 4, the mass matrix is symmetric, and one can find a unitary matrix U such that

$$U^T \mathcal{M} U = D_\nu , \quad (11.5)$$

where the matrix on the right hand side is diagonal, with non-negative diagonal elements which are identified as the mass eigenvalues. The physical neutrino states are then defined through the relation

$$\nu_{\ell\text{L}} = \sum_{\alpha} U_{\ell\alpha} \nu_{\alpha\text{L}} . \quad (11.6)$$

The matrix U is the mixing matrix, also called the PMNS matrix [4].

Of course the form of the mixing matrix depends on the entries in the mass matrix $\mathcal{M}$. As shown in Ch. 7, in the CP-conserving case the matrix U can be written in the form

$$U = OK, \quad (11.7)$$

where O is an orthogonal matrix which diagonalizes $\mathcal{M}$ through a similarity transformation, but the diagonal elements of $O^T \mathcal{M} O$ are not necessarily positive. A diagonal matrix K can then produce positive entries, which are the physical masses. The amount of mixing between neutrinos is really governed by the matrix O .

Since solar and atmospheric neutrino data tell us that neutrinos have large mixings among them, one looks for various special values of the angles $\theta_{\alpha\beta}$ which imply large mixings. For three generation of fermions, the following matrices have been widely discussed in the literature:

$$O_d = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}, \quad (11.8)$$

$$O_b = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \end{pmatrix}. \quad (11.9)$$

The mixing pattern demonstrated in O_d is called democratic [5], whereas that in O_b is called bimaximal [6]. Another pattern, called maximal mixing, will be discussed later. Such simple forms may not represent the actual mixing matrix, but they can still be obeyed as a first approximation.

There is a degree of arbitrariness in defining the mixing matrix, which should be realized at this stage. The mixing matrix is defined through Eq. (11.6). The elements of the mixing matrix thus depend on how we order the flavor states ν_ℓ and the mass eigenstates ν_α . The flavor states will always be represented by ν_e, ν_μ, ν_τ , in that order. However, the ordering of the mass eigenstates is still arbitrary. We fix this arbitrariness by adopting the convention that

$$|\Delta m_{21}^2| \ll |\Delta m_{32}^2|, \quad (11.10)$$

so that Δm_{32}^2 must be the quantity occurring in atmospheric neutrino oscillations, whereas Δm_{21}^2 is responsible for the solar neutrino oscillations.

Present experimental data favors a mixing matrix close to the bi-maximal form. This can be seen as follows. For three generations of fermions, the general form of the orthogonal matrix was written in Eq. (5.20) on page 83, using three angles θ_{12} , θ_{13} and θ_{23} . In §5.1.2, we derived the oscillation probabilities in various limiting cases. For atmospheric neutrinos, the energies and distances are such that the oscillatory terms involving Δm_{21}^2 are effectively frozen, so that we can use Eq. (5.19) for the oscillation probabilities. Plugging in the appropriate matrix elements, we find that the $\nu_\mu \rightarrow \nu_\tau$ conversion probability is given by

$$\begin{aligned} P_{\nu_\mu \nu_\tau}(x) &= 4(U_{\mu 3}U_{\tau 3})^2 \sin^2 \left(\frac{\Delta m_{32}^2}{4E} x \right) \\ &= \cos^4 \theta_{13} \sin^2 2\theta_{23} \sin^2 \left(\frac{\Delta m_{32}^2}{4E} x \right). \end{aligned} \quad (11.11)$$

As mentioned in §5.3, the atmospheric neutrino oscillations are consistent with the maximal case, i.e., $\theta_{\text{atm}} = \pi/4$ is allowed by data. If we assume that these oscillations are indeed maximal, this means that $\cos^4 \theta_{13} \sin^2 2\theta_{23} = 1$ in Eq. (11.11). This can be ensured by taking

$$\theta_{13} = 0, \quad \theta_{23} = \pi/4. \quad (11.12)$$

The general mixing matrix of Eq. (5.20) then reduces to the form

$$O = \begin{pmatrix} c_{12} & -s_{12} & 0 \\ \frac{s_{12}}{\sqrt{2}} & \frac{c_{12}}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{s_{12}}{\sqrt{2}} & \frac{c_{12}}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}. \quad (11.13)$$

We still have one undetermined angle, θ_{12} . But this form already shows that the democratic mixing matrix is ruled out if the atmospheric neutrino oscillations are maximal. Moreover, we see that if θ_{12} equaled $\pi/4$ as well, we would have obtained exactly the bi-maximal form for the mixing matrix. However, solar neutrino data discussed in Ch. 6 show that this angle, though not small, is not maximal either. Therefore the mixing matrix can only be approximately bi-maximal.

11.3.2 Example of the Zee model

To see how such forms occur in interesting models of neutrino mass, let us take the example of the Zee model discussed in §7.3.2. The neutrino mass matrix that arises in this model was given by

$$\mathcal{M} = m_0 \begin{pmatrix} 0 & \sigma & c \\ \sigma & 0 & s \\ c & s & 0 \end{pmatrix}, \quad (11.14)$$

where $c = \cos \alpha$, $s = \sin \alpha$, with α and σ defined in Eq. (7.42). If any one of the parameters σ , c and s is very small, first order perturbation in that parameter yields the following eigenvalues of the mass matrix:

$$\begin{aligned} m_1 &= -m_\star K_1^2(1 - \sigma cs), \\ m_2 &= m_\star K_2^2(1 + \sigma cs), \\ m_3 &= -2m_\star K_3^2 \sigma cs, \end{aligned} \quad (11.15)$$

where K_1, K_2, K_3 are the elements of the matrix K appearing in Eq. (11.7), with K_α^2 equal to either $+1$ or -1 in order to make the mass eigenvalues positive. The constant $m_\star$ bears a simple relation with m_0 : if the small parameter happens to be σ , then $m_\star = m_0$; otherwise, $m_\star = m_0 \sqrt{1 + \sigma^2}$ in first order perturbation in the small parameter.

Note that we have ordered the mass eigenvalues so as to obey the convention of Eq. (11.10). However, contrary to the naïve expectation, the small difference occurs between the two largest eigenvalues. This pattern is called the inverted mass hierarchy, which is discussed in more detail in §11.3.3.

Clearly, the form of the mixing matrix does not depend on $m_\star$. However, unlike the mass hierarchy, it depends on which one of the parameters σ , c and s is small. We argued in §7.3.2 that the parameter σ is expected to be small, because it involves the ratio m_μ^2/m_τ^2 . However, if this is the case, the Zee model fails miserably because, to the approximation $\sigma = 0$, the mixing matrix is given by

$$O = \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{2}s & c & c \\ -\sqrt{2}c & s & s \\ 0 & -1 & 1 \end{pmatrix}. \quad (11.16)$$

This is untenable in the light of present experimental data. If we want to have a maximal or even near-maximal mixing in the atmospheric

neutrino sector, we should have $s \approx 1$. But in that case $c \approx 0$ so that the eigenstate ν_1 is almost purely ν_e . This is not consistent with the solar neutrino mixing angle being also substantial.

Let us then consider the other extreme [7], i.e., $s \ll 1$ in the mass matrix. In this case, it is more convenient to rewrite the mass matrix as

$$\mathcal{M} = m'_0 \begin{pmatrix} 0 & \sin \beta & \cos \beta \\ \sin \beta & 0 & \epsilon \\ \cos \beta & \epsilon & 0 \end{pmatrix}, \quad (11.17)$$

with suitably defined parameters m'_0 , β and ϵ . To first order in ϵ , the mixing matrix is now given by

$$O = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 + \frac{1}{2}\epsilon cs & -1 + \frac{1}{2}\epsilon cs & -\sqrt{2}\epsilon(c^2 - s^2) \\ s - \epsilon c(1 - \frac{3}{2}s^2) & s + \epsilon c(1 - \frac{3}{2}s^2) & -\sqrt{2}c \\ c - \epsilon s(1 - \frac{3}{2}c^2) & c + \epsilon s(1 - \frac{3}{2}c^2) & \sqrt{2}s \end{pmatrix}, \quad (11.18)$$

where in this equation c stands for $\cos \beta$ and s for $\sin \beta$. Clearly, in the limit $\epsilon = 0$, this gives the bimaximal matrix if $\beta = \pi/4$.

But even this form is inconsistent with present data. The mixing angle for solar neutrinos is now given by

$$\sin^2 2\theta_{12} = 1 - \frac{1}{32} \left(\frac{\Delta m_{21}^2}{\Delta m_{32}^2} \right)^2 = 1 - \frac{1}{32} \left(\frac{\Delta m_{\odot}^2}{\Delta m_{\text{atm}}^2} \right)^2. \quad (11.19)$$

This is too close to 1 to be acceptable within the allowed range of mixing angles dictated by the solar neutrino data.

□ **Exercise 11.1** Derive Eq. (11.18).

□ **Exercise 11.2** Verify Eq. (11.19).

It should be remarked that this inconsistency of the Zee model should not be seen as a failure of the near-bimaximal form of the mixing matrix. On the contrary, a near-bimaximal form seems to agree quite well with experimental data. In the Zee model, however, the mass matrix has only three independent parameters, the diagonal elements being zero. The three mass eigenvalues are related to the three mixing angles, as exemplified by Eq. (11.19). These are the relations which are in conflict with experimental data.

11.3.3 Patterns for mass matrices

We can now ask the general question of how a particular form for the mixing matrix might arise from a mass matrix. This is easy. In fact, using Eqs. (11.5) and (11.7), we can easily write

$$\mathcal{M} = ODK^2O^\top. \quad (11.20)$$

If O denotes the bimaximal matrix shown in Eq. (11.9), it is easily seen that the mass matrix should be of the form

$$\mathcal{M} = \begin{pmatrix} A+B & C & C \\ C & A & B \\ C & B & A \end{pmatrix} \quad (11.21)$$

Obviously, in the limit $\alpha = 0$ and $\beta = \pi/4$, the mass matrix in the Zee model has this form with $A = B = 0$. That is why the mixing matrix of Eq. (11.18) becomes bimaximal in this limit, as remarked earlier.

□ **Exercise 11.3** With the mass matrix of Eq. (11.21), verify that $O_b^\top \mathcal{M} O_b$ is diagonal, given by

$$O_b^\top \mathcal{M} O_b = \text{diag} \left(A+B+\sqrt{2}C, A+B-\sqrt{2}C, A-B \right). \quad (11.22)$$

□ **Exercise 11.4** Show that the democratic mixing matrix is obtained for the following general form of the mass matrix:

$$\mathcal{M} = \begin{pmatrix} A & B & \sqrt{2}B \\ B & A-2C & \sqrt{2}C \\ \sqrt{2}B & \sqrt{2}C & A-C \end{pmatrix}. \quad (11.23)$$

In order to obtain a bimaximal mixing pattern to a leading approximation, the task of a model-builder would then reduce to obtaining a mass matrix of the form given in Eq. (11.21). However, apart from the mixing pattern, one should also pay attention to the pattern of the mass eigenvalues. In this regard, much less is known from experiments. We know that the mass squared difference in the solar neutrino oscillations is much smaller than that in the atmospheric neutrino oscillations. We have fixed our convention for mass differences in Eq. (11.10), so that

$$\Delta m_\odot^2 = \Delta m_{21}^2, \quad \Delta m_{\text{atm}}^2 = \Delta m_{32}^2. \quad (11.24)$$

This still does not fix the eigenvalues, or even their order. Present data allow the relative ordering of the masses to fall into many different types described below and schematically represented in Fig. 11.1.

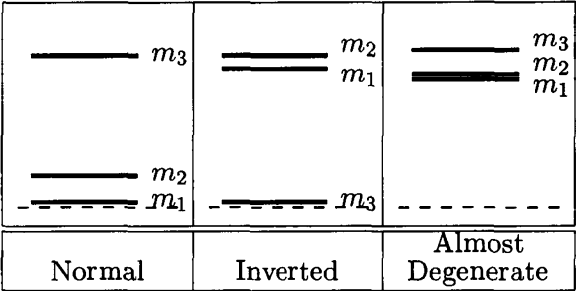


Figure 11.1: Schematic picture of different possible hierarchies for neutrino mass. The thin dashed line in each case is supposed to represent zero mass.

1. Normal hierarchy ($m_1 \ll m_2 \ll m_3$): Since the solar neutrino oscillations must involve ν_e , the dominant component of ν_e is ν_1 , which is the lightest. This hierarchy pattern is similar to that for quarks and charged leptons.
2. Inverted hierarchy ($m_3 \ll m_1 \approx m_2$): In this case [8] the dominant component of ν_e is not the eigenstate with the smallest mass. The smallest mass squared difference occurs between two eigenstates which have the largest masses.
3. Approximately degenerate pattern ($m_1 \approx m_2 \approx m_3$): This represents a case [9] where all mass eigenvalues are large compared to their differences.

One then has to take one form of mass hierarchy as an ansatz, choose a mixing matrix to obtain a mass matrix in the flavor basis which we have chosen throughout. For example, suppose we want to obtain a bimaximal mixing pattern, which is quite close to the mixing pattern favored by data. As already discussed, the mass matrix for this should have the general form as given in Eq. (11.21). The eigenvalues of this matrix are given in Eq. (11.22). Apart from the signs, these are the mass eigenvalues.

Suppose we want to have a normal hierarchy of masses. This can be obtained if we take $A \approx -B$ and C small. Putting $A = -B$ and $C = 0$ as a first approximation, we obtain to the desired the mass matrix in

the form

$$\mathcal{M} = A \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}. \quad (11.25)$$

With this matrix, the lower two eigenvalues would be degenerate and the mixing would be strictly bimaximal. Slight departures from this form would split the degeneracy and make the mixing matrix slightly different from bimaximal. So finally, the mass matrix that we want would be of the form

$$\mathcal{M} = m_0 \begin{pmatrix} \epsilon^{1+n_1} & \epsilon^{1+n_2} & \epsilon \\ \epsilon^{1+n_2} & 1 + \epsilon & -1 \\ \epsilon & -1 & 1 \end{pmatrix}, \quad (11.26)$$

with $n_{1,2} \geq 1$. We will call this a “Type A” matrix.

On the other hand, suppose we want the inverted hierarchy of masses. The two largest masses now should be nearly equal, so to a first approximation we should choose $A = B = 0$, which produces two degenerate eigenstates and a massless one. Slight departures from this form breaks the degeneracy and produces a mass for the lightest one. So in this case, the desired form of the mass matrix would be

$$\mathcal{M} = m_0 \begin{pmatrix} \epsilon & 1 & 1 \\ 1 & \epsilon & \epsilon \\ 1 & \epsilon & \epsilon \end{pmatrix}. \quad (11.27)$$

This will be called a “Type B” matrix. The different elements denoted by ϵ need not be equal. They have to be small compared to 1. Clearly, it reduces to the matrix in Eq. (11.21) when all small elements are set to zero.

Finally, it is trivial to see that the degenerate mass pattern can also emerge from the matrix in Eq. (11.21). The eigenvalues are exactly degenerate if $B = C = 0$. Thus, if the magnitudes of B and C are much smaller compared to that of A , we will obtain an almost degenerate solution for neutrino masses.

The entire discussion above assumes the simple case where we have ignored CP violation. The presence of CP phases allows for some variations. We would like to look for clues to a fundamental understanding of neutrino mixings by studying these mass matrices. An important point to emphasize is that these matrices are in the basis where the charged lepton masses are diagonal.

- **Exercise 11.5** Show that the mass matrix in Eq. (11.23) gives normal hierarchy of masses if C dominates, the inverted hierarchy if B dominates, and the almost degenerate pattern if A dominates.

11.4 Symmetries and neutrino mass textures

11.4.1 Two generation example with a discrete symmetry

An elementary way to proceed in this directions is to start with ways to understand the maximal mixing using only two flavors. It is well known that if we have a matrix of the form

$$\mathcal{M} = \begin{pmatrix} A & B \\ B & A \end{pmatrix}, \quad (11.28)$$

then its eigenstates are maximal admixtures of the original flavor states. The eigenvalues are $A+B$ and $A-B$. It is now clear that if we want one of the eigenvalues to be much less than the other, we must have $A \approx B$. One must therefore satisfy two conditions, viz, a symmetric matrix of the above form to get maximal mixing plus almost equal entries in all of them to get the solar mass splitting to be small. These together become very restrictive conditions.

The next question is how this matrix might arise from a reasonable model. This is part of the art of model building and can be done in different ways and the usual expectation is that different ways of implementing would lead to different predictions that will then finally settle the issue of fundamental theory. Here we give an example of such a model.

The model is an extension of the standard model which includes a triplet Higgs field Δ which has a large mass term M arising from some high scale theory. Then, as discussed in Ch. 8, the neutral component of this multiplet will have a vev of order $v_L \simeq v_{\text{wk}}^2/M$, which can be in the eV range. Now suppose the high scale theory containing Δ has a permutation symmetry S_2 under which the two lepton doublets of the standard model go into each other. In this case, $\psi_{1L} \pm \psi_{2L} \equiv \psi_{\pm L}$ are even and odd under the S_2 symmetry. We also include in the theory two standard model Higgs doublets ϕ_+ and ϕ_- depending on whether they are even or odd under the S_2 symmetry. All other fields are assumed to be even under the S_2 symmetry. One can now write the Yukawa

couplings:

$$\begin{aligned}\mathcal{L}_Y = & f_+ \psi_{+L} \psi_{+L} \Delta + f_- \psi_{-L} \psi_{-L} \Delta + \bar{\psi}_{+L} \phi_+ (h_{1+} e_R + h_{2+} \mu_R) \\ & + \bar{\psi}_{-L} \phi_- (h_{1-} e_R + h_{2-} \mu_R) + \text{h.c.}\end{aligned}\quad (11.29)$$

Elementary algebra shows that in the ψ_1 - ψ_2 basis this Lagrangian leads to the symmetric form for neutrino mass matrix after symmetry breaking while leaving no trace of the symmetry in the charged lepton mass matrix, as required.

Thus this is an example where a high scale symmetry via see-saw indeed explains why neutrino mixings are so different from those of the quarks. This example is purely for illustrative purposes but provides a kind of an existence theorem for this idea.

- **Exercise 11.6** Retrace the argument for obtaining a mass matrix of the form of Eq. (11.28) by starting with the mass terms written in the ψ_1 - ψ_2 basis.

11.4.2 Three generation case with continuous symmetries

As discussed in Ch. 2, the generational lepton numbers L_e , L_μ and L_τ are conserved in absence of neutrino mass. Neutrino mass terms break these symmetries in general.

However, in realistic patterns of neutrino mass, some generational symmetries remain intact to a good extent. In §11.3.3, we discussed some realistic forms of neutrino mass matrix. For example, if the neutrino masses obey normal hierarchy, we argued that we should have a matrix of Type A, shown in Eq. (11.26). If we neglect the small terms, the matrix reduces to that in Eq. (11.25), which clearly shows that L_e is still a conserved quantum number. In other words, if L_e is violated only slightly in a model, we will obtain the normal hierarchy of neutrino mass.

Similar arguments can be given for the inverted hierarchy of masses. In the Type B matrix in Eq. (11.27), if we neglect the small terms, we find that the combination $L_e - L_\mu - L_\tau$ is conserved [10]. Starting with a model which has this symmetry, we can then arrive at the inverted mass hierarchy by breaking this symmetry by a small amount. If the inverted mass pattern is confirmed by future experiments, this symmetry will provide an important clue to new neutrino related physics beyond the standard model [11].

A cautionary remark should be made here. The continuous symmetries mentioned above do indeed guarantee that certain terms will be zero in the symmetric limits. However, they do not guarantee any relation between the non-zero terms in general. For example, the $L_e - L_\mu - L_\tau$ symmetry does not explain why the 12 and the 13 elements of the Type B mass matrix should be equal. To ensure the relation between them, one might need some discrete symmetries, e.g., permutation symmetries of the type described in the two generation example given earlier.

11.4.3 Maximal mixing matrix

For the mass matrix of Eq. (11.28), the mixing matrix can be written as:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (11.30)$$

We will call this matrix “maximal mixing matrix”.

This can be easily generalized to higher number of generations. In fact, if $\mathcal{N}$ is a prime number, the maximal mixing matrix for $\mathcal{N}$ generations is given by

$$U_{\mathcal{N}} = \frac{1}{\sqrt{\mathcal{N}}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & z & z^2 & \cdots & \cdots \\ 1 & z^2 & z^4 & \cdots & \cdots \\ 1 & z^3 & z^6 & \cdots & \cdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \cdots & \cdots & \cdots & \cdots \end{pmatrix}, \quad (11.31)$$

where $z = \exp(2\pi i/\mathcal{N})$ is the $\mathcal{N}$ -th root of unity and the rows and columns extend in an obvious manner to make the matrix $\mathcal{N} \times \mathcal{N}$. The general form of an arbitrary element is given by

$$(U_{\mathcal{N}})_{ab} = z^{(a-1)(b-1)}. \quad (11.32)$$

When the number of generations is not a prime number, we can make the prime decomposition of this number in the form $\mathcal{N} = p_1 p_2 p_3 \cdots$. In this case,

$$U_{\mathcal{N}} = U_{p_1} \otimes U_{p_2} \otimes U_{p_3} \otimes \cdots \quad (11.33)$$

where “ $\otimes$ ” stands for a direct product. For example for the case of $\mathcal{N} = 4$, we have for the maximal mixing matrix $U_4 = U_2 \otimes U_2$ which is given by

$$U_4 = \frac{1}{\sqrt{2}} \begin{pmatrix} U_2 & U_2 \\ U_2 & -U_2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \quad (11.34)$$

The 4×4 case turns out to be interesting since by decoupling a linear combination of the states from the 4×4 system, we can get the exact bimaximal form [12]. The corresponding 3×3 mass matrix can also be obtained by starting from this discussion. To see how one gets that, note that the mass matrix in Eq. (11.28) for the 2×2 case has a S_2 symmetry. The corresponding symmetry in the 4×4 case is then $S_2 \times S_2$. Using this property and decoupling one linear combination of the fields, we can get the mass matrix for the 3×3 case [12] that is exactly of the form given in Eq. (11.21), which leads to bimaximal mixing.

There are many examples in the literature as to how one can understand large mixings. Another mass matrix of interest [13] has the form

$$\mathcal{M} = \begin{pmatrix} D & C & C \\ C & A & B \\ C & B & A \end{pmatrix} \quad (11.35)$$

which is a four parameter mass matrix which introduces a new parameter that can be varied to obtain the desired solar neutrino mixing. The mixing matrix that results [13] from this mass matrix is exactly of the form given in Eq. (11.13).

- **Exercise 11.7** Diagonalize the mass matrix of Eq. (11.35) with the mixing matrix of Eq. (11.13). Show that the mixing angle θ_{12} is given by

$$\tan 2\theta_{12} = \frac{2\sqrt{2}C}{D - A - B}. \quad (11.36)$$

Show that we get the normal mass hierarchy $C, D \ll A, B$ and the inverted pattern for $C \gg A, B, D$. Indicate the kind of relation between the matrix elements which gives an almost degenerate mass pattern.

11.5 Radiative corrections and large mixings

While the mass matrix approach to understanding large mixings is promising and allows for a large variety of possible models, there also exists an alternative approach which works in some settings. The idea here is that the neutrino mass matrix is a high scale phenomenon but observations are made at the weak scale energies. Therefore, it is reasonable to expect that there will be corrections to the high scale mass matrix as we extrapolate it down to the weak scale. The scheme for extrapolation [14] is the well known renormalization group approach.

In the see-saw models for neutrino masses, the neutrino mass arises from the effective operator

$$\mathcal{O}_\nu = -\frac{1}{4} \sum_{\ell, \ell'} \kappa_{\ell\ell'} \frac{\psi_{\ell L} \phi \psi_{\ell' L} \phi}{M}. \quad (11.37)$$

where ψ and ϕ are the leptonic and the Higgs doublets respectively, ℓ, ℓ' denote the weak flavor index. After symmetry breaking $\langle \phi_0 \rangle \neq 0$, so that the matrix κ becomes the neutrino mass matrix after symmetry breaking. This operator is defined at the scale M since it arises after the heavy field N_R is integrated out. On the other hand, in conventional oscillation experiments, the neutrino masses and mixings being probed are at the weak scale. One must therefore extrapolate the operator down from the see-saw scale M to the weak scale M_Z .

The form of the renormalization group extrapolation of course depends on the details of the theory. For simplicity we will consider only the supersymmetric theories, where the only contributions come from the wave function renormalization and is therefore easy to calculate. The equation governing the extrapolation of the $\kappa_{\alpha\beta}$ matrix is given in the case of MSSM by [14]:

$$\frac{d\kappa}{dt} = \left[-3g_2^2 + 6\text{Tr}(Y_u^\dagger Y_u) \right] \kappa + \frac{1}{2} \left[\kappa Y_\ell^\dagger Y_\ell + Y_\ell^\dagger Y_\ell \kappa \right], \quad (11.38)$$

where $t = (1/16\pi^2) \ln(\mu/M_Z)$, where μ denotes the mass scale where κ is being evaluated. We note two kinds of effects on the neutrino mass matrix from the above formula: one that is flavor independent and a part that is flavor specific. If we work in a basis where the charged leptons are diagonal, then the resulting correction to the neutrino mass

matrix is given by:

$$\mathcal{M}_\nu(M_Z) = (1 + \xi)\mathcal{M}(M_{B-L})(1 + \xi) \quad (11.39)$$

where ξ is a diagonal matrix with matrix elements

$$\xi_{\ell\ell'} \simeq -\frac{m_\ell^2 \tan^2 \beta}{16\pi^2 v^2} \delta_{\ell\ell'}, \quad (11.40)$$

where $\delta_{\ell\ell'}$ is the Kronecker delta. In more complicated theories, the corrections will be different.

Let us now study some implications of this corrections. First note that in the MSSM, this effect can be sizable if $\tan \beta$ is large (of order 10 or bigger). This leads to one possible suggestion that perhaps the mixing angles in both quark and lepton sectors are similar at some high scale; but due to renormalization effects, the mixing angles in the leptonic sector may become magnified at low scales. It was shown [15] that this indeed happens if the neutrino spectrum is almost degenerate.

This can be seen in a simple way in two-generation case [15]. The two charged leptons will be denoted by ℓ_1 and ℓ_2 , the latter being the heavier. To keep matters simple, we assume that CP is conserved and that all Majorana neutrinos have the same CP eigenvalue. In that case, the flavor basis and the mass basis are related by an orthogonal transformation. Denoting the neutrino mass eigenvalues by m_1 and m_2 , we can then write down the mass matrix in the flavor basis as follows:

$$\mathcal{M} = \begin{pmatrix} c_\theta & s_\theta \\ -s_\theta & c_\theta \end{pmatrix} \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} c_\theta & -s_\theta \\ s_\theta & c_\theta \end{pmatrix}. \quad (11.41)$$

Due to the presence of radiative corrections to m_1 and m_2 , the matrix $\mathcal{M}$ gets modified to

$$\mathcal{M} \rightarrow \overline{\mathcal{M}} = \begin{pmatrix} 1 + \xi_{\ell_1} & 0 \\ 0 & 1 + \xi_{\ell_2} \end{pmatrix} \mathcal{M} \begin{pmatrix} 1 + \xi_{\ell_1} & 0 \\ 0 & 1 + \xi_{\ell_2} \end{pmatrix}. \quad (11.42)$$

The mixing angle $\bar{\theta}$ that now diagonalizes the matrix $\mathcal{M}_\mathcal{F}$ at the low scale μ (after radiative corrections) can be related to the old mixing angle θ through the following expression:

$$\tan 2\bar{\theta} = \frac{1 + \xi_{\ell_1} + \xi_{\ell_2}}{\lambda} \tan 2\theta, \quad (11.43)$$

where

$$\lambda = 1 + \frac{2\xi_{\ell_2}(m_1 s_\theta^2 + m_2 c_\theta^2) - 2\xi_{\ell_1}(m_1 c_\theta^2 + m_2 s_\theta^2)}{(m_2 - m_1)c_{2\theta}}. \quad (11.44)$$

If

$$(m_1 - m_2)c_{2\theta} = 2\xi_{\ell_2}(m_1 s_\theta^2 + m_2 c_\theta^2) - 2\xi_{\ell_1}(m_1 c_\theta^2 + m_2 s_\theta^2), \quad (11.45)$$

then $\lambda = 0$ or equivalently $\bar{\theta} = \pi/4$, which means maximal mixing. Given the mass hierarchy of the charged leptons: $m_{l_\alpha} \ll m_{l_\beta}$, we expect $|\xi_{\ell_1}| \ll |\xi_{\ell_2}|$, which reduces Eq. (11.45) to a simpler form:

$$2|\xi_{\ell_2}| = \frac{(m_2 - m_1)c_{2\theta}}{m_1 s_\theta^2 + m_2 c_\theta^2} \quad (11.46)$$

In the case of MSSM, the radiative magnification condition can be satisfied provided

$$h_\tau(\text{MSSM}) \approx \sqrt{\frac{8\pi^2 |\Delta m^2(\Lambda)| c_{2\theta}}{m^2 \ln(\Lambda/\mu)}}, \quad (11.47)$$

where h_τ is the tau Yukawa coupling and is related to the tau lepton mass by $h_\tau v \cos \beta = m_\tau$, and $\tan \beta$ is the ratio of the vevs of the up and down Higgs doublets in the supersymmetric extension of the standard model. For $\Delta m^2 \simeq \Delta m_{\text{atm}}^2$, this condition can be satisfied for a very wide range of values of $\tan \beta$.

It is important to emphasize that this magnification occurs only if at the see-saw scale the neutrino masses are nearly degenerate. The idea can be extended to the three generation case as well [16].

11.6 Sum-rules and large mixings

Another possible way to understand large mixings may be to look for useful sum-rules between the quark, charged lepton and neutrino mass matrices. An example of this type has been discussed in recent literature [17, 18]:

$$M_\nu = c(M_d - M_\ell) \quad (11.48)$$

Now quark-lepton symmetry implies that for the second and third generation, the $M_{d,\ell}$ have the following generic form:

$$M_d = \begin{pmatrix} \epsilon_1 & \epsilon_2 \\ \epsilon_2 & m_b \end{pmatrix} \quad (11.49)$$

and

$$M_\ell = \begin{pmatrix} \epsilon'_1 & \epsilon'_2 \\ \epsilon'_2 & m_\tau \end{pmatrix} \quad (11.50)$$

where $\epsilon_i \ll m_{b,\tau}$ as is required by low energy observations. It is well known that in supersymmetric theories, when low energy quark and lepton masses are extrapolated to the GUT scale, one gets approximately that $m_b \simeq m_\tau$. One then sees from the above sum-rule for neutrino masses that all entries for the neutrino mass matrix are of the same order leading very naturally to the atmospheric mixing angle to be large. Thus one has a natural understanding of the large atmospheric neutrino mixing angle. No extra symmetries are assumed for this purpose.

For this model to be a viable one for three generations, one must show that the same sum-rule also explains the large solar angle while giving small θ_{13} . This has been shown in [18]. To see how this comes about, note that for the case of three generations, the quark and lepton mass matrices at the GUT scale have typically the form:

$$M_{d,\ell} \approx m_{b,\tau} \begin{pmatrix} \lambda^3 & \lambda^3 & \lambda^3 \\ \lambda^3 & \lambda^2 & \lambda^2 \\ \lambda^3 & \lambda^2 & 1 \end{pmatrix} \quad (11.51)$$

where $\lambda \sim 0.22$ and the matrix elements are supposed to give only the approximate order of magnitude. As we extrapolate the quark masses to the GUT scale, due to the fact that $m_b - m_\tau \approx m_\tau \lambda^2$ for some value of $\tan \beta$, the neutrino mass matrix $M_\nu = c(M_d - M_\ell)$ takes roughly the form

$$M_\nu = c(M_d - M_\ell) \approx m_0 \begin{pmatrix} \lambda^3 & \lambda^3 & \lambda^3 \\ \lambda^3 & \lambda^2 & \lambda^2 \\ \lambda^3 & \lambda^2 & \lambda^2 \end{pmatrix} \quad (11.52)$$

It is then easy to see from this mass matrix that both the θ_{12} (solar angle) and θ_{23} (the atmospheric angle) are now large. The detailed magnitudes of these angles of course depend on the details of the quark masses at the GUT scale [18].